FIT5196 Week 1 Quiz - Gemini 自动转录版

> 使用 Google Gemini Flash 自动转录
> 总题数: 164

按 Mock Quiz 重排（前 6 周）

> 这份 real quiz 实际覆盖的是前 6 周知识点。我按 `Mock Quiz.pdf` / `mock-quiz.md` 的专题分类重排，并补上了参考答案。
> 分类优先级：以 mock quiz 的 week 标签为准；因此部分 text pre-processing 题被归到 Week 4，metadata / cataloging / profiling 题被归到 Week 5。

前 6 周 Roadmap

Week 1

- 熟悉 Jupyter Notebook 的 code cell / markdown cell。
- 掌握 Markdown 基础语法：标题、列表、加粗、斜体。
- 建立最基础的 Python/notebook 使用能力。

Week 2

- 理解 Data Wrangling 的目标、挑战与完整流程。
- 能区分 discovery、collection、pre-processing、cleaning、transformation、enrichment、validation、storing。
- 会做基础 pandas 操作，并把具体操作映射回 wrangling task。

Week 3

- 系统学习 regular expressions。
- 掌握 character set、group、anchor、quantifier、lookaround。
- 能写出用于匹配、提取、验证的常见 pattern。

Week 4

- 做 EDA 与 pre-processing：识别数据类型、看分布、看关系。
- 掌握 numerical / categorical / text data 的常见处理。
- 练习 CSV / Excel / JSON 读取，以及 tokenisation、stop words、lemmatisation、vectorisation。

Week 5

- 学习 data discovery 与 data collection 的目标和流程。
- 区分 primary / secondary、structured / unstructured collection method。
- 掌握 metadata、cataloging、profiling、ethics、privacy / security / compliance。

Week 6

- 学习 data structuring。
- 理解 structured / unstructured / graph / time-series 等结构。
- 知道不同结构适合什么场景，以及为什么要先把数据组织好。

逐题分类索引

Week 1 (6 题)

- 范围：Jupyter Notebook、Markdown、Python basics
- Q1 How do you make text italic in Markdown?
- Q6 To create a bullet list in Markdown, which symbol can you use?
- Q20 To emphasise text with bold in Markdown, you can use:
- Q26 For numbered lists in Markdown, what character follows the number? (Select all that apply)
- Q116 To emphasise text with bold in Markdown, you can use:
- Q155 To emphasise text with bold in Markdown, you can use:

Week 2 (48 题)

- 范围：Data Wrangling Process & Tasks + Week 2 pandas applied session
- Q8 How do you select rows 10 to 20 from a DataFrame df? (Select all that apply)
- Q9 Which of the following are considered tasks in the Data Wrangling process? (Select all that apply)
- Q12 Data simplification can involve: (Select all that apply)
- Q14 The goals of data enrichment include: (Select all that apply)
- Q17 Data cleaning may involve: (Select all that apply)
- Q22 How can you rename the column 'A' to 'B' in a DataFrame df?
- Q33 How do you check for missing values in a DataFrame df? (Select all that apply)
- Q34 Scalability in data wrangling is essential for:
- Q36 Data standardization helps ensure data is:
- Q39 What process involves scaling data to a small, specified range?
- Q41 In data quality management, what is crucial for maintaining high-quality data?
- Q43 Data enrichment may involve:
- Q45 Machine Learning enhances data quality management through:
- Q46 Data Integration aims to:
- Q47 Which type of data anomalies refers to a single data point deviating significantly from the rest?
- Q48 Which of the following is NOT a goal of data validation?
- Q52 In data wrangling, data validation ensures: (Select all that apply)
- Q53 How can you rename the column 'A' to 'B' in a DataFrame df?

- Q59 Which of the following are common goals of data wrangling? (Select all that apply)
- Q67 What role does machine learning play in data quality management?
- Q70 Data cleaning may involve: (Select all that apply)
- Q78 Which method would you use to sort a DataFrame df by the column 'Name'?
- Q79 Which is NOT a feature of Big Data?
- Q80 What does simplifying access to data aim to achieve?
- Q82 What is the primary purpose of Data Wrangling?
- Q83 What does data enrichment not involve?
- Q86 The process of identifying and correcting inconsistencies in data is called:
- Q87 Facilitating data integration helps with:
- Q92 Which of the following is NOT a goal of data validation?
- Q93 A major challenge in data wrangling is:
- Q94 Machine Learning enhances data quality management through:
- Q101 Data Integration and Consolidation require: (Select all that apply)
- Q103 How do you select a single column named 'Age' from a DataFrame df? (Select al that apply)
- Q104 Which of the following are common goals of data wrangling? (Select all that apply)
- Q108 In data wrangling, data validation ensures: (Select all that apply)
- Q109 To drop rows with any missing values in the data frame `df`, you use:
- Q114 Data cleaning may involve: (Select all that apply)
- Q118 Which function is used to concatenate two DataFrames vertically?
- Q119 Effective data validation checks are: (Select all that apply)
- Q121 Data integration challenges include:
- Q125 Which type of data anomalies refers to a single data point deviating significantly from the rest?
- Q126 Data duplication is an example of:
- Q129 Which is not a challenge in data wrangling?
- Q130 Which is NOT a challenge in maintaining data quality?
- Q138 The purpose of data aggregation is to:
- Q143 What is a 'Point Anomaly'?
- Q152 Which method would you use to sort a DataFrame df by the column 'Name'?
- Q160 What process involves scaling data to a small, specified range?

Week 3 (39 题)

- 范围：Regular Expressions
- Q3 For matching IPv4 addresses, which regex is accurate and avoids matching numbers beyond the valid range of 0-255?
- Q7 What does a regular expression describe?

- Q11 In the regex (?:foo)(foo), what is the difference between (?:foo) and (foo)?
- Q13 What does the regex `(?<=\b20)\d{2}\b` match in "Class of 2024"?
- Q15 Which of the following is NOT a character set used in regular expressions?
- Q18 How would you match "http" at the beginning of a string?
- Q19 What will the regex (?<=\b20)[0-9]{2}\b match in "Graduating in 2025"?
- Q21 Which character escapes special characters in regular expressions?
- Q29 Which of the following is a valid way to match either 'color' or 'colour'?
- Q30 Given the string 'hello123', what does the regex (?<=\D)(?=\d) locate?
- Q54 Which regex finds all instances of "not" that are not immediately followed by "good"?
- Q56 What does the regex `^[\w.+-]+@([\w-]+\.)+[a-zA-Z]{2,7}$` not allow in the email username part?
- Q58 What does the regex (?<=\b20)\d{2}\b match in "Class of 2024"?
- Q61 Which metacharacter matches the beginning of a line?
- Q62 How would you match "cat" or "bat"?
- Q64 In regex, what does (?:abc) represent?
- Q69 What does the + quantifier do?
- Q73 What does the {3,6} quantifier mean?
- Q74 In regular expressions, what does \A match?
- Q76 What does the regex `^(?!.*\bfoo\b).*$` ensure about the matched strings?
- Q95 Which regex matches a string if it starts with "http://" and does not have "www" immediately after?
- Q97 What does the regex \A(?:[0-9a-fA-F]{2}:){5}[0-9a-fA-F]{2}\z match?
- Q99 What does the regex \d{2,}(?=\D*$) match in "Item 23 is 42 years old"?
- Q100 What is a regular expression?
- Q102 What does the regex \b\d{1,3}(,\d{3})*\b match?
- Q106 Given the string "hello123", what does the regex (?<=\D)(?=\d) locate?
- Q112 What does the regex (?<!cat)dog ensure?
- Q113 What would the regex ^(\w+)(\s+\1)+$ match?
- Q115 Which character escapes special characters in regular expressions?
- Q117 What is the function of the ? quantifier?
- Q141 What does the * quantifier do in regular expressions?
- Q142 What does (?<=>)\d+(?=<\/?) match in a string like "

  123

  "?
- Q144 Which regex captures text between double quotes, including escaped quotes within the text?
- Q146 What would the regex `\b(?!\d+\b)\w+\b` exclude in its matches?
- Q147 Which regex finds all instances of "not" that are not immediately followed by "good"?

- Q148 Which of the following is NOT a character set used in regular expressions?
- Q150 Which quantifier would match exactly 3 occurrences of the letter 'a'?
- Q156 How would you match any line that does not contain the exact phrase "regex"?
- Q157 What will (\b[a-zA-Z]{3}\b)\s+\1 match?

Week 4 (41 题)

- 范围：EDA & Pre-processing + Week 4 parsing + mock quiz text pre-processing block
- Q2 What challenges are presented by textual content in EDA for character and text data? (Select all that apply)
- Q4 What factors influence the choice of data pre-processing techniques? (Select all that apply)
- Q5 What measures of central tendency are commonly used in EDA for numerical data? (Select all that apply)
- Q16 Which method is used to get a summary of a DataFrame's structure and column types?
- Q23 For what reasons might case normalisation not always be needed in text data pre-processing? (Select all that apply)
- Q24 For what reasons might case normalisation not always be needed in text data pre-processing? (Select all that apply)
- Q25 The process that simplifies text by converting it to its base form is called:
- Q27 Why is understanding the structure of language (syntax) important in EDA for character and text data? (Select all that apply)
- Q28 Which Pandas function is used to read a CSV file?
- Q32 Which parameter in `pd.read_csv()` specifies the delimiter?
- Q35 What does 'Character Data' consist of?
- Q37 Which technique is used to handle categorical data by creating a binary column for each category?
- Q40 In the context of text data pre-processing, what is the purpose of case normalisation?
- Q42 Exploratory Data Analysis (EDA) is used for:
- Q55 Why is it important to remove stop words in text data pre-processing? (Select all that apply)
- Q57 To read an Excel file into a DataFrame, which function is used?
- Q63 How do you set a DataFrame's column 'ID' as its index?
- Q66 What does the pd.read_json() function do?
- Q71 Which techniques are essential for handling numerical data in EDA? (Select all that apply)
- Q75 Which parameter in read_csv() is used to specify which rows should be used as header?
- Q77 What does the `skiprows=3` parameter in `pd.read_csv()` do?
- Q81 What is the middle value after all values are ordered in a data set called?

- Q88 Which of the following is crucial for preparing text for NLP tasks?
- Q90 Which measure of central tendency is the arithmetic average of a set of values?
- Q96 For what reasons might case normalisation not always be needed in text data pre-processing? (Select all that apply)
- Q105 Which technique is used to handle categorical data by creating a binary column for each category?
- Q110 Which data types are primarily analysed during EDA? (Select all that apply)
- Q111 Which steps are involved in Lemmatisation for text data pre-processing? (Select all that apply)
- Q120 Which method is used to get a summary of a DataFrame's structure and column types?
- Q122 Which parameter in pd.read_csv() specifies the delimiter?
- Q123 What is the aim of using Descriptive Statistics in EDA for Numerical Data?
- Q124 What does continuous data represent?
- Q131 Exploratory Data Analysis (EDA) is used for:
- Q135 Which data visualisation method is primarily used for numerical data to understand its spread and central tendency?
- Q145 Which of the following are benefits of correctly identifying data types during EDA? (Select that all apply)
- Q149 What challenges are presented by textual content in EDA for character and text data? (Select all that apply)
- Q151 What are the steps involved in Count Vectorisation? (Select all that apply)
- Q153 Which of the following are goals of Exploratory Data Analysis (EDA)? (Select all that apply)
- Q158 For what reasons might case normalisation not always be needed in text data pre-processing? (Select all that apply)
- Q163 What is the aim of using Descriptive Statistics in EDA for Numerical Data?
- Q164 What is the significance of identifying data types during EDA?

Week 5 (25 题)

- 范围：Data Discovery & Collection
- Q10 What aspects does Data Cataloging and Metadata Management cover? (Select all that apply)
- Q31 Qualitative Data is:
- Q44 Data Cataloging aids in:
- Q49 Which is NOT a method of collecting structured data?
- Q50 Which of the following is NOT a part of the Data Discovery Process?
- Q68 Factors to consider during the data collection phase include: (Select all that apply)
- Q72 Which are considered primary sources of data? (Select all that apply)
- Q84 Which is NOT a method of collecting structured data?

- Q85 Which phase comes immediately after data discovery?
- Q89 Data Security measures are essential for:
- Q91 Data Cataloging aids in:
- Q98 What roles does data play in decision-making? (Select all that apply)
- Q107 Ethical considerations in Data Collection include: (Select all that apply)
- Q127 Qualitative Data is:
- Q128 The main advantage of using existing data is:
- Q132 Metadata in Data Cataloging includes:
- Q133 What role does data play in modern business decision-making?
- Q134 Metadata Collection is critical for understanding:
- Q136 Ensuring data privacy and security is crucial due to:
- Q137 Which of the following is NOT a part of the Data Discovery Process?
- Q139 Text Mining and Natural Language Processing (NLP) are methods for collecting:
- Q154 What are the roles of Data in modern business decision-making? (Select all that apply)
- Q159 Data Security measures are essential for:
- Q161 Data Privacy measures are intended to:
- Q162 Metadata Collection is critical for understanding:

Week 6 (5 题)

- 范围：Data Structuring
- Q38 Graph data models are ideal for:
- Q51 What does Structured Data refer to?
- Q60 Data Profiling and Assessment involve understanding which types of data structure? (Select all that apply)
- Q65 Which data structures are essential for Data Discovery and Collection? (Select all that apply)
- Q140 Time-series data is indexed in:

歧义题提醒

- Q8：参考按 `a,d`；如果老师按 label-based `loc` 来解释，也可能接受 `b`。
- Q26：参考按 `c`；部分 Markdown 方言也接受 `1)`。
- Q29：参考按 `c`；`d` 从 regex 语义上也能匹配。
- Q65：题干把 data structures / sources 混在一起，参考先按 `a,b,c,d,e` 记。
- Q95：参考按 `c`；`b` 也能正确匹配前缀条件。
- Q126：duplicates 既可视为 anomaly，也常被归为 data quality issue；这里按 `b`。
- Q144：选项本身不够严谨，参考先按 `c`。

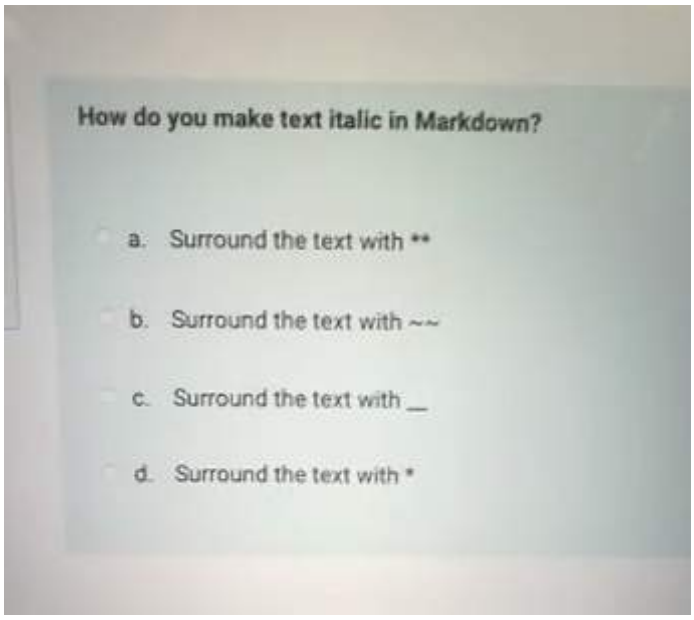
原始题库（按索引分类合并）

> 下方题目顺序已与上面的 Week 1-6 索引完全一致，题号保留原编号。

Week 1 (6 题)

- 范围：Jupyter Notebook、Markdown、Python basics

Question 1



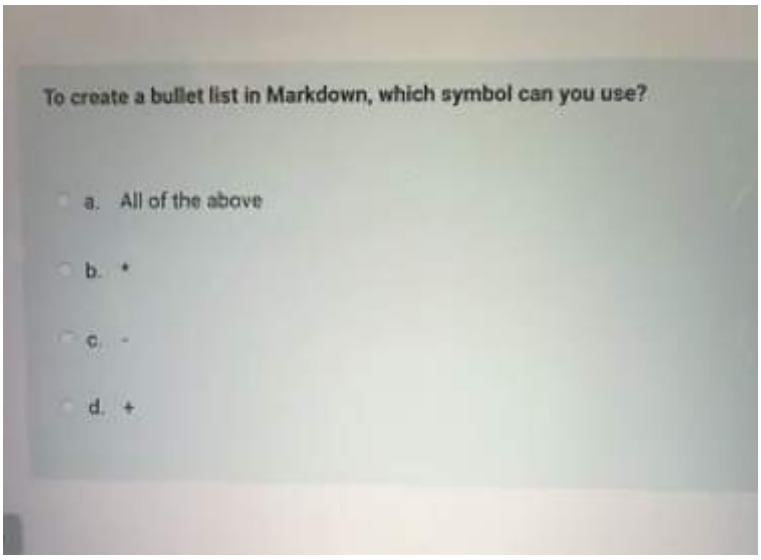
How do you make text italic in Markdown? (Select all that apply)

a. Surround the text with **
b. Surround the text with ~~
c. Surround the text with _
d. Surround the text with *

参考答案： `c, d`

解析： 斜体在 Markdown 里通常写成 `*text*` 或 `_text_`，不是 `**`（粗体）也不是 `~~`（删除线）。

Question 6



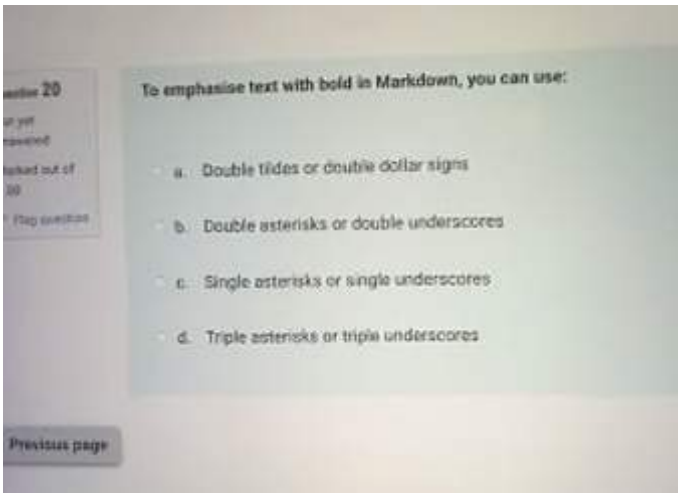
To create a bullet list in Markdown, which symbol can you use?

a. All of the above
b. *
c. -
d. +

参考答案： `a`

解析： 正确项是 `a`: All of the above。它最符合这道题在课程里的定义。

Question 20



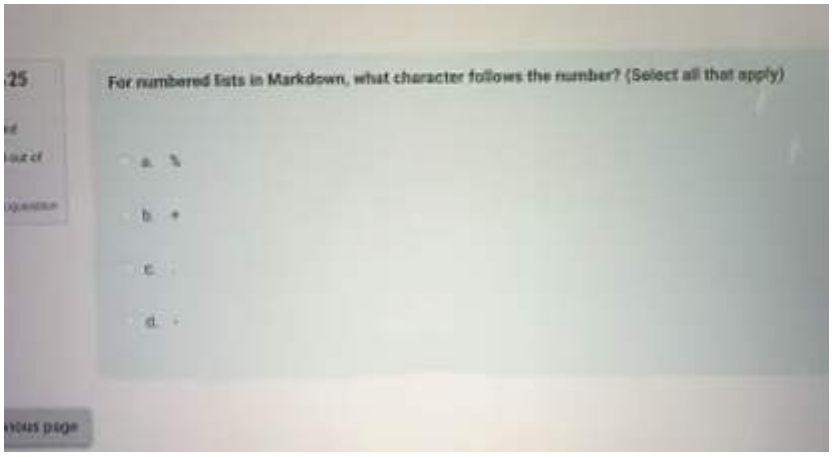
To emphasise text with bold in Markdown, you can use:

a. Double tildes or double dollar signs
b. Double asterisks or double underscores
c. Single asterisks or single underscores
d. Triple asterisks or triple underscores

参考答案： `b`

解析： Markdown 粗体用双星号或双下划线，即 `**text**` 或 `__text__`。

Question 26



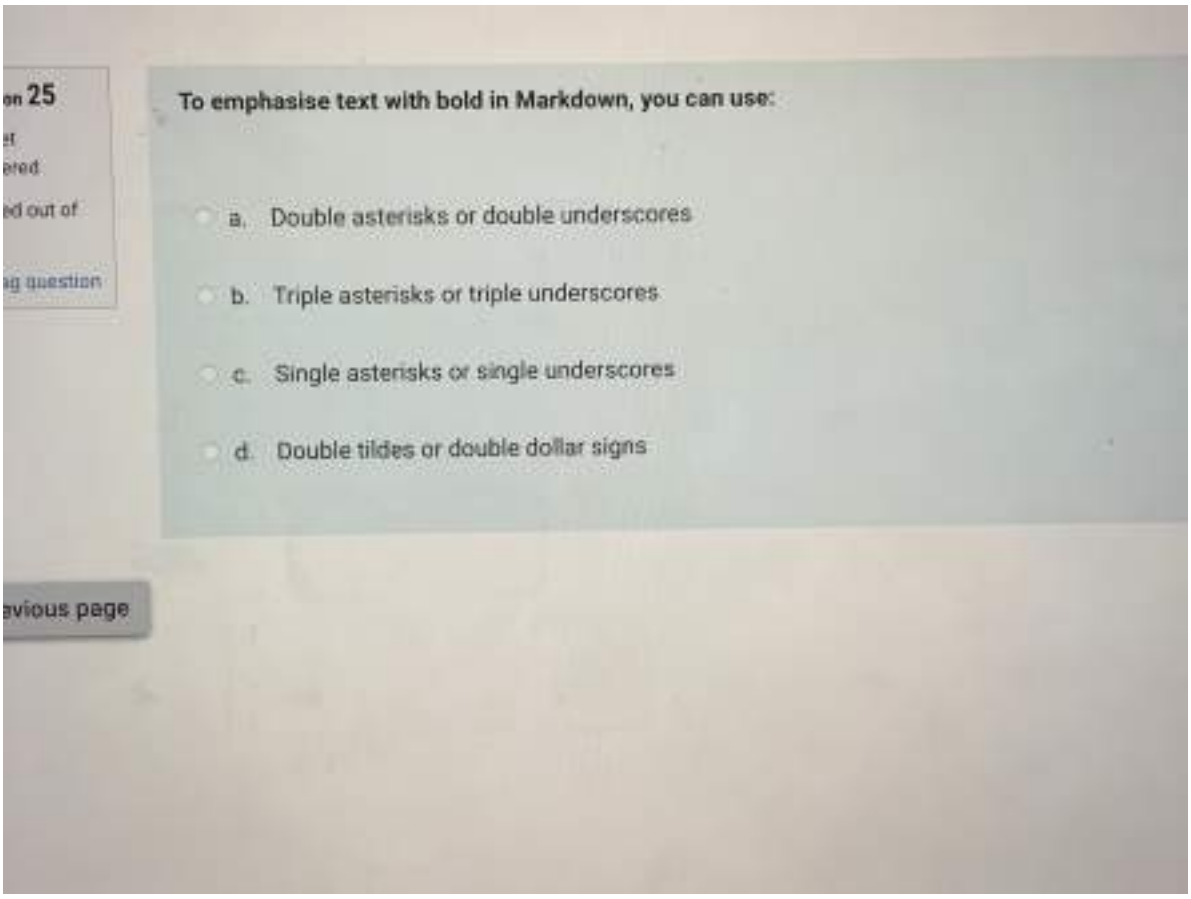
For numbered lists in Markdown, what character follows the number? (Select all that apply)

a.)
b. *
c. .
d. -

参考答案： `c`

解析： 课程默认的 Markdown 有序列表写法是 `1.`，所以答案按 `.` 记。

Question 116



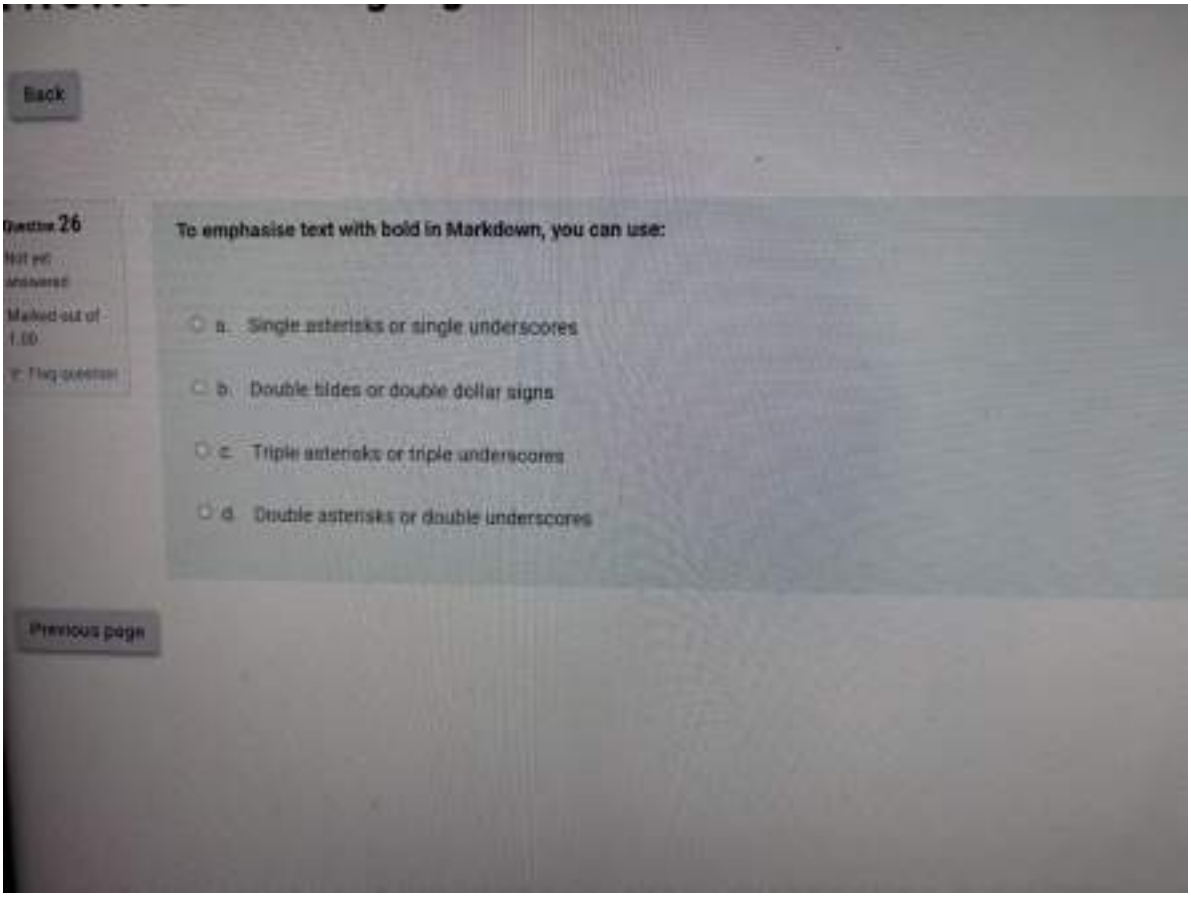
To emphasise text with bold in Markdown, you can use:

a. Double asterisks or double underscores
b. Triple asterisks or triple underscores
c. Single asterisks or single underscores
d. Double tildes or double dollar signs

参考答案： `a`

解析： 和 Q20 同义，粗体是双星号或双下划线。

Question 155



To emphasise text with bold in Markdown, you can use:

a. Single asterisks or single underscores
b. Double tildes or double dollar signs
c. Triple asterisks or triple underscores
d. Double asterisks or double underscores

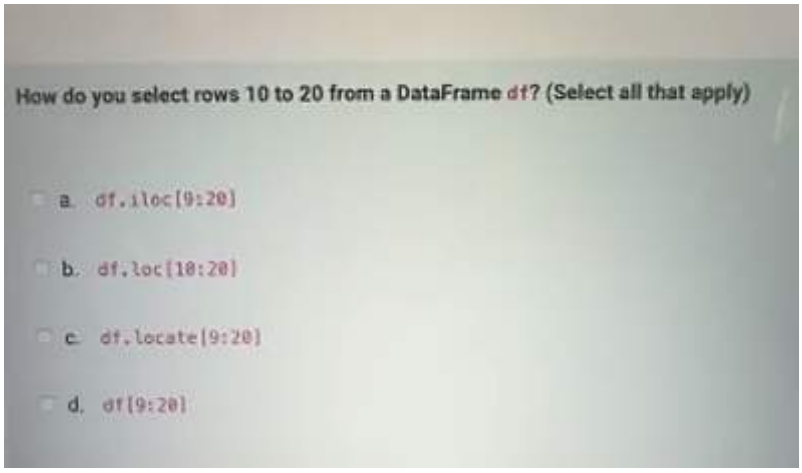
参考答案： `d`

解析： 和 Q20/Q116 同义，粗体仍是双星号或双下划线。

Week 2 (48 题)

- 范围：Data Wrangling Process & Tasks + Week 2 pandas applied session

Question 8



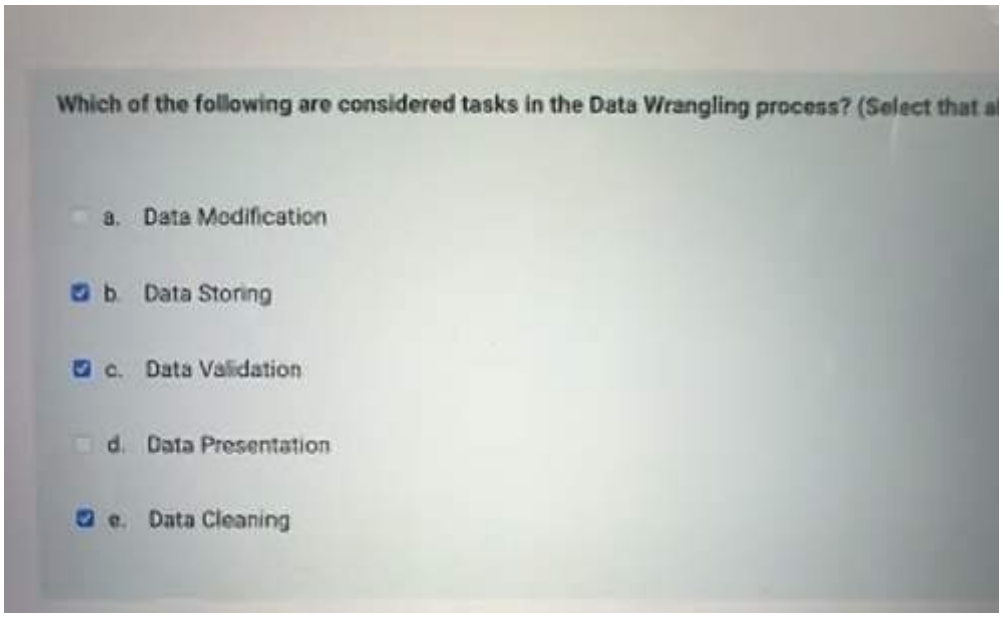
How do you select rows 10 to 20 from a DataFrame df? (Select all that apply)

a. df.iloc[9:20]
b. df.loc[10:20]
c. df.locate[9:20]
d. df[9:20]

参考答案： `a, d`

解析： `iloc[9:20]` 取的是第 10 到第 20 行（位置索引，右端不含）；`df[9:20]` 也会按切片取对应行。`loc[10:20]` 依赖标签索引，课内通常不作为标准答案。

Question 9



Which of the following are considered tasks in the Data Wrangling process? (Select all that apply)

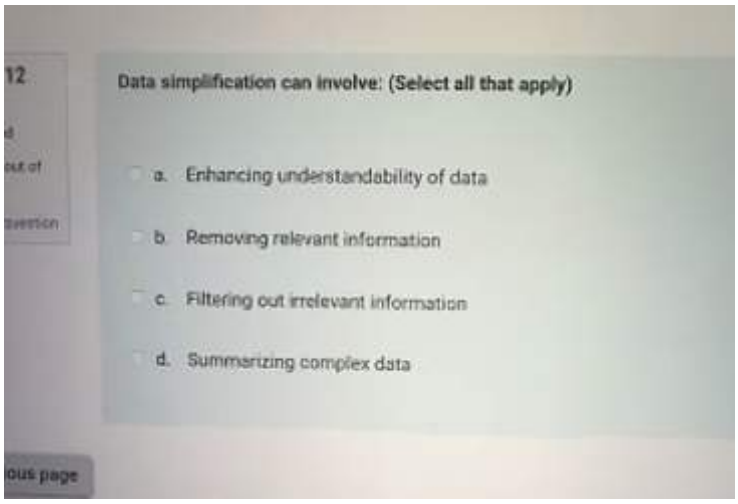
a. Data Modification
b. Data Storing
c. Data Validation
d. Data Presentation

e. Data Cleaning

参考答案： `b, c, e`

解析： 正确项是 `b`: Data Storing, `c`: Data Validation, `e`: Data Cleaning。这些选项共同对应课程里的正确定义。

Question 12



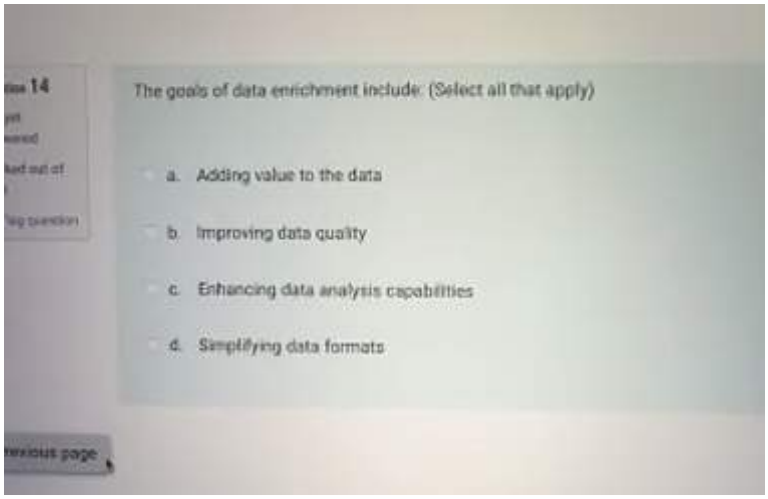
Data simplification can involve: (Select all that apply)

a. Enhancing understandability of data
b. Removing relevant information
c. Filtering out irrelevant information
d. Summarizing complex data

参考答案： `a, c, d`

解析： 正确项是 `a`: Enhancing understandability of data, `c`: Filtering out irrelevant information, `d`: Summarizing complex data。这些选项共同对应课程里的正确定义。

Question 14



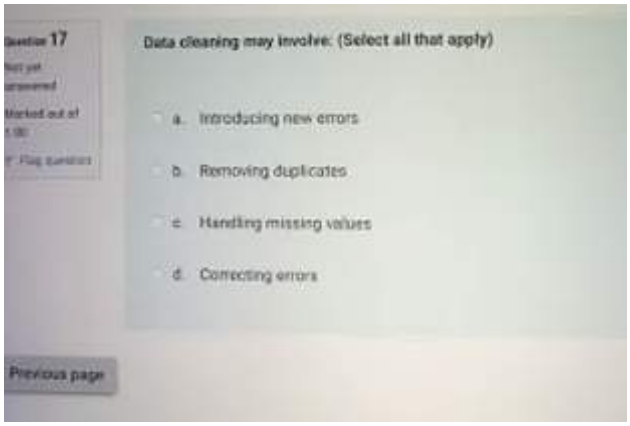
The goals of data enrichment include: (Select all that apply)

a. Adding value to the data
b. Improving data quality
c. Enhancing data analysis capabilities
d. Simplifying data formats

参考答案： `a, b, c`

解析： data enrichment 的目标是加上下文、提升价值、改善可分析性；“simplifying data formats” 更像 pre-processing / transformation。

Question 17



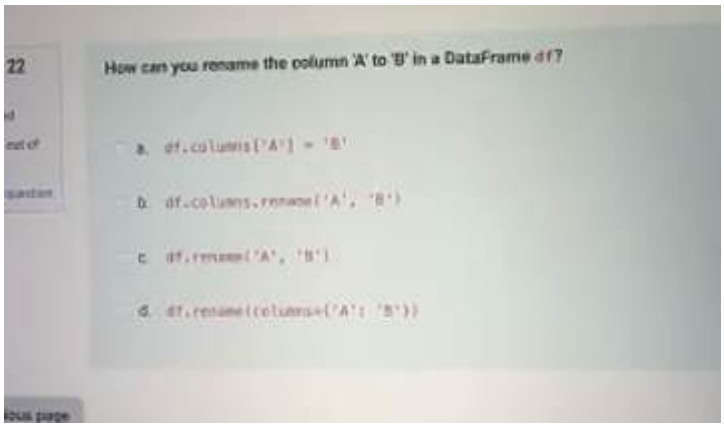
Data cleaning may involve: (Select all that apply)

a. Introducing new errors
b. Removing duplicates
c. Handling missing values
d. Correcting errors

参考答案： `b, c, d`

解析： 正确项是 `b`: Removing duplicates, `c`: Handling missing values, `d`: Correcting errors。这些选项共同对应课程里的正确定义。

Question 22



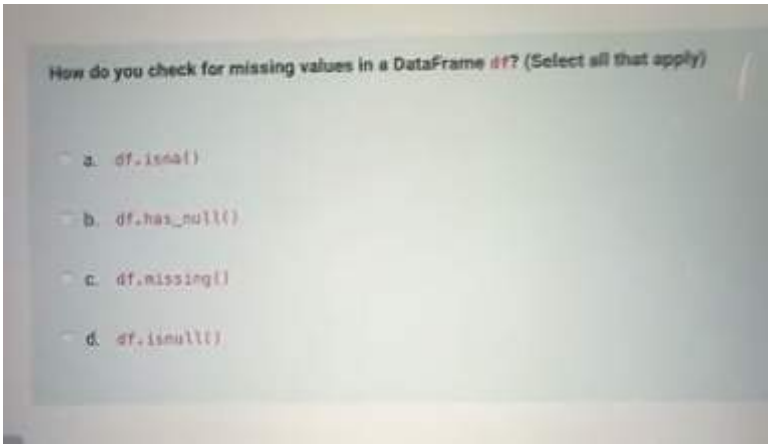
How can you rename the column 'A' to 'B' in a DataFrame df?

a. df.columns['A'] = 'B'
b. df.columns.rename('A', 'B')
c. df.rename('A', 'B')
d. df.rename(columns={'A': 'B'})

参考答案： `d`

解析： Pandas 改列名标准写法是 `df.rename(columns={"A": "B"})`。

Question 33



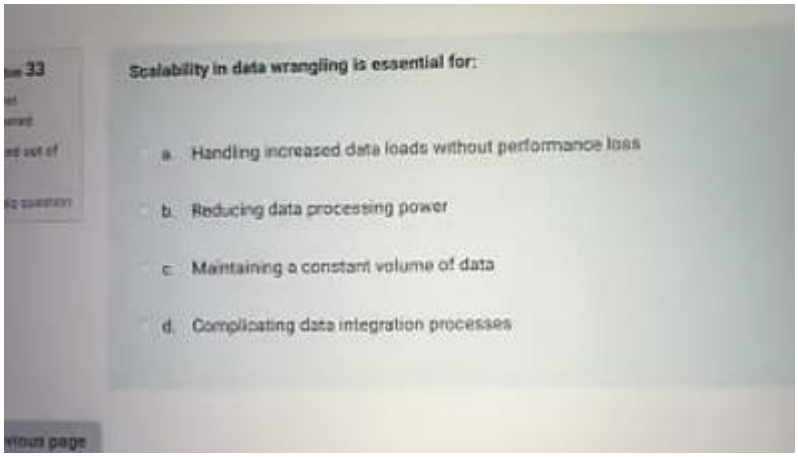
How do you check for missing values in a DataFrame df? (Select all that apply)

a. df.isna()
b. df.has_null()
c. df.missing()
d. df.isnull()

参考答案： `a, d`

解析： Pandas 检查缺失值常用 `isna()` 和 `isnull()`。

Question 34



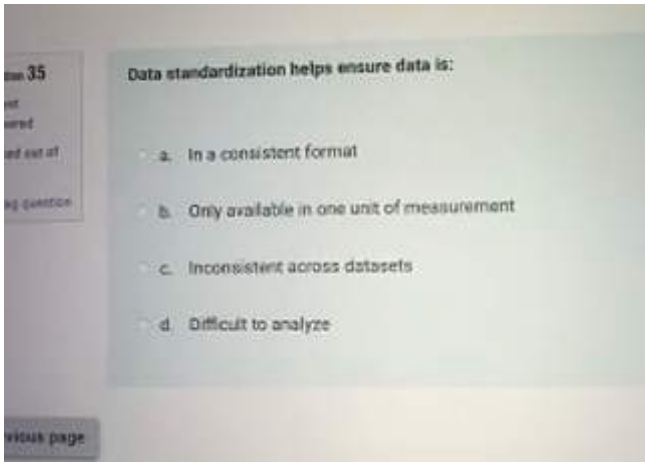
Scalability in data wrangling is essential for:

a. Handling increased data loads without performance loss
b. Reducing data processing power
c. Maintaining a constant volume of data
d. Complicating data integration processes

参考答案： a

解析： scalability 指数据量上来后仍能处理，不是把数据量固定不变。

Question 36



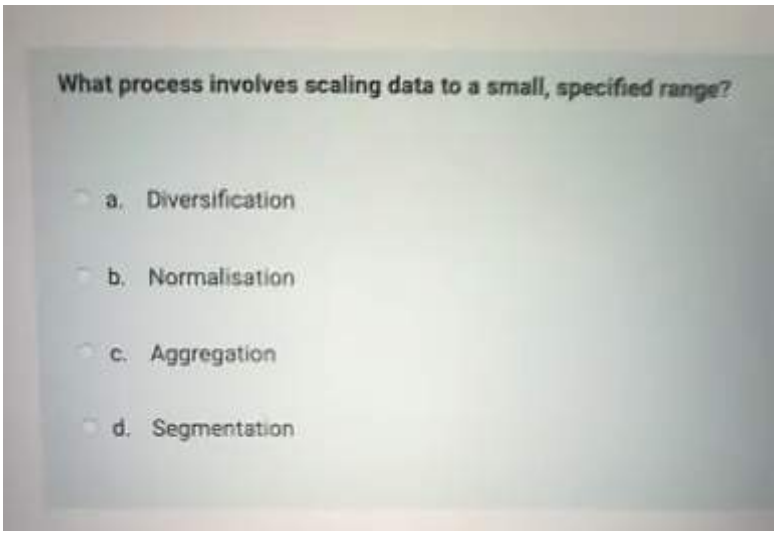
Data standardization helps ensure data is:

a. In a consistent format
b. Only available in one unit of measurement
c. Inconsistent across datasets
d. Difficult to analyze

参考答案： a

解析： 这里的 standardization 更偏“统一标准/格式”，目标是让数据表示保持一致。

Question 39



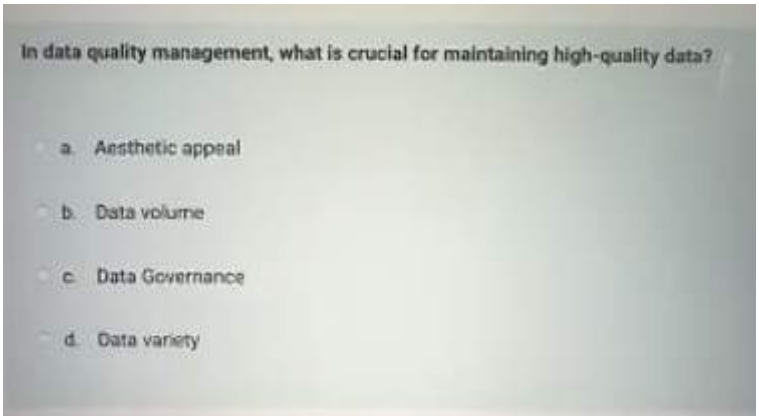
What process involves scaling data to a small, specified range?

a. Diversification
b. Normalisation
c. Aggregation
d. Segmentation

参考答案： `b`

解析： 把数值缩到一个小范围（如 `[0,1]`）是 normalisation。

Question 41



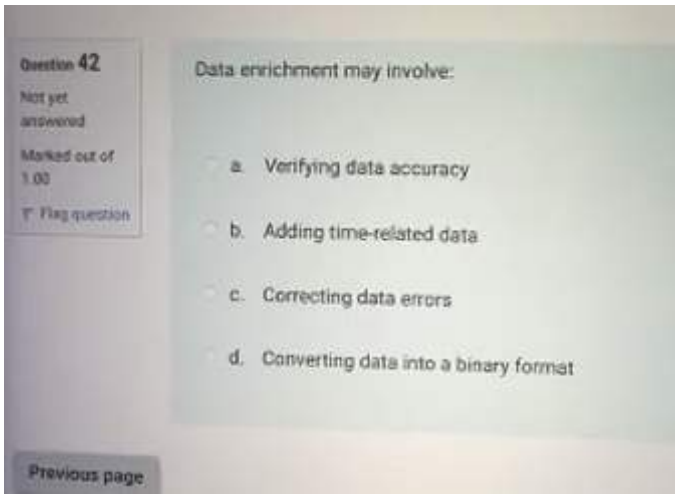
In data quality management, what is crucial for maintaining high-quality data?

a. Aesthetic appeal
b. Data volume
c. Data Governance
d. Data variety

参考答案： `c`

解析： 持续高质量数据离不开 data governance，因为要靠规则、责任和流程来维护。

Question 43



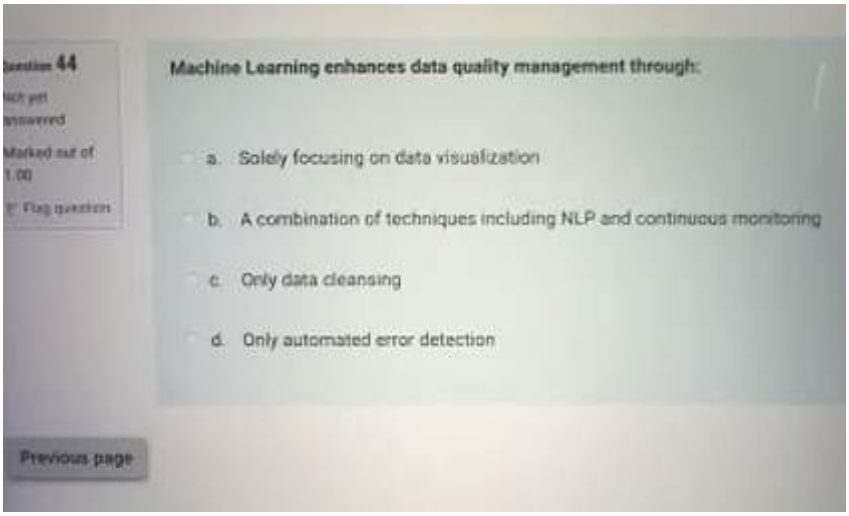
Data enrichment may involve:

a. Verifying data accuracy
b. Adding time-related data
c. Correcting data errors
d. Converting data into a binary format

参考答案： b

解析： enrichment 常见做法是补充时间、地理、人口等额外上下文。

Question 45



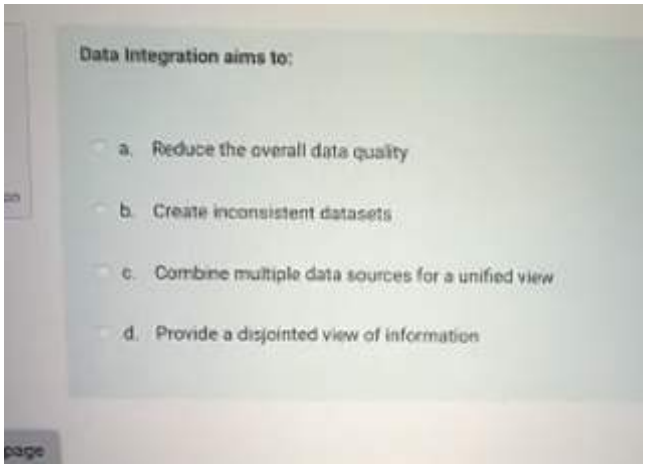
Machine Learning enhances data quality management through:

a. Solely focusing on data visualization
b. A combination of techniques including NLP and continuous monitoring
c. Only data cleansing
d. Only automated error detection

参考答案： b

解析： 机器学习在 data quality management 里常用于自动识别问题、持续监控和 NLP 等组合方法。

Question 46



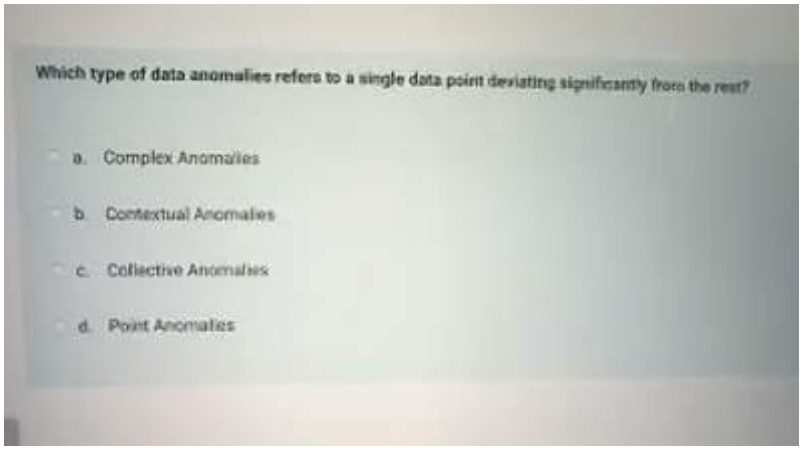
Data Integration aims to:

a. Reduce the overall data quality
b. Create inconsistent datasets
c. Combine multiple data sources for a unified view
d. Provide a disjointed view of information

参考答案： `c`

解析： data integration 的目标是把多个来源统一成一个一致视图。

Question 47



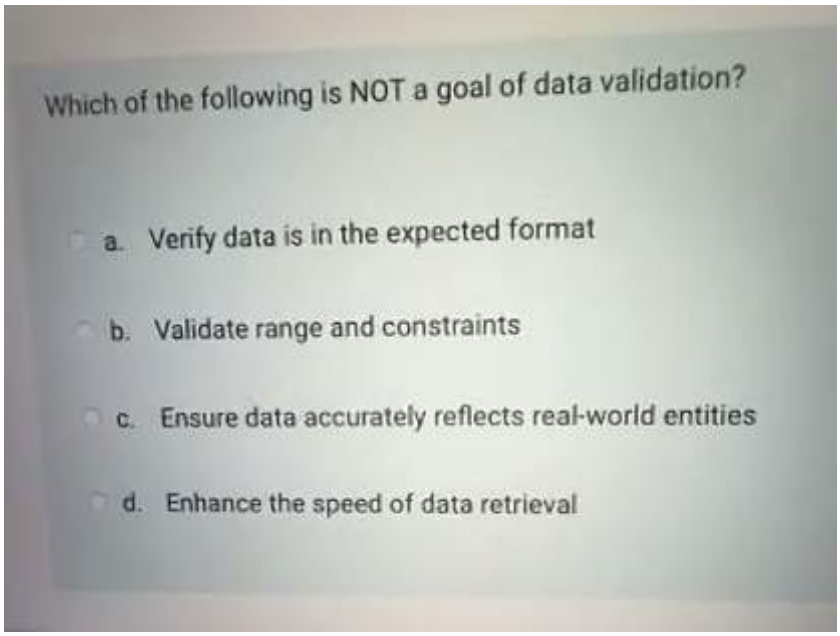
Which type of data anomalies refers to a single data point deviating significantly from the rest?

a. Complex Anomalies
b. Contextual Anomalies
c. Collective Anomalies
d. Point Anomalies

参考答案： `d`

解析： 单个点明显偏离整体，就是 point anomaly。

Question 48



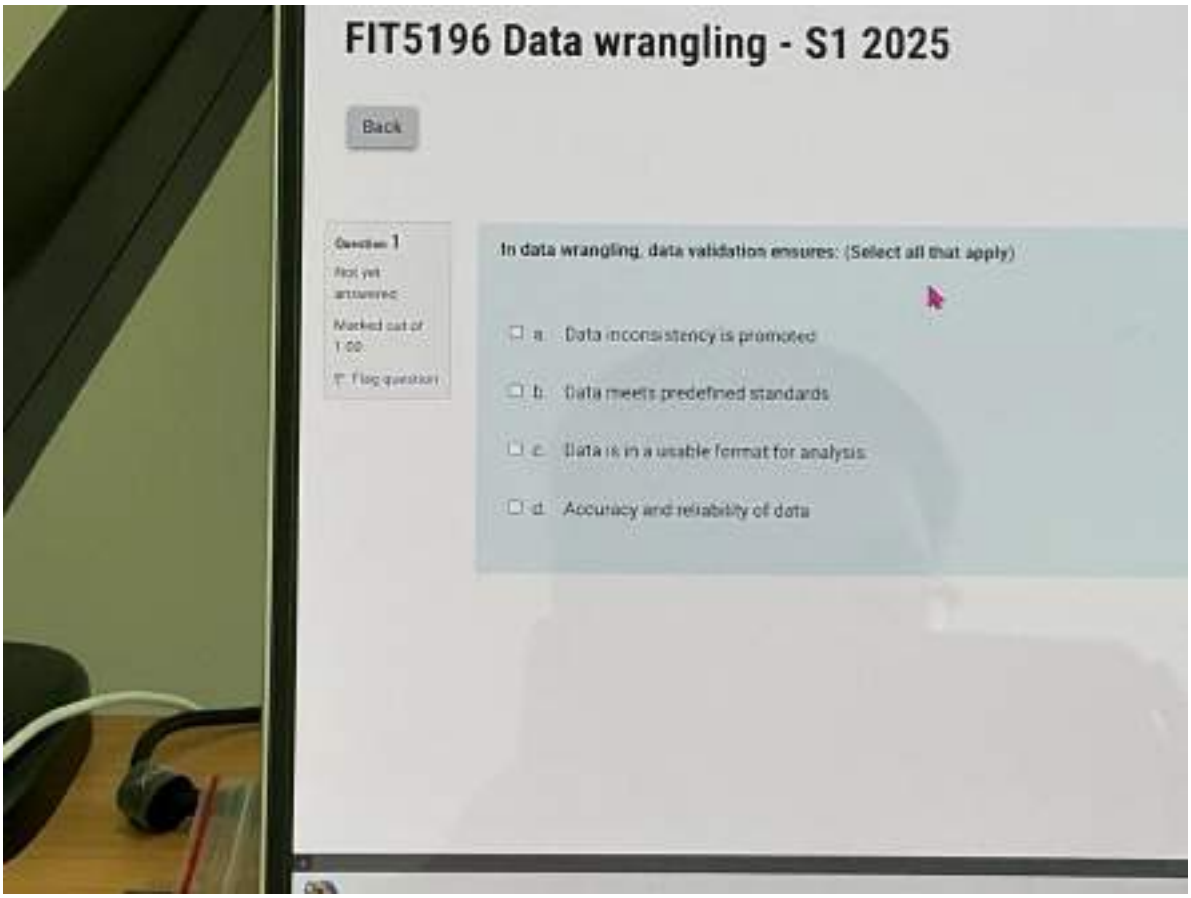
Which of the following is NOT a goal of data validation?

a. Verify data is in the expected format
b. Validate range and constraints
c. Ensure data accurately reflects real-world entities
d. Enhance the speed of data retrieval

参考答案： `d`

解析： validation 是检查格式、范围、逻辑是否合理，不是提升读取速度。

Question 52



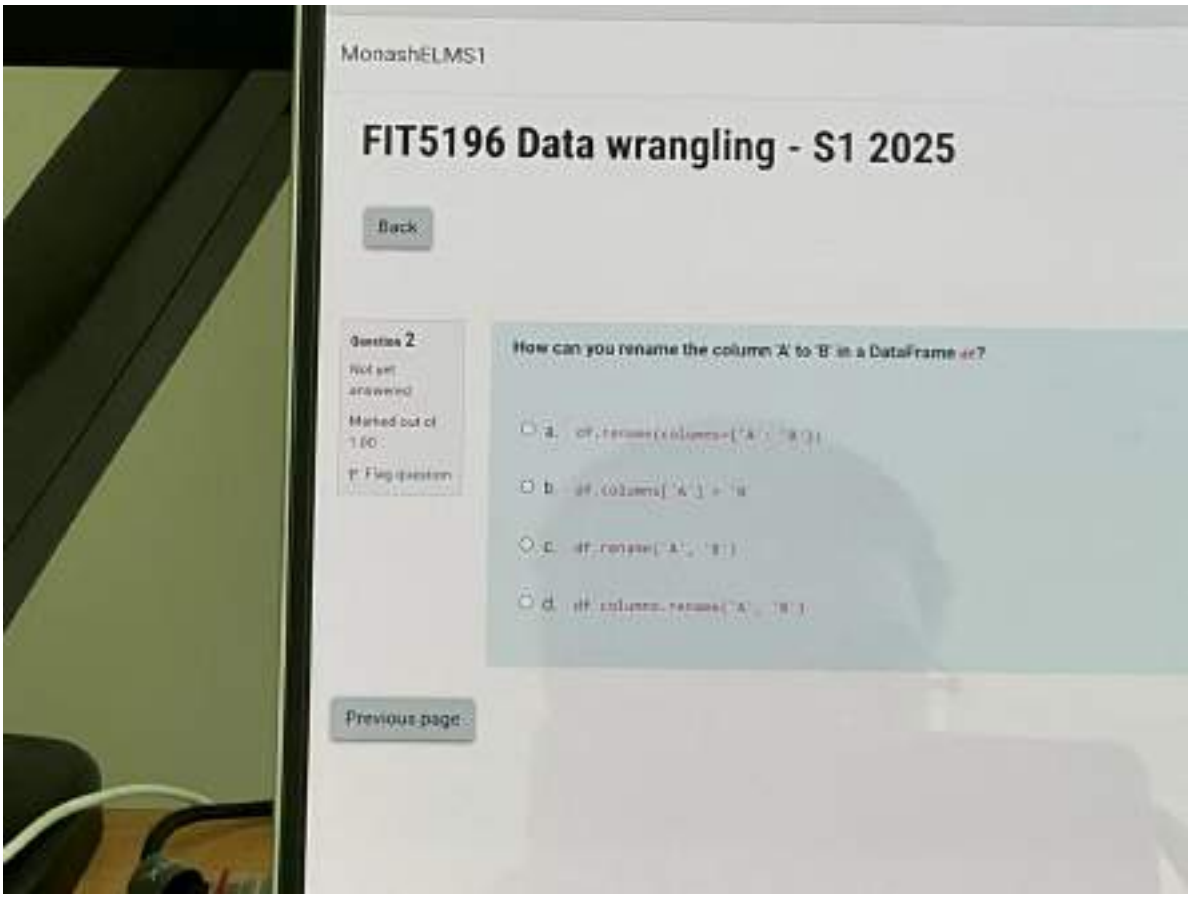
In data wrangling, data validation ensures: (Select all that apply)

a. Data inconsistency is promoted
b. Data meets predefined standards
c. Data is in a usable format for analysis
d. Accuracy and reliability of data

参考答案： `b, c, d`

解析： validation 要保证数据符合预设标准、可靠且可用于后续分析。

Question 53



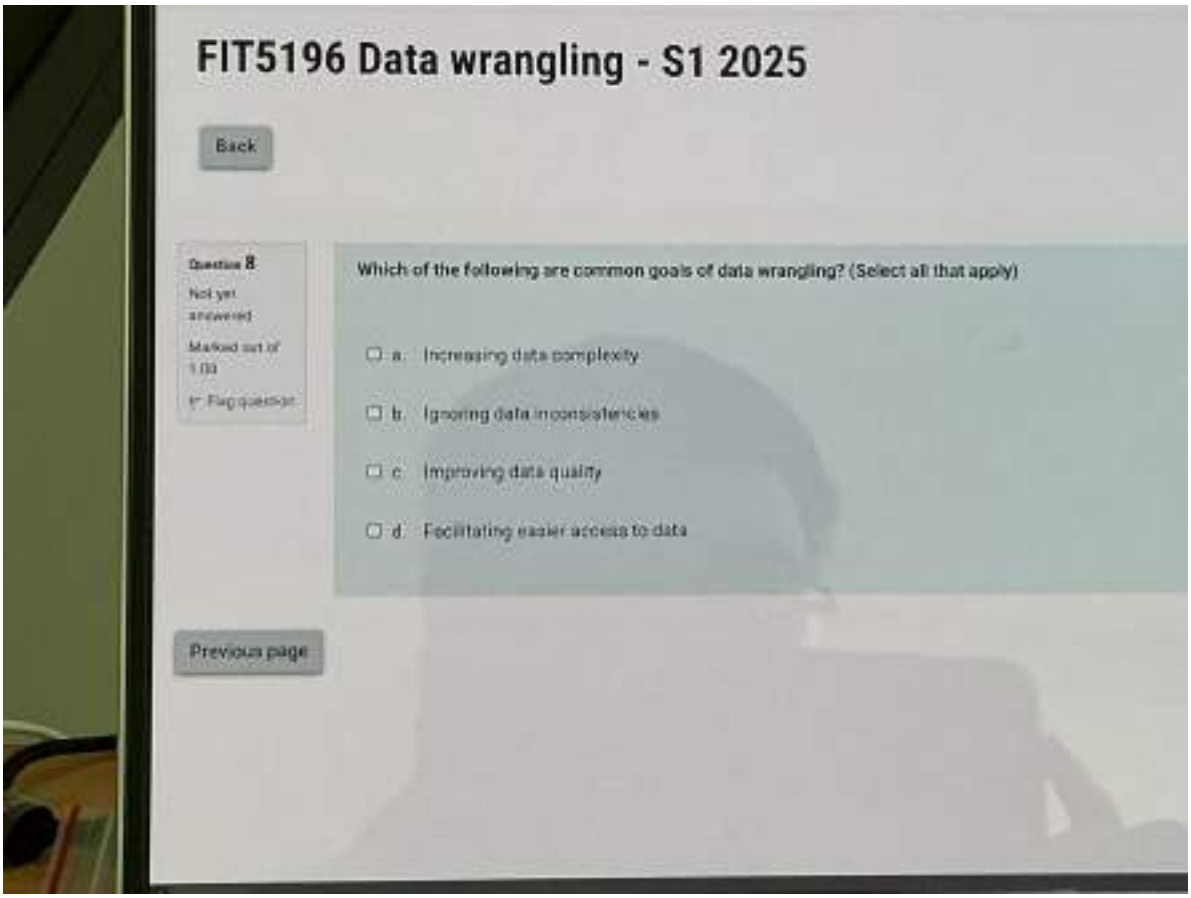
How can you rename the column 'A' to 'B' in a DataFrame df?

a. `df.rename(columns={'A': 'B'})`
b. `df.columns['A'] = 'B'`
c. `df.rename('A', 'B')`
d. `df.columns.rename('A', 'B')`

参考答案： `a`

解析： 这题和 Q22 同义，标准写法仍是 `df.rename(columns={"A": "B"})`。

Question 59



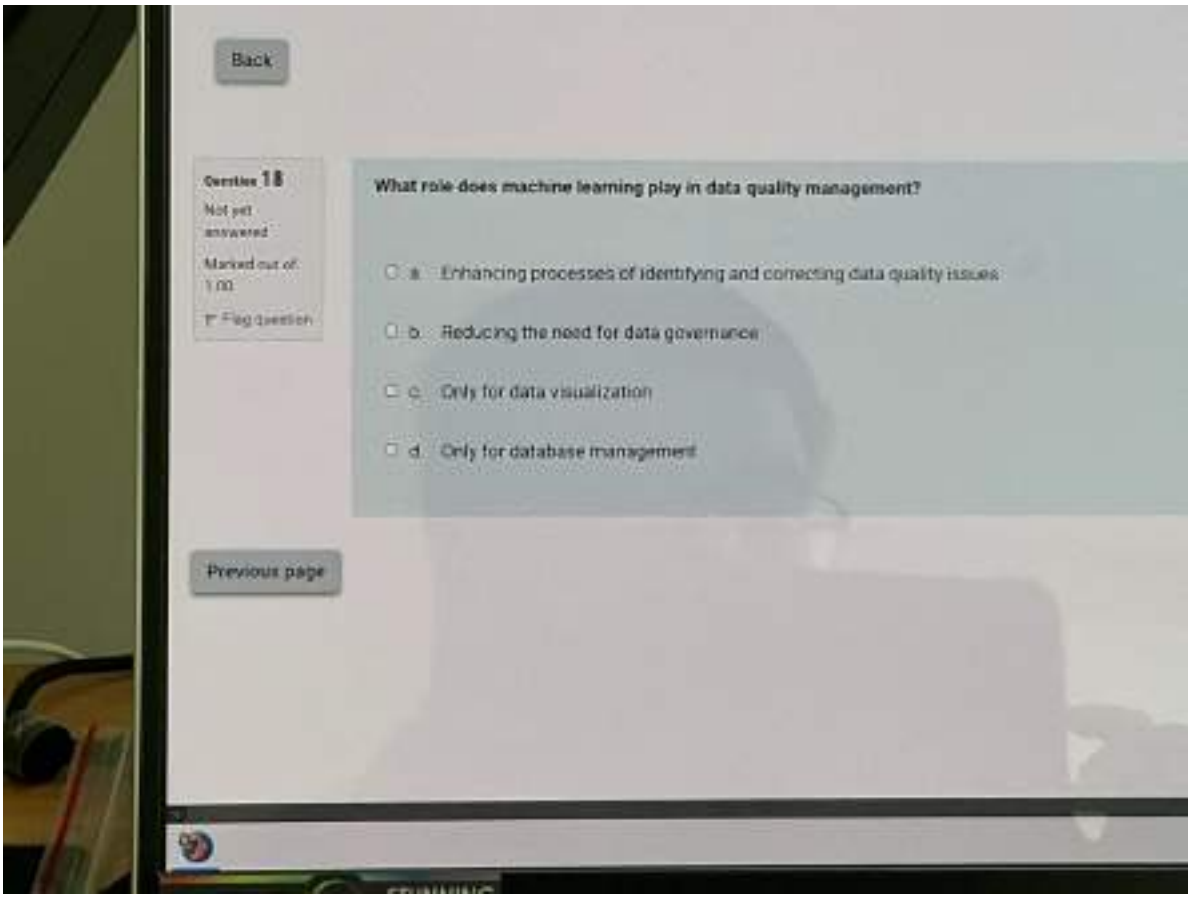
Which of the following are common goals of data wrangling? (Select all that apply)

a. Increasing data complexity
b. Ignoring data inconsistencies
c. Improving data quality
d. Facilitating easier access to data

参考答案： `c, d`

解析： wrangling 的常见目标是提高质量、提升可访问性，不是增加复杂度。

Question 67



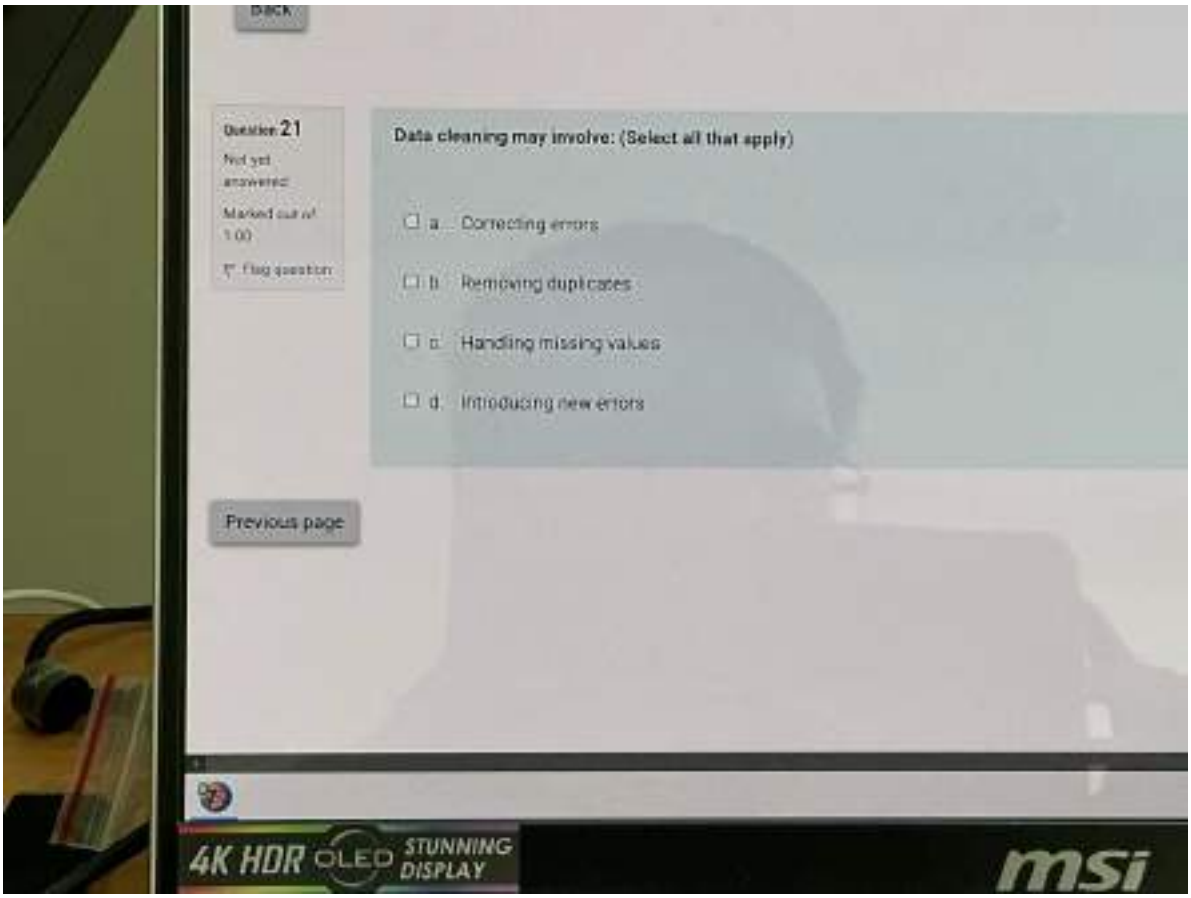
What role does machine learning play in data quality management?

a. Enhancing processes of identifying and correcting data quality issues
b. Reducing the need for data governance
c. Only for data visualization
d. Only for database management

参考答案： a

解析： 机器学习的作用是帮助发现、识别并修正数据质量问题。

Question 70



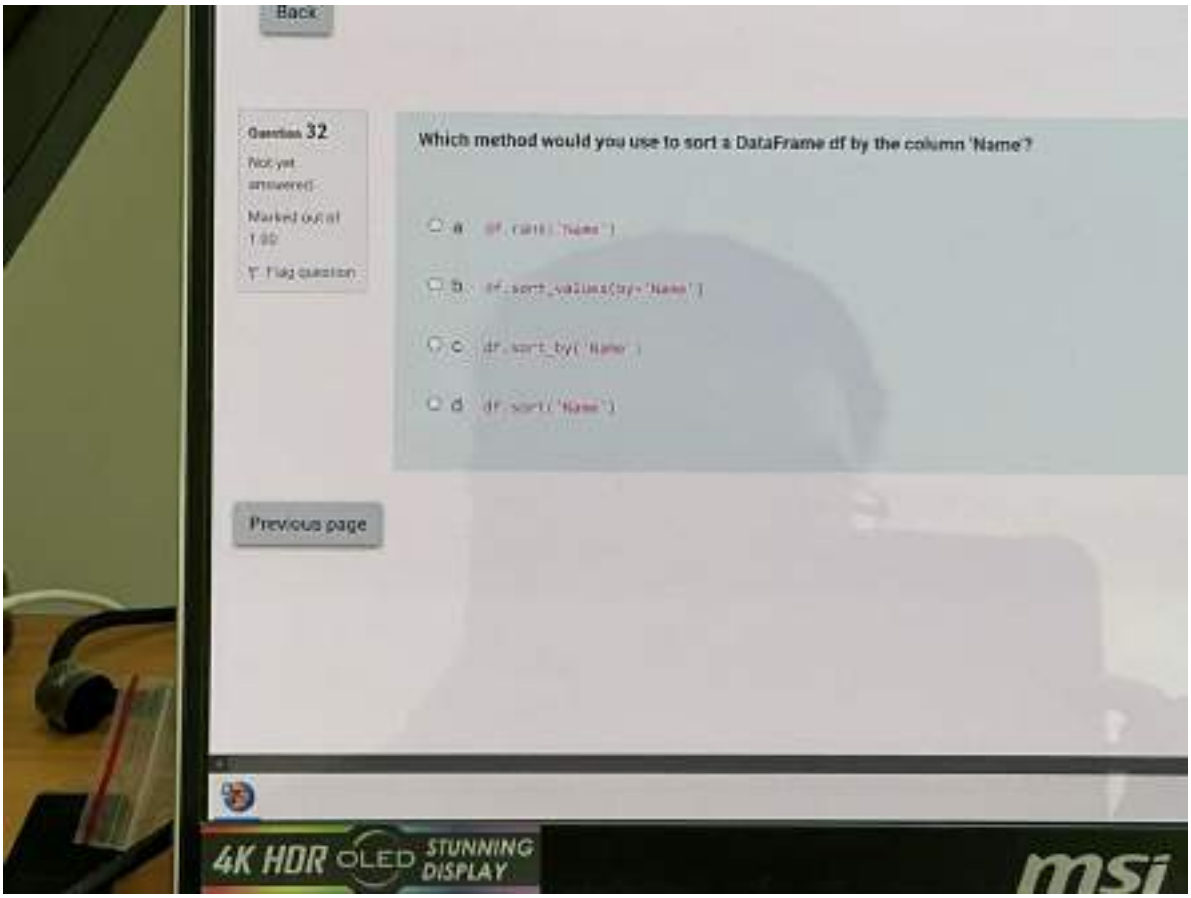
Data cleaning may involve: (Select all that apply)

a. Correcting errors
b. Removing duplicates
c. Handling missing values
d. Introducing new errors

参考答案： `a, b, c`

解析： cleaning 包括纠错、去重、处理缺失，不会“引入新错误”。

Question 78



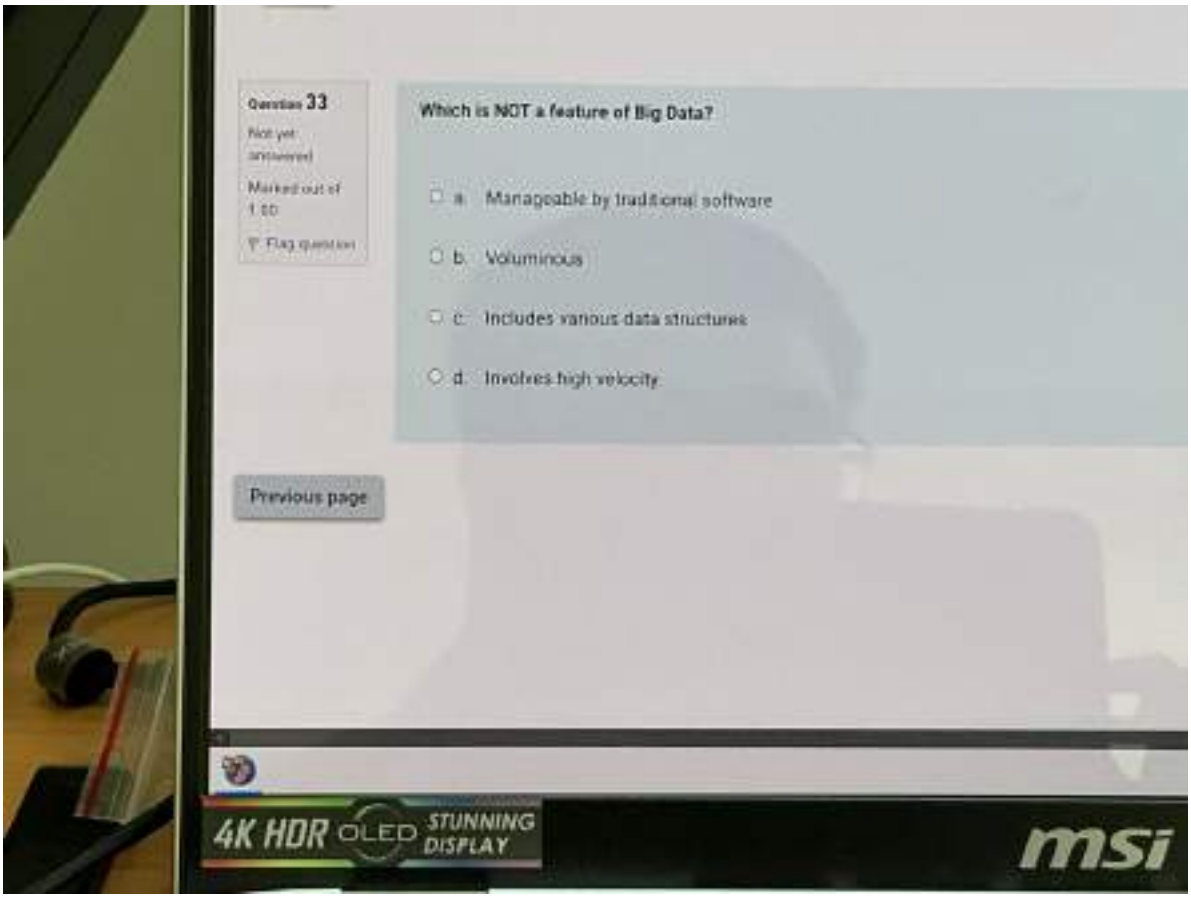
Which method would you use to sort a DataFrame df by the column 'Name'?

a. `df.rank('Name')`
b. `df.sort_values(by='Name')`
c. `df.sort_by('Name')`
d. `df.sort('Name')`

参考答案： `b`

解析： 按列排序标准写法是 `df.sort_values(by="Name")`。

Question 79



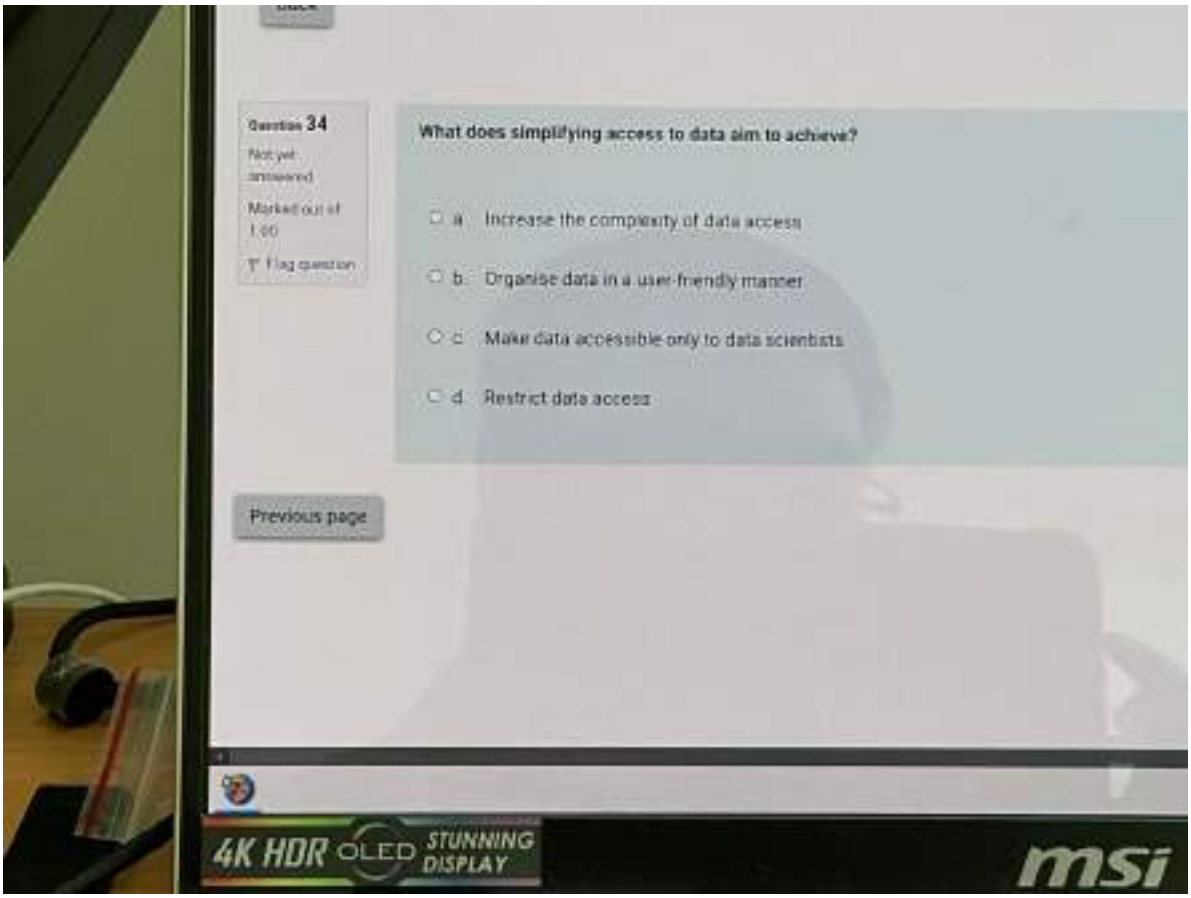
Which is NOT a feature of Big Data?

a. Manageable by traditional software
b. Voluminous
c. Includes various data structures
d. Involves high velocity

参考答案： a

解析： Big Data 的典型特征是 volume、variety、velocity，通常不能被传统软件轻松处理。

Question 80



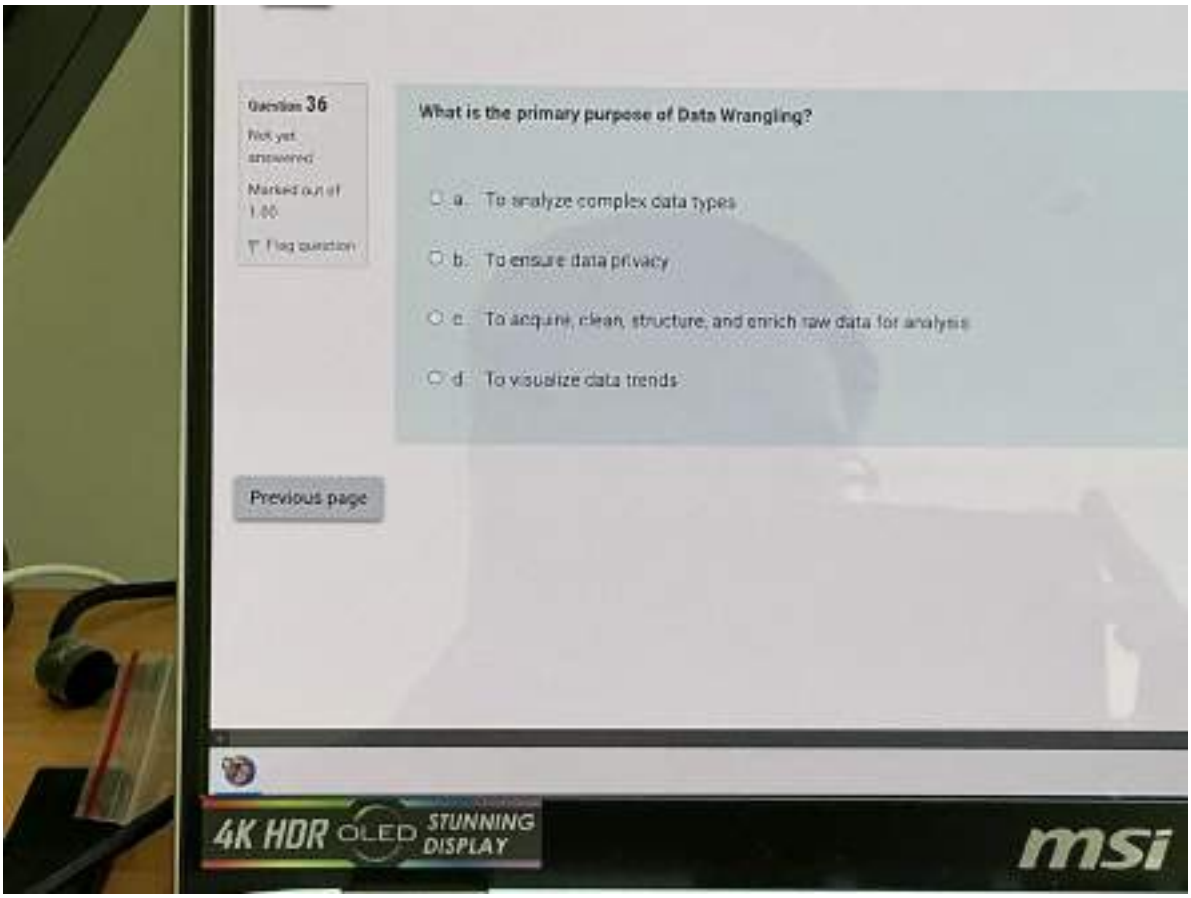
What does simplifying access to data aim to achieve?

a. Increase the complexity of data access
b. Organise data in a user-friendly manner
c. Make data accessible only to data scientists
d. Restrict data access

参考答案： `b`

解析： simplifying access 的目标是让数据更容易找、更容易用。

Question 82



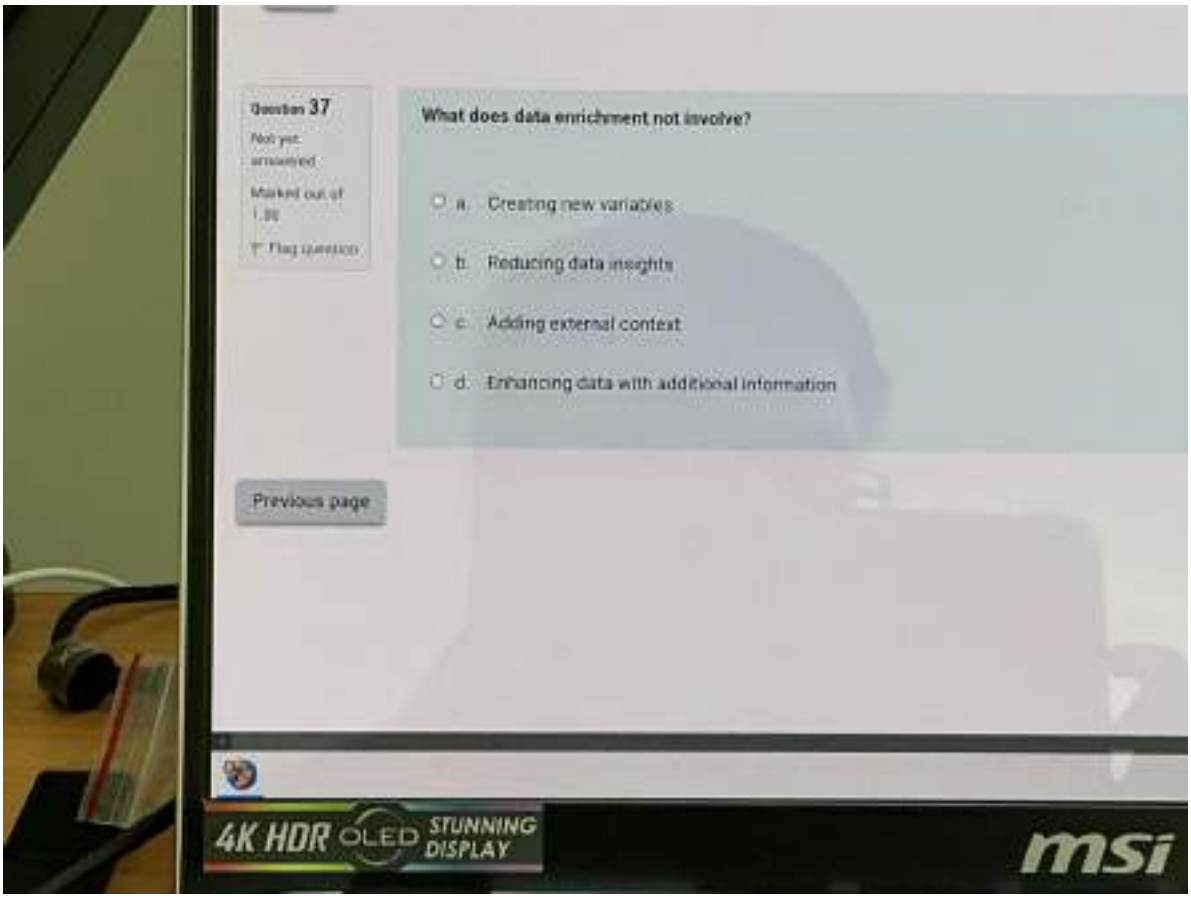
What is the primary purpose of Data Wrangling?

a. To analyze complex data types
b. To ensure data privacy
c. To acquire, clean, structure, and enrich raw data for analysis
d. To visualize data trends

参考答案： `c`

解析： data wrangling 的核心就是把原始数据整理成可分析数据。

Question 83



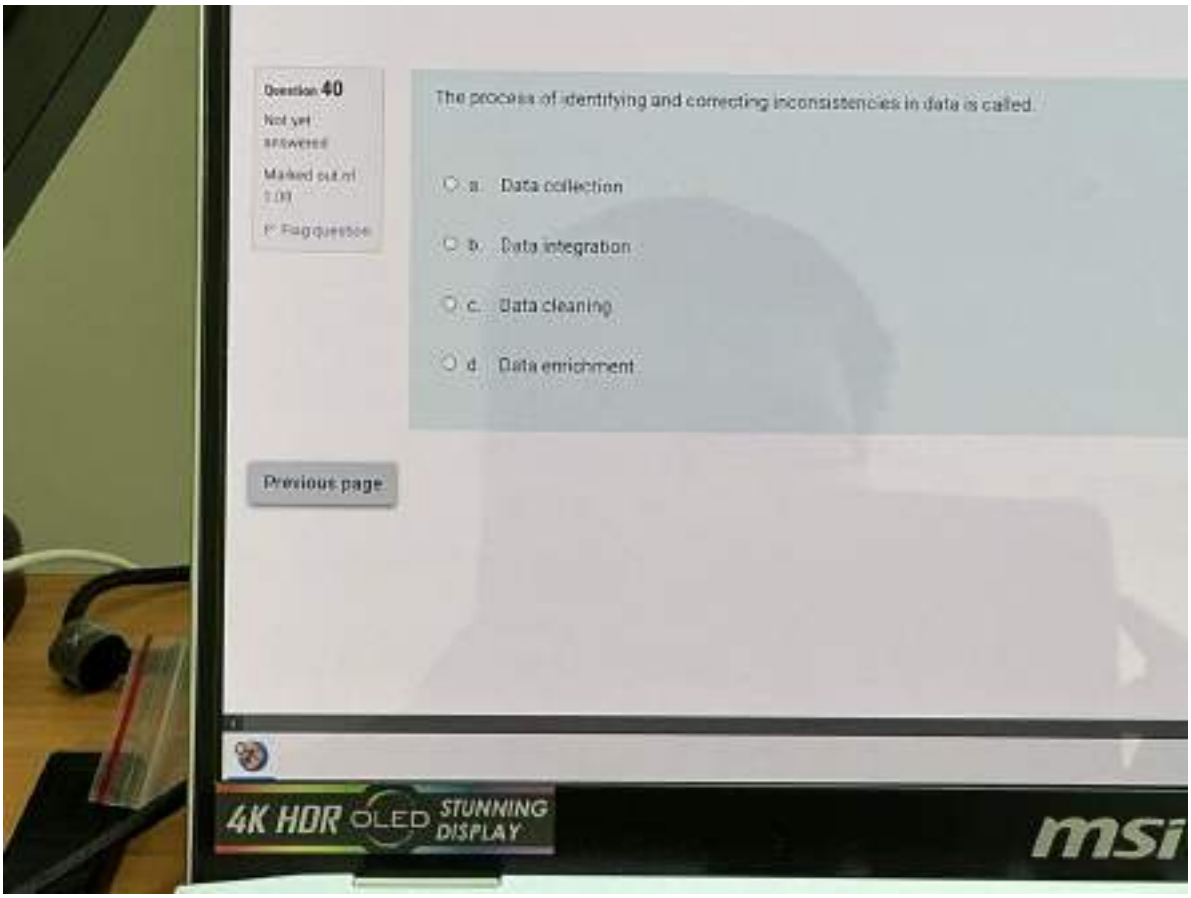
What does data enrichment not involve?

a. Creating new variables
b. Reducing data insights
c. Adding external context
d. Enhancing data with additional information

参考答案： `b`

解析： enrichment 是增加信息与洞察，不是减少 insights。

Question 86



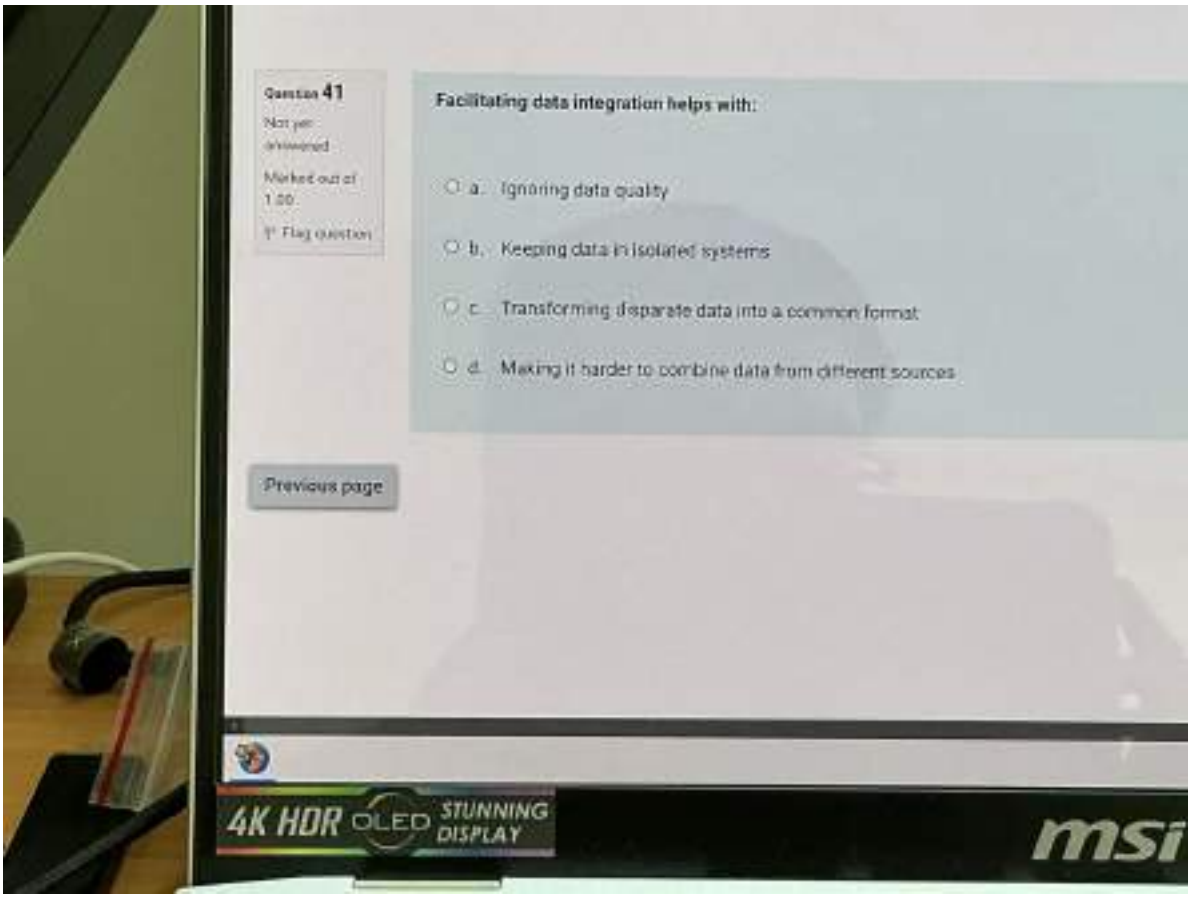
The process of identifying and correcting inconsistencies in data is called:

a. Data collection
b. Data integration
c. Data cleaning
d. Data enrichment

参考答案： `c`

解析： 识别并修正不一致，属于 data cleaning。

Question 87



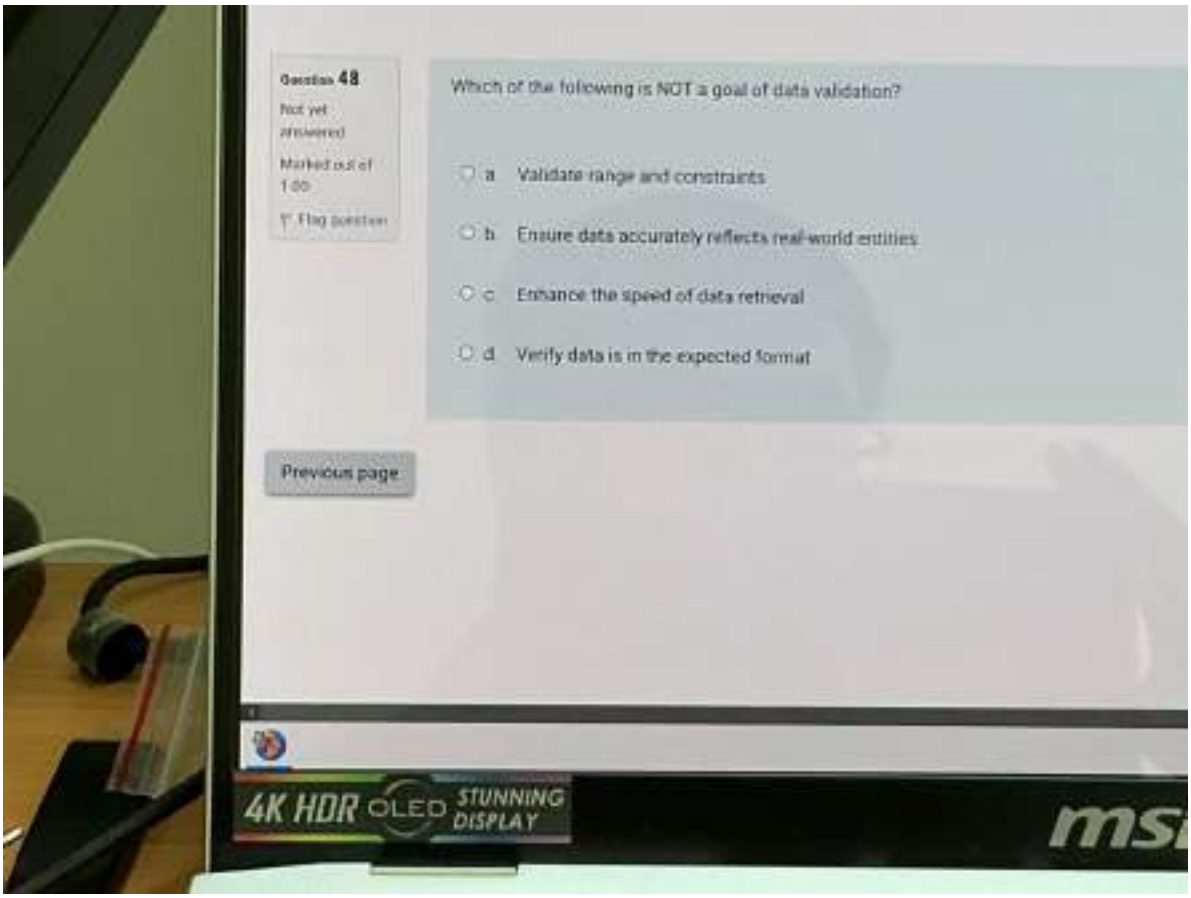
Facilitating data integration helps with:

a. Ignoring data quality
b. Keeping data in isolated systems
c. Transforming disparate data into a common format
d. Making it harder to combine data from different sources

参考答案： `c`

解析： integration 的一个关键作用就是把异构数据转成共同格式。

Question 92



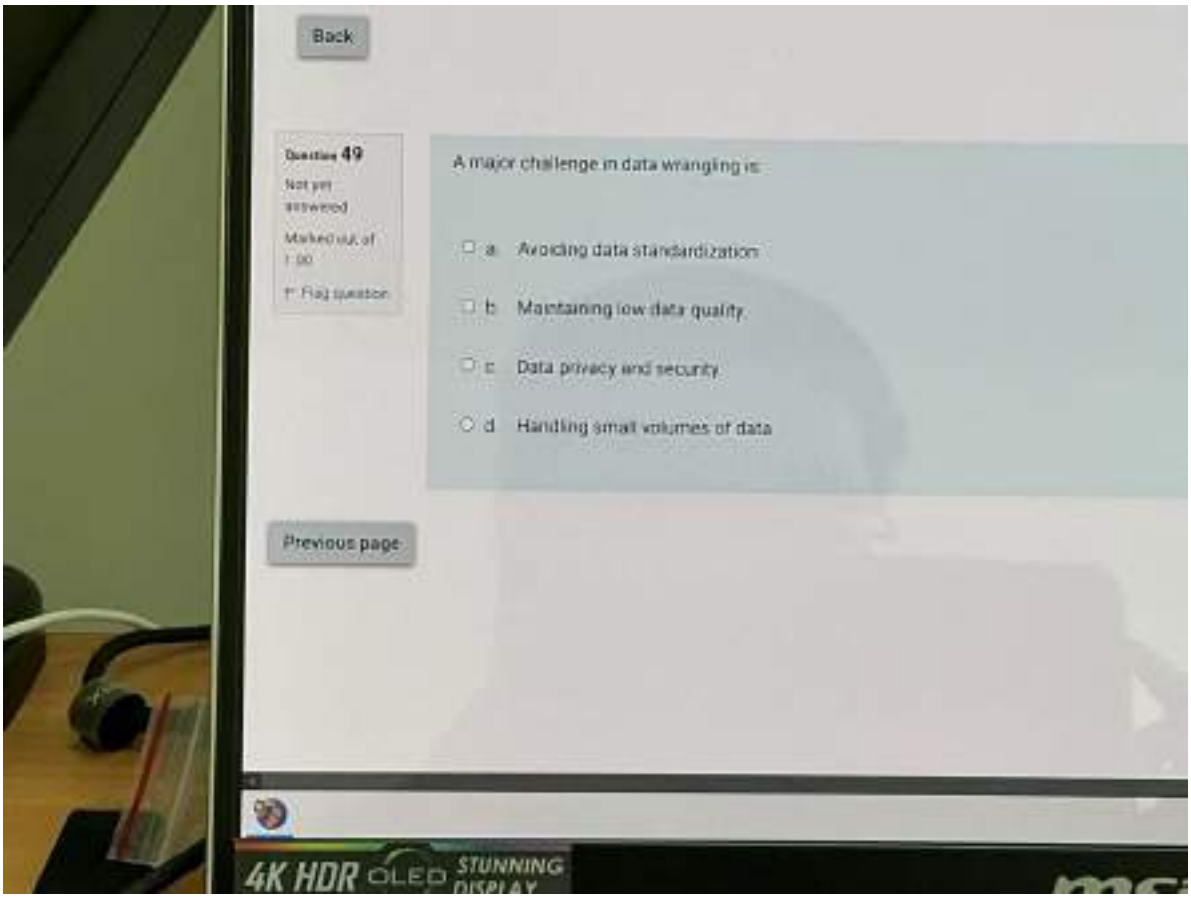
Which of the following is NOT a goal of data validation?

a. Validate range and constraints
b. Ensure data accurately reflects real-world entities
c. Enhance the speed of data retrieval
d. Verify data is in the expected format

参考答案： `c`

解析： 和 Q48 同理，validation 不以提升检索速度为目标。

Question 93



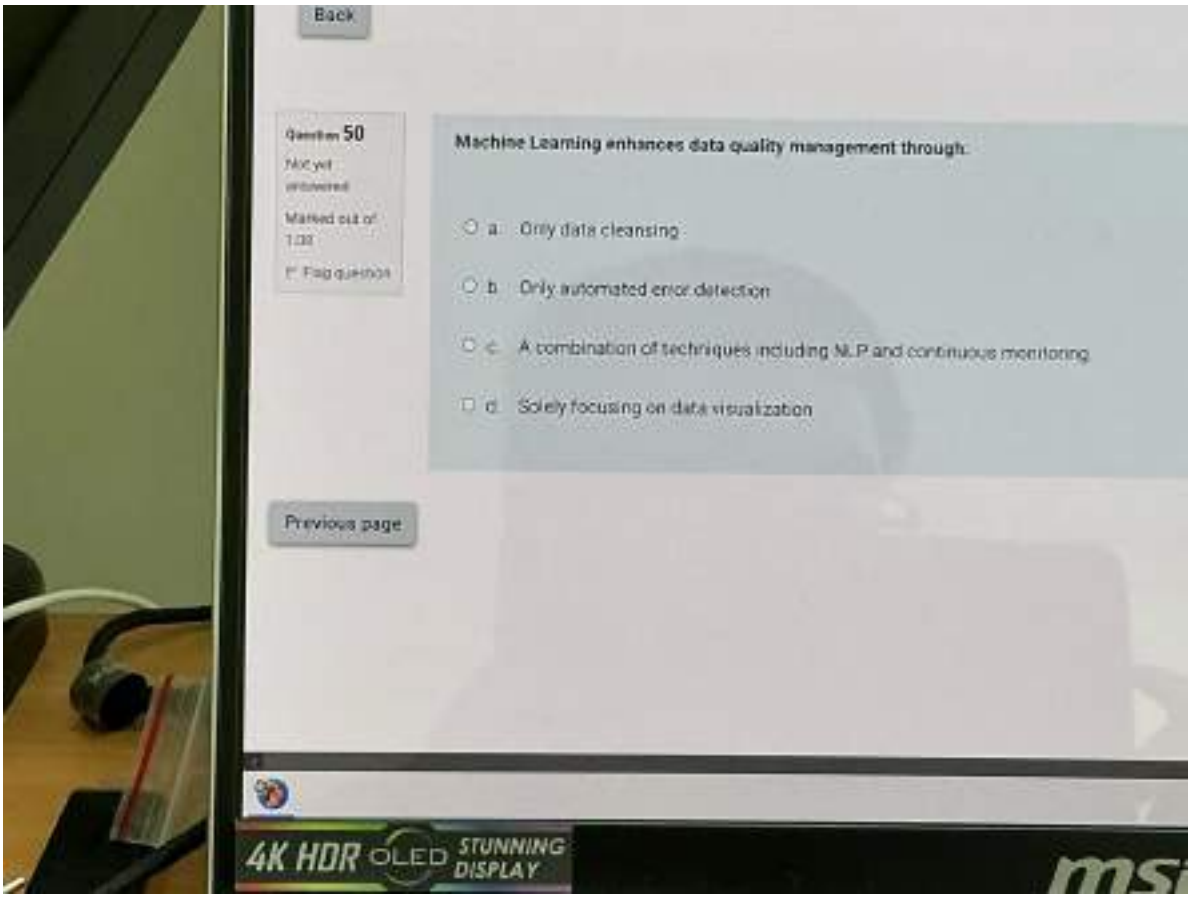
A major challenge in data wrangling is:

a. Avoiding data standardization
b. Maintaining low data quality
c. Data privacy and security
d. Handling small volumes of data

参考答案： `c`

解析： data privacy and security 是 data wrangling 的典型难点之一。

Question 94



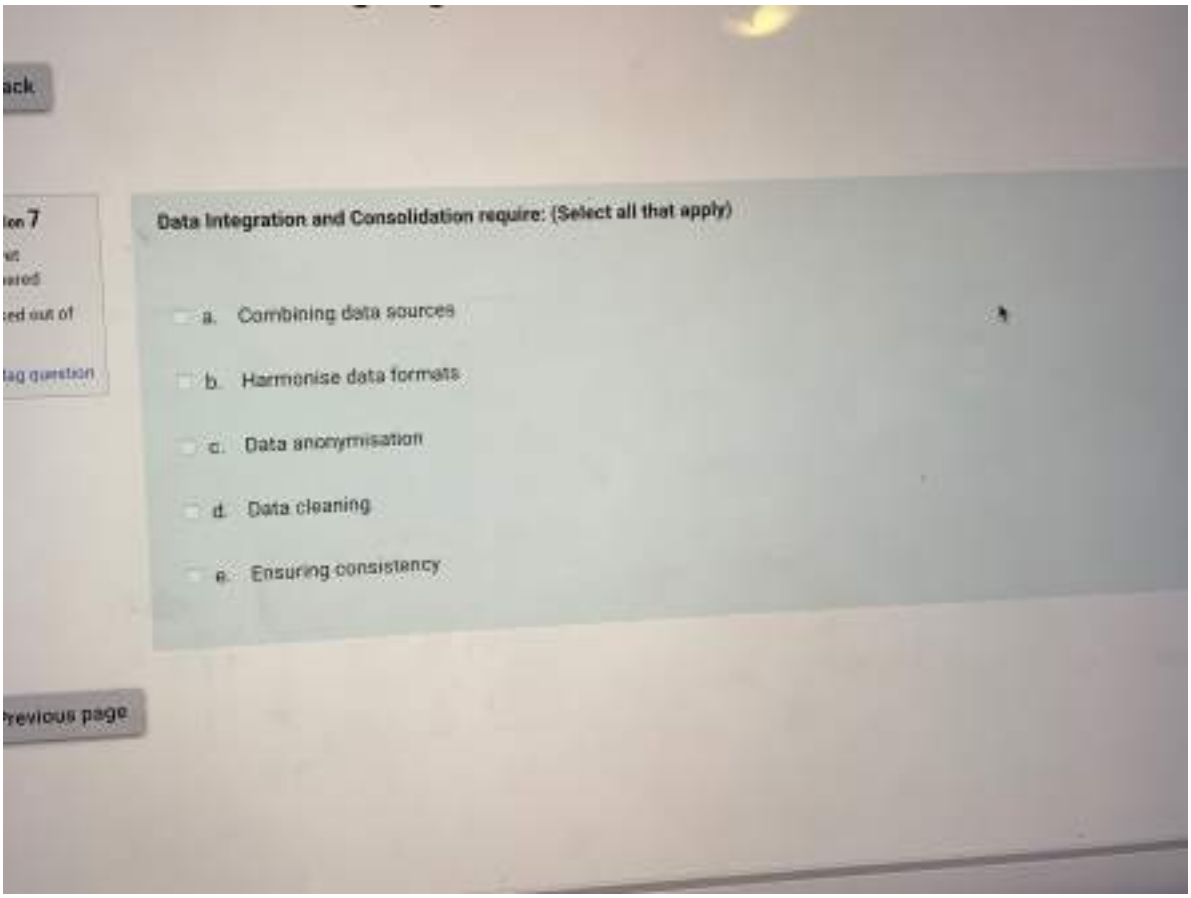
Machine Learning enhances data quality management through:

a. Solely focusing on data visualization
b. A combination of techniques including NLP and continuous monitoring
c. Only data cleansing
d. Only automated error detection

参考答案： `b`

解析： 这题和 Q45 同义，关键是 ML 能结合多种自动化方法提升质量管理。

Question 101



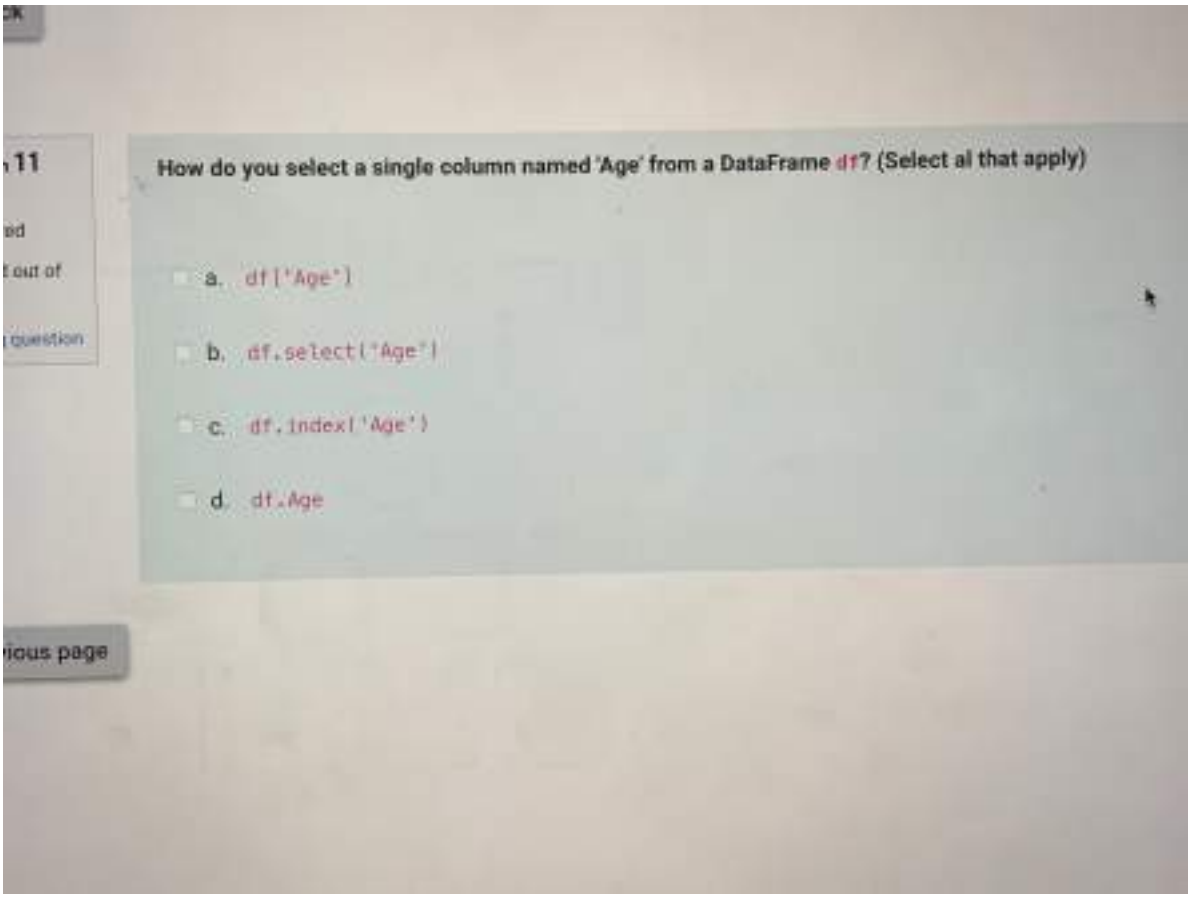
Data Integration and Consolidation require: (Select all that apply)

a. Combining data sources
b. Harmonise data formats
c. Data anonymisation
d. Data cleaning
e. Ensuring consistency

参考答案： `a, b, d, e`

解析： integration / consolidation 要合并来源、统一格式、清洗并保证一致性。

Question 103



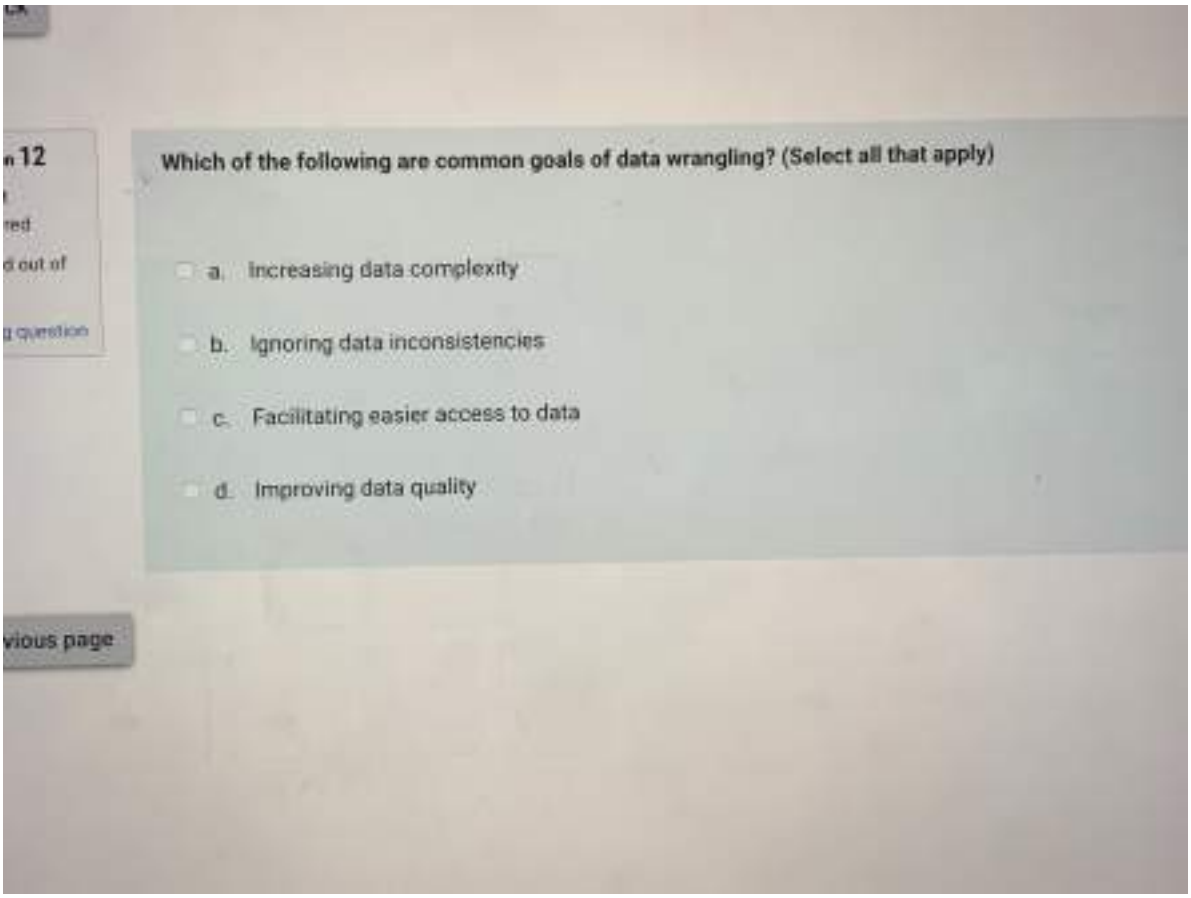
How do you select a single column named 'Age' from a DataFrame df? (Select al that apply)

a. `df['Age']`
b. `df.select('Age')`
c. `df.index('Age')`
d. `df.Age`

参考答案： `a, d`

解析： 单列可用 `df["Age"]` 或属性访问 `df.Age`。

Question 104



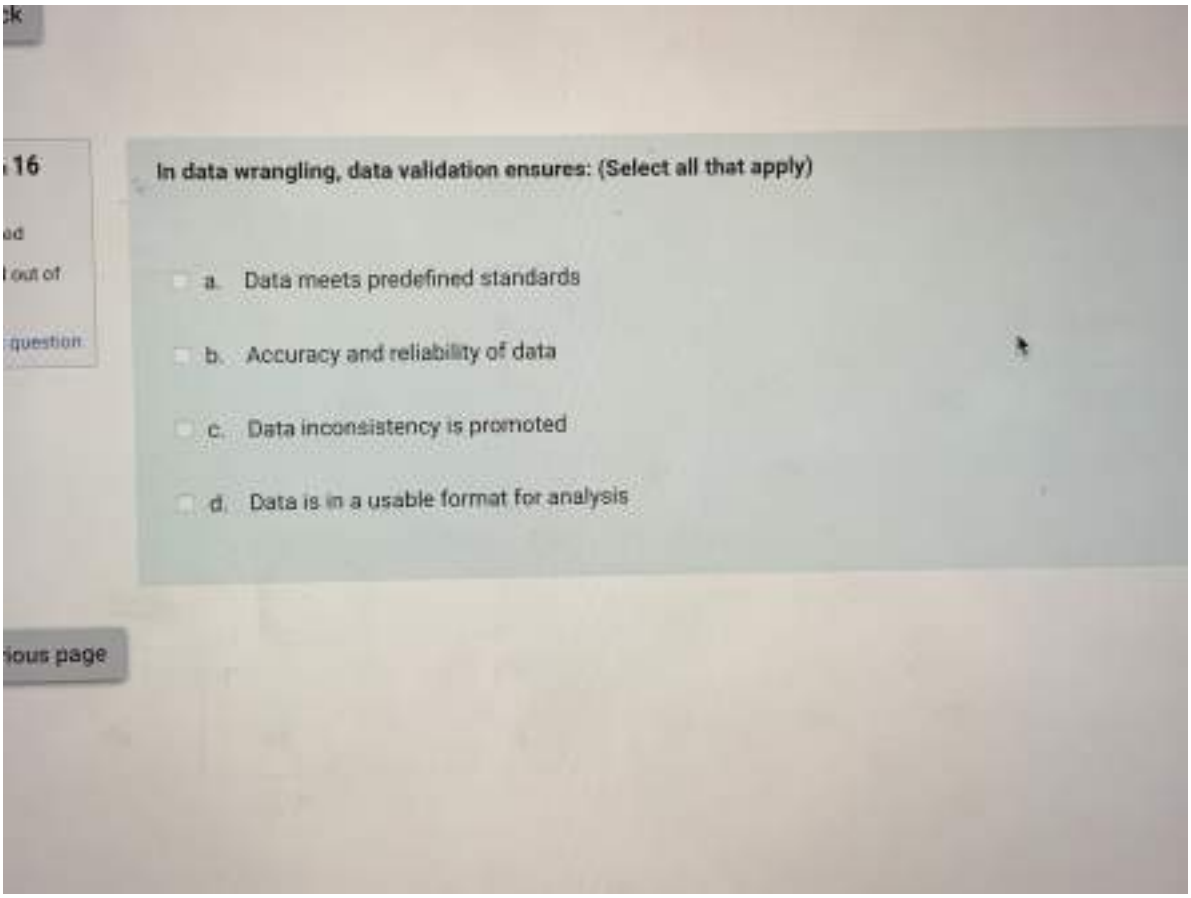
Which of the following are common goals of data wrangling? (Select all that apply)

a. Increasing data complexity
b. Ignoring data inconsistencies
c. Facilitating easier access to data
d. Improving data quality

参考答案： `c, d`

解析： 和 Q59 同义，目标仍是提升质量与可访问性。

Question 108



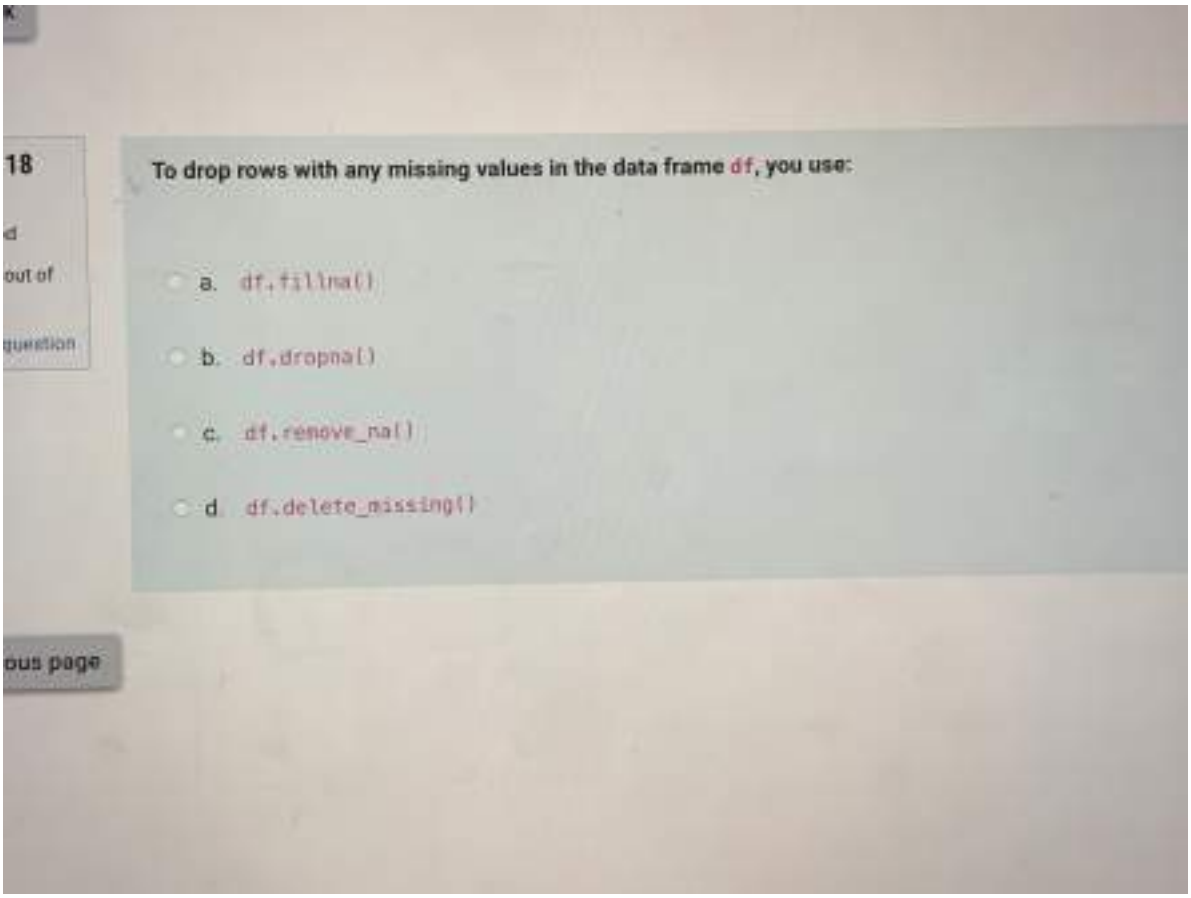
In data wrangling, data validation ensures: (Select all that apply)

a. Data meets predefined standards
b. Accuracy and reliability of data
c. Data inconsistency is promoted
d. Data is in a usable format for analysis

参考答案： `a, b, d`

解析： 和 Q52 同义，validation 关注标准、可靠性和可用性。

Question 109



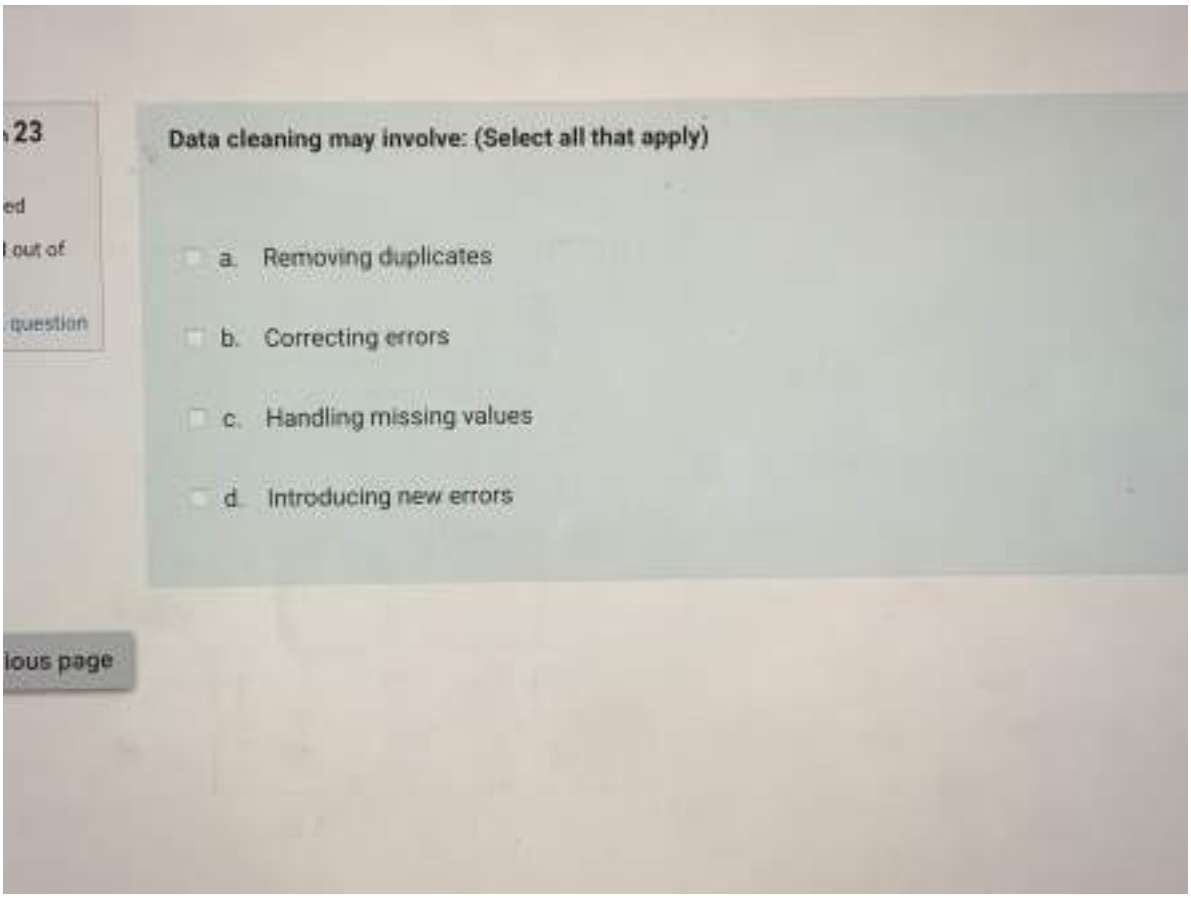
To drop rows with any missing values in the data frame `df`, you use:

a. `df.fillna()`
b. `df.dropna()`
c. `df.remove_na()`
d. `df.delete_missing()`

参考答案： `b`

解析： 删除含缺失值的行用 `df.dropna()`。

Question 114



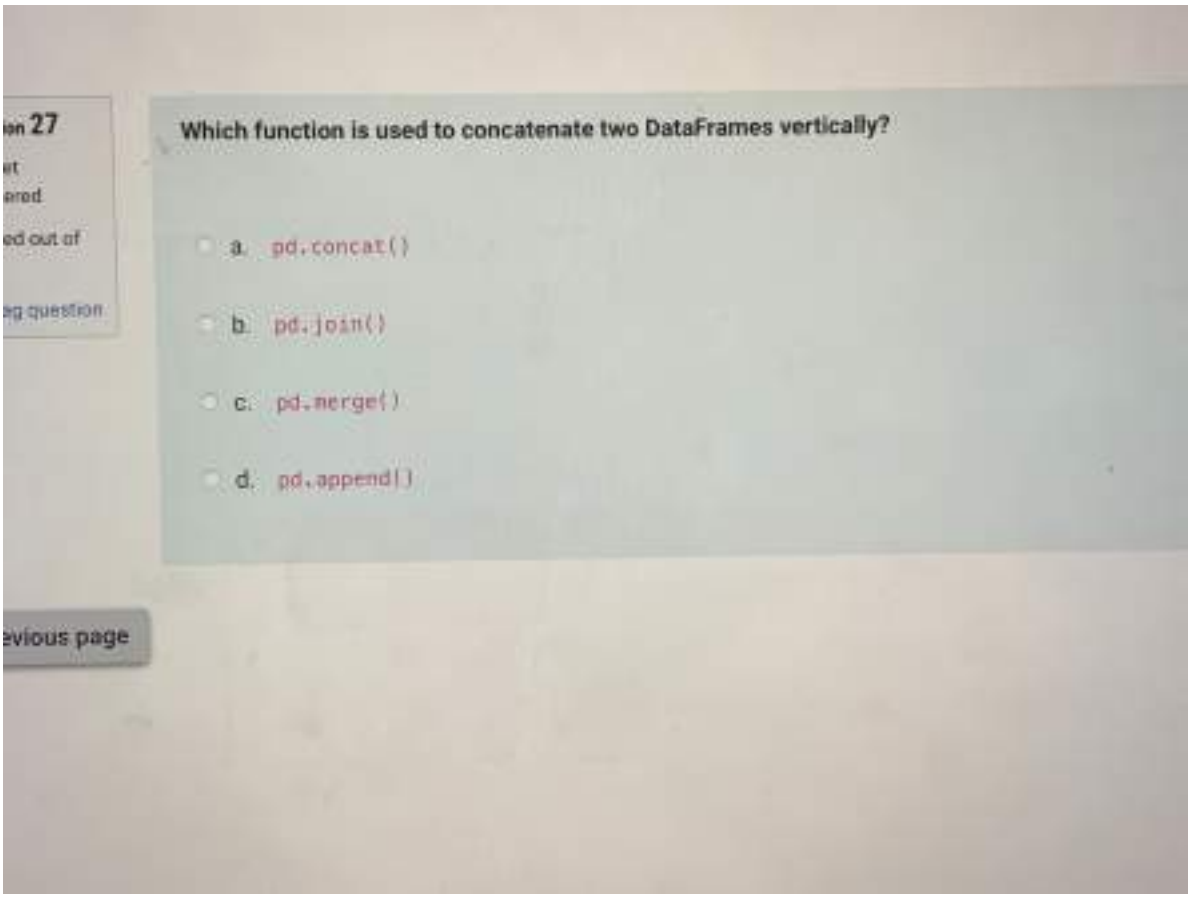
Data cleaning may involve: (Select all that apply)

a. Removing duplicates
b. Correcting errors
c. Handling missing values
d. Introducing new errors

参考答案： `a, b, c`

解析： 和 Q17/Q70 同义，cleaning 还是去重、纠错、处理缺失。

Question 118



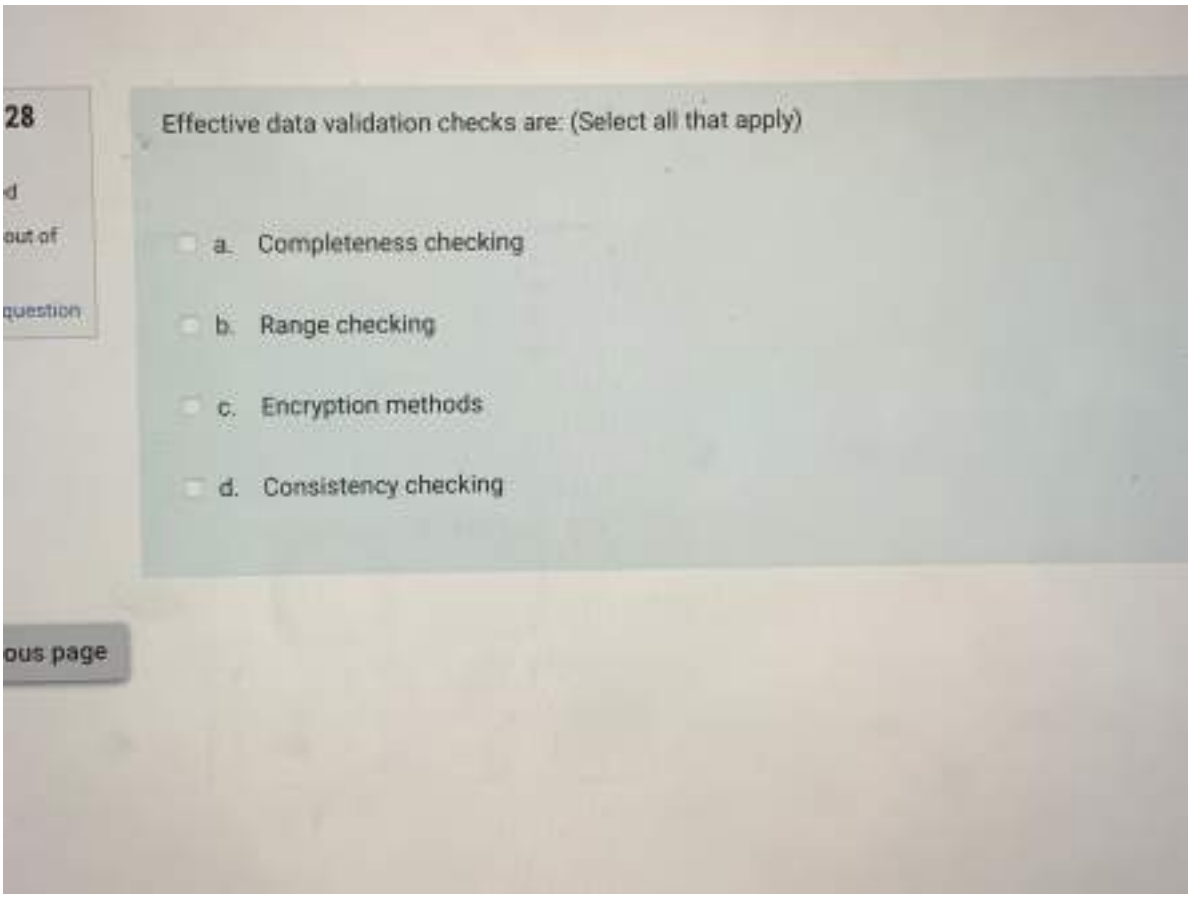
Which function is used to concatenate two DataFrames vertically?

a. `pd.concat()`
b. `pd.join()`
c. `pd.merge()`
d. `pd.append()`

参考答案： `a`

解析： 纵向拼接两个 DataFrame 通常用 `pd.concat()`。

Question 119



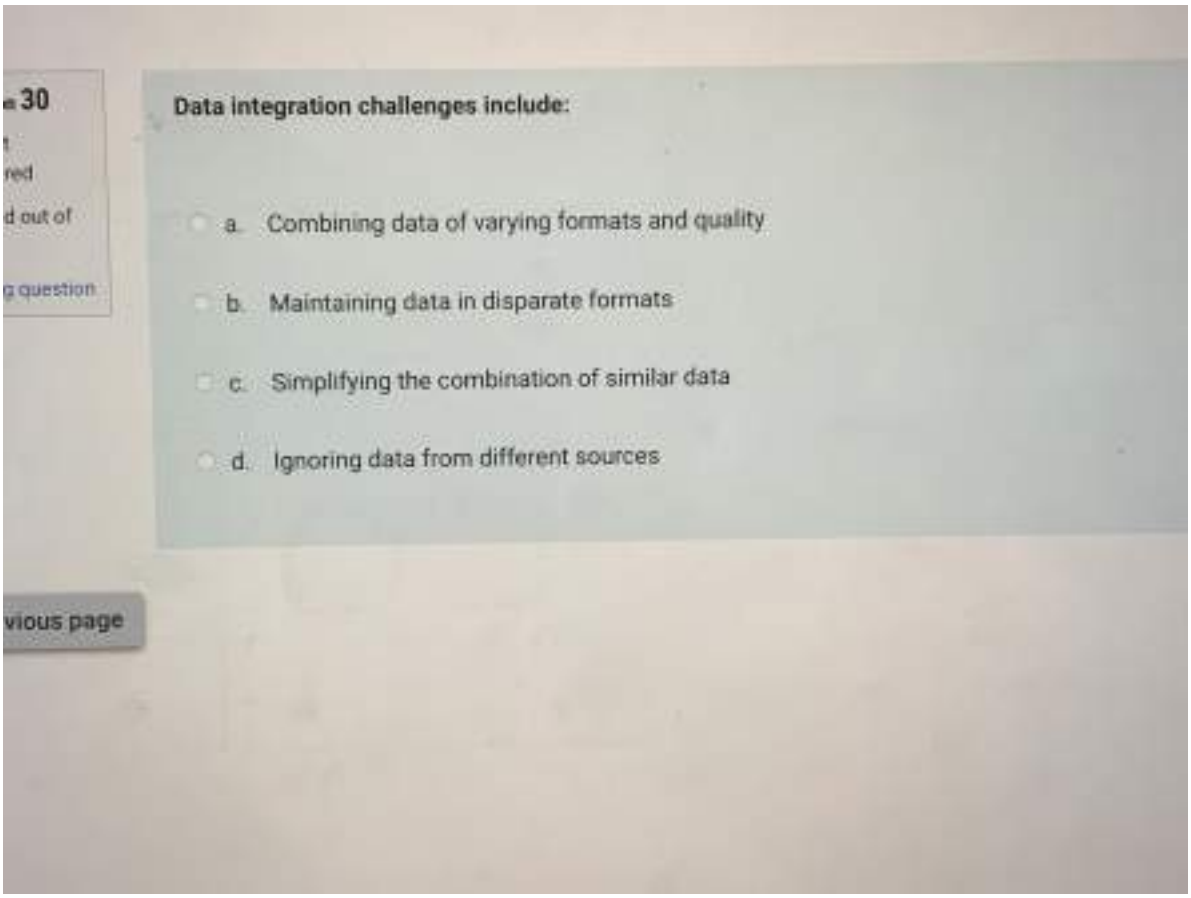
Effective data validation checks are: (Select all that apply)

a. Completeness checking
b. Range checking
c. Encryption methods
d. Consistency checking

参考答案： `a, b, d`

解析： 有效 validation check 常见有 completeness、range、consistency；encryption 不是 validation check。

Question 121



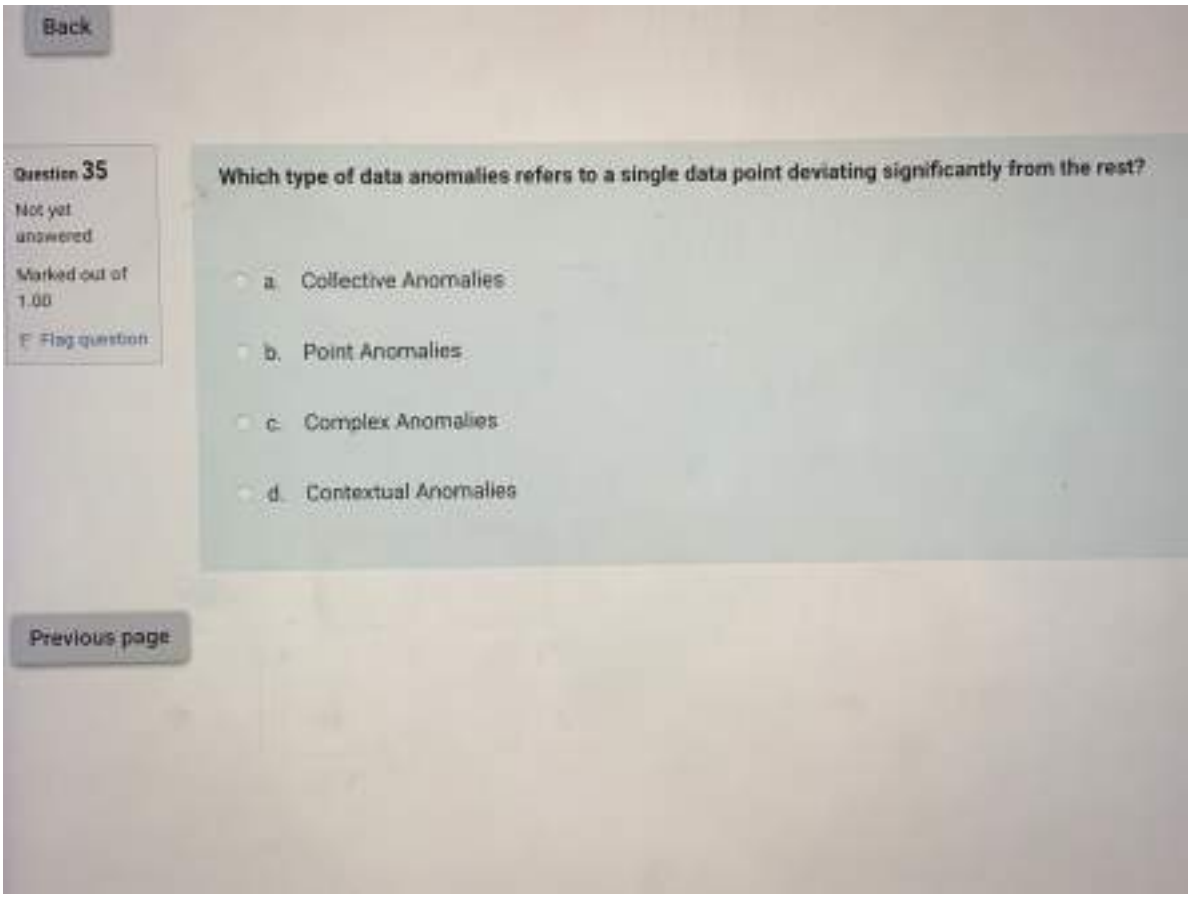
Data integration challenges include:

a. Combining data of varying formats and quality
b. Maintaining data in disparate formats
c. Simplifying the combination of similar data
d. Ignoring data from different sources

参考答案： a

解析： integration 真正难在格式和质量不一致，不是“忽略不同来源”。

Question 125



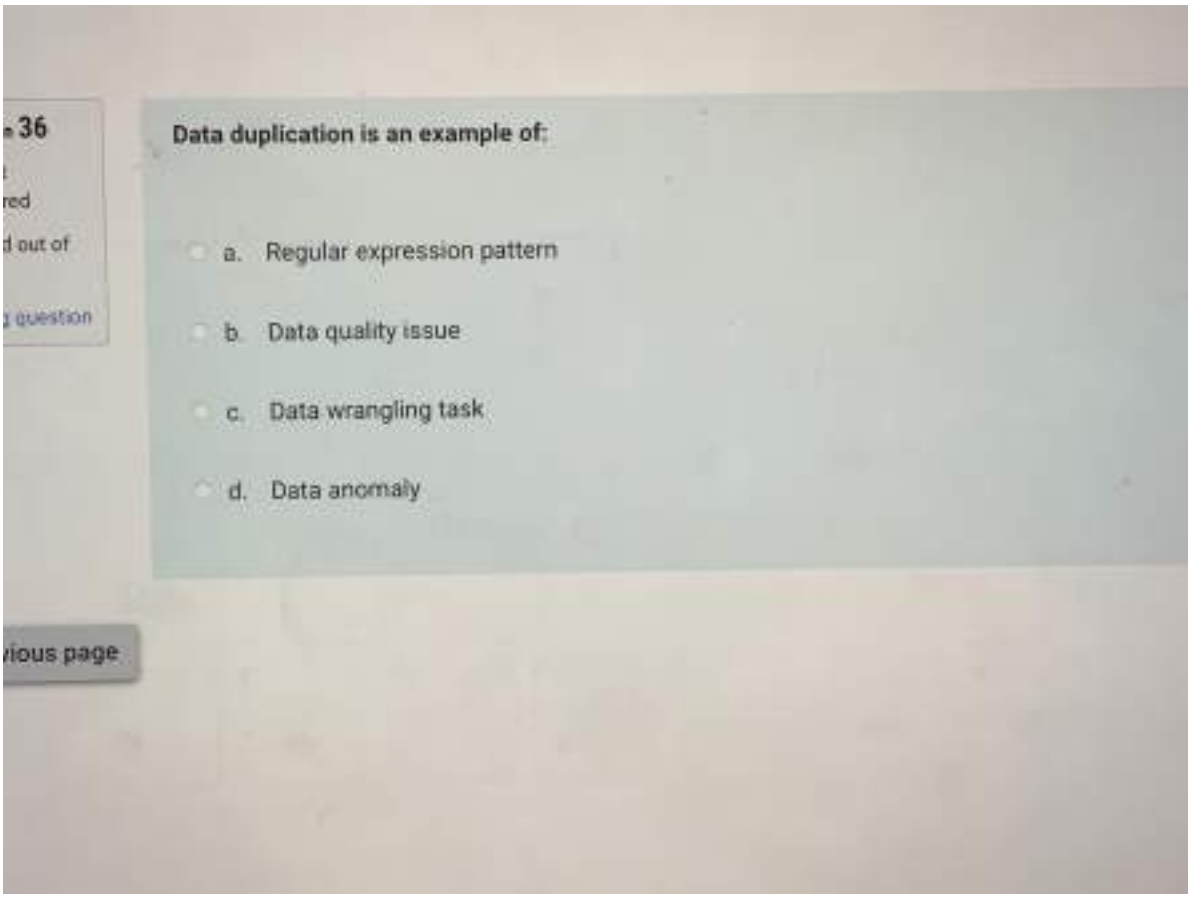
Which type of data anomalies refers to a single data point deviating significantly from the rest?

a. Collective Anomalies
b. Point Anomalies
c. Complex Anomalies
d. Contextual Anomalies

参考答案： `b`

解析： 和 Q47 同义，这里 point anomaly 的选项顺序变了。

Question 126



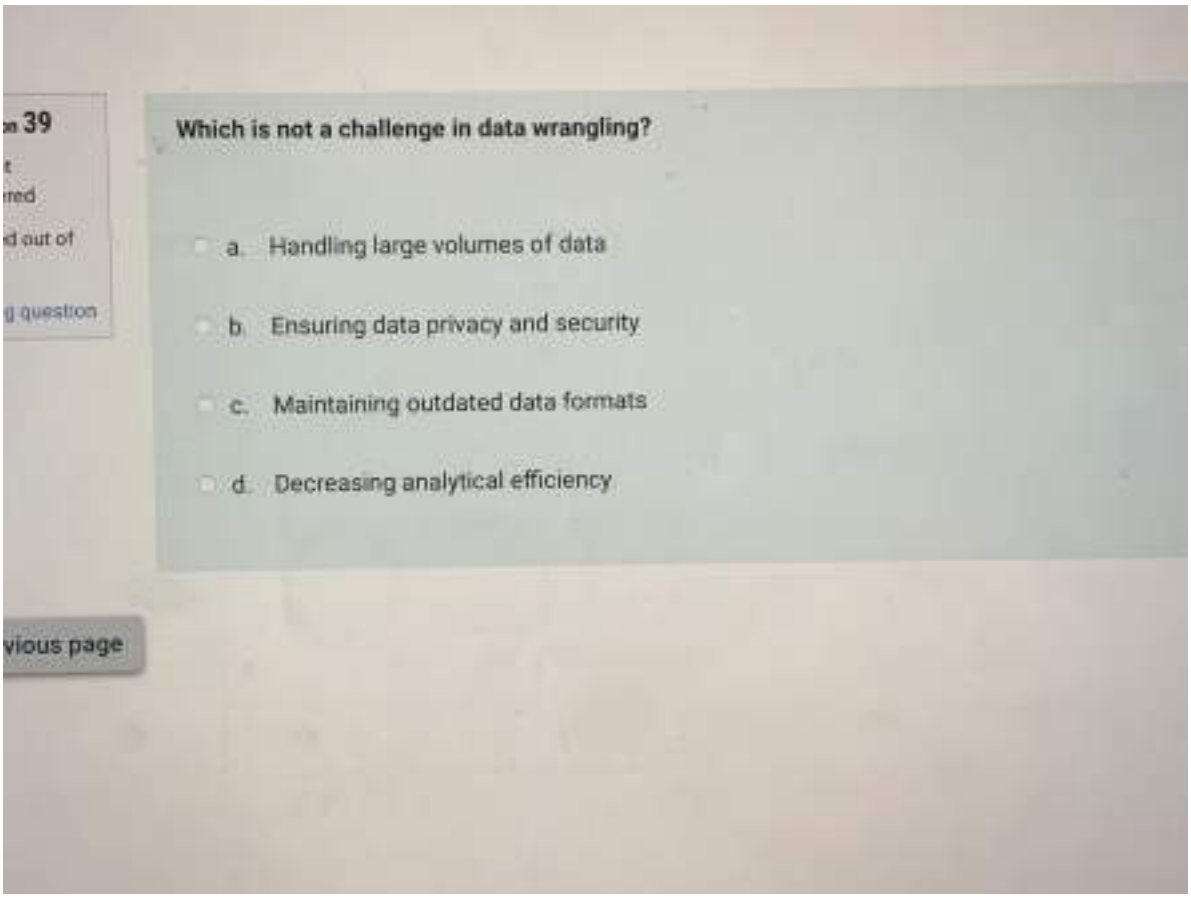
Data duplication is an example of:

a. Regular expression pattern
b. Data quality issue
c. Data wrangling task
d. Data anomaly

参考答案： `b`

解析： 重复数据首先是 data quality issue；虽然也常被看作 anomaly，但复习时先按课程答案记。

Question 129



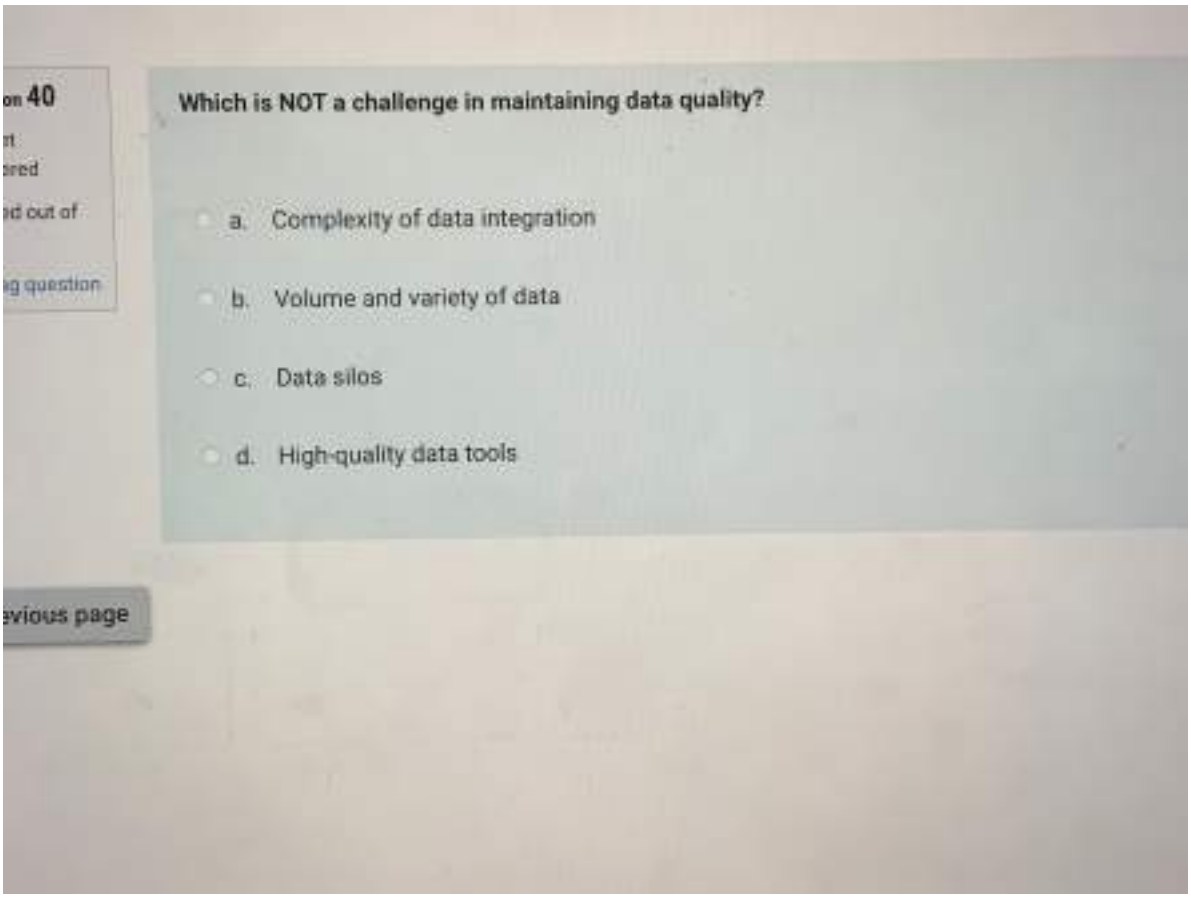
Which is not a challenge in data wrangling?

a. Handling large volumes of data
b. Ensuring data privacy and security
c. Maintaining outdated data formats
d. Decreasing analytical efficiency

参考答案： `d`

解析： 前三项都是挑战；“decreasing analytical efficiency” 不是挑战表述，而是负面结果说法。

Question 130



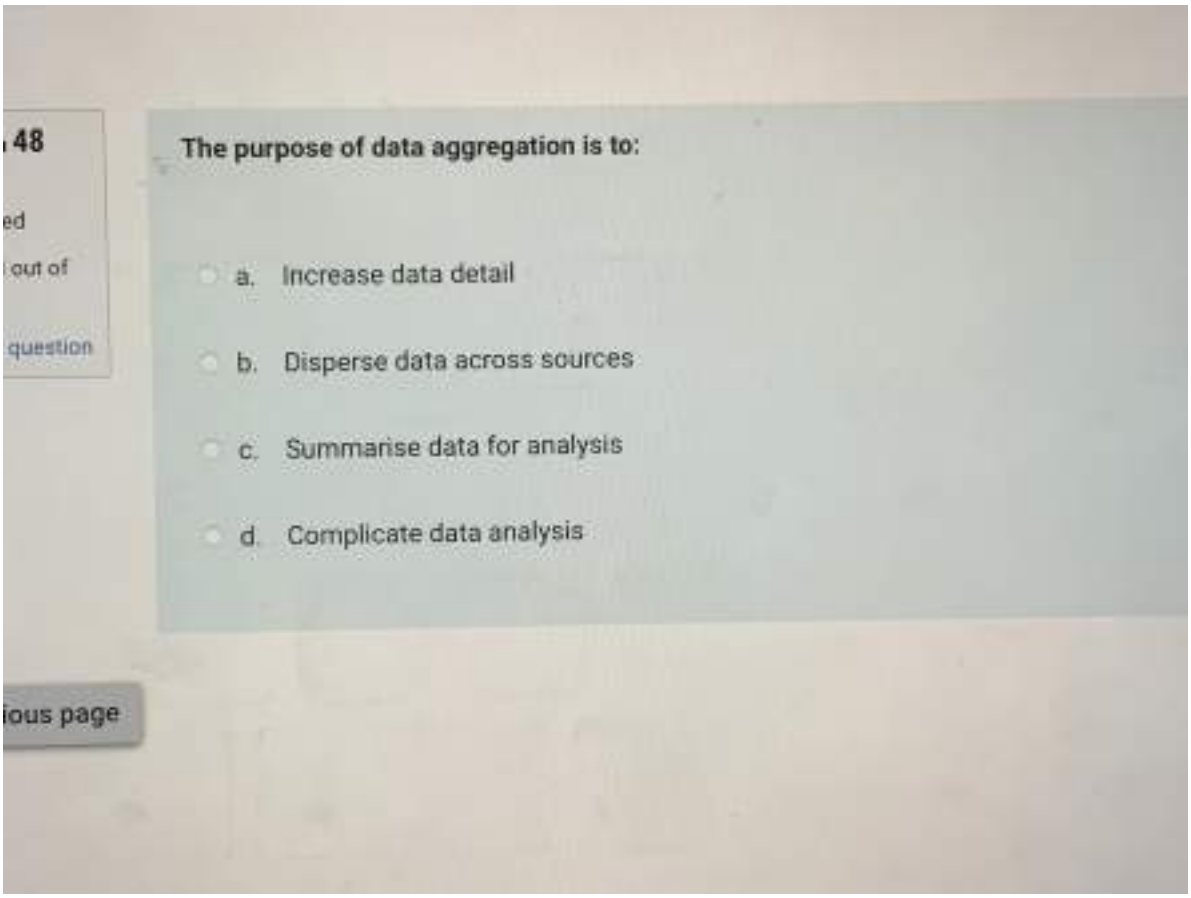
Which is NOT a challenge in maintaining data quality?

a. Complexity of data integration
b. Volume and variety of data
c. Data silos
d. High-quality data tools

参考答案： `d`

解析： 真正的挑战是集成复杂、量大、数据孤岛；高质量工具本身不是挑战。

Question 138



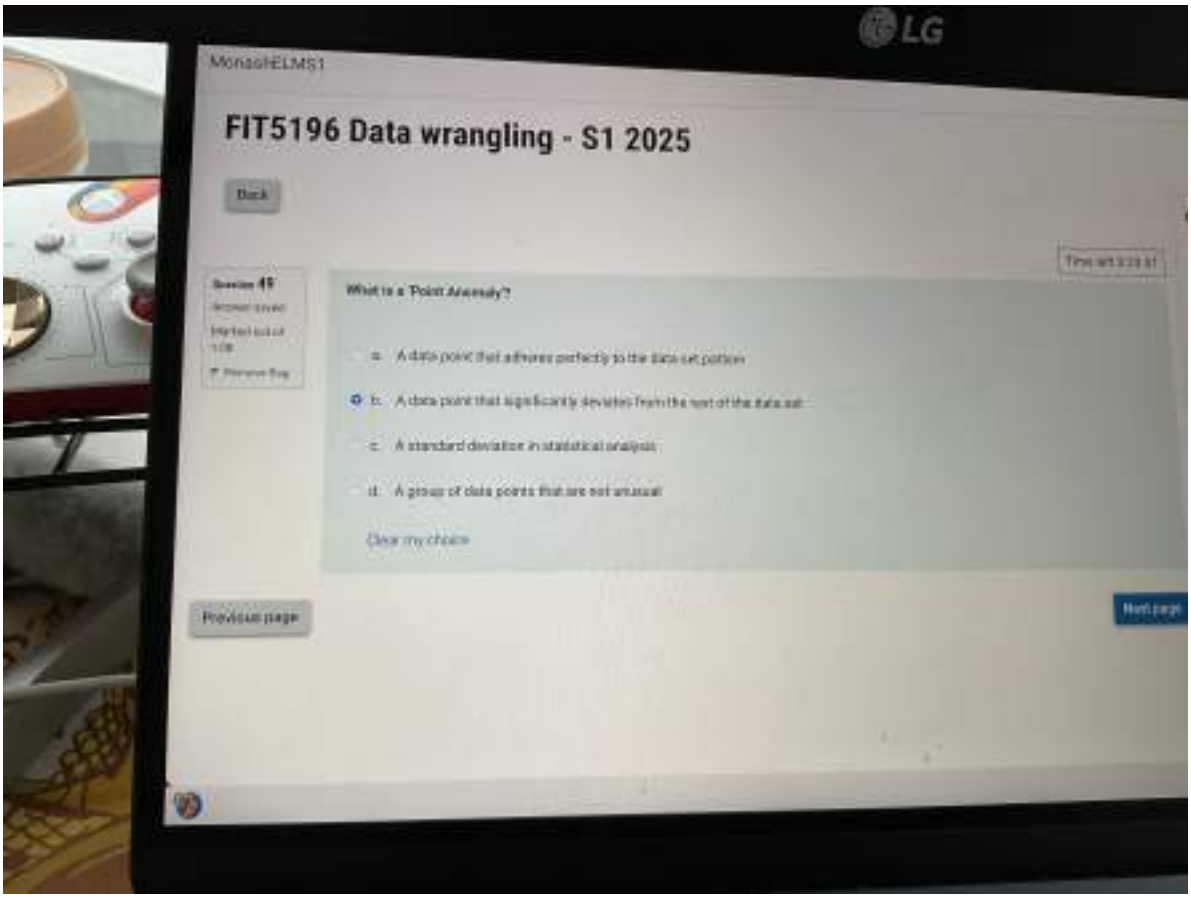
The purpose of data aggregation is to:

a. Increase data detail
b. Disperse data across sources
c. Summarise data for analysis
d. Complicate data analysis

参考答案： `c`

解析： aggregation 的目的就是把明细汇总成更适合分析的结果。

Question 143



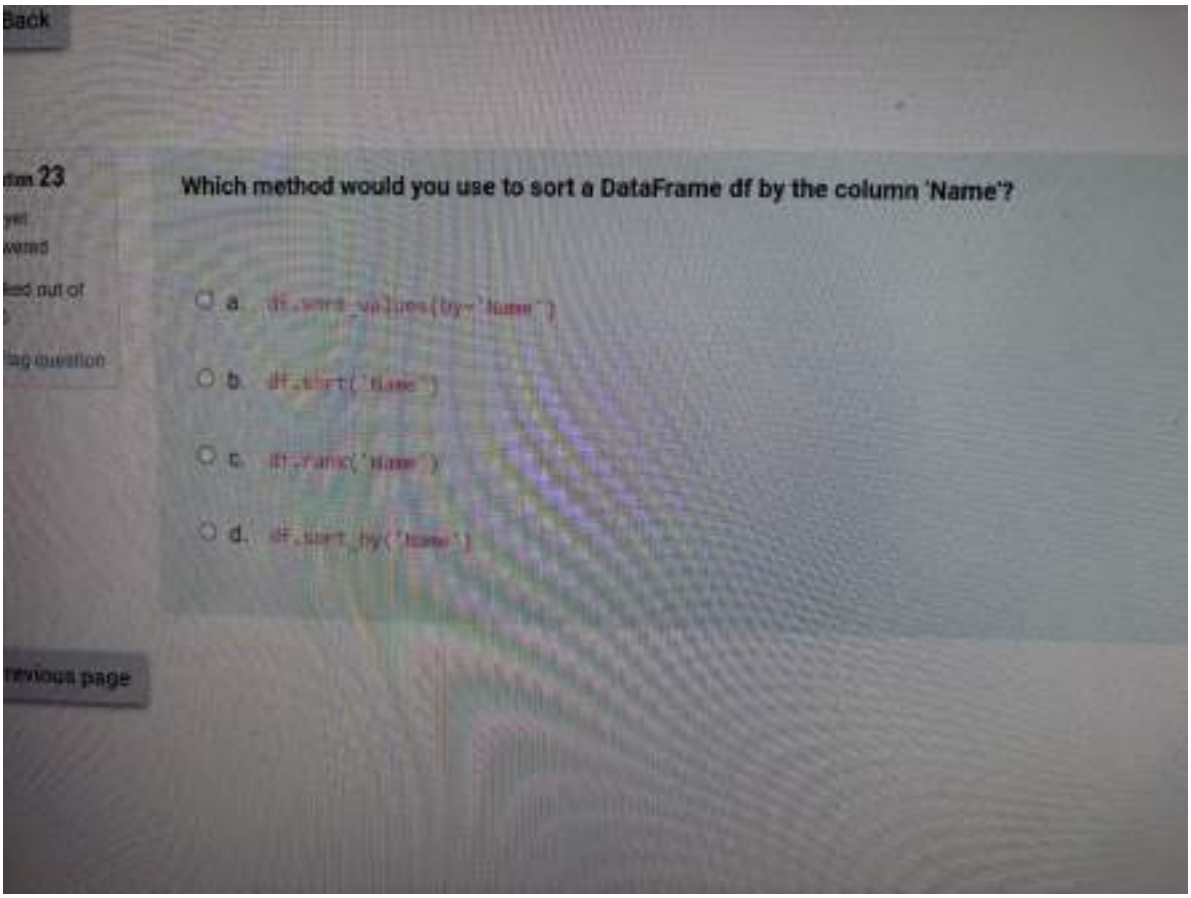
What is a 'Point Anomaly'?

a. A data point that adheres perfectly to the data set pattern
b. A data point that significantly deviates from the rest of the data set
c. A standard deviation in statistical analysis
d. A group of data points that are not unusual

参考答案： b

解析： point anomaly 就是单个点显著偏离整体。

Question 152



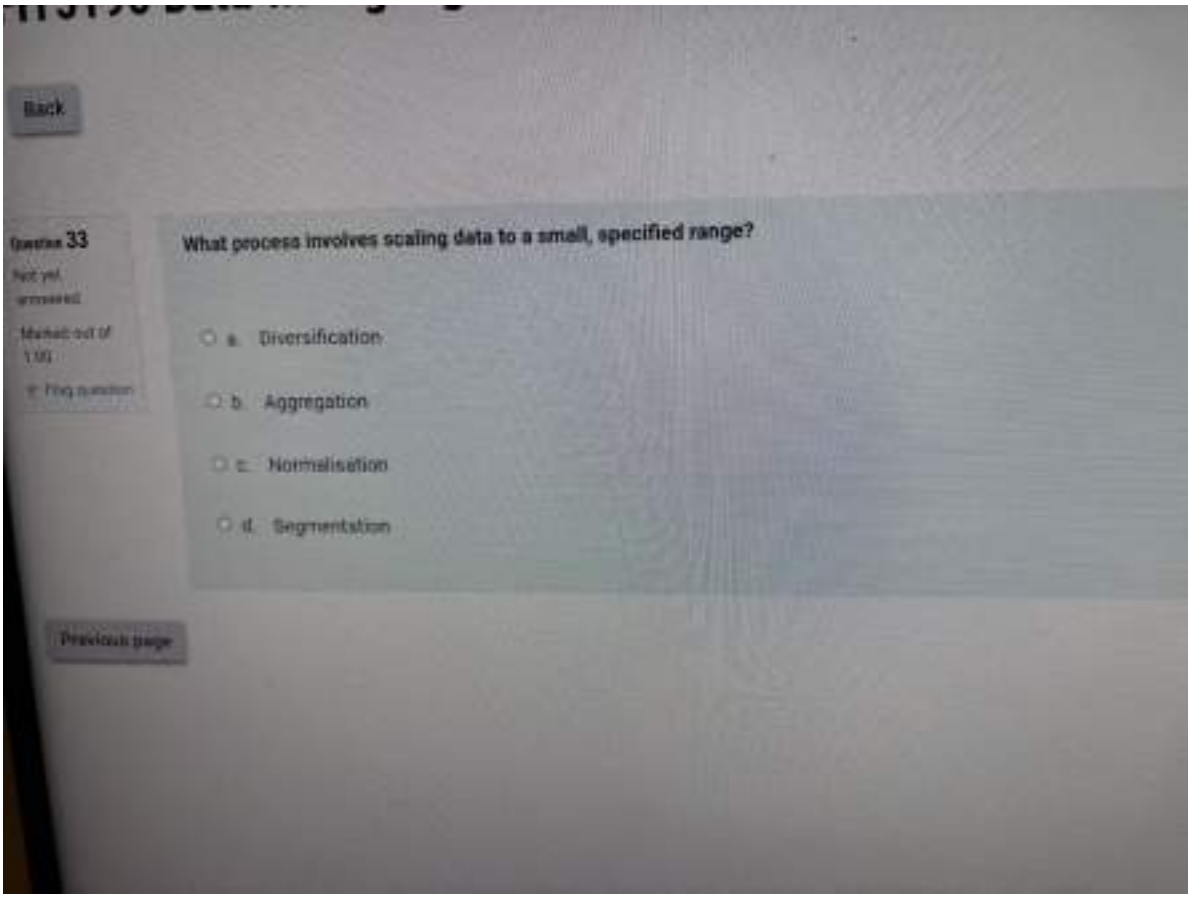
Which method would you use to sort a DataFrame df by the column 'Name'?

a. `df.sort_values(by='Name')`
b. `df.sort('Name')`
c. `df.rank('Name')`
d. `df.sort_by('Name')`

参考答案： `a`

解析： 和 Q78 同义，只是正确项顺序换成了 `a`。

Question 160



What process involves scaling data to a small, specified range?

a. Diversification
b. Aggregation
c. Normalisation
d. Segmentation

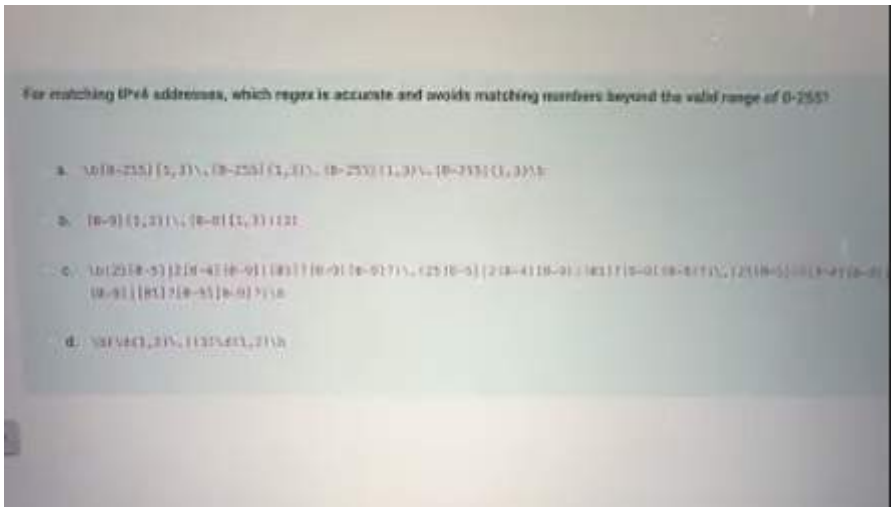
参考答案： `c`

解析： 和 Q39 同义，缩放到小范围依然是 normalisation。

Week 3 (39 题)

- 范围：Regular Expressions

Question 3



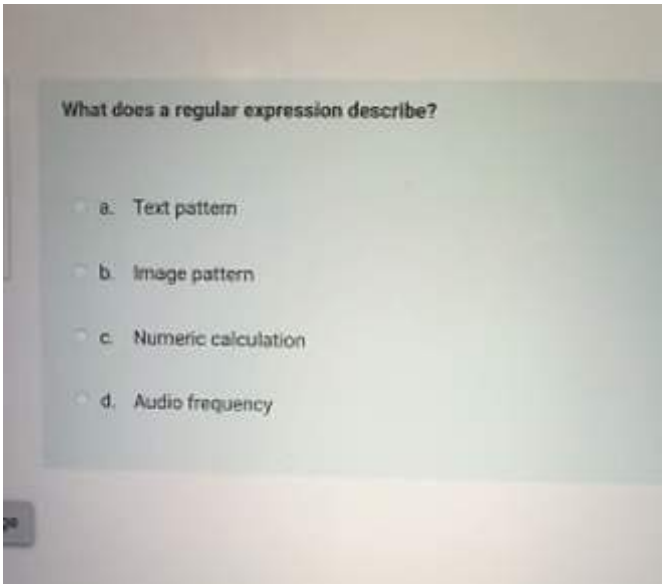
For matching IPv4 addresses, which regex is accurate and avoids matching numbers beyond the valid range of 0-255?

a. \b(0-255){1,3}.(0-255){1,3}.(0-255){1,3}.(0-255){1,3}\b
b. [0-9]{1,3}.([0-9]{1,3}){3}
c. \b(25[0-5]|2[0-4][0-9]|[01]?[0-9][0-9]?).(25[0-5]|2[0-4][0-9]|[01]?[0-9][0-9]?).(25[0-5]|2[0-4][0-9]|[01]?[0-9][0-9]?).(25[0-5]|2[0-4][0-9]|[01]?[0-9][0-9]?)\b
d. \b(\d{1,3}.){3}\d{1,3}\b

参考答案： `c`

解析： 只有 `c` 把每一段都限制在 `0-255` 范围内；`b` 和 `d` 只限制位数，不限制数值大小。

Question 7



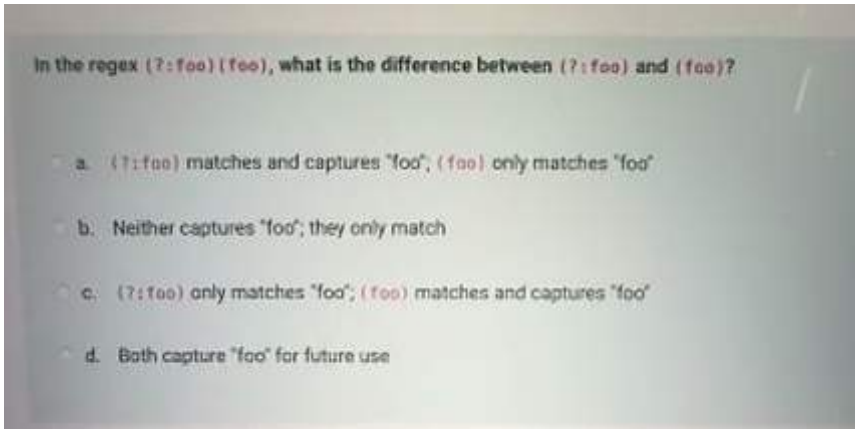
What does a regular expression describe?

a. Text pattern
b. Image pattern
c. Numeric calculation
d. Audio frequency

参考答案： `a`

解析： 正确项是 `a`: Text pattern。它最符合这道题在课程里的定义。

Question 11



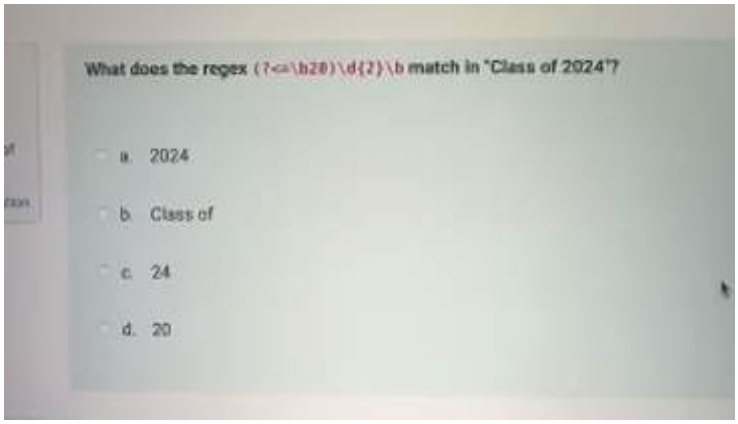
In the regex (?:foo)(foo), what is the difference between (?:foo) and (foo)?

a. (?:foo) matches and captures "foo"; (foo) only matches "foo"
b. Neither captures "foo"; they only match
c. (?:foo) only matches "foo"; (foo) matches and captures "foo"
d. Both capture "foo" for future use

参考答案： `c`

解析： `(?:foo)` 是非捕获组，只分组不保存；`(foo)` 是捕获组，会把匹配结果保存下来供后续引用。

Question 13



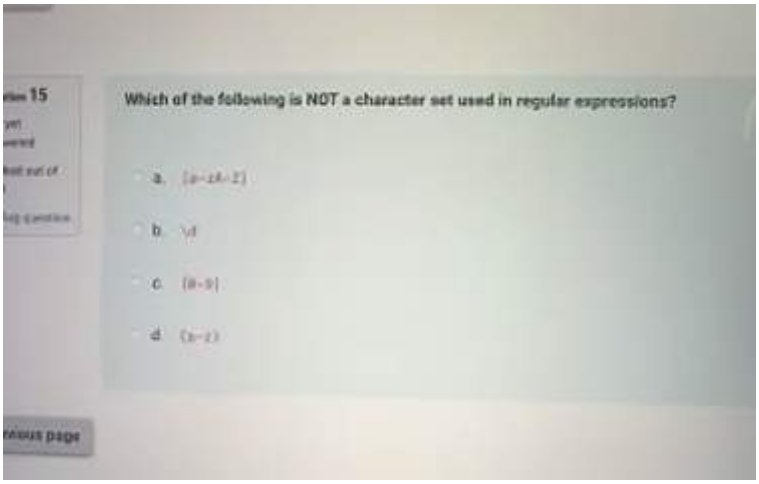
What does the regex `(?<=\b20)\d{2}\b` match in "Class of 2024"?

a. 2024
b. Class of
c. 24
d. 20

参考答案： `c`

解析： lookbehind `(?<=\b20)` 要求前面紧跟 `20`，因此真正被匹配的是后面的两位数字 `24`。

Question 15



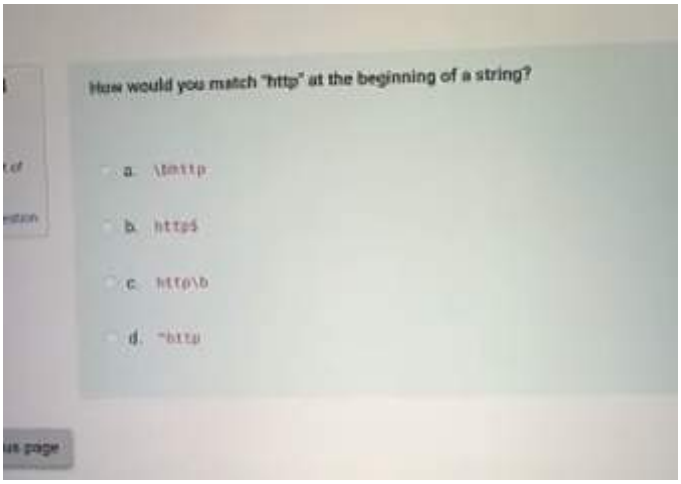
Which of the following is NOT a character set used in regular expressions?

a. [a-zA-Z]
b. \d
c. [0-9]
d. (a-z)

参考答案： `d`

解析： 正确项是 `d`: (a-z)。它最符合这道题在课程里的定义。

Question 18



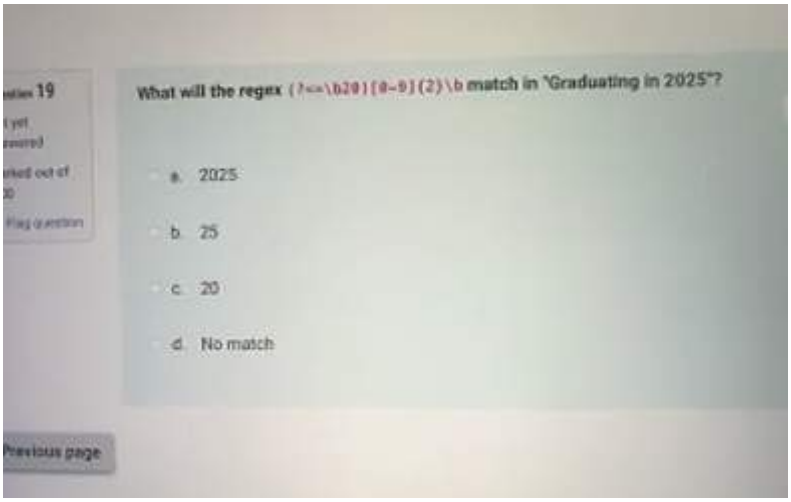
How would you match "http" at the beginning of a string?

a. \bhttp
b. http$
c. http\b
d. ^http

参考答案： `d`

解析： `^` 匹配字符串开头，所以要写 `^http`。

Question 19



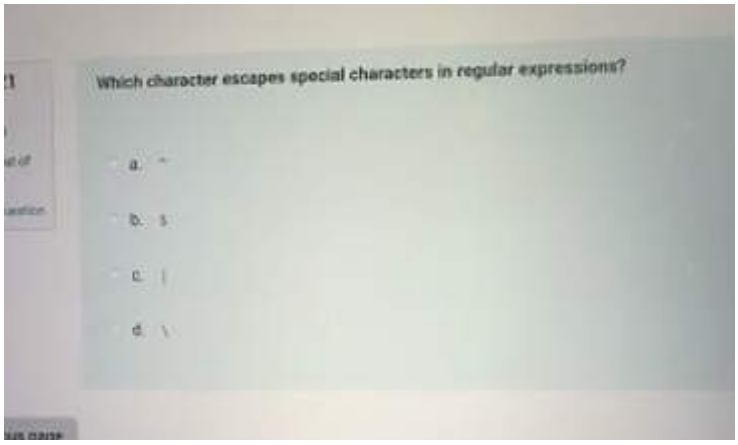
What will the regex (?<=\b20)[0-9]{2}\b match in "Graduating in 2025"?

a. 2025
b. 25
c. 20
d. No match

参考答案： `b`

解析： 和 Q13 同理，`(?<=\b20)` 只把 `20` 后面的两位数取出来，所以是 `25`。

Question 21



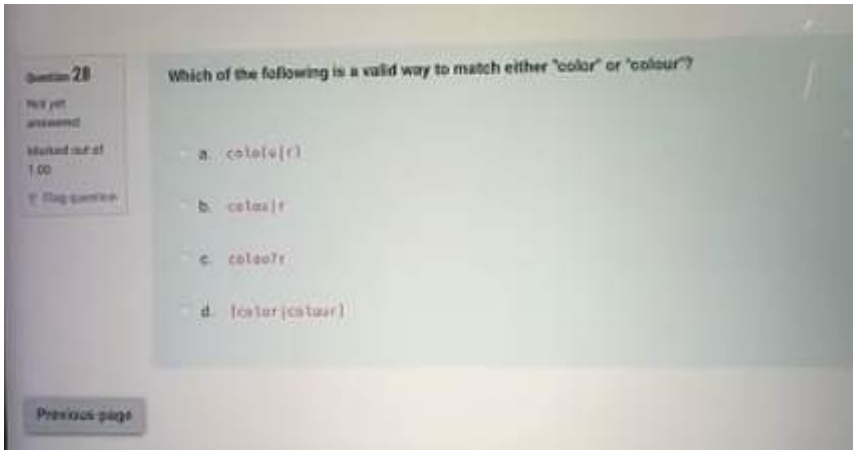
Which character escapes special characters in regular expressions?

a. ^
b. $
c. |
d. \

参考答案： `d`

解析： 正确项是 `d`: \。它最符合这道题在课程里的定义。

Question 29



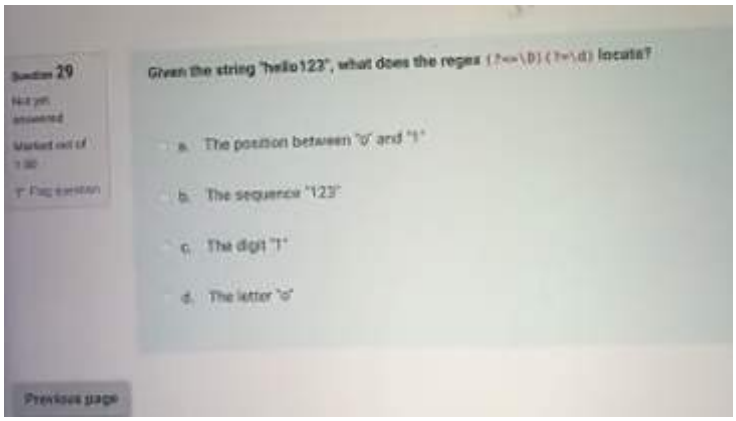
Which of the following is a valid way to match either 'color' or 'colour'?

a. colo(u|r)
b. colou|r
c. colou?r
d. (color|colour)

参考答案： `c`

解析： `colou?r` 中 `u?` 表示 `u` 可有可无，因此能同时匹配 `color` 和 `colour`。`(color|colour)` 也能匹配，但课里通常把 `c` 当标准答案。

Question 30



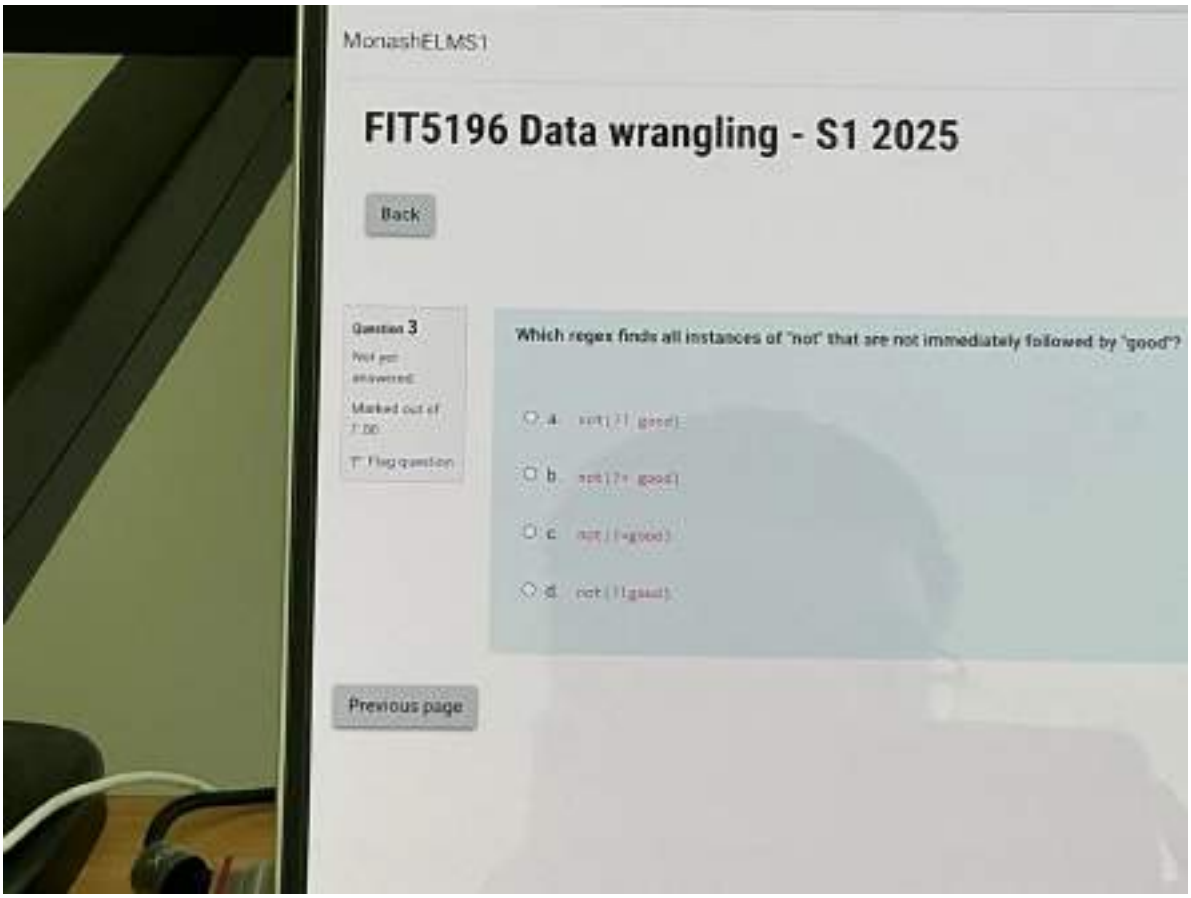
Given the string 'hello123', what does the regex (?<=\D)(?=\d) locate?

a. The position between 'o' and '1'
b. The sequence "123"
c. The digit "1"
d. The letter 'o'

参考答案： `a`

解析： 这个零宽断言定位的是“非数字后面紧跟数字”的那个位置，也就是 `o` 和 `1` 中间。

Question 54



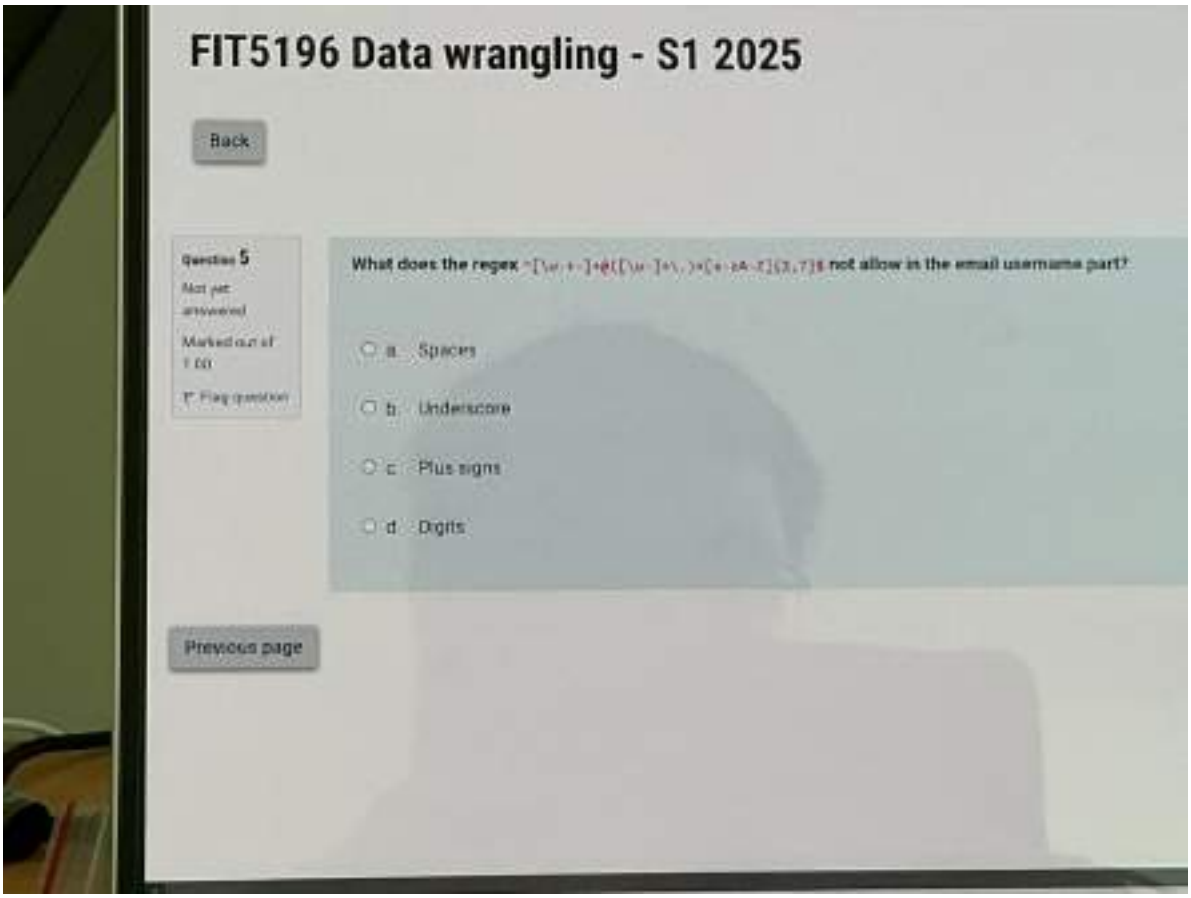
Which regex finds all instances of "not" that are not immediately followed by "good"?

a. not(?! good)
b. not(?= good)
c. not(?=good)
d. not(?!good)

参考答案： `d`

解析： 负向前瞻 `(?!good)` 表示后面不能立刻跟 `good`，所以选 `not(?!good)`。

Question 56



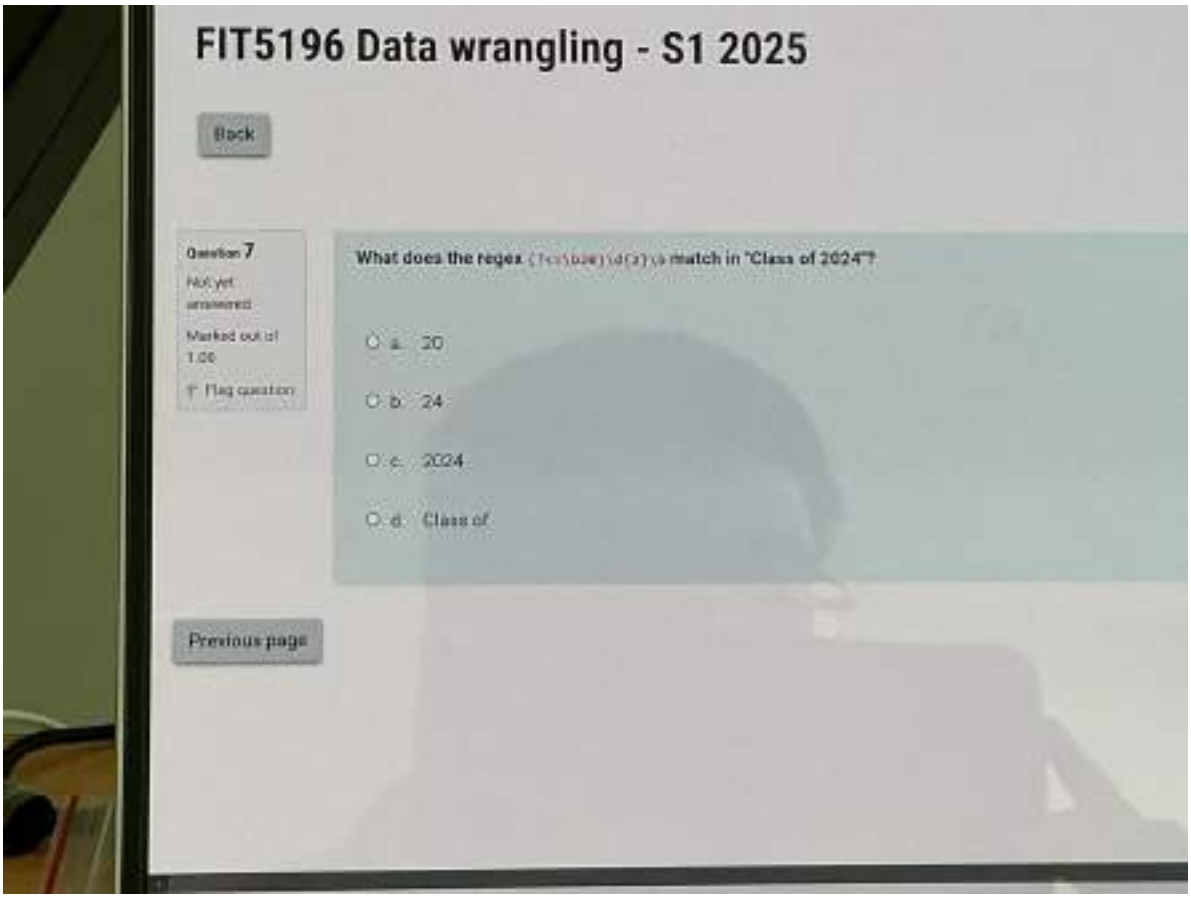
What does the regex `^[\w.+-]+@([\w-]+\.)+[a-zA-Z]{2,7}$` not allow in the email username part?

a. Spaces
b. Underscore
c. Plus signs
d. Digits

参考答案： a

解析： 用户名部分的字符类里不包含空格，所以空格不允许。

Question 58



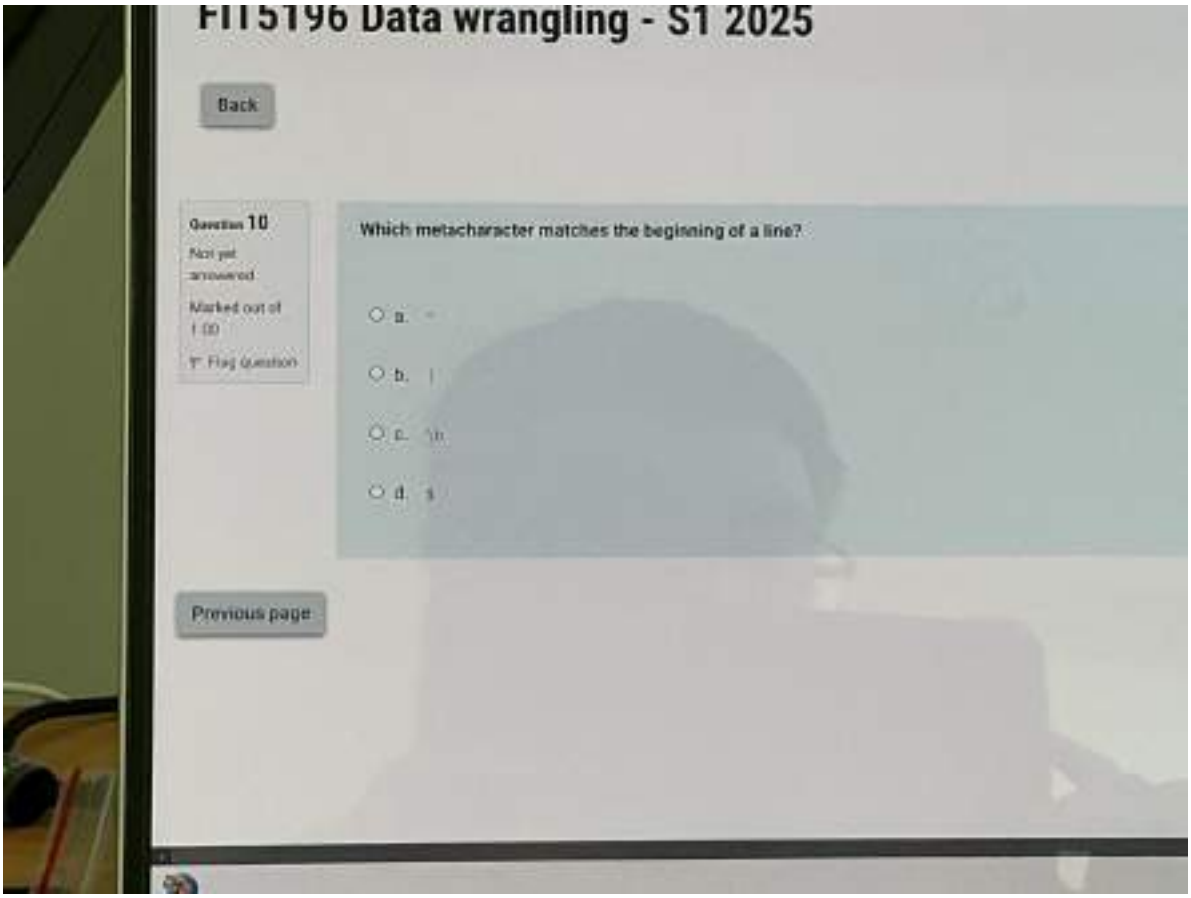
What does the regex (?<=\b20)\d{2}\b match in "Class of 2024"?

a. 20
b. 24
c. 2024
d. Class of

参考答案： `b`

解析： 和 Q13/Q19 同理，真正被匹配到的是 `20` 后面的两位，即 `24`。

Question 61



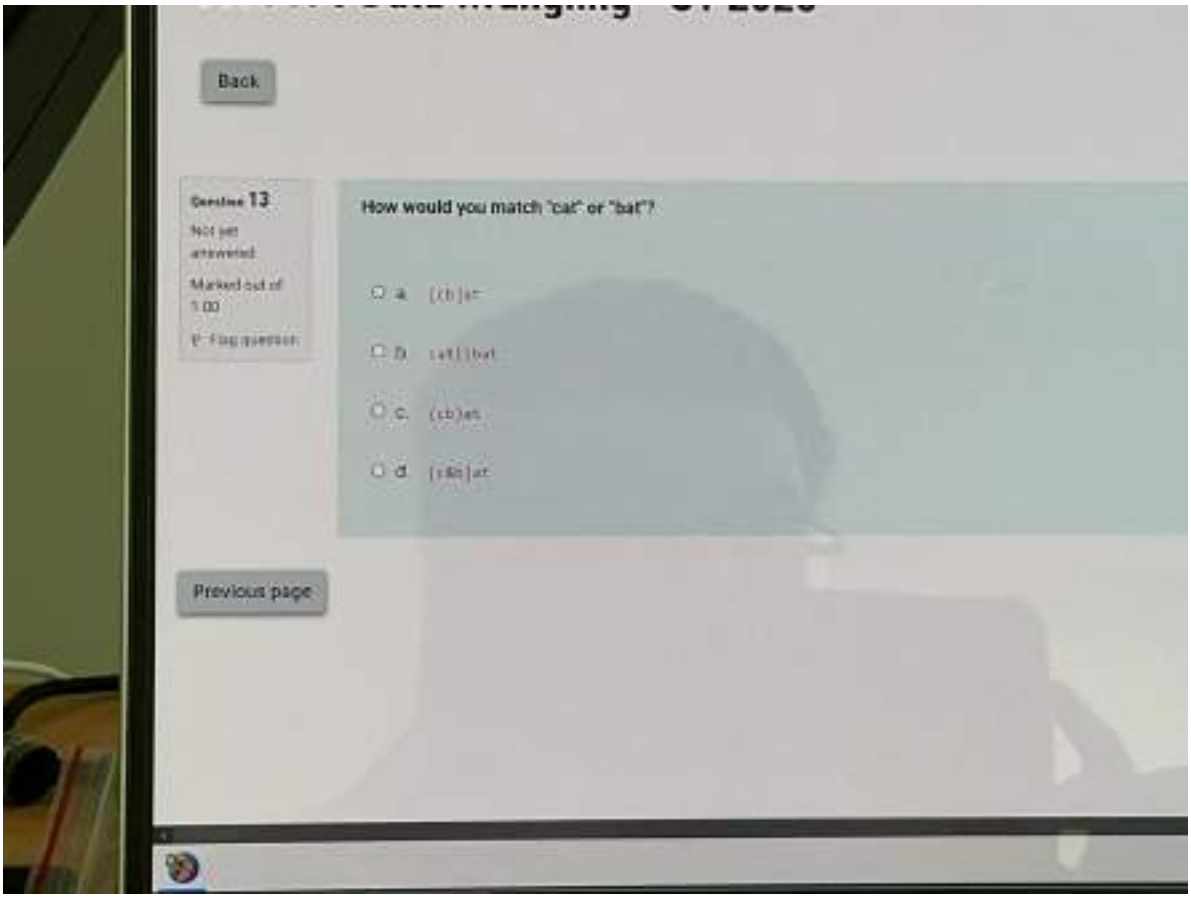
Which metacharacter matches the beginning of a line?

a. ^
b. |
c. \b
d. $

参考答案： `a`

解析： 行首锚点是 `^`。

Question 62



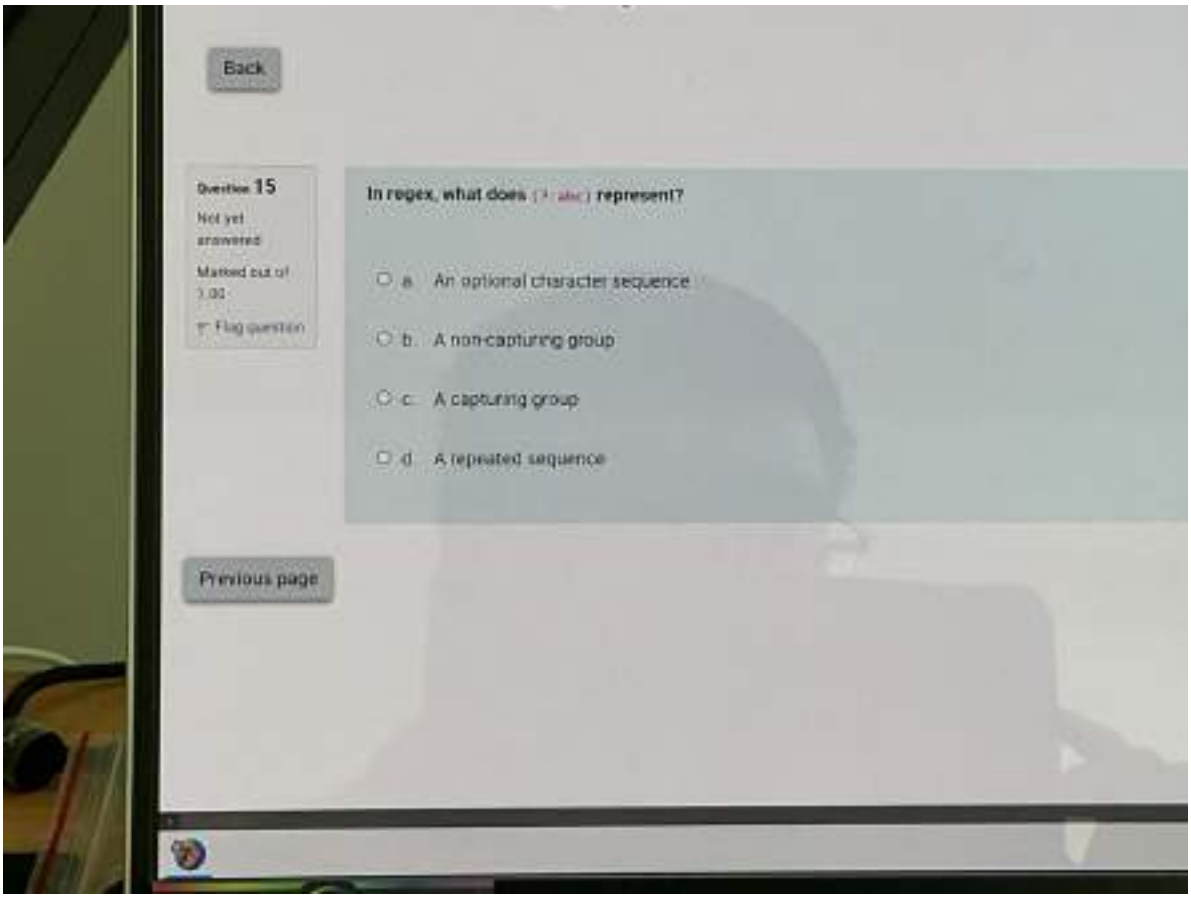
How would you match "cat" or "bat"?

a. [cb]at
b. cat||bat
c. (cb)at
d. [c&b]at

参考答案： `a`

解析： `[cb]at` 表示首字母可以是 `c` 或 `b`。

Question 64



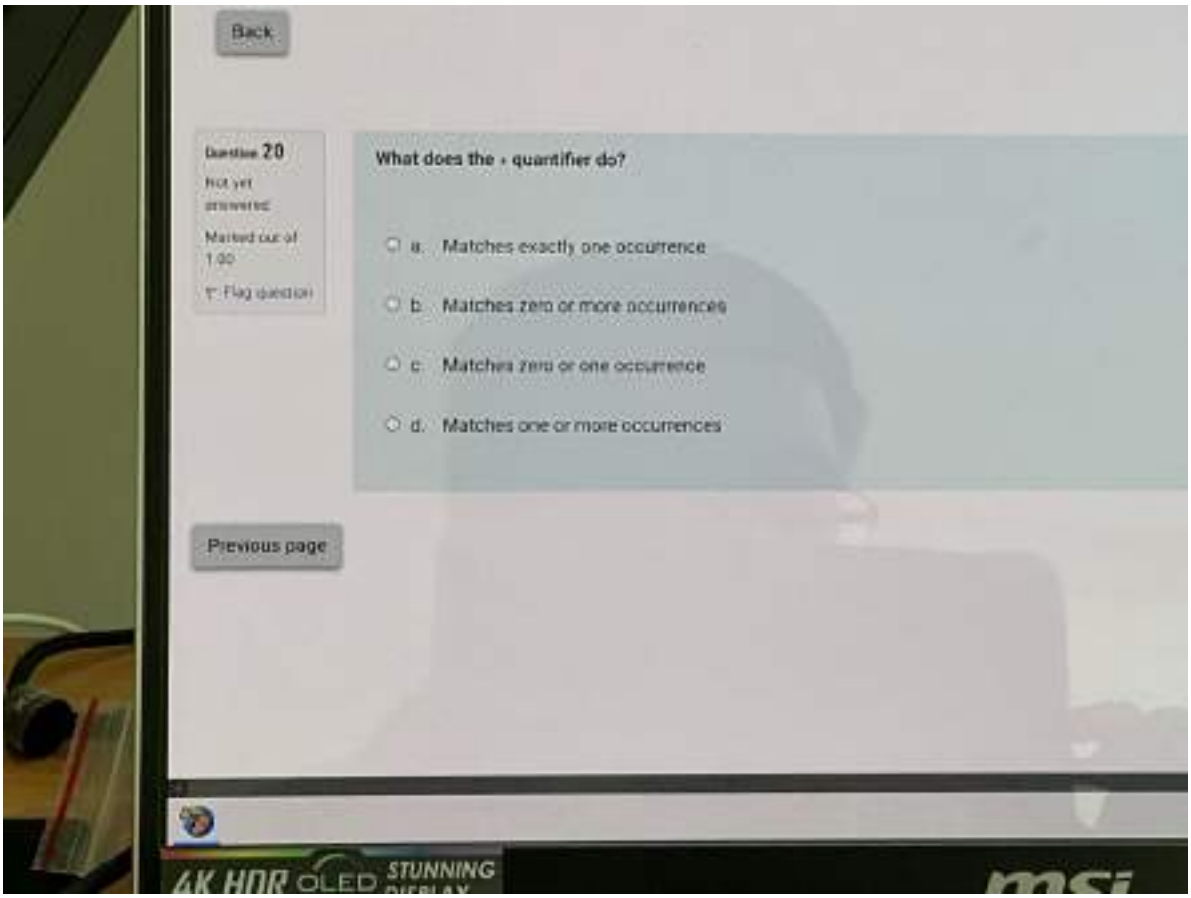
In regex, what does (?:abc) represent?

a. An optional character sequence
b. A non-capturing group
c. A capturing group
d. A repeated sequence

参考答案： `b`

解析： `(?:abc)` 是 non-capturing group。

Question 69



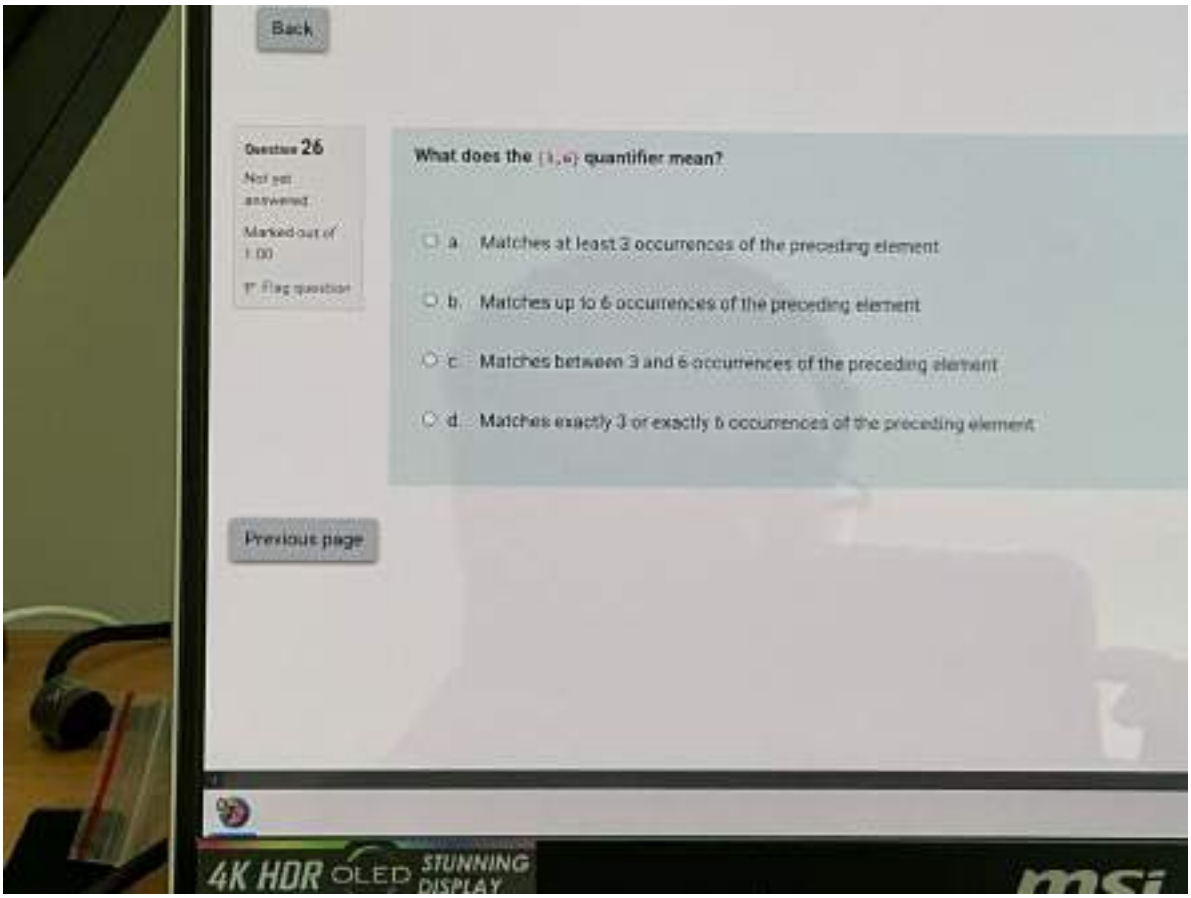
What does the + quantifier do?

a. Matches exactly one occurrence
b. Matches zero or more occurrences
c. Matches zero or one occurrence
d. Matches one or more occurrences

参考答案： `d`

解析： `+` 表示前一项出现 1 次或多次。

Question 73



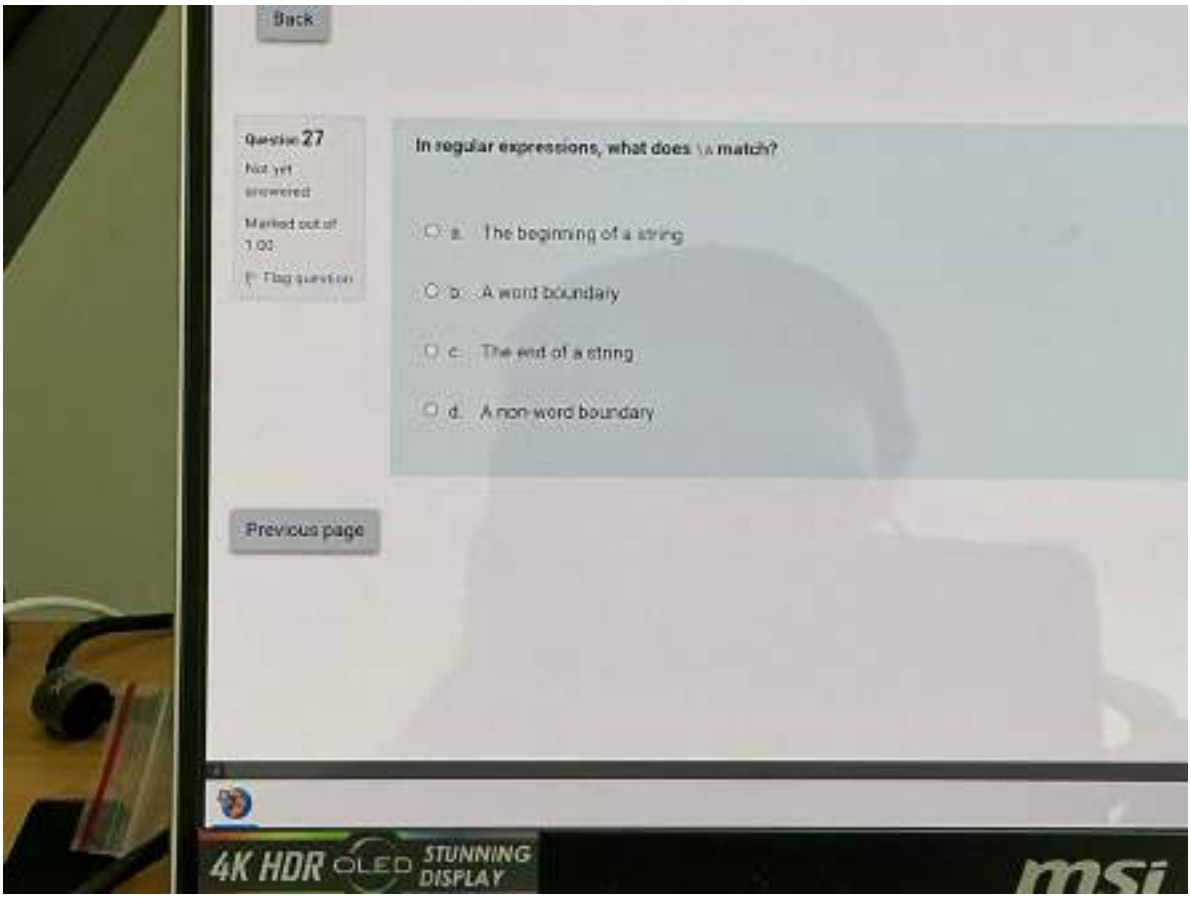
What does the {3,6} quantifier mean?

a. Matches at least 3 occurrences of the preceding element
b. Matches up to 6 occurrences of the preceding element
c. Matches between 3 and 6 occurrences of the preceding element
d. Matches exactly 3 or exactly 6 occurrences of the preceding element

参考答案： `c`

解析： `{3,6}` 就是前一项出现 3 到 6 次。

Question 74



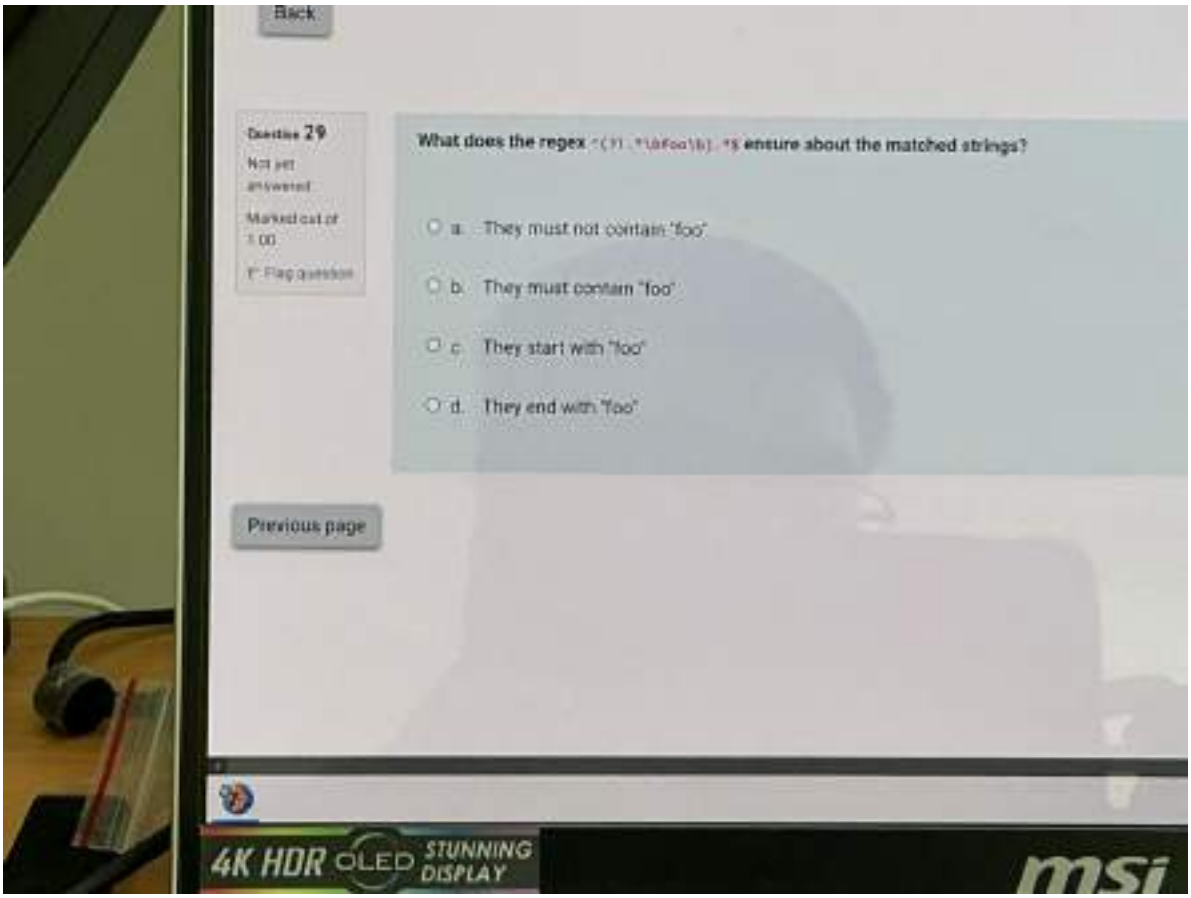
In regular expressions, what does \A match?

a. The beginning of a string
b. A word boundary
c. The end of a string
d. A non-word boundary

参考答案： `a`

解析： `\A` 匹配整个字符串开头。

Question 76



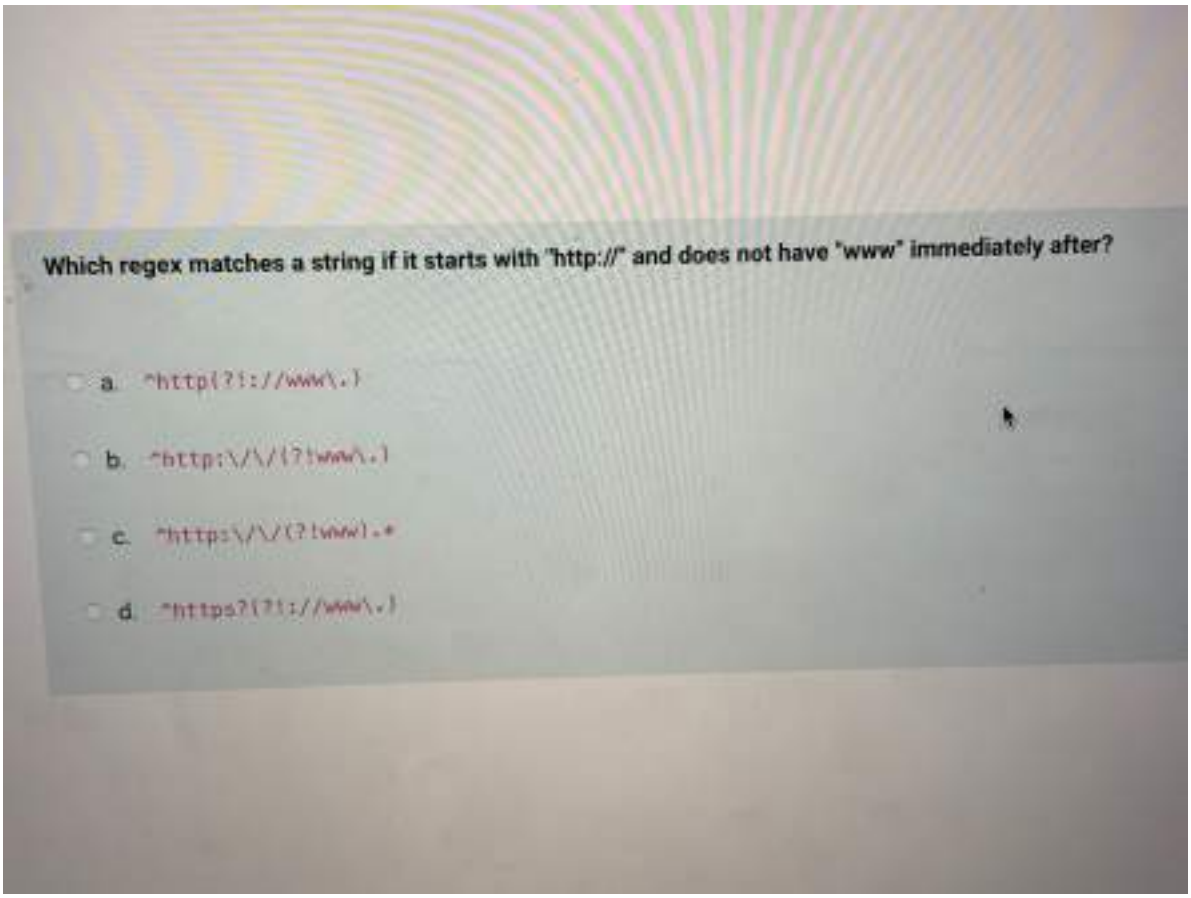
What does the regex `^(?!.*\bfoo\b).*$` ensure about the matched strings?

a. They must not contain "foo"
b. They must contain "foo"
c. They start with "foo"
d. They end with "foo"

参考答案： `a`

解析： 整条表达式用负向前瞻排除了包含独立单词 `foo` 的字符串。

Question 95



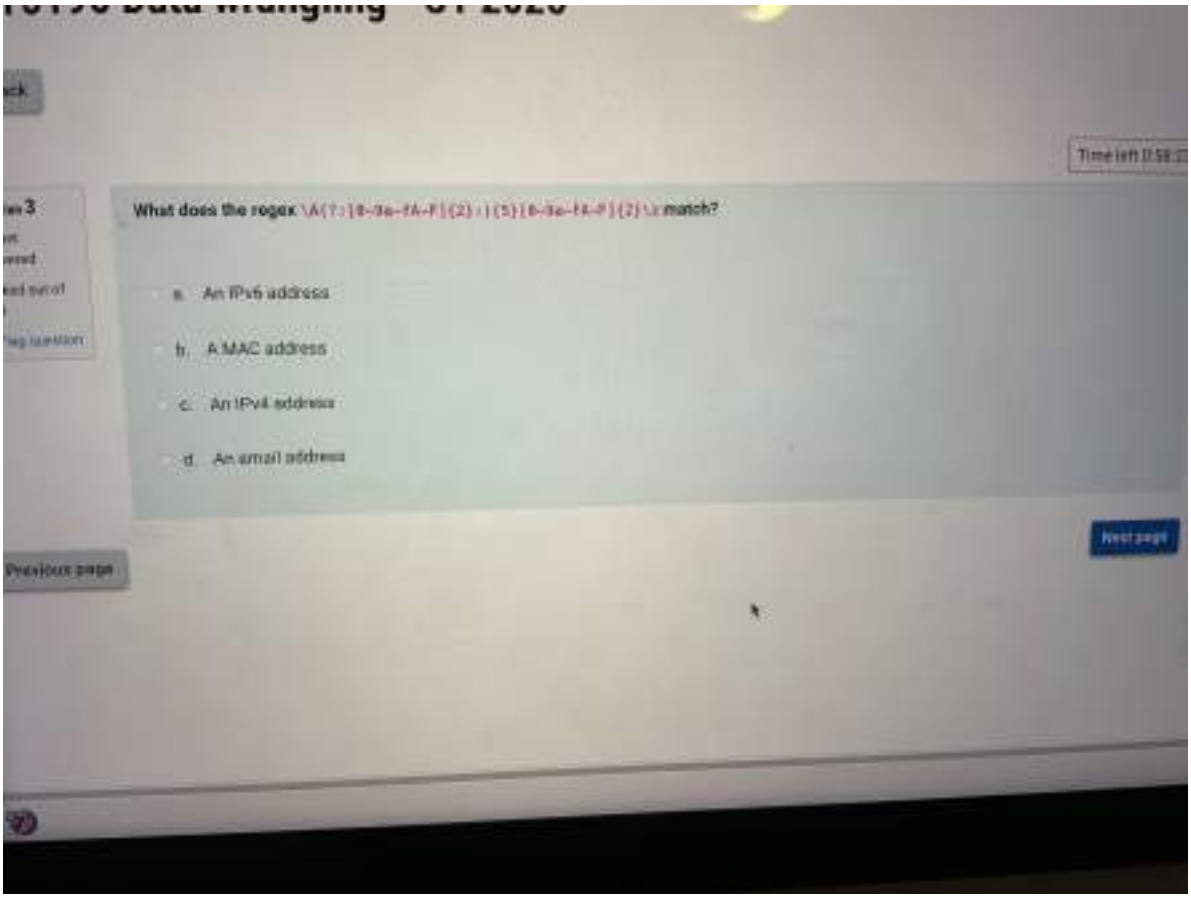
Which regex matches a string if it starts with "http://" and does not have "www" immediately after?

a. ^http(?!://www.)
b. ^http:\/\/(?!www.)
c. ^http:\/\/(?!www).*
d. ^https?(?!://www.)

参考答案： `c`

解析： `^http://(?!www).*` 的思路最直接：以 `http://` 开头，且后面不是 `www`。`b` 也能匹配，但这里先按 `c` 记。

Question 97



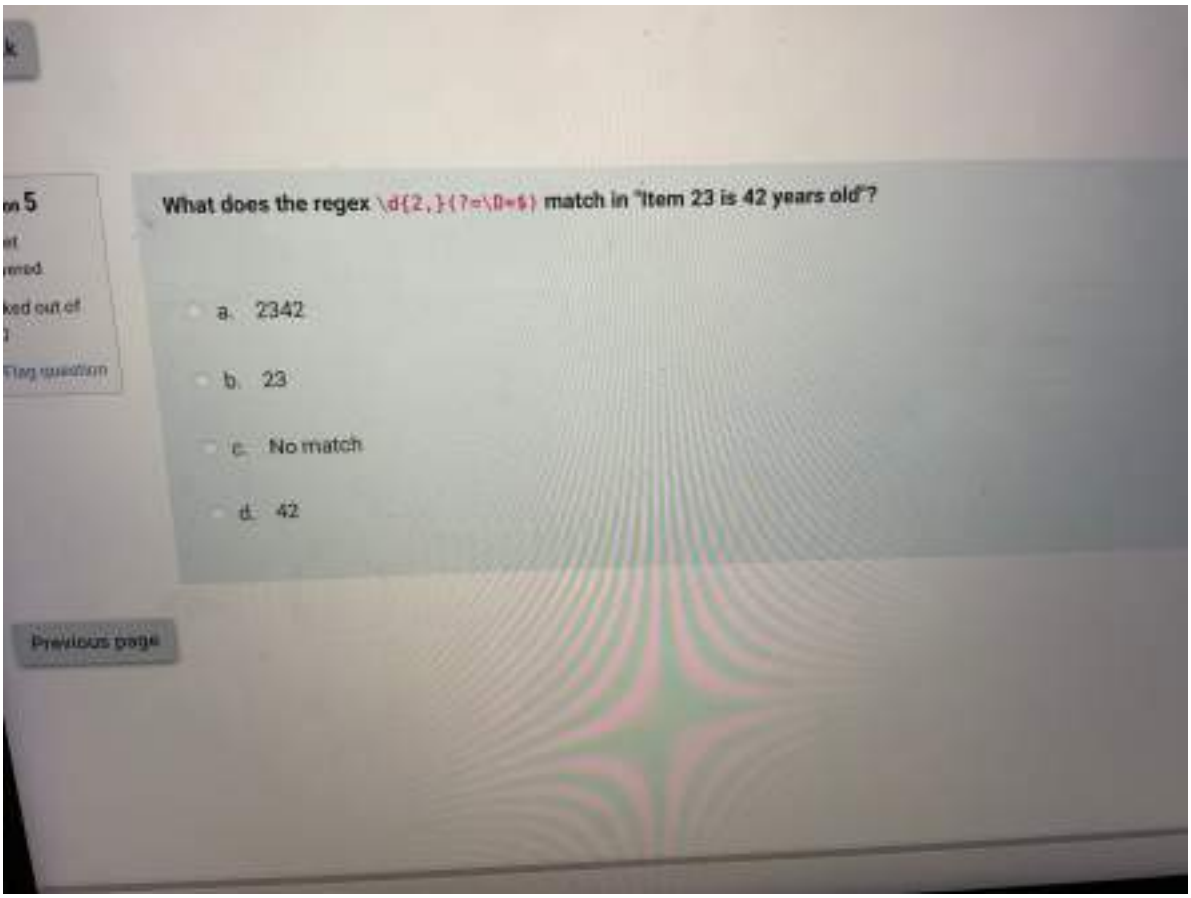
What does the regex \A(?:[0-9a-fA-F]{2}:){5}[0-9a-fA-F]{2}\z match?

a. An IPv6 address
b. A MAC address
c. An IPv4 address
d. An email address

参考答案： `b`

解析： 六组两位十六进制加冒号分隔，是典型 MAC address。

Question 99



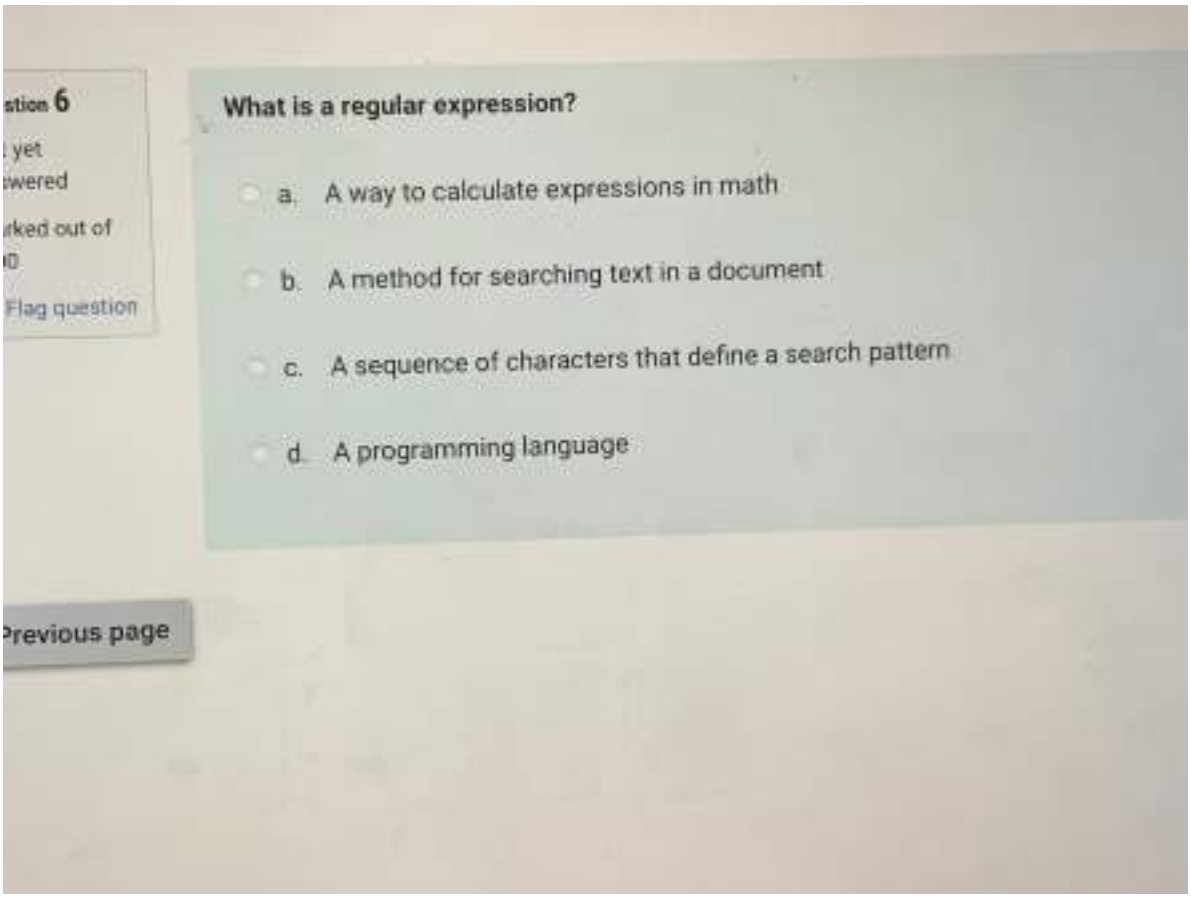
What does the regex \d{2,}(?=\D*$) match in "Item 23 is 42 years old"?

a. 2342
b. 23
c. No match
d. 42

参考答案： `d`

解析： 正则要找的是“最后一个连续数字片段”，所以会匹配到句尾前的 `42`。

Question 100



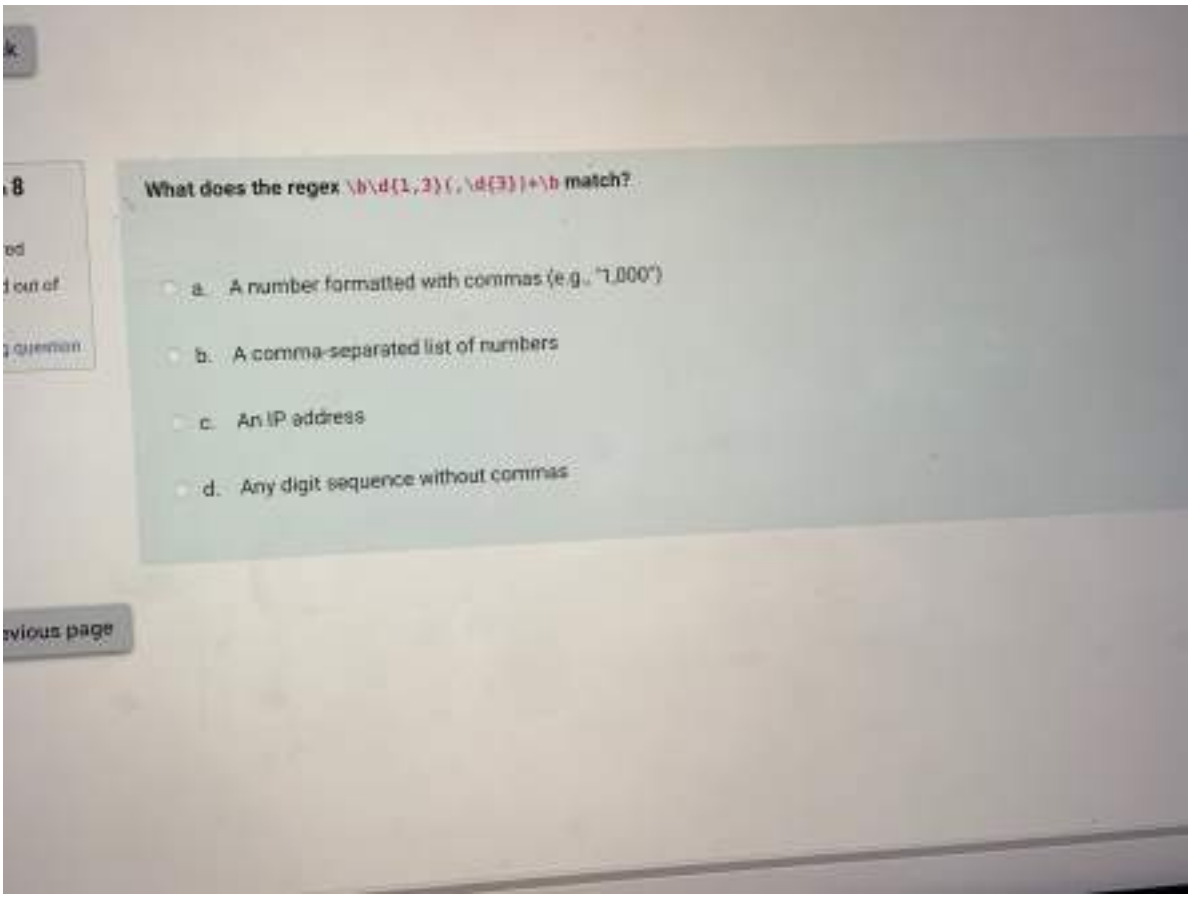
What is a regular expression?

a. A way to calculate expressions in math
b. A method for searching text in a document
c. A sequence of characters that define a search pattern
d. A programming language

参考答案： `c`

解析： regular expression 本质上就是定义搜索模式的一串字符。

Question 102



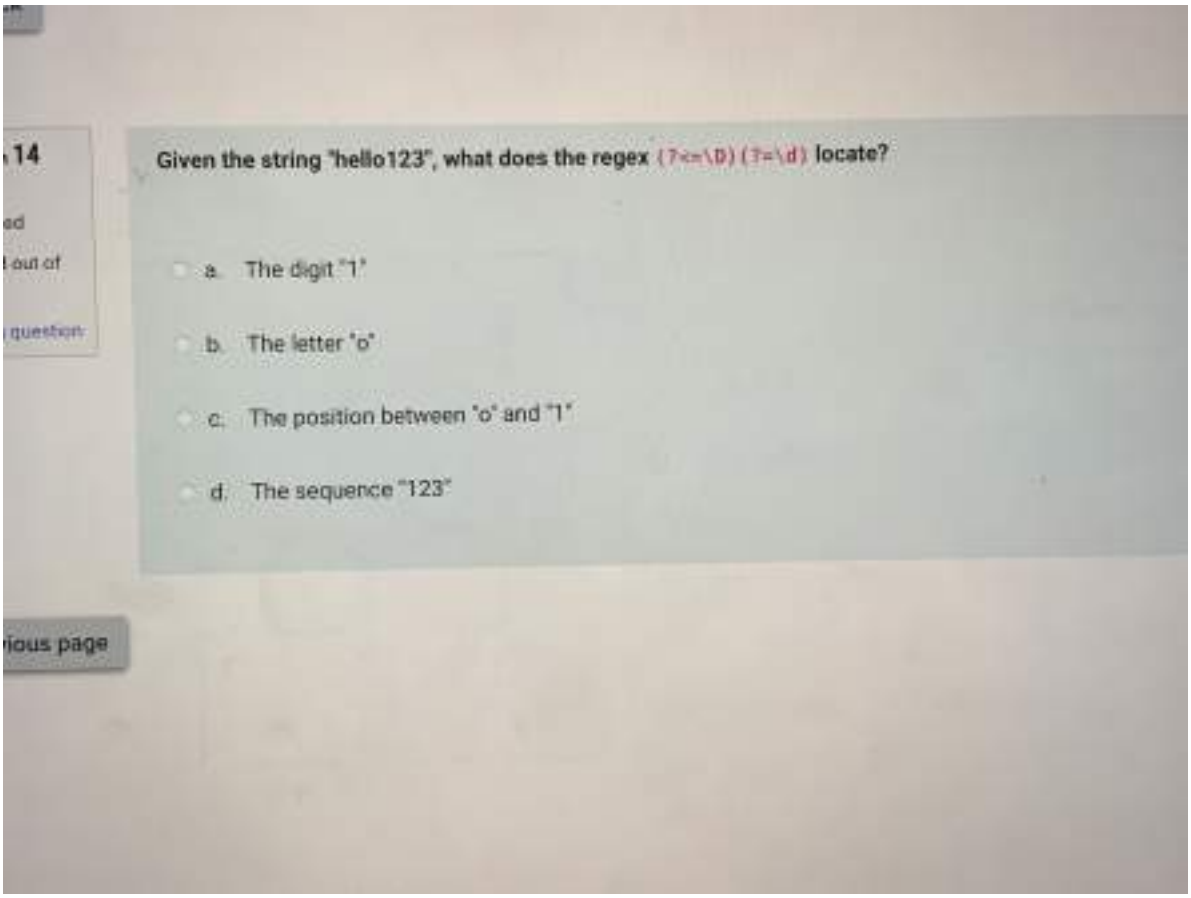
What does the regex \b\d{1,3}(,\d{3})*\b match?

a. A number formatted with commas (e.g., "1,000")
b. A comma-separated list of numbers
c. An IP address
d. Any digit sequence without commas

参考答案： `a`

解析： 这个模式匹配像 `1,000`、`12,345,678` 这样的千分位数字格式。

Question 106



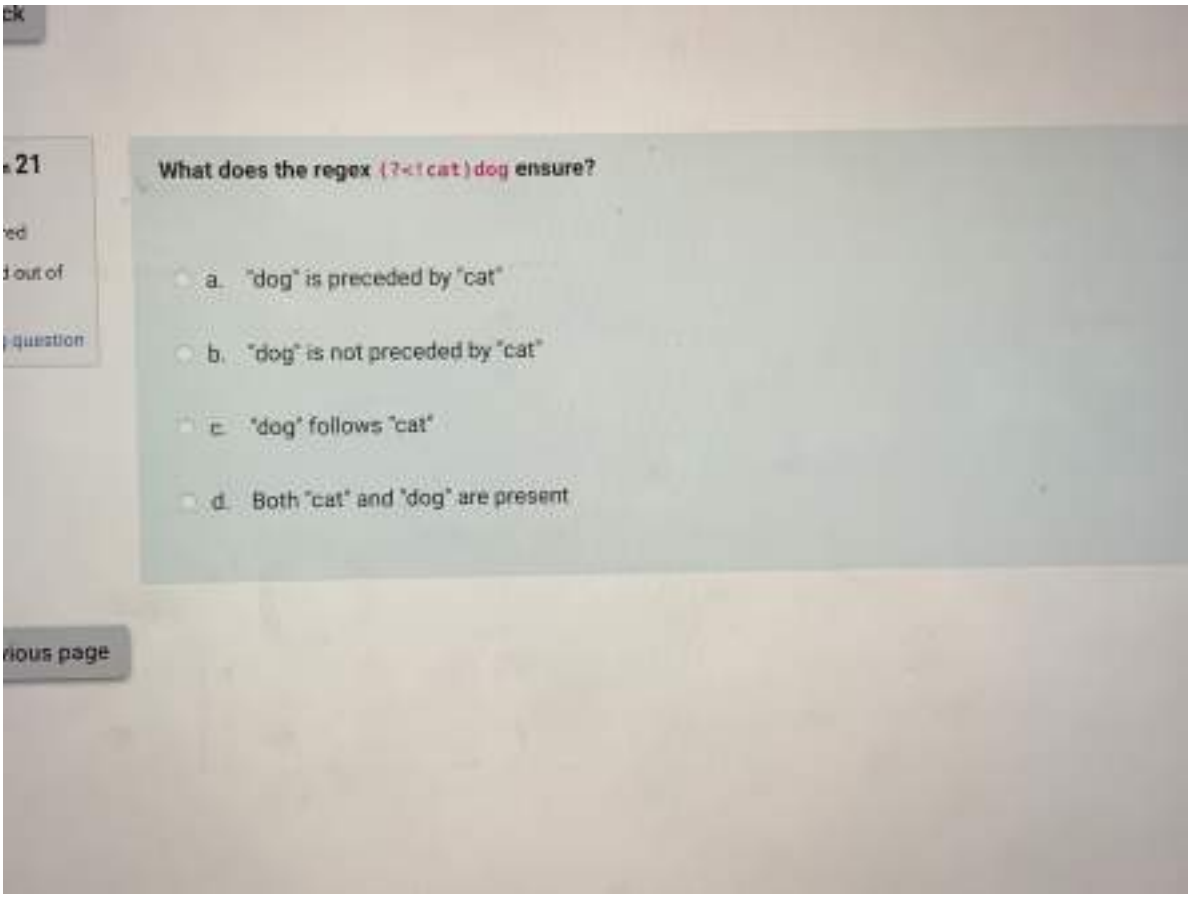
Given the string "hello123", what does the regex (?<=\D)(?=\d) locate?

a. The digit "1"
b. The letter "o"
c. The position between "o" and "1"
d. The sequence "123"

参考答案： `c`

解析： 和 Q30 同义，定位的是 `o` 与 `1` 之间的位置。

Question 112



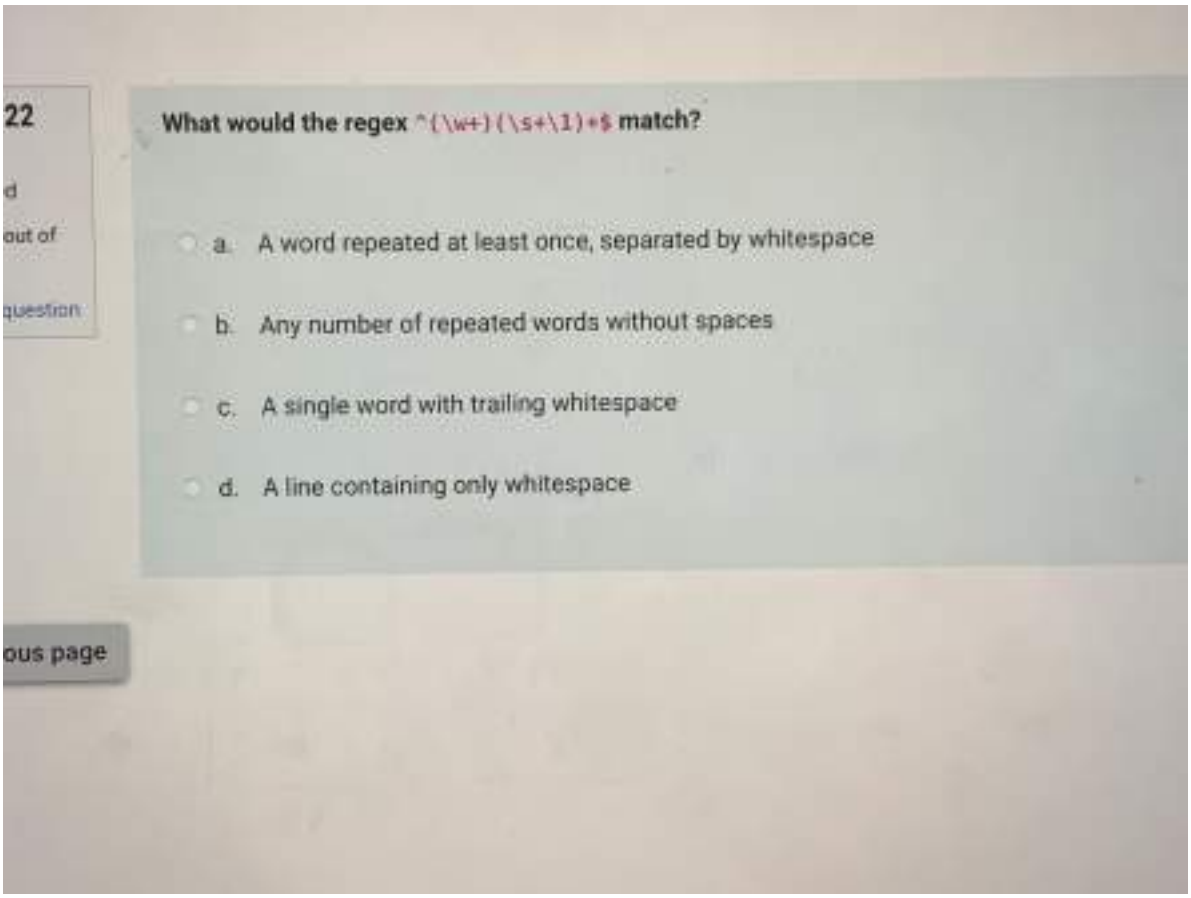
What does the regex (?<!cat)dog ensure?

a. "dog" is preceded by "cat"
b. "dog" is not preceded by "cat"
c. "dog" follows "cat"
d. Both "cat" and "dog" are present

参考答案： `b`

解析： 负向后顾 `(?<!cat)` 要求 `dog` 前面不能是 `cat`。

Question 113



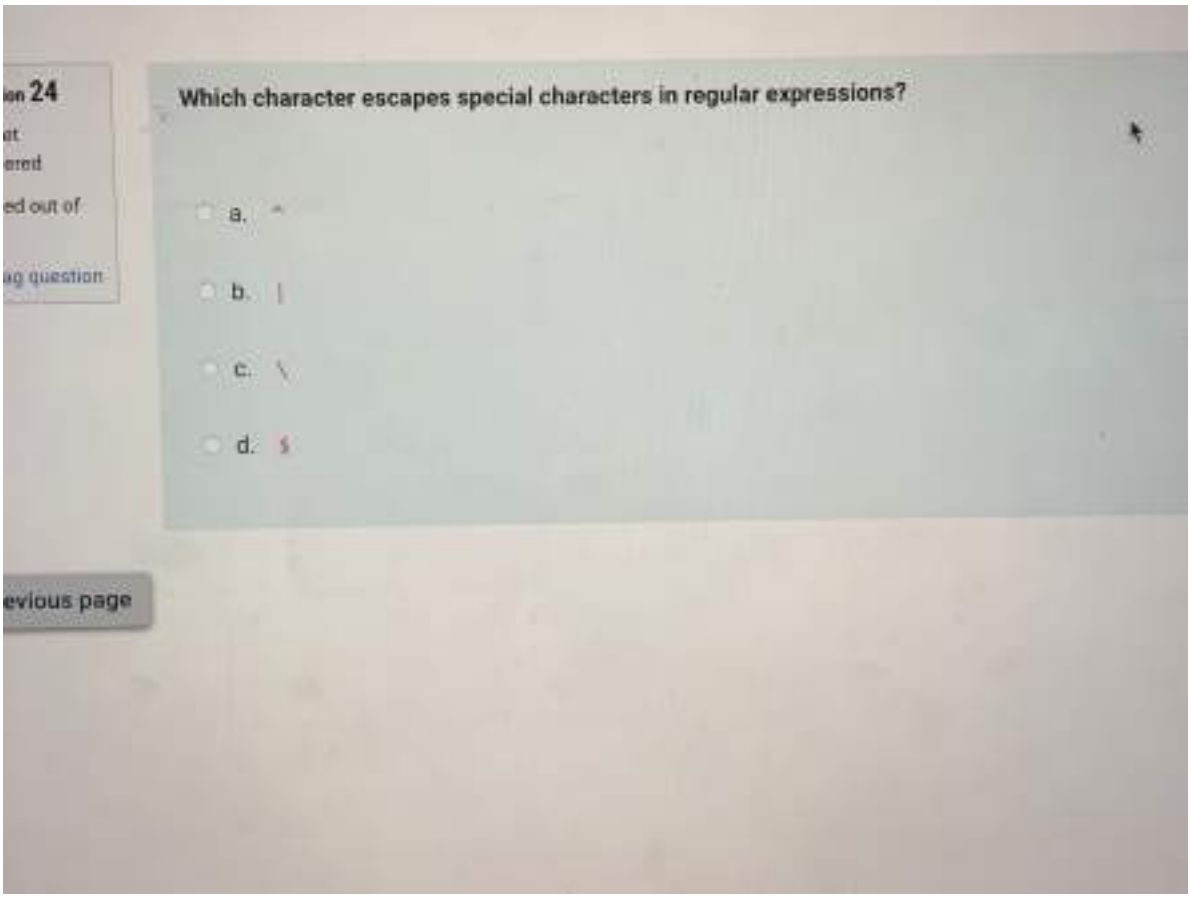
What would the regex ^(\w+)(\s+\1)+$ match?

a. A word repeated at least once, separated by whitespace
b. Any number of repeated words without spaces
c. A single word with trailing whitespace
d. A line containing only whitespace

参考答案： `a`

解析： 这个表达式通过反向引用 `\1` 匹配“同一个词重复出现”。

Question 115



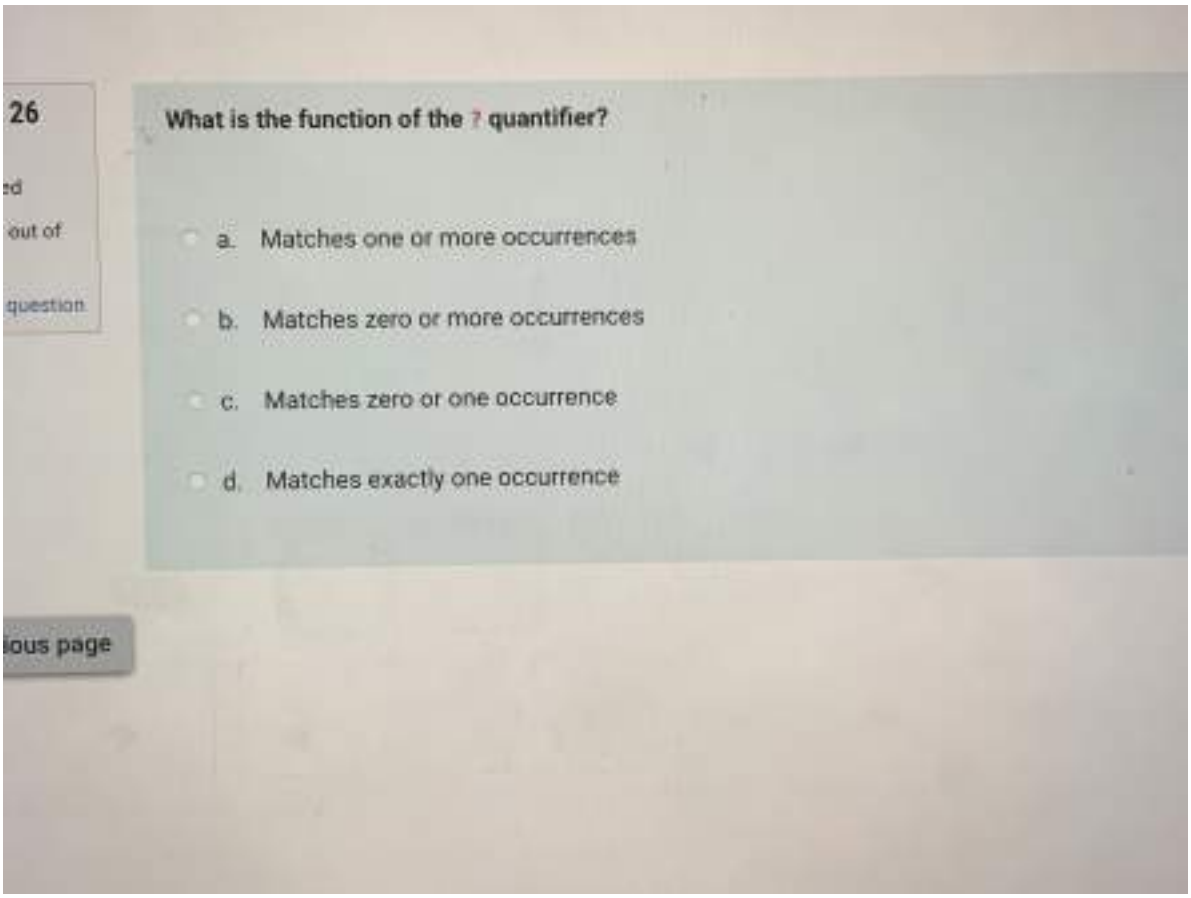
Which character escapes special characters in regular expressions?

a. ^
b. |
c. \
d. $

参考答案： c

解析： 和 Q21 同义，转义特殊字符靠反斜杠。

Question 117



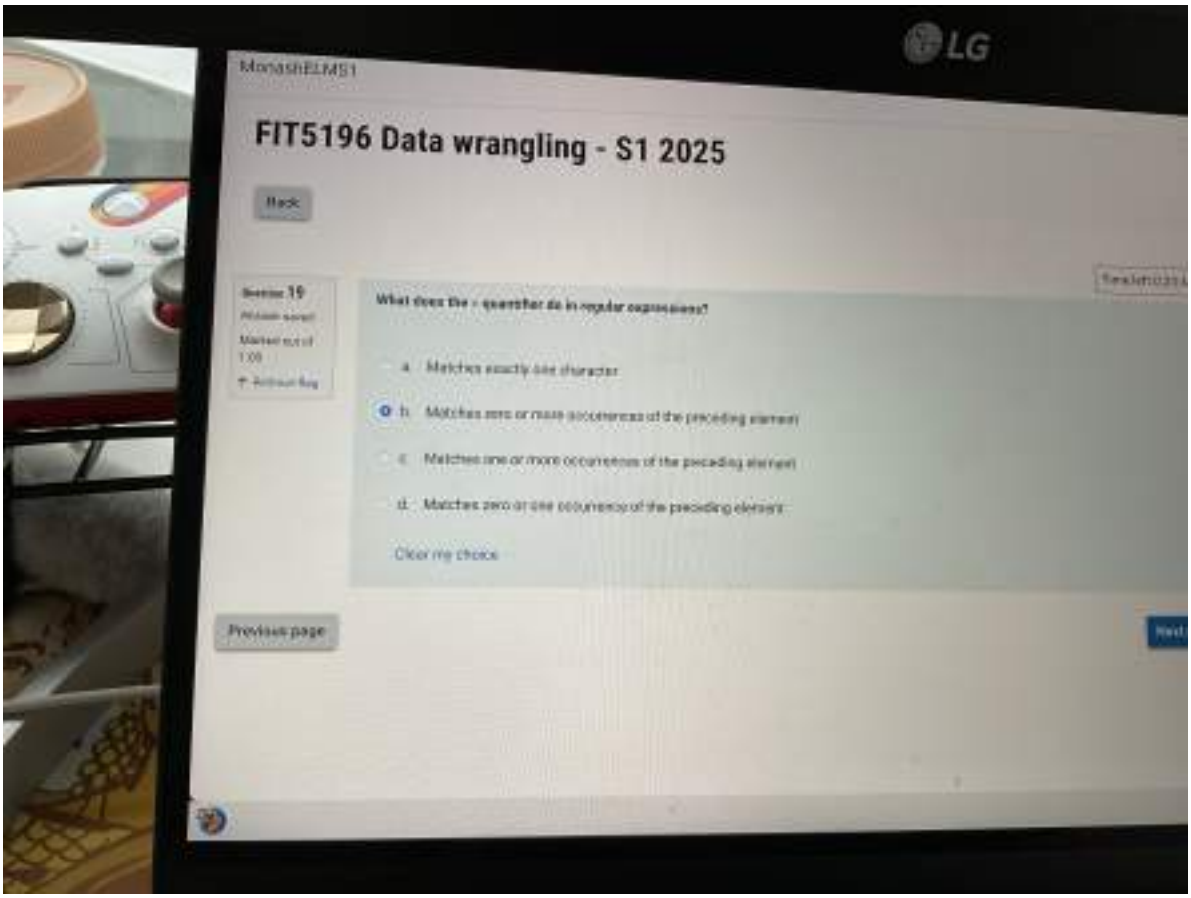
What is the function of the ? quantifier?

a. Matches one or more occurrences
b. Matches zero or more occurrences
c. Matches zero or one occurrence
d. Matches exactly one occurrence

参考答案： `c`

解析： `?` 表示前一项出现 0 次或 1 次。

Question 141



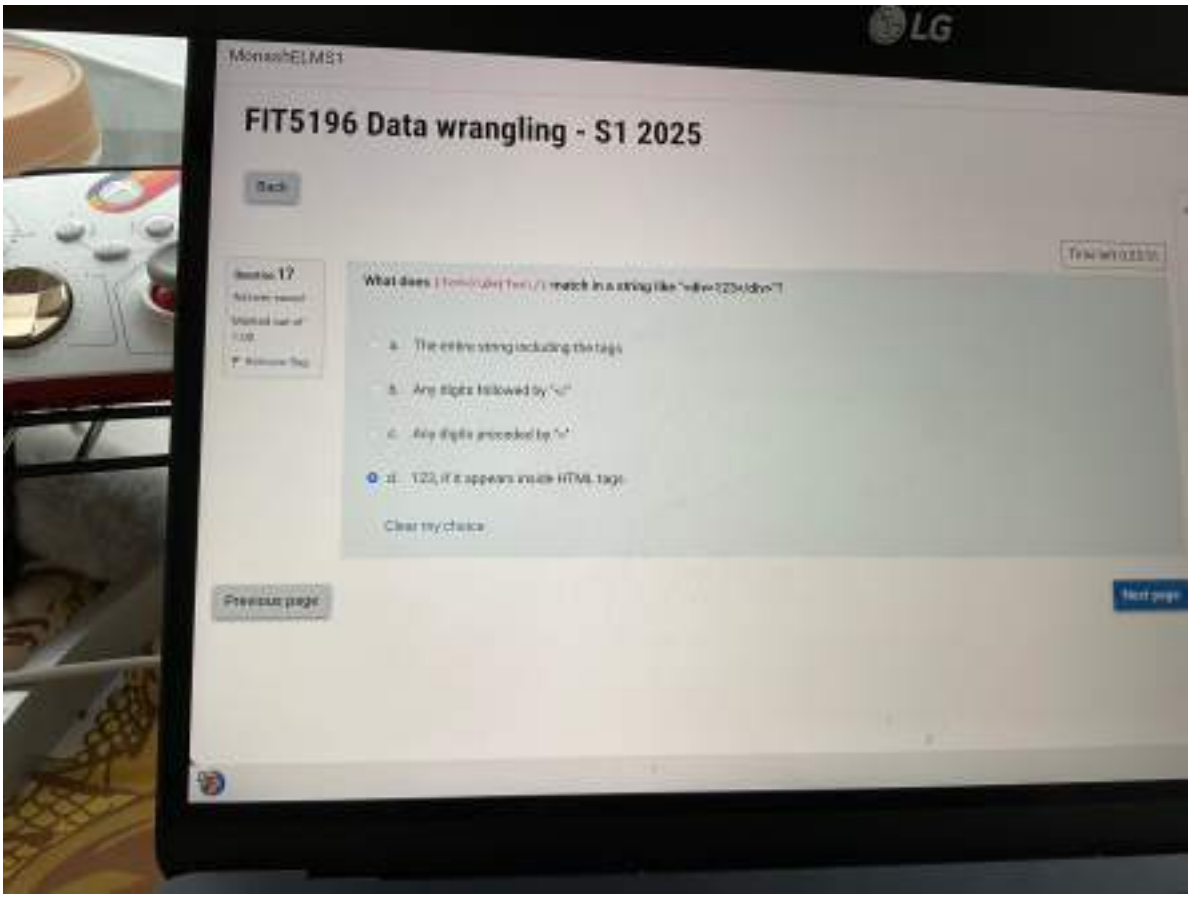
What does the * quantifier do in regular expressions?

a. Matches exactly one character
b. Matches zero or more occurrences of the preceding element
c. Matches one or more occurrences of the preceding element
d. Matches zero or one occurrence of the preceding element

参考答案： `b`

解析： `*` 表示前一项出现 0 次或多次。

Question 142



What does (?<=>)\d+(?=<\/?) match in a string like "

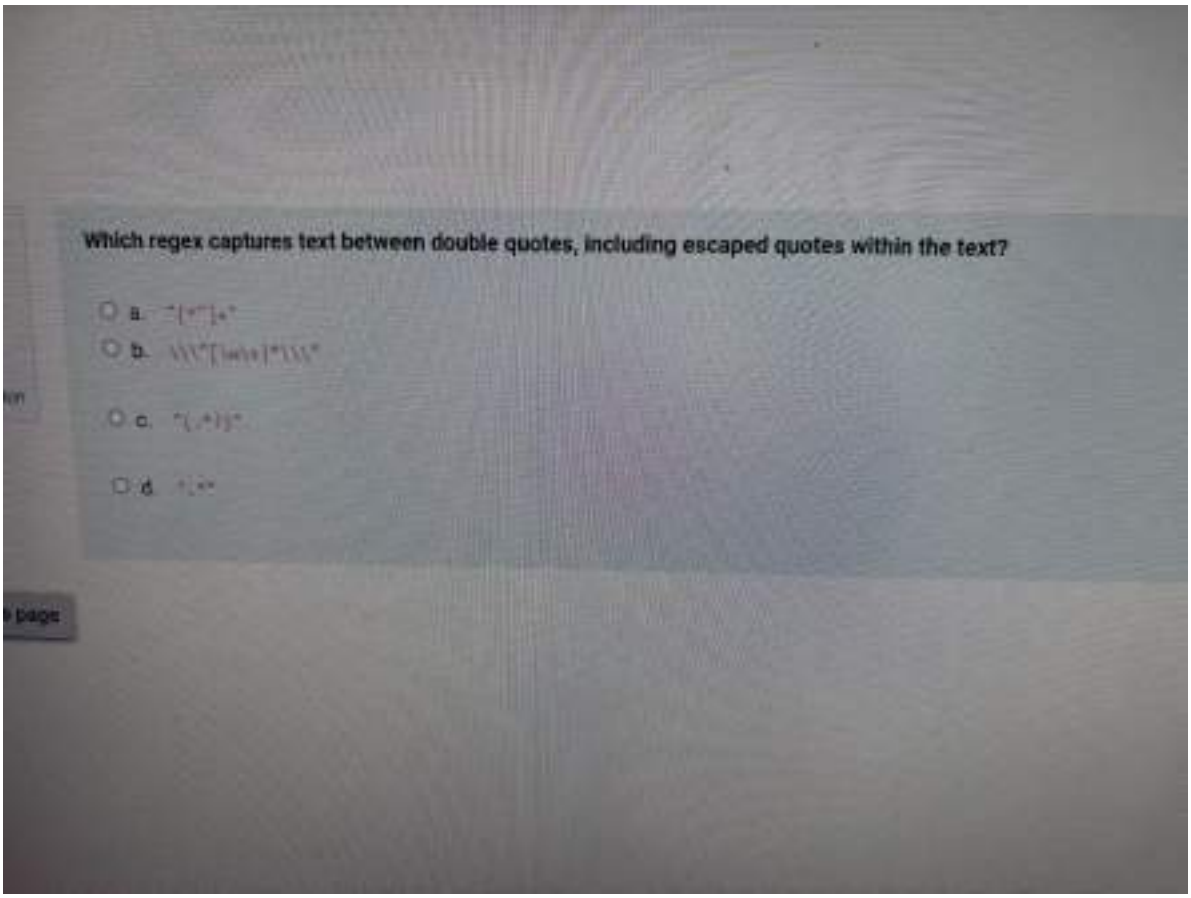
123
"?

a. The entire string including the tags
b. Any digits followed by "</"
c. Any digits preceded by ">"
d. 123, if it appears inside HTML tags

参考答案： `d`

解析： 这个模式会把 HTML 标签之间的数字内容 `123` 抠出来。

Question 144



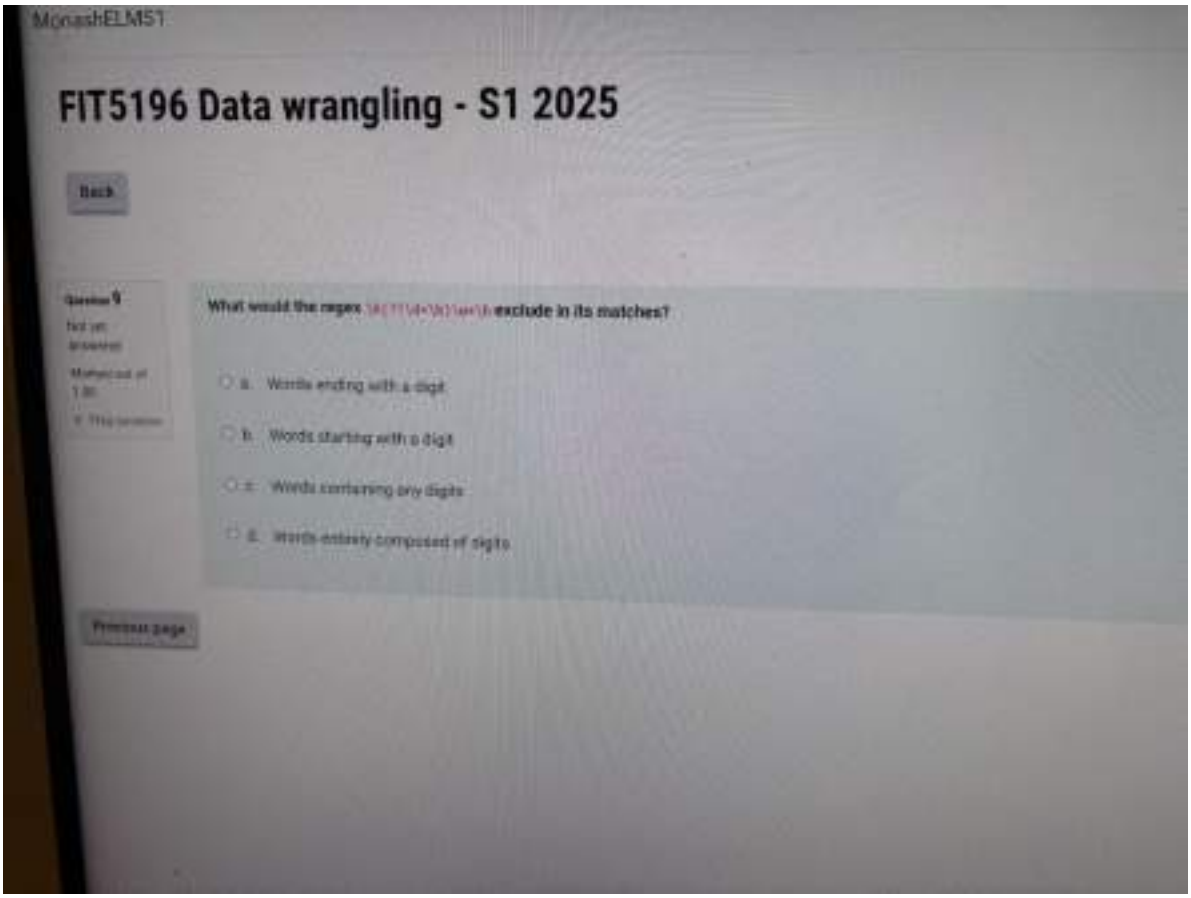
Which regex captures text between double quotes, including escaped quotes within the text?

a. `"[^"]+"`
b. `\\\"[\w\s]*\\\"`
c. `"(.*?)"`
d. `".*"`

参考答案： `c`

解析： 在给出的选项里，`"(.*?)"` 最接近“抓取双引号之间内容”的写法；但这题选项本身不够严谨。

Question 146



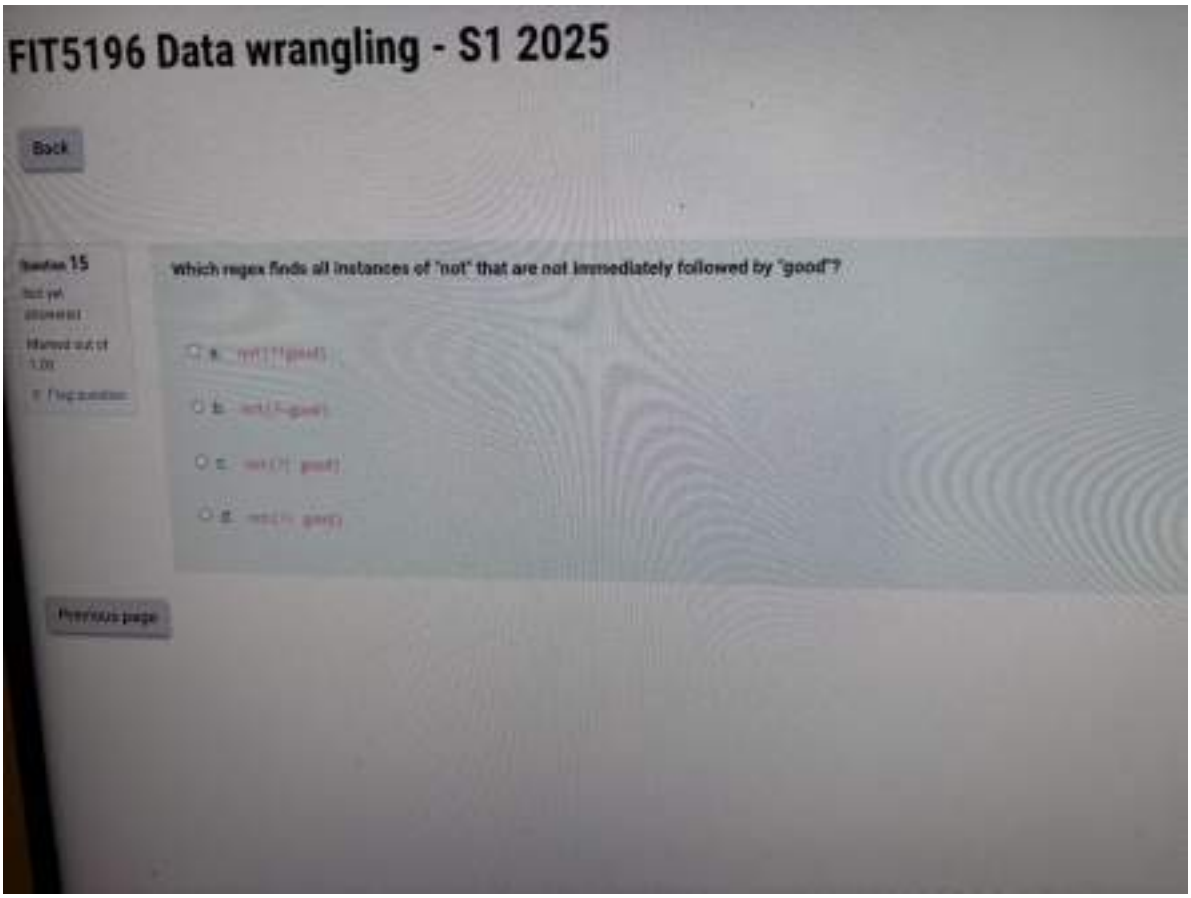
What would the regex `\b(?!\d+\b)\w+\b` exclude in its matches?

a. Words ending with a digit
b. Words starting with a digit
c. Words containing any digits
d. Words entirely composed of digits

参考答案： `d`

解析： 负向前瞻 `(?!\d+\b)` 排除了“整个词就是纯数字”的情况。

Question 147



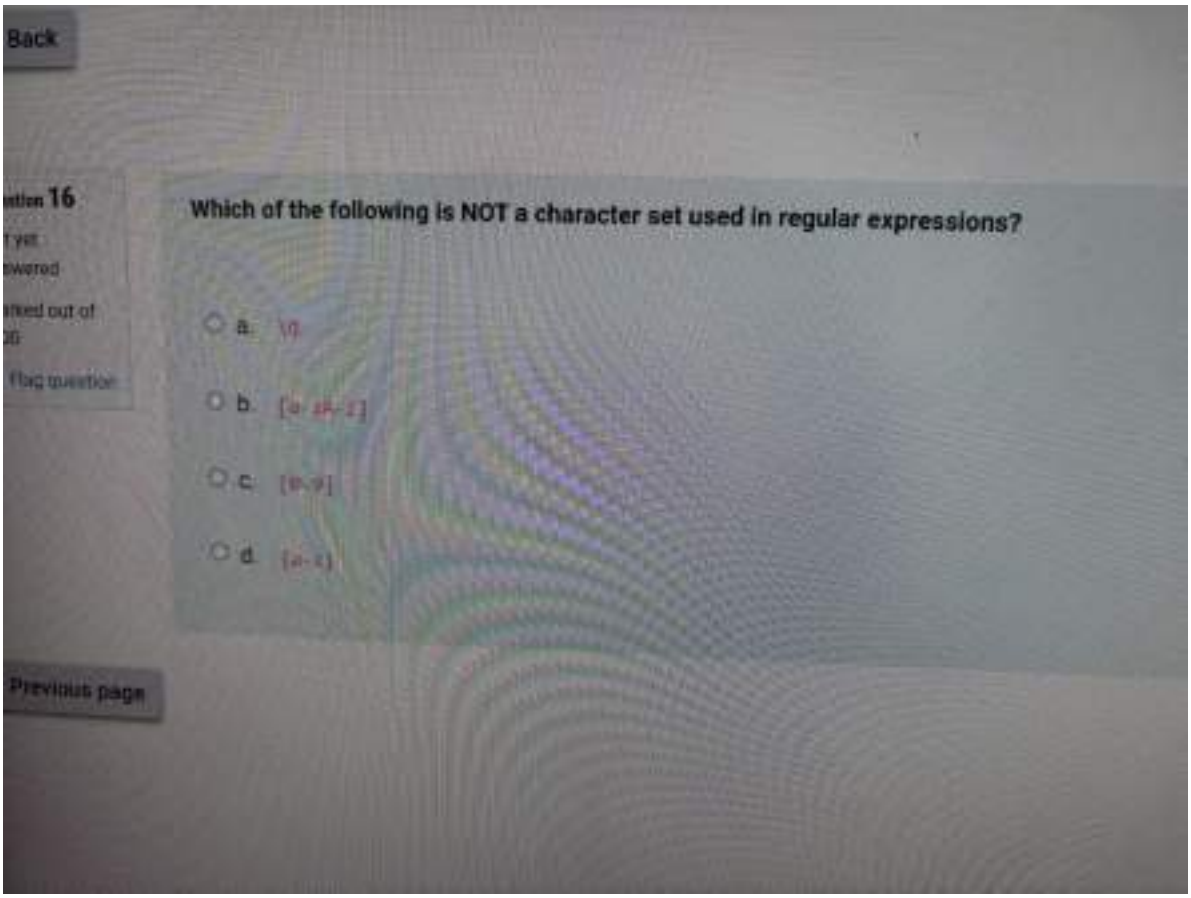
Which regex finds all instances of "not" that are not immediately followed by "good"?

a. `not(?!good)`
b. `not(?=good)`
c. `not(?! good)`
d. `not(?= good)`

参考答案： `a`

解析： 和 Q54 同义，这里标准答案写成了带反引号的 `not(?!good)`。

Question 148



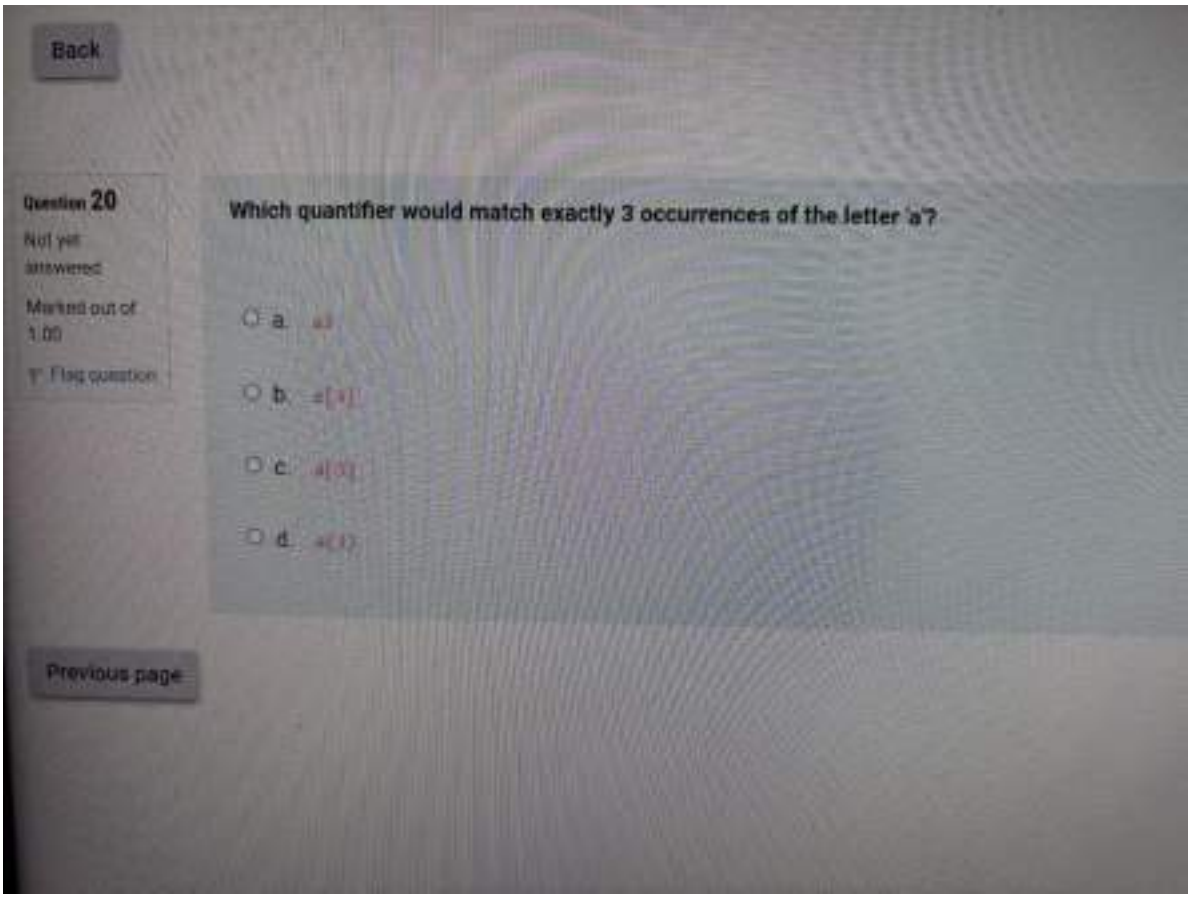
Which of the following is NOT a character set used in regular expressions?

a. \d
b. [a-zA-Z]
c. [0-9]
d. {a-z}

参考答案： `d`

解析： `{a-z}` 不是字符集写法；字符集要用方括号。

Question 150



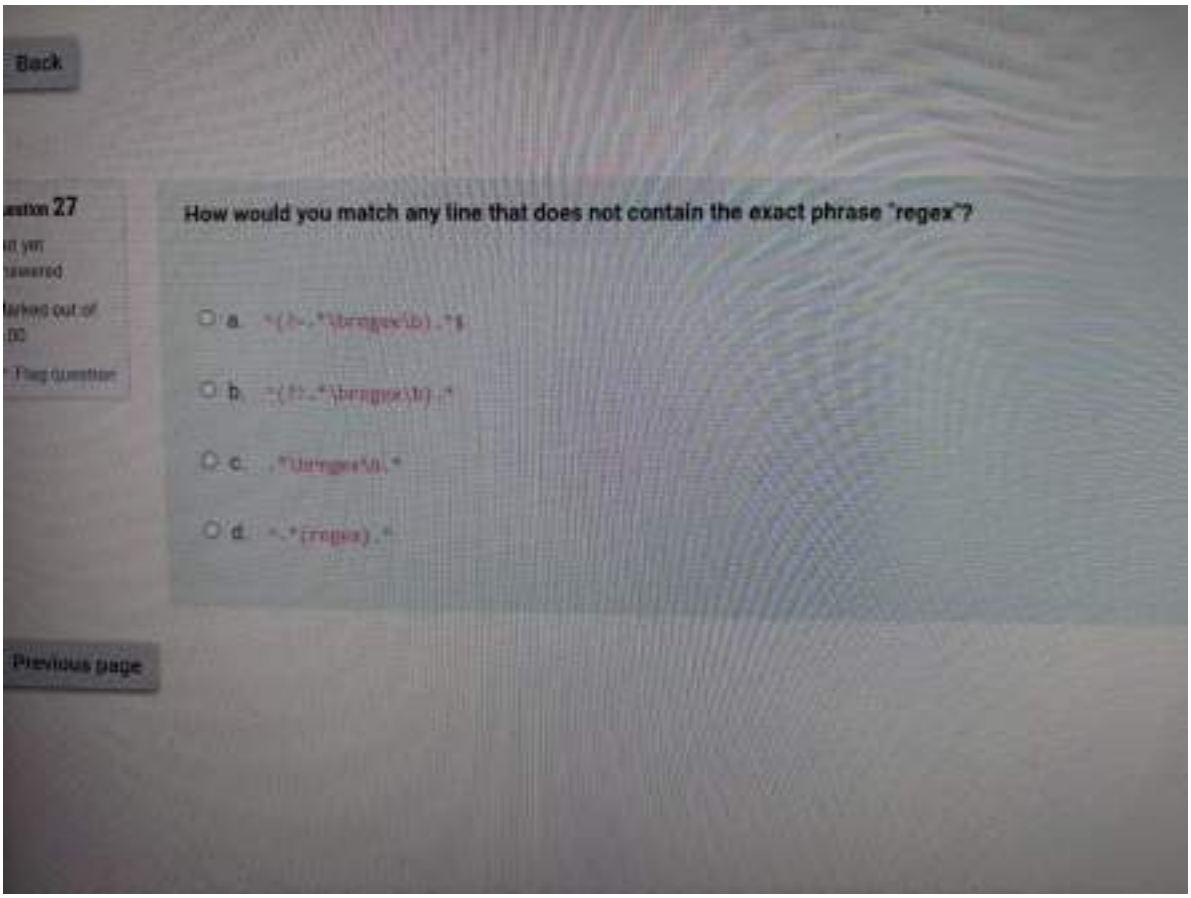
Which quantifier would match exactly 3 occurrences of the letter 'a'?

a. a3
b. a{3}
c. a[3]
d. a(3)

参考答案： `b`

解析： 恰好 3 次要用 `{3}`，所以是 `a{3}`。

Question 156



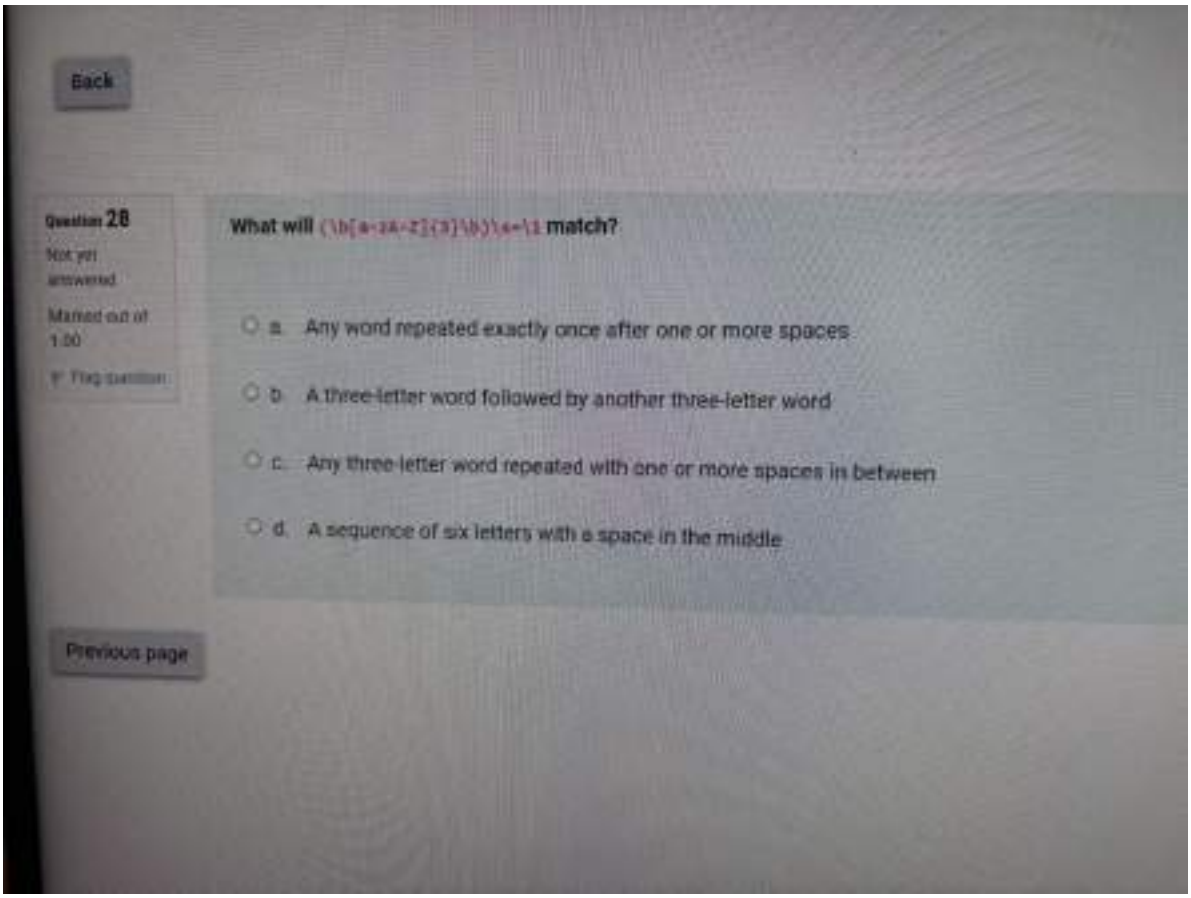
How would you match any line that does not contain the exact phrase "regex"?

a. ^(?=.*\bregex\b).*$
b. ^(?!.*\bregex\b).*
c. .*\bregex\b.*
d. ^.*(regex).*

参考答案： `b`

解析： 负向前瞻 `^(?!.*\bregex\b).*` 用来排除包含 `regex` 的整行。

Question 157



What will (\b[a-zA-Z]{3}\b)\s+\1 match?

a. Any word repeated exactly once after one or more spaces
b. A three-letter word followed by another three-letter word
c. Any three-letter word repeated with one or more spaces in between
d. A sequence of six letters with a space in the middle

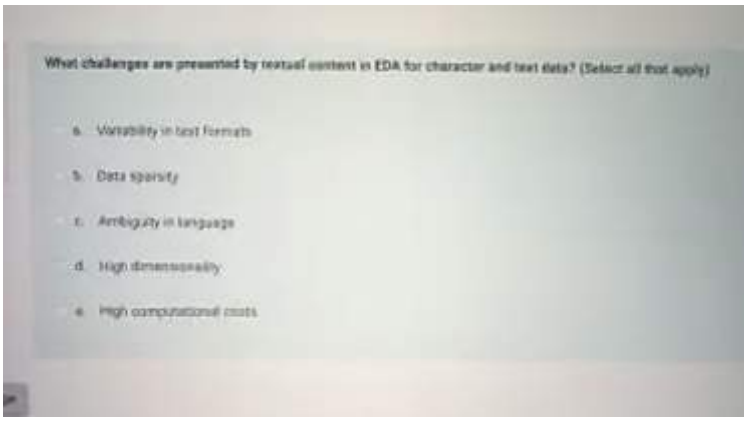
参考答案： `c`

解析： 这里要求“三个字母的单词重复出现”，且中间有空格。

Week 4 (41 题)

- 范围：EDA & Pre-processing + Week 4 parsing + mock quiz text pre-processing block

Question 2



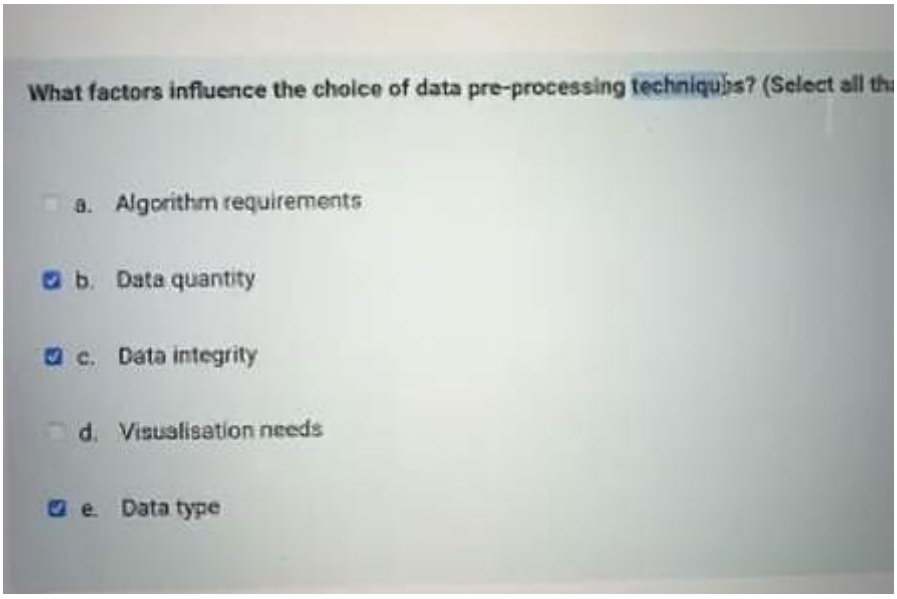
What challenges are presented by textual content in EDA for character and text data? (Select all that apply)

a. Variability in text formats
b. Data sparsity
c. Ambiguity in language
d. High dimensionality
e. High computational costs

参考答案： `a, b, c, d, e`

解析： 正确项是 `a`: Variability in text formats, `b`: Data sparsity, `c`: Ambiguity in language, `d`: High dimensionality, `e`: High computational costs。这些选项共同对应课程里的正确定义。

Question 4



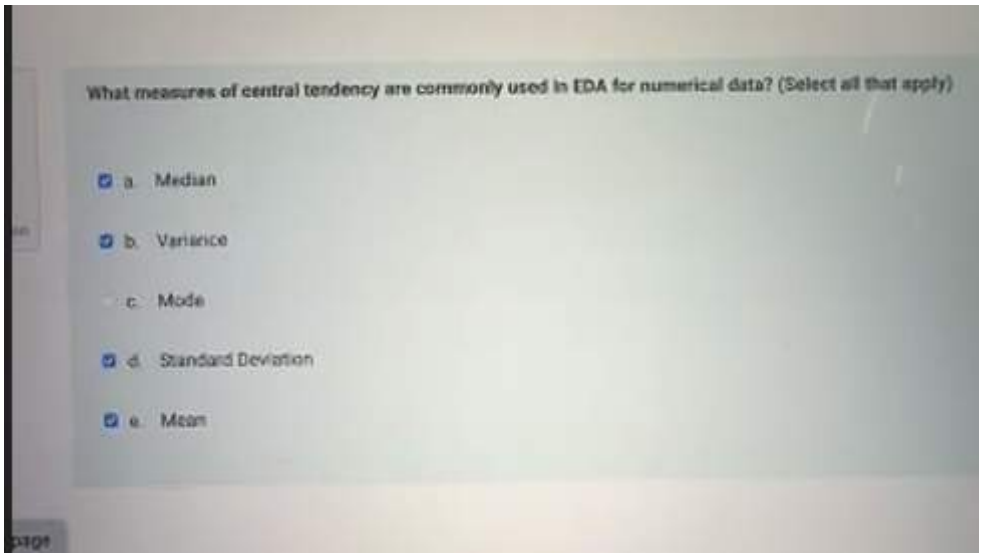
What factors influence the choice of data pre-processing techniques? (Select all that apply)

a. Algorithm requirements
b. Data quantity
c. Data integrity
d. Visualisation needs
e. Data type

参考答案： `a, b, c, d, e`

解析： 正确项是 `a`: Algorithm requirements, `b`: Data quantity, `c`: Data integrity, `d`: Visualisation needs, `e`: Data type。这些选项共同对应课程里的正确定义。

Question 5



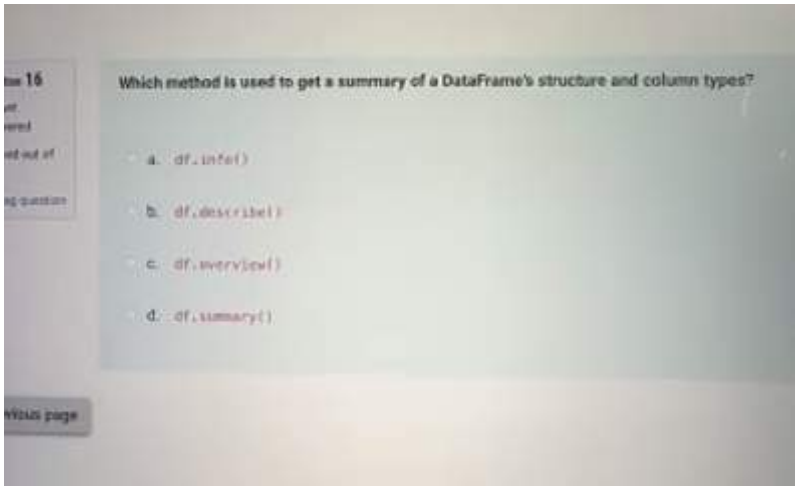
What measures of central tendency are commonly used in EDA for numerical data? (Select all that apply)

a. Median
b. Variance
c. Mode
d. Standard Deviation
e. Mean

参考答案： `a, c, e`

解析： 中心趋势看的是数据“中心”在哪里，常见是 mean / median / mode；variance 和 standard deviation 属于离散程度。

Question 16



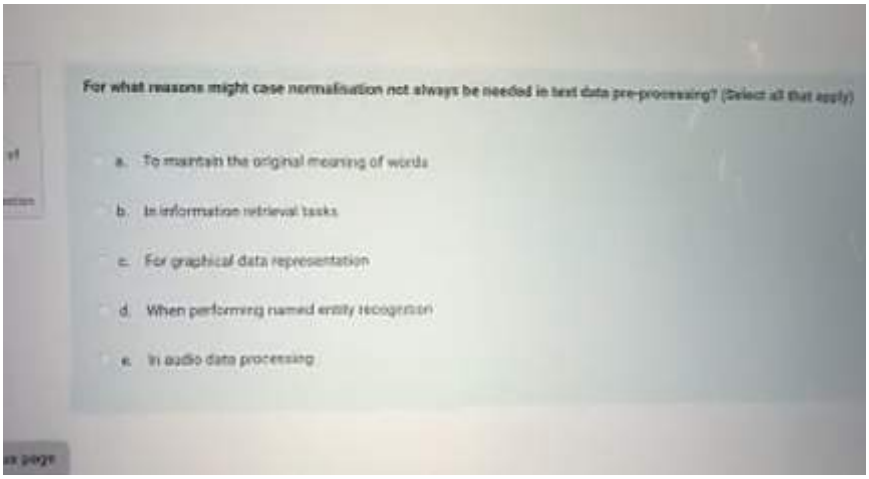
Which method is used to get a summary of a DataFrame's structure and column types?

a. df.info()
b. df.describe()
c. df.overview()
d. df.summary()

参考答案： `a`

解析： `df.info()`` 会给出列名、非空数、dtype 等结构信息；describe()` 更偏统计摘要。

Question 23



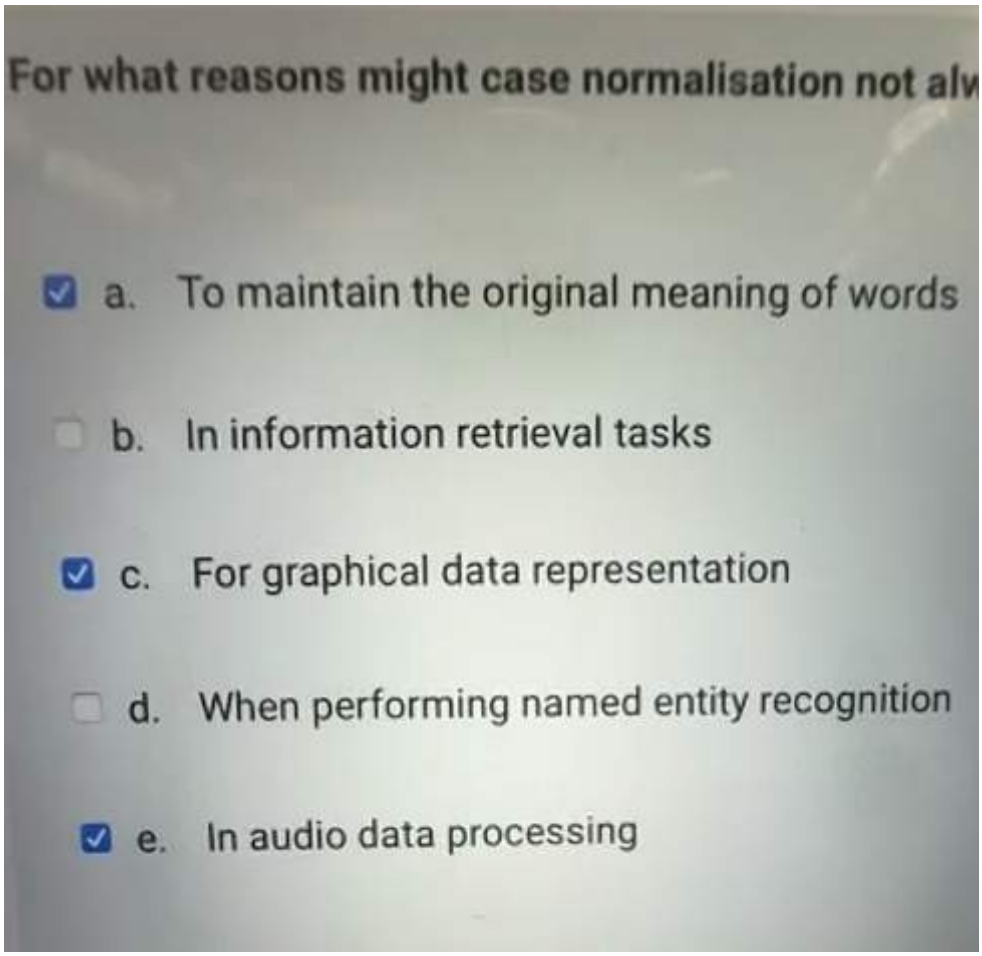
For what reasons might case normalisation not always be needed in text data pre-processing? (Select all that apply)

a. To maintain the original meaning of words
b. In information retrieval tasks
c. For graphical data representation
d. When performing named entity recognition
e. In audio data processing

参考答案： `a, b, d`

解析： case normalisation 并不总是必须；像 named entity recognition 或需要保留原大小写信息时，大小写本身就有语义。

Question 24



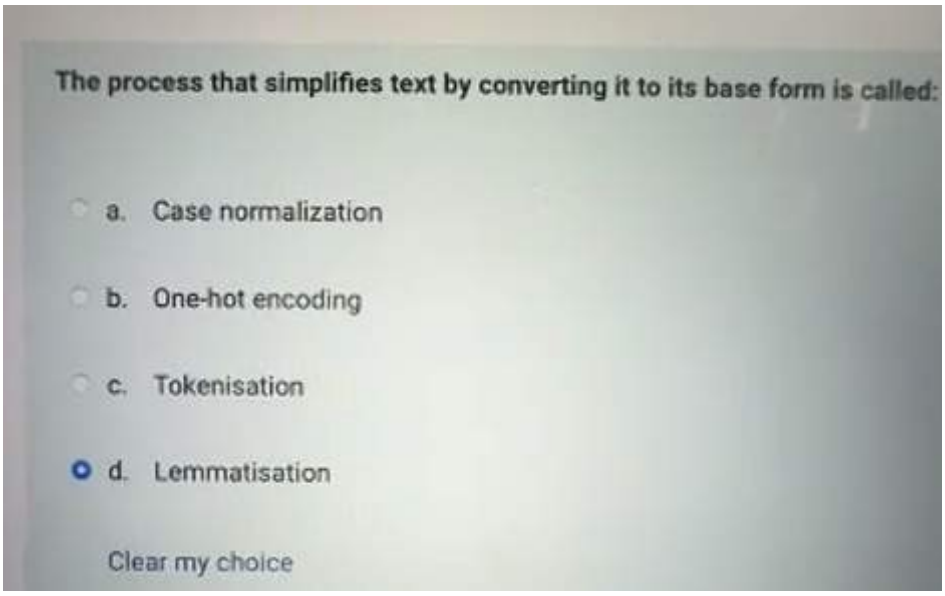
For what reasons might case normalisation not always be needed in text data pre-processing? (Select all that apply)

a. To maintain the original meaning of words
b. In information retrieval tasks
c. For graphical data representation
d. When performing named entity recognition
e. In audio data processing

参考答案： `a, b, d`

解析： 这题和 Q23 同义，核心也是“某些任务里大小写信息本身有用，所以不一定统一转小写”。

Question 25



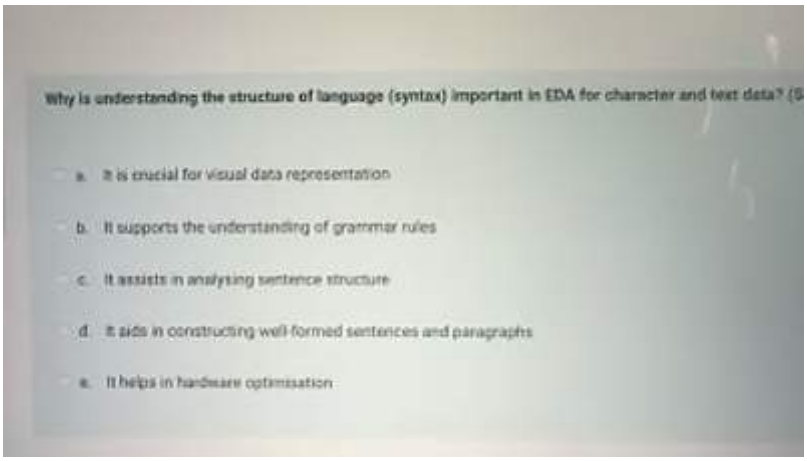
The process that simplifies text by converting it to its base form is called:

a. Case normalization
b. One-hot encoding
c. Tokenisation
d. Lemmatisation

参考答案： `d`

解析： 把词还原成词典中的基本形式是 lemmatisation。

Question 27



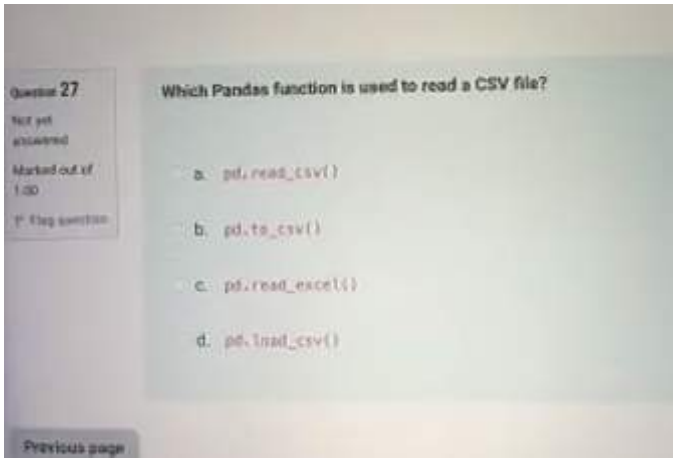
Why is understanding the structure of language (syntax) important in EDA for character and text data? (Select all that apply)

a. It is crucial for visual data representation
b. It supports the understanding of grammar rules
c. It assists in analysing sentence structure
d. It aids in constructing well-formed sentences and paragraphs
e. It helps in hardware optimisation

参考答案： `b, c, d`

解析： syntax 关注语言结构与语法规则，有助于分析句子结构并生成/识别 well-formed text。

Question 28



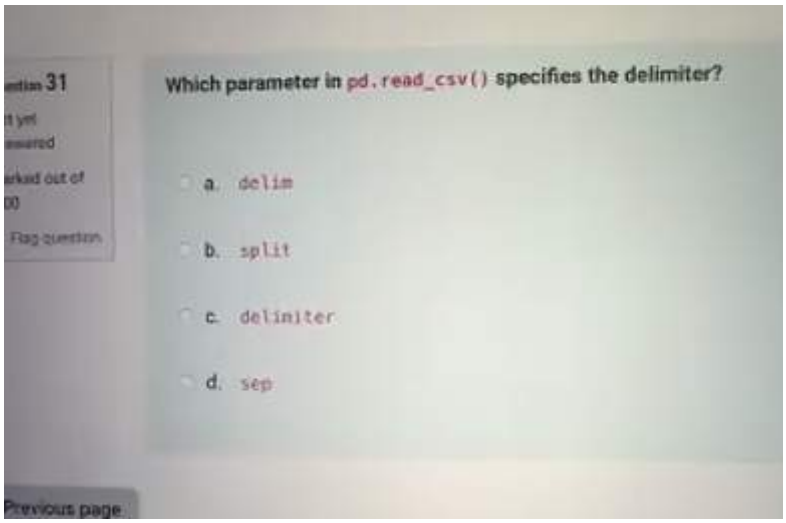
Which Pandas function is used to read a CSV file?

a. pd.read_csv()
b. pd.to_csv()
c. pd.read_excel()
d. pd.load_csv()

参考答案： `a`

解析： 读取 CSV 用 `pd.read_csv()`。

Question 32



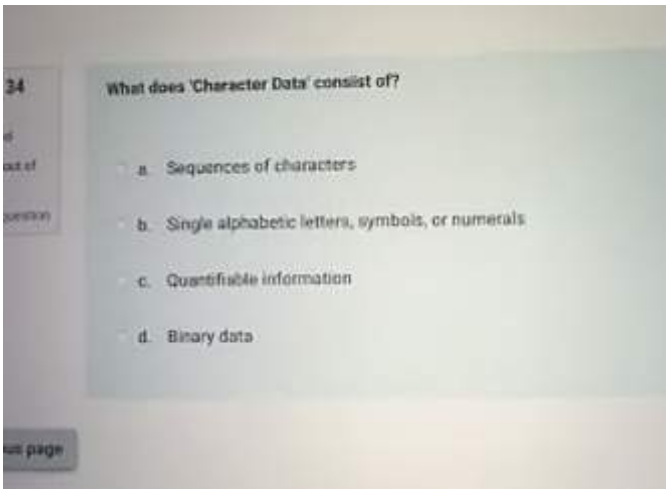
Which parameter in `pd.read_csv()` specifies the delimiter?

a. `delim`
b. `split`
c. `delimiter`
d. `sep`

参考答案： `d`

解析： `pd.read_csv()` 指定分隔符最常用的是 `sep`。

Question 35



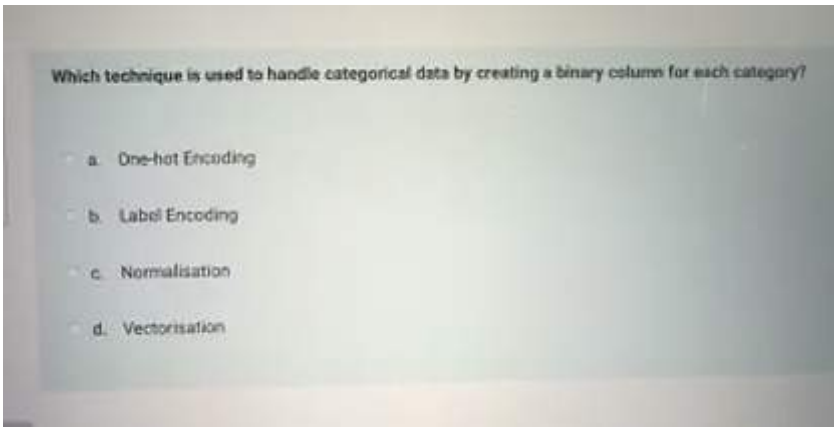
What does 'Character Data' consist of?

a. Sequences of characters
b. Single alphabetic letters, symbols, or numerals
c. Quantifiable information
d. Binary data

参考答案： `b`

解析： character data 指单个字符层面的数据，如字母、数字、符号。

Question 37



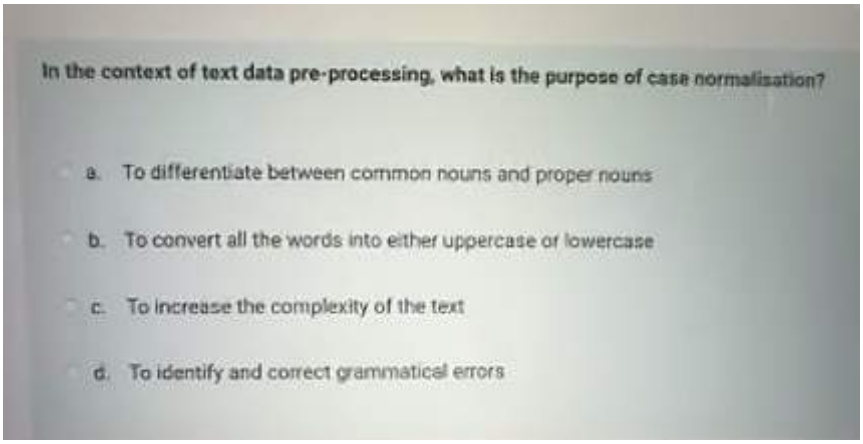
Which technique is used to handle categorical data by creating a binary column for each category?

a. One-hot Encoding
b. Label Encoding
c. Normalisation
d. Vectorisation

参考答案： `a`

解析： 为每个类别开一列 0/1 特征，就是 one-hot encoding。

Question 40



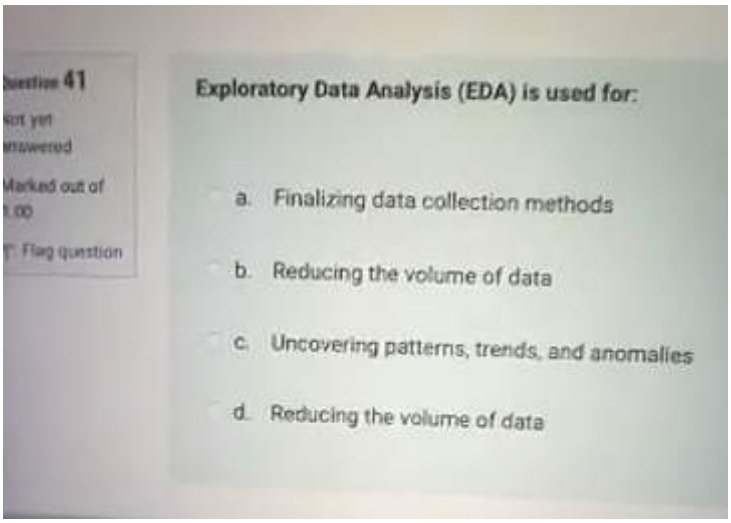
In the context of text data pre-processing, what is the purpose of case normalisation?

a. To differentiate between common nouns and proper nouns
b. To convert all the words into either uppercase or lowercase
c. To increase the complexity of the text
d. To identify and correct grammatical errors

参考答案： `b`

解析： case normalisation 就是把文本统一成大写或小写，减少同词不同写法带来的碎片化。

Question 42



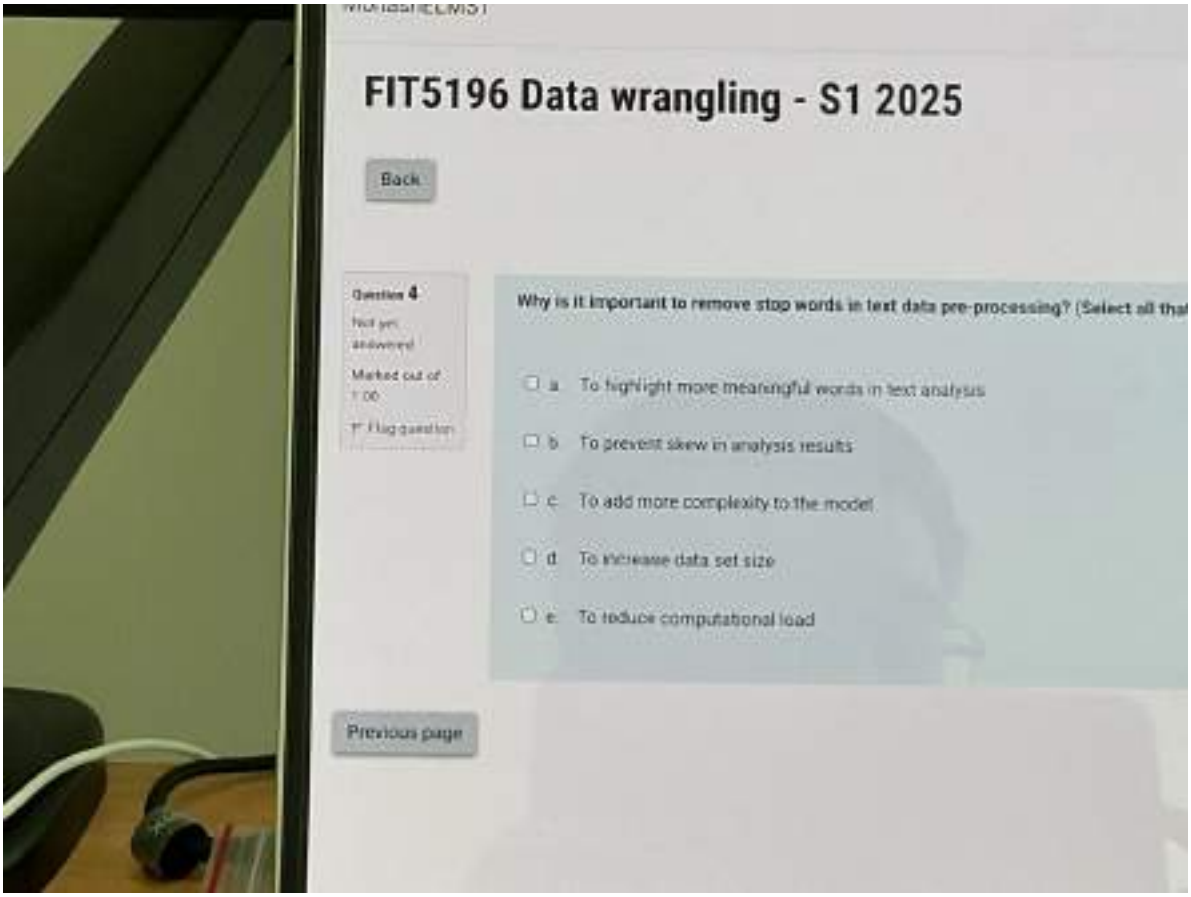
Exploratory Data Analysis (EDA) is used for:

a. Finalizing data collection methods
b. Reducing the volume of data
c. Uncovering patterns, trends, and anomalies
d. Reducing the volume of data

参考答案： `c`

解析： EDA 的目标是发现模式、趋势和异常，不是定稿数据采集方案。

Question 55



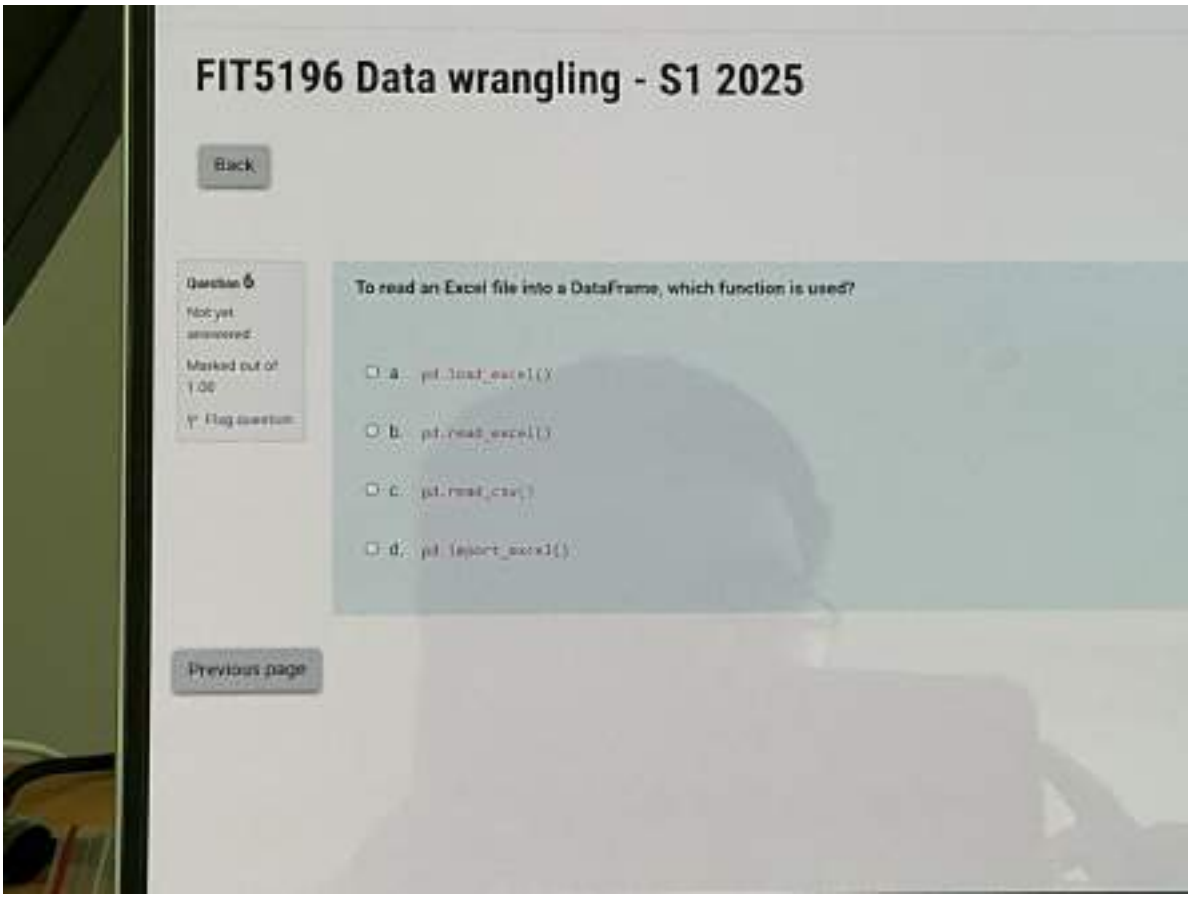
Why is it important to remove stop words in text data pre-processing? (Select all that apply)

a. To highlight more meaningful words in text analysis
b. To prevent skew in analysis results
c. To add more complexity to the model
d. To increase data set size
e. To reduce computational load

参考答案： `a, b, e`

解析： 去停用词是为了突出更有信息量的词、减少噪声并降低计算负担。

Question 57



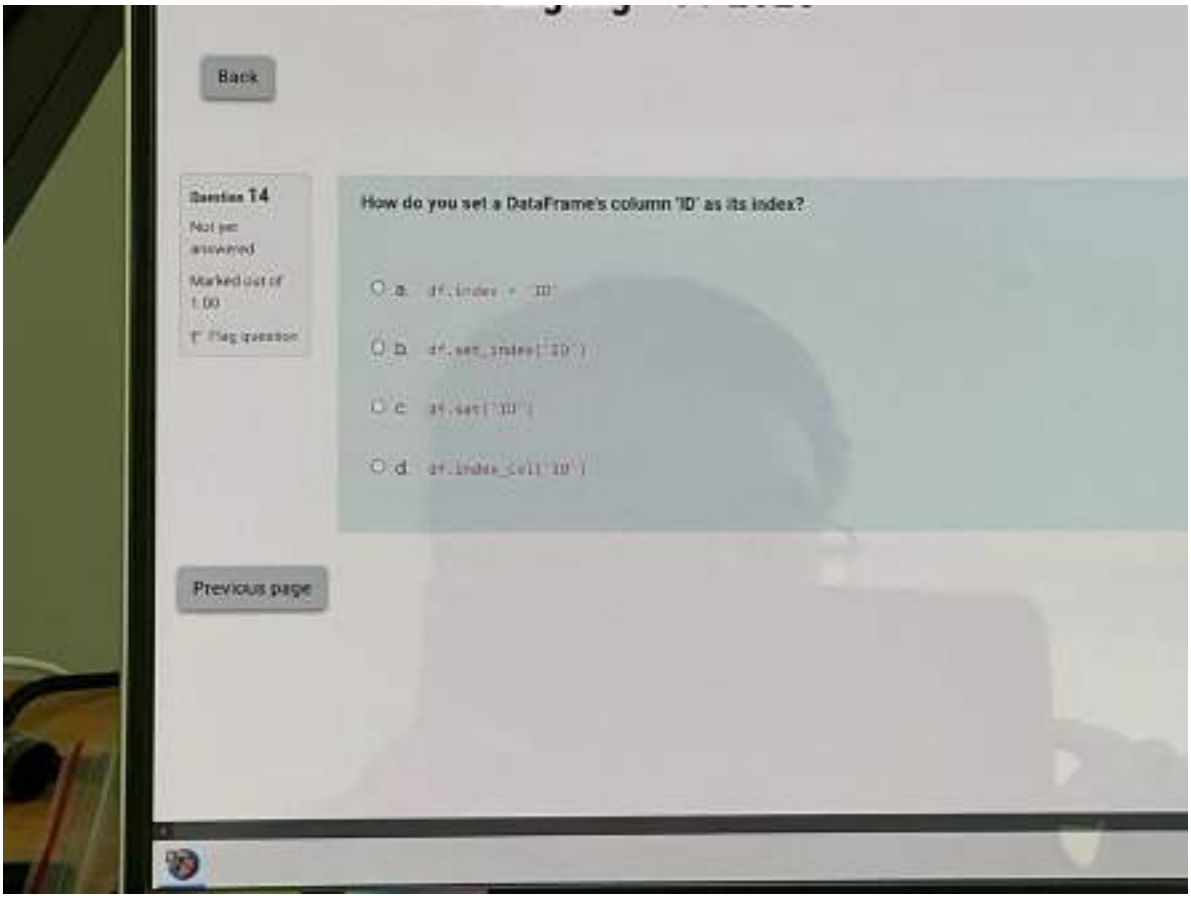
To read an Excel file into a DataFrame, which function is used?

a. `pd.load_excel()`
b. `pd.read_excel()`
c. `pd.read_csv()`
d. `pd.import_excel()`

参考答案： `b`

解析： 读取 Excel 文件用 `pd.read_excel()`。

Question 63



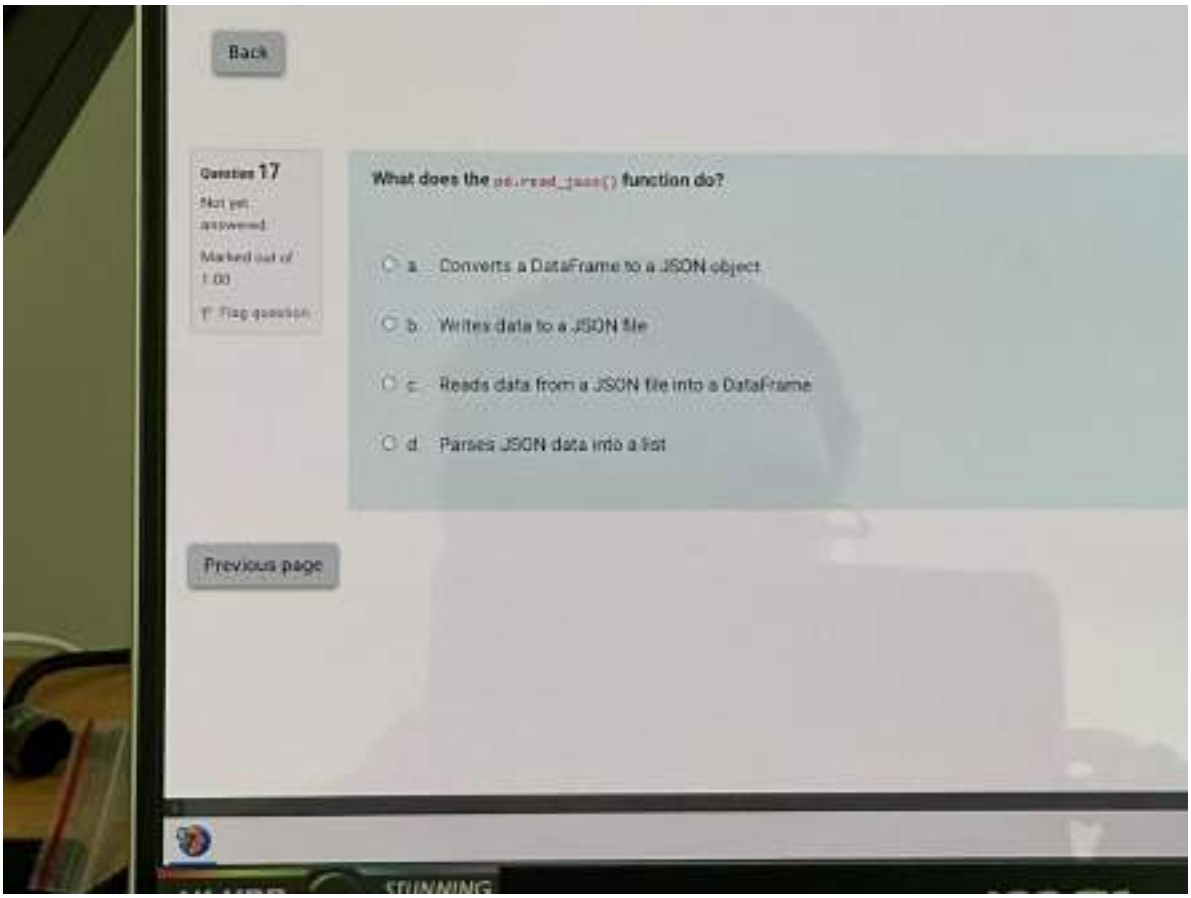
How do you set a DataFrame's column 'ID' as its index?

a. `df.index = 'ID'`
b. `df.set_index('ID')`
c. `df.set('ID')`
d. `df.index_col('ID')`

参考答案： `b`

解析： 把某列设成索引常用 `df.set_index("ID")`。

Question 66



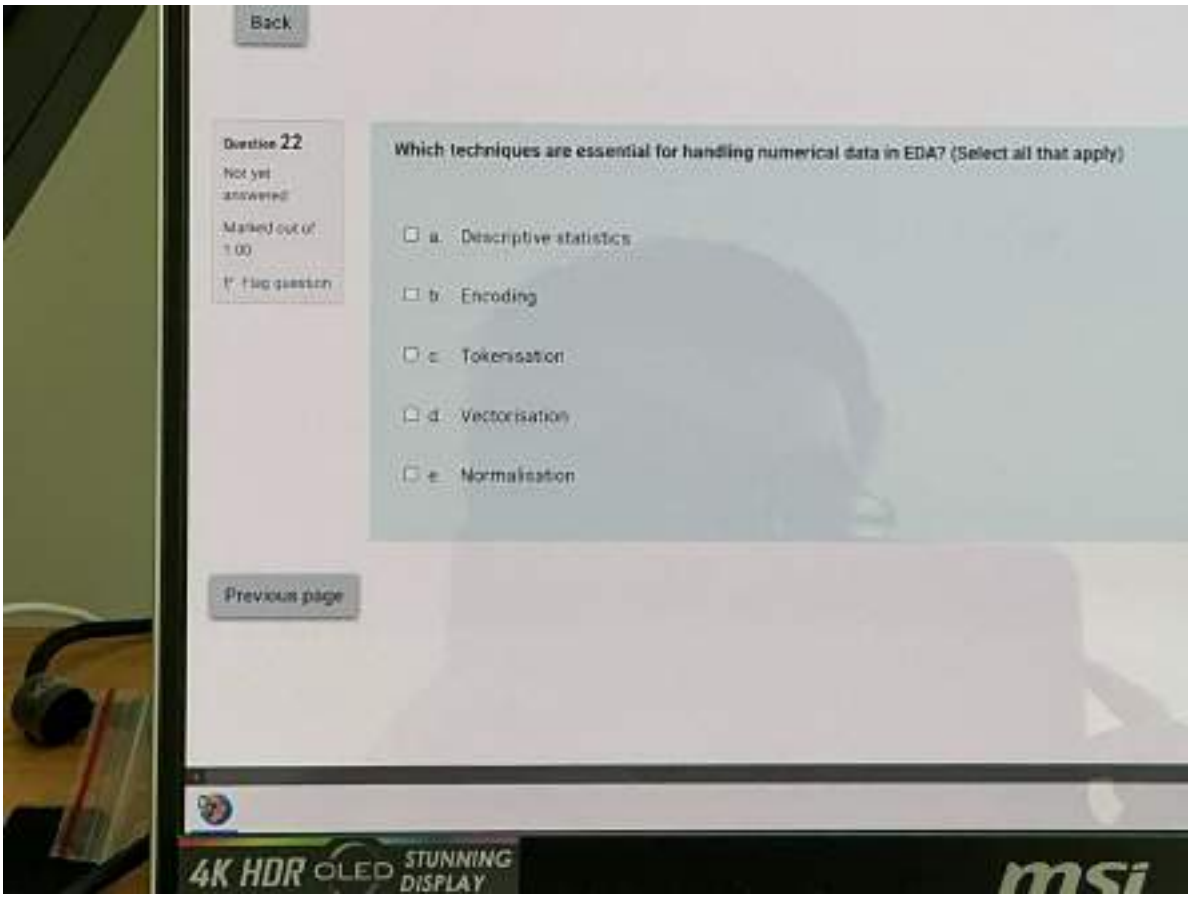
What does the pd.read_json() function do?

a. Converts a DataFrame to a JSON object
b. Writes data to a JSON file
c. Reads data from a JSON file into a DataFrame
d. Parses JSON data into a list

参考答案： `c`

解析： `pd.read_json()` 会把 JSON 读进 DataFrame。

Question 71



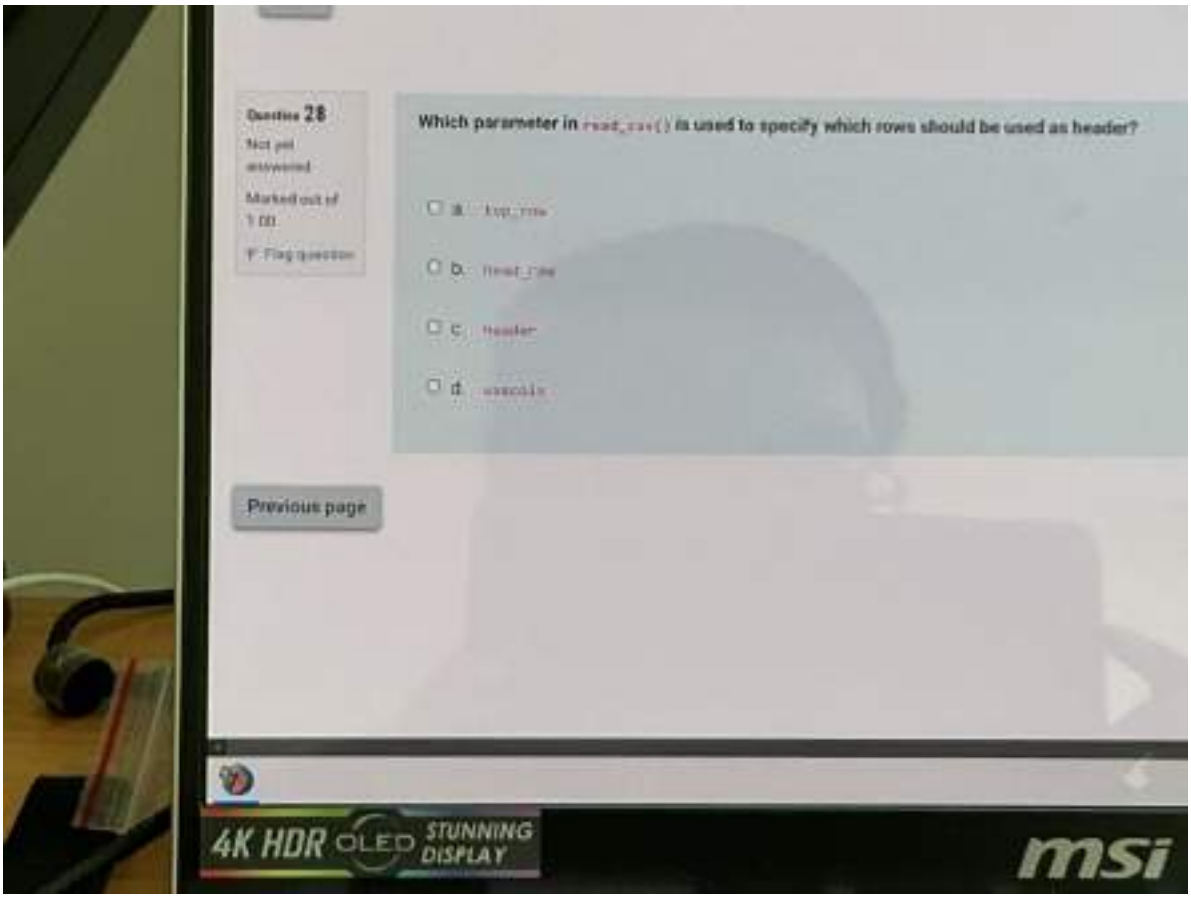
Which techniques are essential for handling numerical data in EDA? (Select all that apply)

a. Descriptive statistics
b. Encoding
c. Tokenisation
d. Vectorisation
e. Normalisation

参考答案： `a, e`

解析： 数值型 EDA 常用 descriptive statistics，也可能做 normalisation；encoding / tokenisation / vectorisation 不是数值数据核心方法。

Question 75



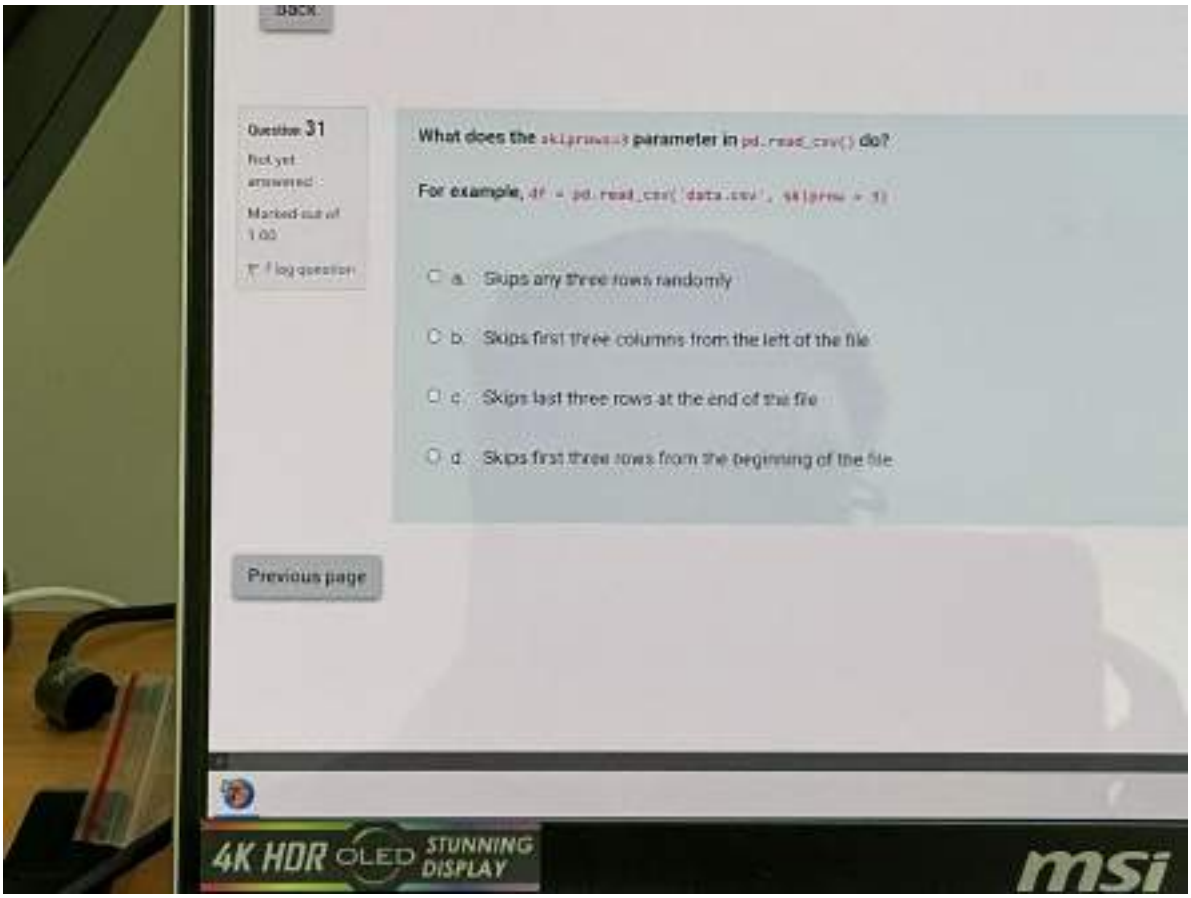
Which parameter in read_csv() is used to specify which rows should be used as header?

a. top_row
b. head_row
c. header
d. usecols

参考答案： `c`

解析： `header` 参数决定哪一行当列名。

Question 77



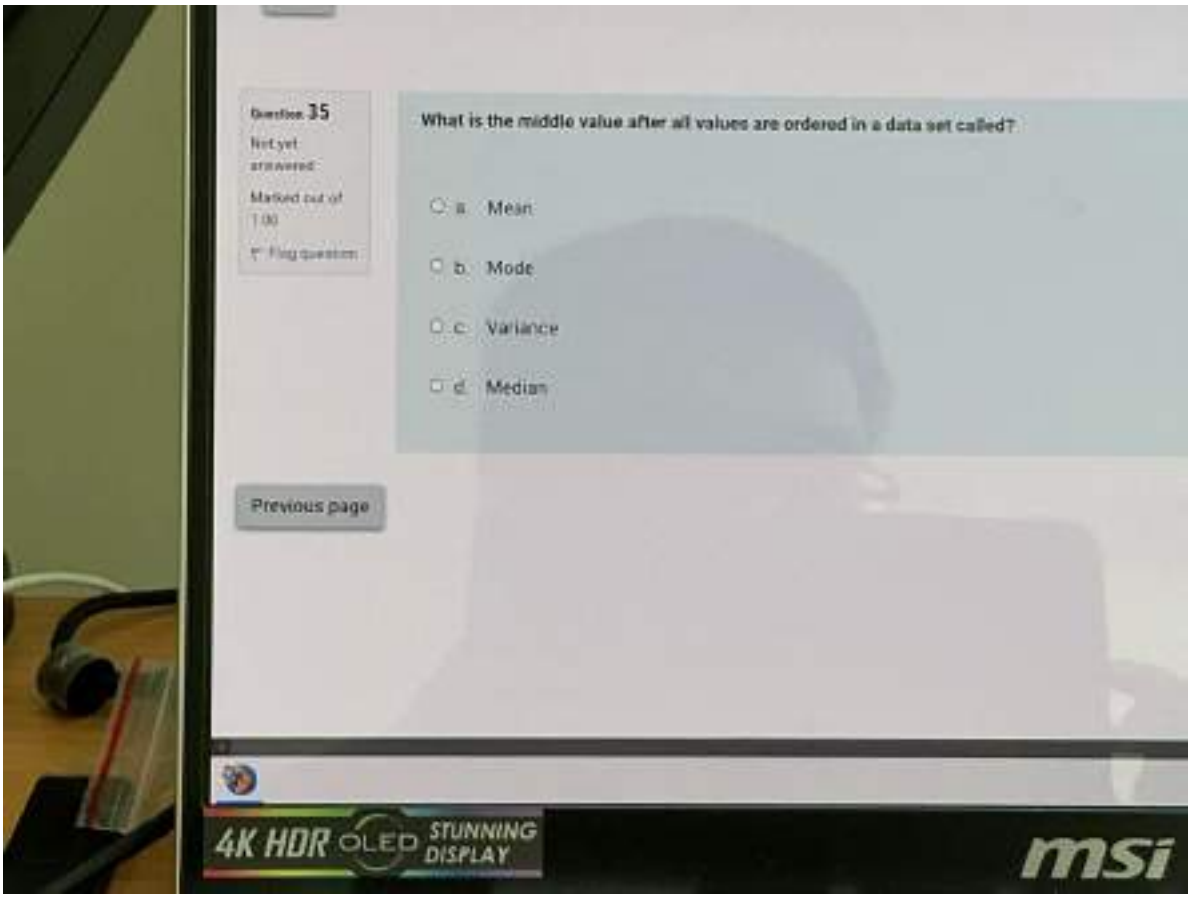
What does the `skiprows=3` parameter in `pd.read_csv()` do?
For example, `df = pd.read_csv('data.csv', skiprow = 3)`

a. Skips any three rows randomly
b. Skips first three columns from the left of the file
c. Skips last three rows at the end of the file
d. Skips first three rows from the beginning of the file

参考答案： `d`

解析： `skiprows=3` 表示从文件开头跳过前三行。

Question 81



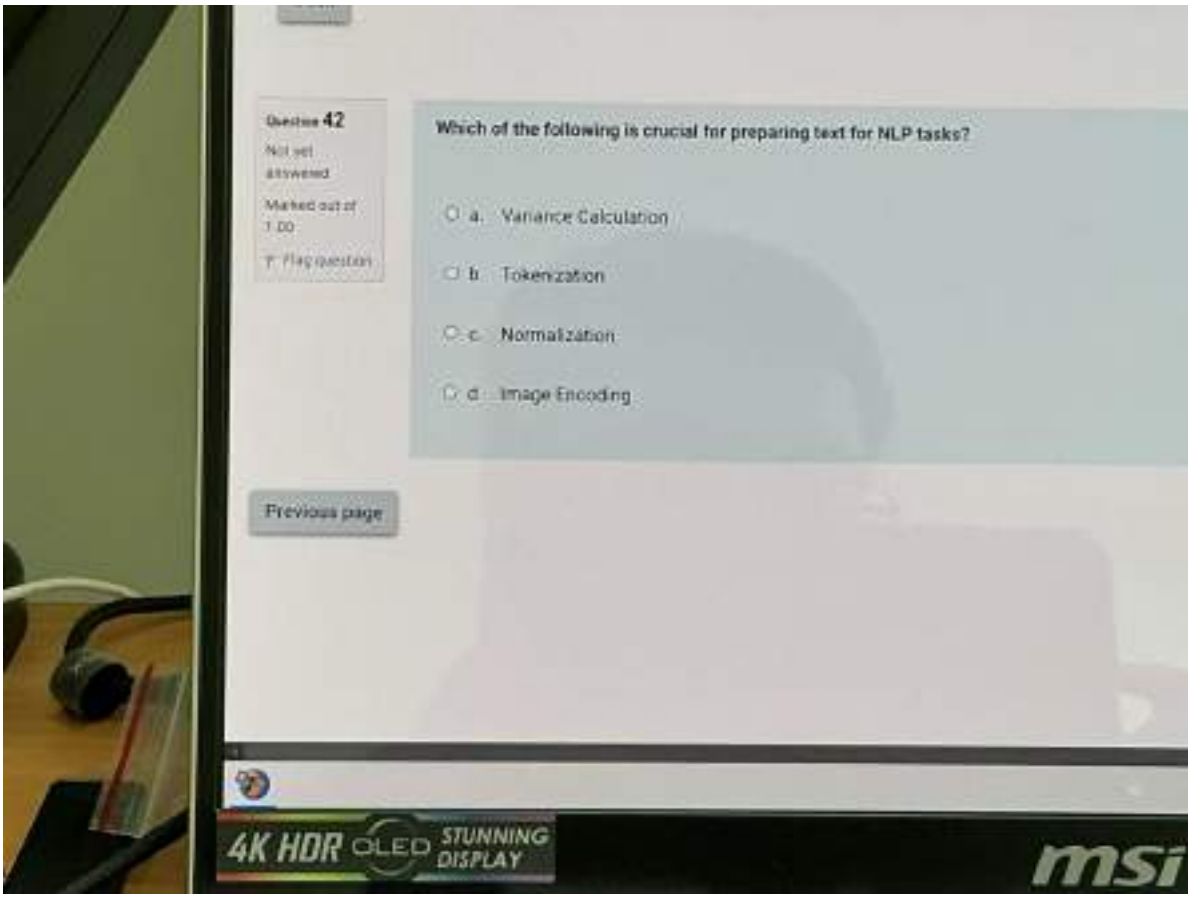
What is the middle value after all values are ordered in a data set called?

a. Mean
b. Mode
c. Variance
d. Median

参考答案： `d`

解析： 有序数据中间那个值叫 median。

Question 88



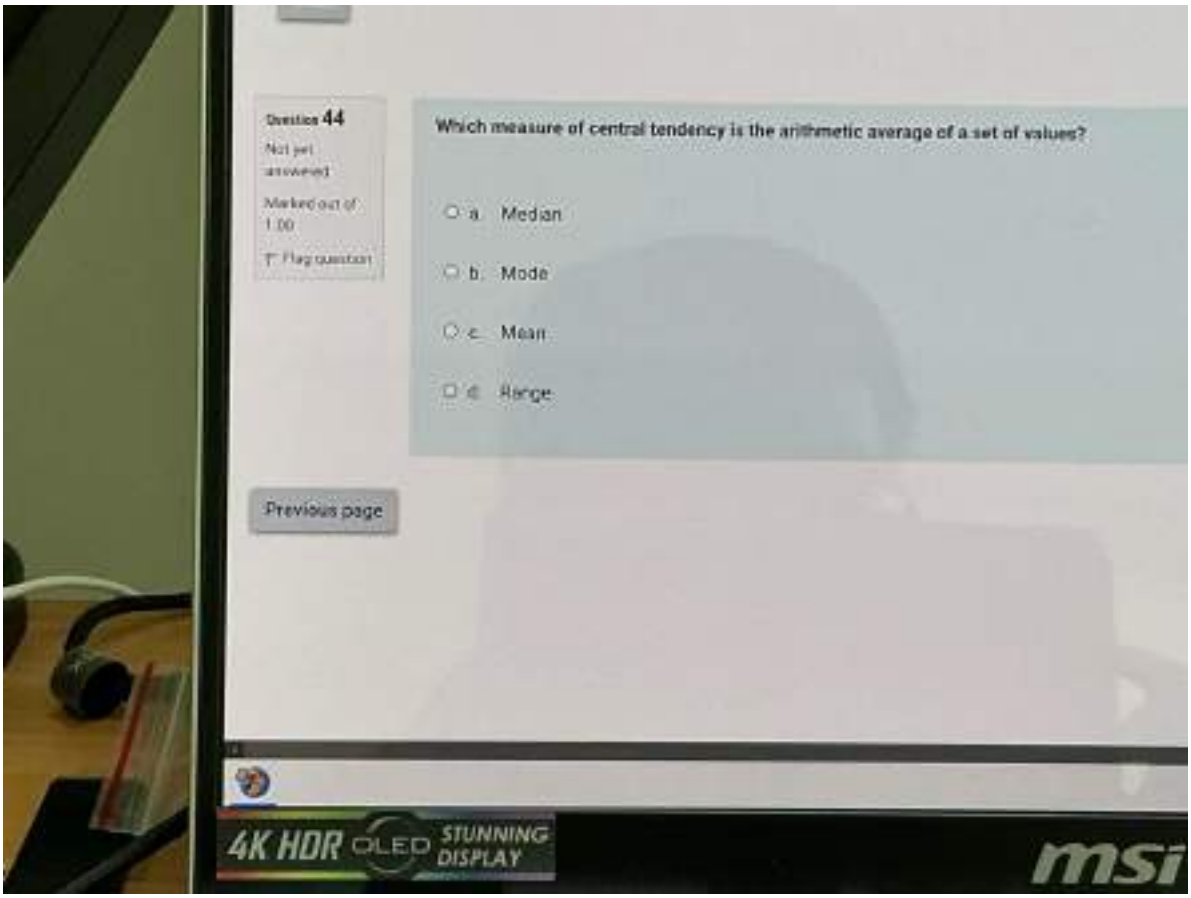
Which of the following is crucial for preparing text for NLP tasks?

a. Variance Calculation
b. Tokenization
c. Normalization
d. Image Encoding

参考答案： `b`

解析： NLP 的第一步通常是 tokenization。

Question 90



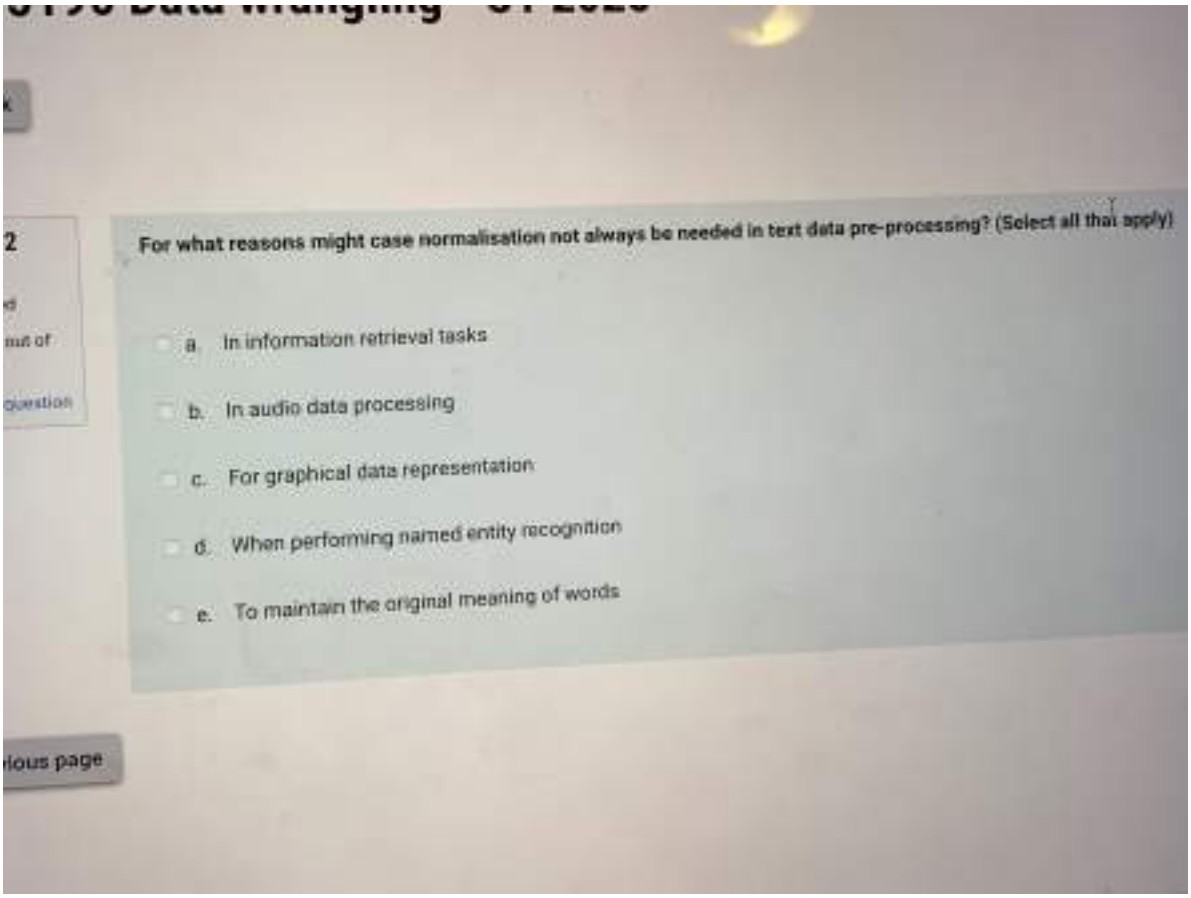
Which measure of central tendency is the arithmetic average of a set of values?

a. Median
b. Mode
c. Mean
d. Range

参考答案： `c`

解析： 算术平均数就是 mean。

Question 96



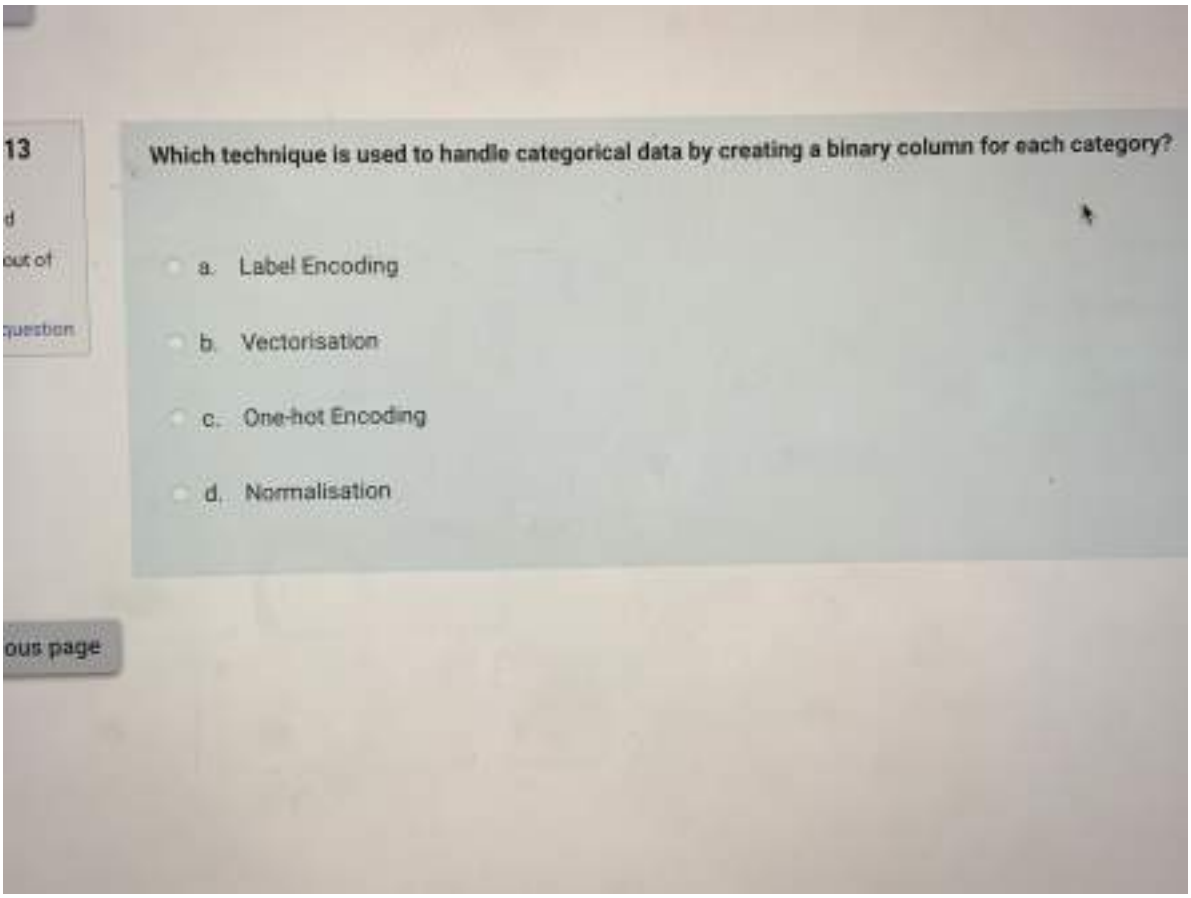
For what reasons might case normalisation not always be needed in text data pre-processing? (Select all that apply)

a. In information retrieval tasks
b. In audio data processing
c. For graphical data representation
d. When performing named entity recognition
e. To maintain the original meaning of words

参考答案： `a, d, e`

解析： 和 Q23/Q24 同理，IR、NER、保留原义等场景不一定要统一大小写。

Question 105



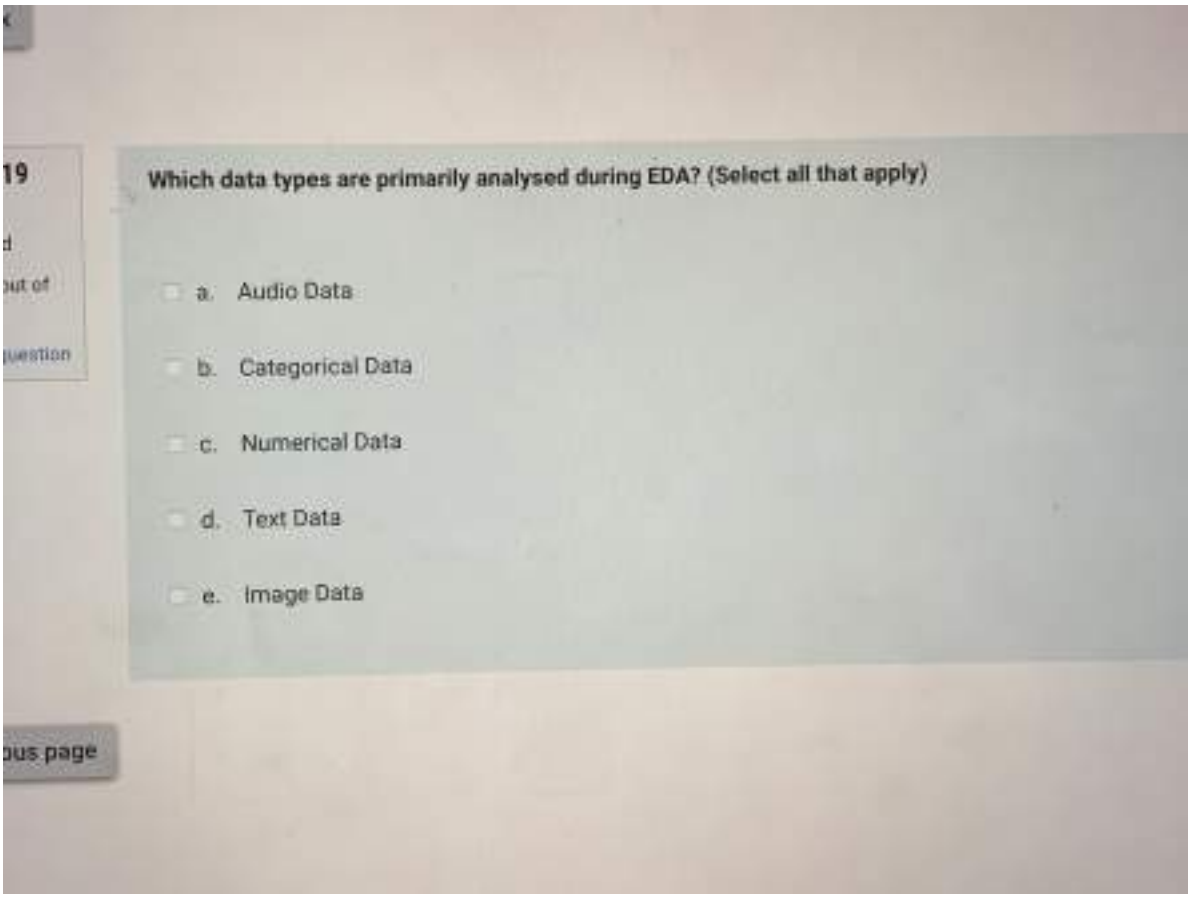
Which technique is used to handle categorical data by creating a binary column for each category?

a. Label Encoding
b. Vectorisation
c. One-hot Encoding
d. Normalisation

参考答案： `c`

解析： 和 Q37 同义，这里正确项是顺序不同后的 one-hot encoding。

Question 110



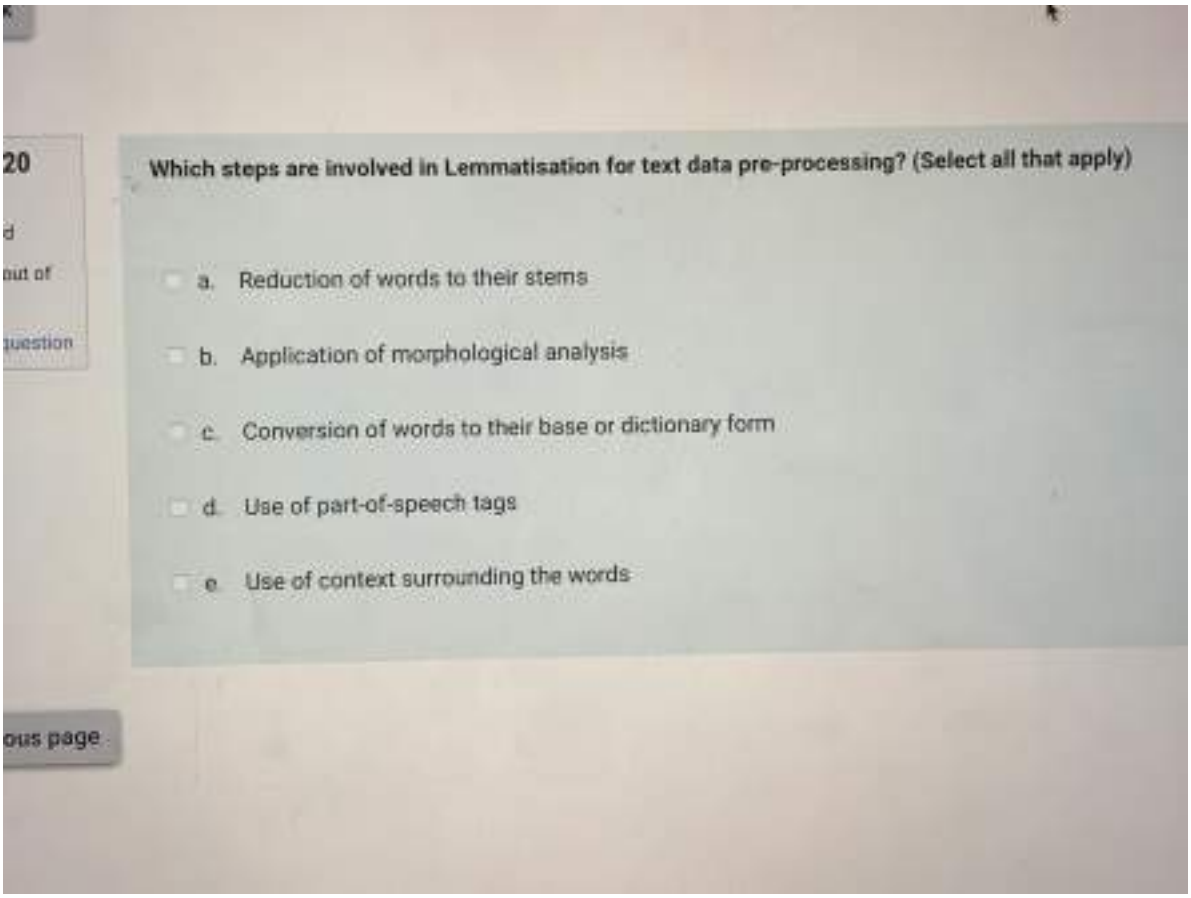
Which data types are primarily analysed during EDA? (Select all that apply)

a. Audio Data
b. Categorical Data
c. Numerical Data
d. Text Data
e. Image Data

参考答案： `b, c, d`

解析： 课程中的 EDA 主要看 categorical、numerical、text 三类。

Question 111



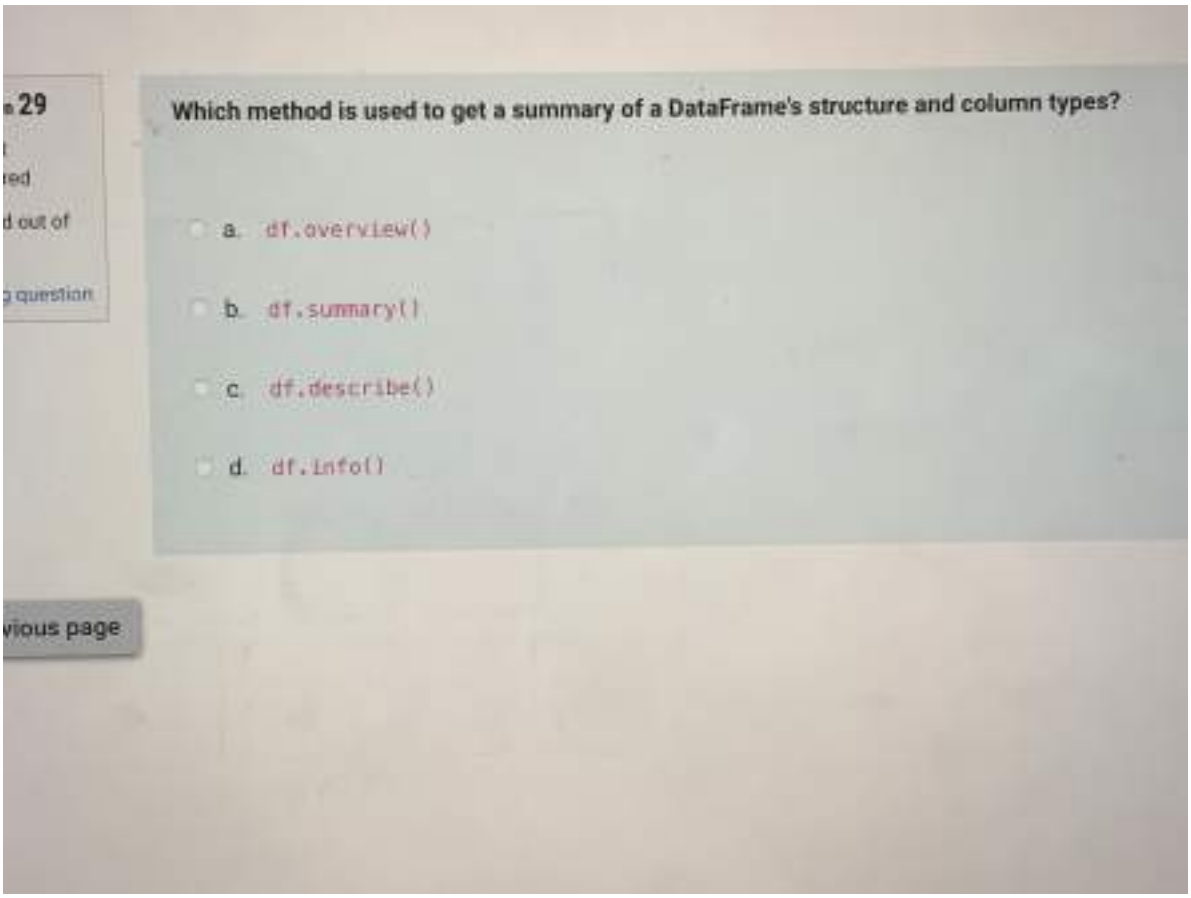
Which steps are involved in Lemmatisation for text data pre-processing? (Select all that apply)

a. Reduction of words to their stems
b. Application of morphological analysis
c. Conversion of words to their base or dictionary form
d. Use of part-of-speech tags
e. Use of context surrounding the words

参考答案： `b, c, d, e`

解析： lemmatisation 依赖词形分析、词性和上下文，把词还原成 dictionary form；“stem” 不是它的核心定义。

Question 120



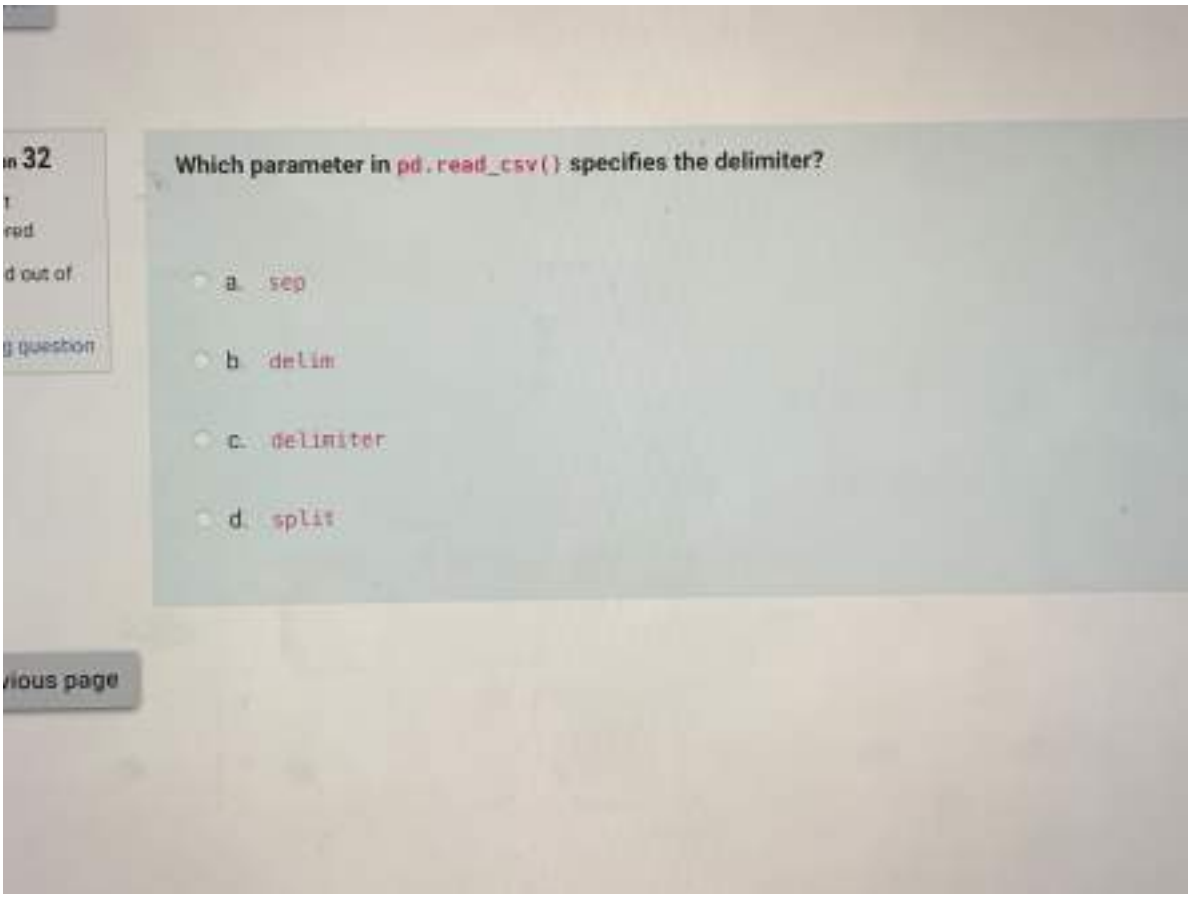
Which method is used to get a summary of a DataFrame's structure and column types?

a. df.overview()
b. df.summary()
c. df.describe()
d. df.info()

参考答案： `d`

解析： 和 Q16 同义，结构摘要用 `df.info()`。

Question 122



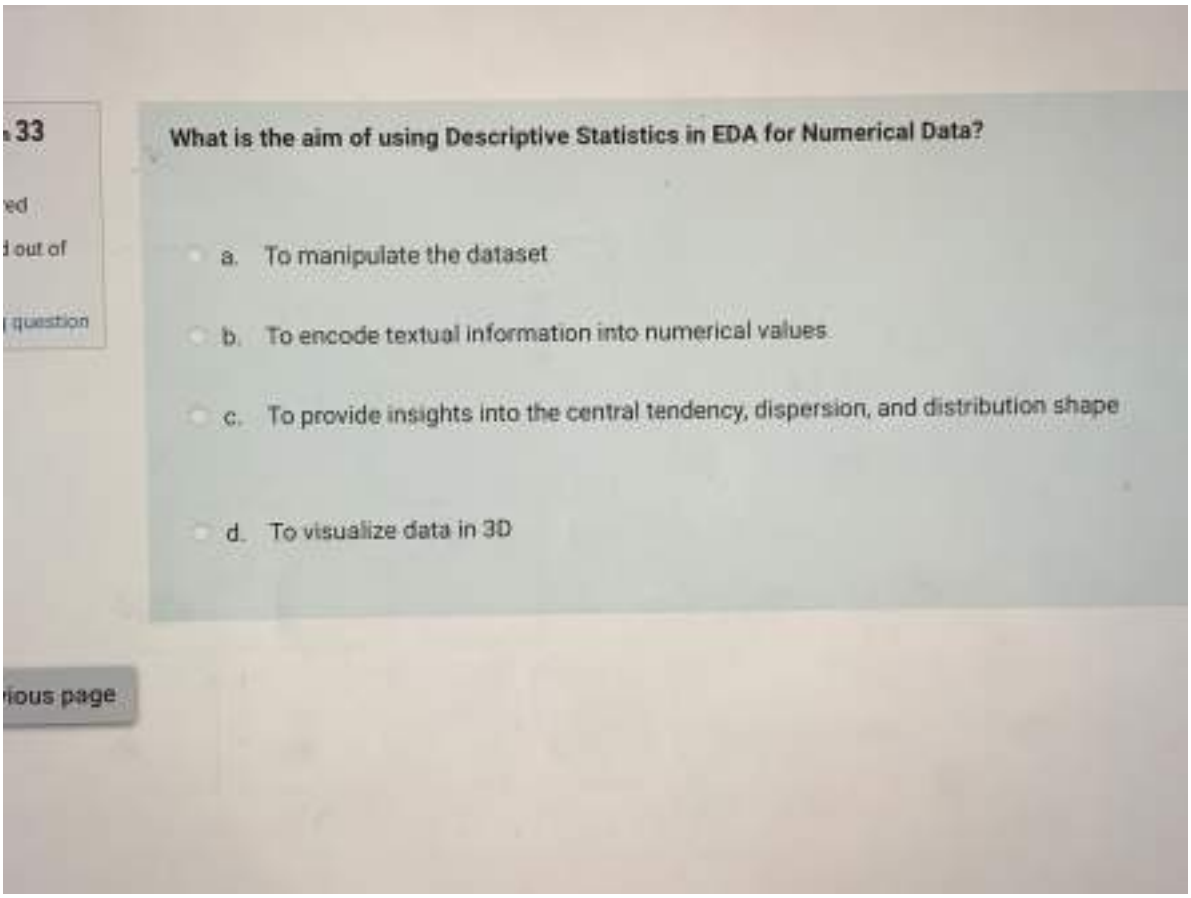
Which parameter in pd.read_csv() specifies the delimiter?

a. sep
b. delim
c. delimiter
d. split

参考答案： `a`

解析： 和 Q32 同义，这里正确项换成了 `sep`。

Question 123



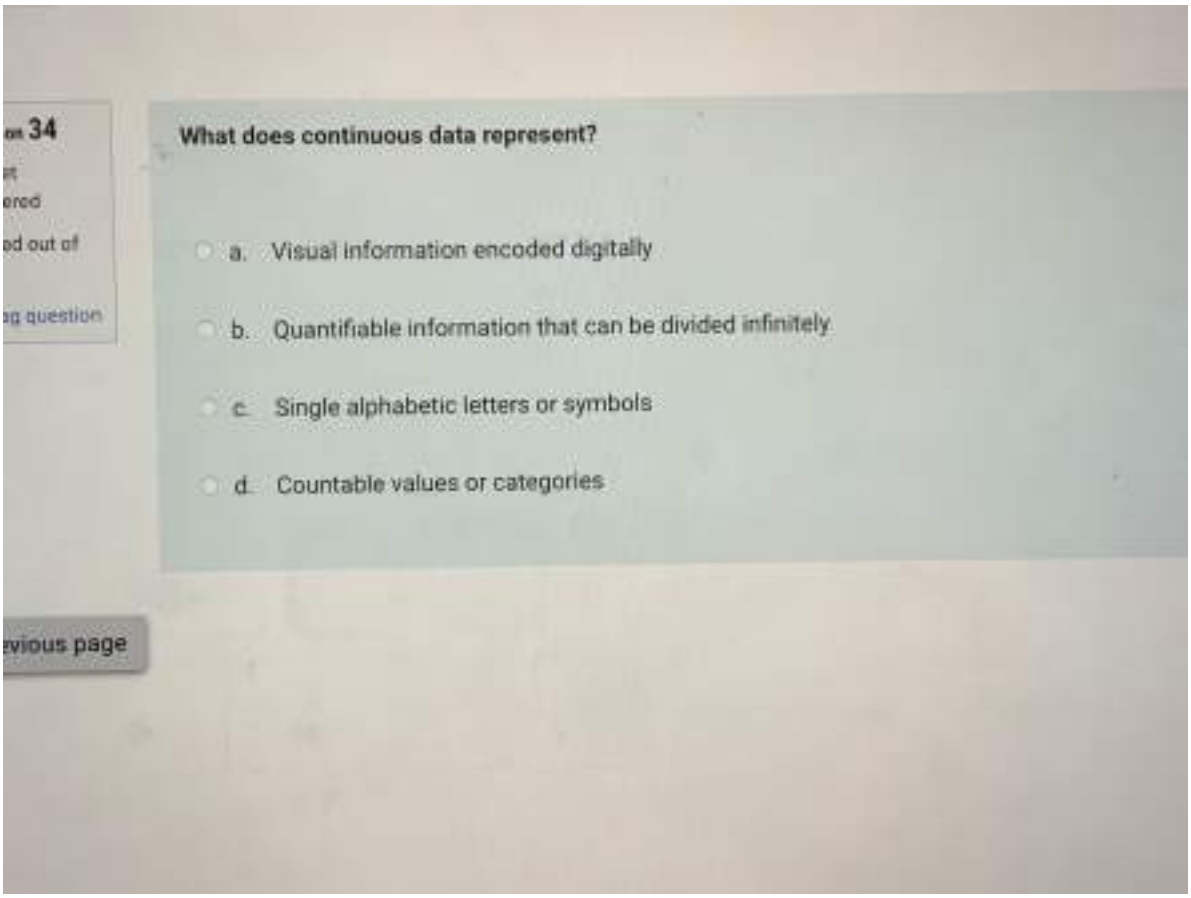
What is the aim of using Descriptive Statistics in EDA for Numerical Data?

a. To manipulate the dataset
b. To encode textual information into numerical values
c. To provide insights into the central tendency, dispersion, and distribution shape
d. To visualize data in 3D

参考答案： `c`

解析： descriptive statistics 的目标是概括中心位置、离散程度和分布形态。

Question 124



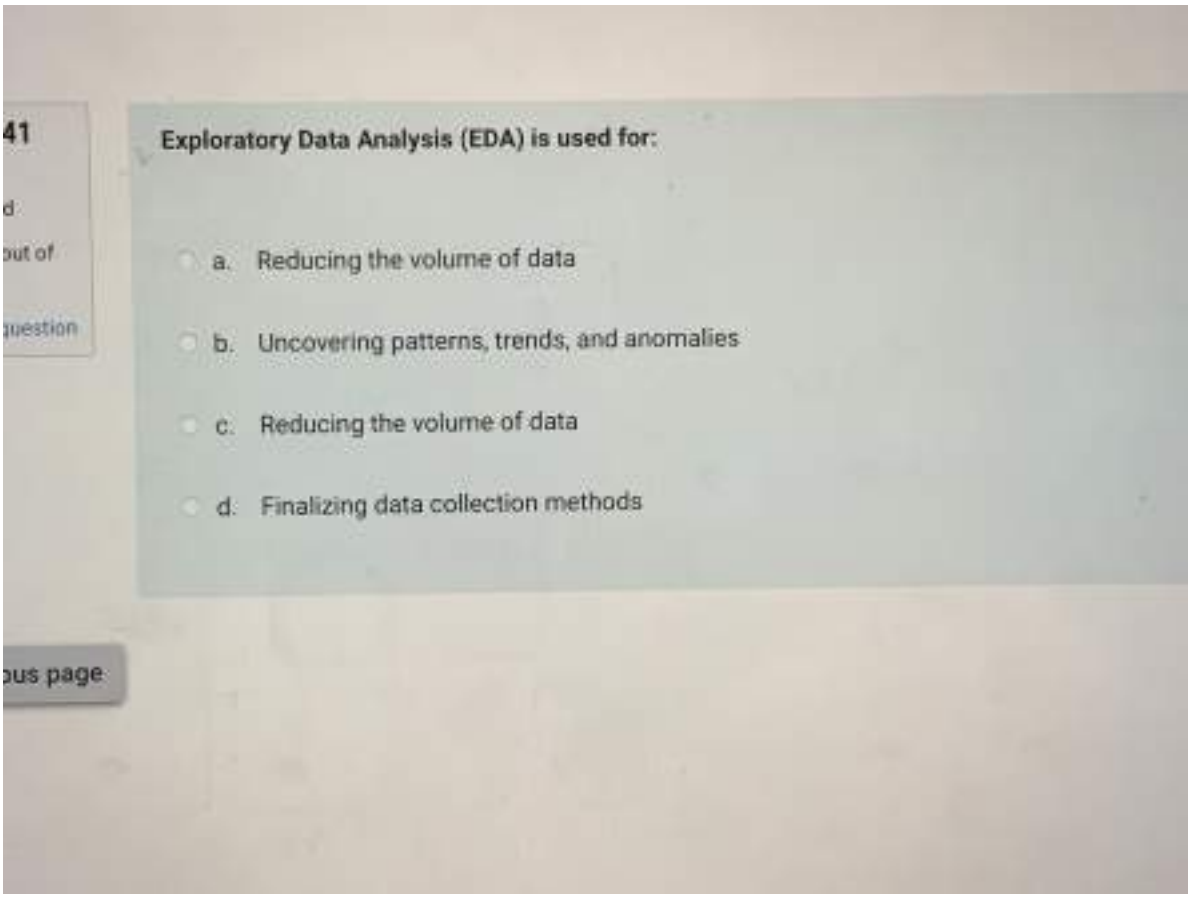
What does continuous data represent?

a. Visual information encoded digitally
b. Quantifiable information that can be divided infinitely
c. Single alphabetic letters or symbols
d. Countable values or categories

参考答案： `b`

解析： continuous data 是可以无限细分的可量化数值。

Question 131



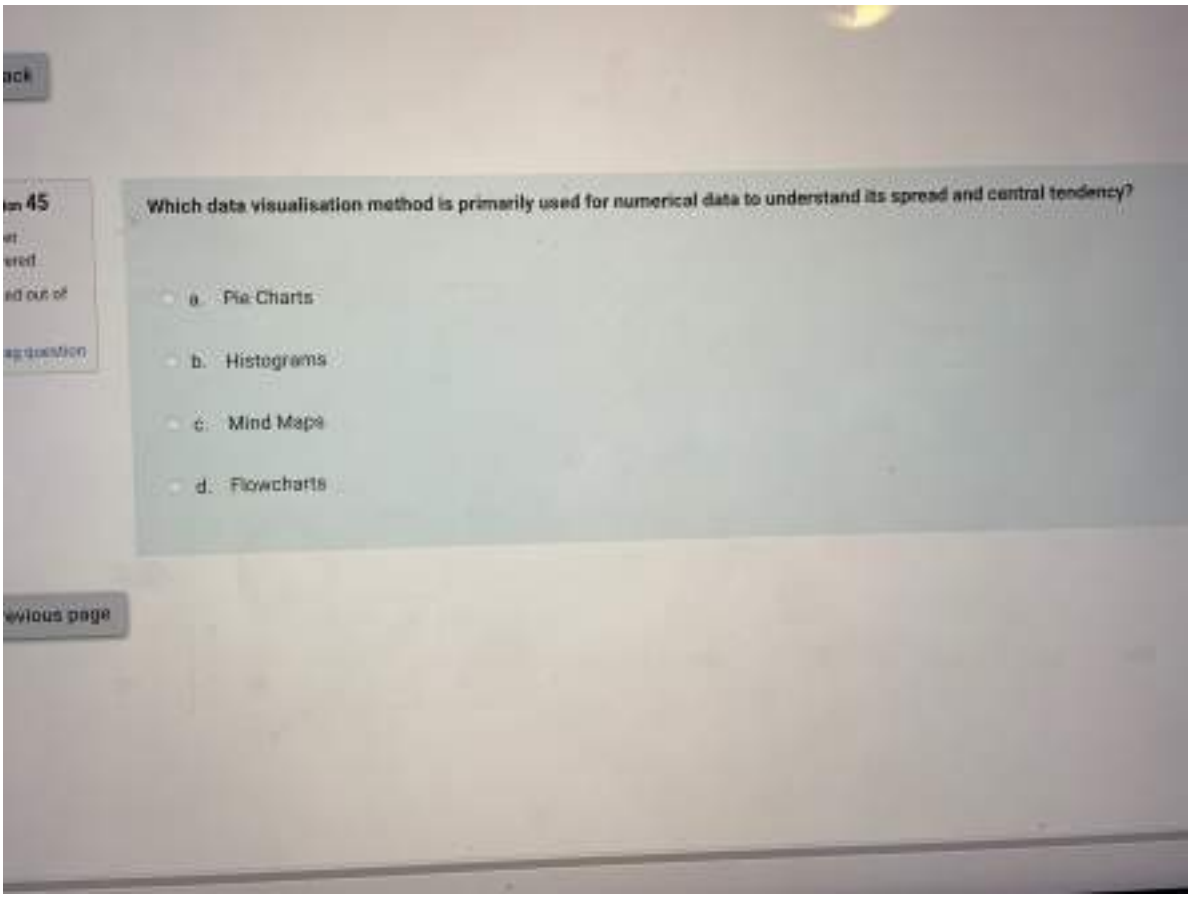
Exploratory Data Analysis (EDA) is used for:

a. Reducing the volume of data
b. Uncovering patterns, trends, and anomalies
c. Reducing the volume of data
d. Finalizing data collection methods

参考答案： `b`

解析： 和 Q42 同义，EDA 用来发现 patterns / trends / anomalies。

Question 135



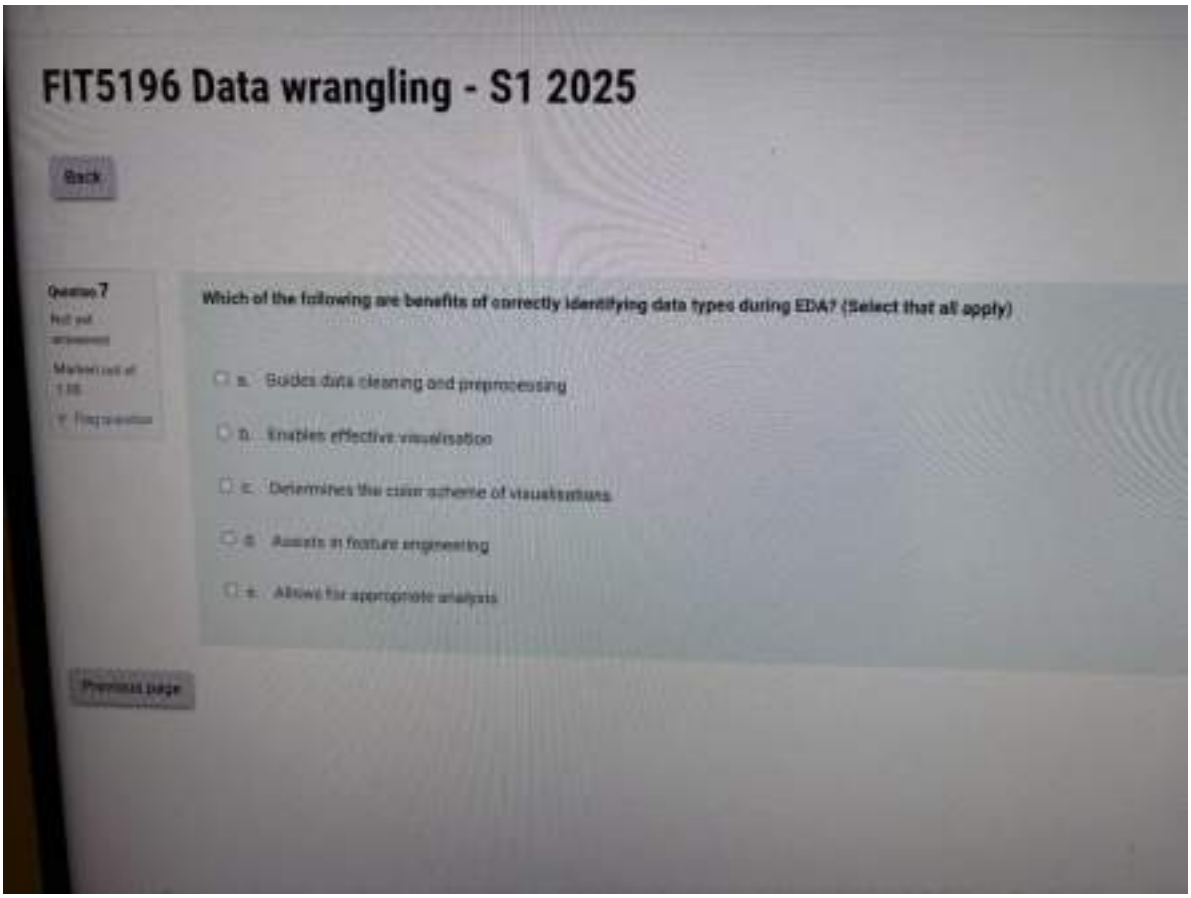
Which data visualisation method is primarily used for numerical data to understand its spread and central tendency?

a. Pie Charts
b. Histograms
c. Mind Maps
d. Flowcharts

参考答案： `b`

解析： 看数值分布与中心位置最常用 histogram。

Question 145



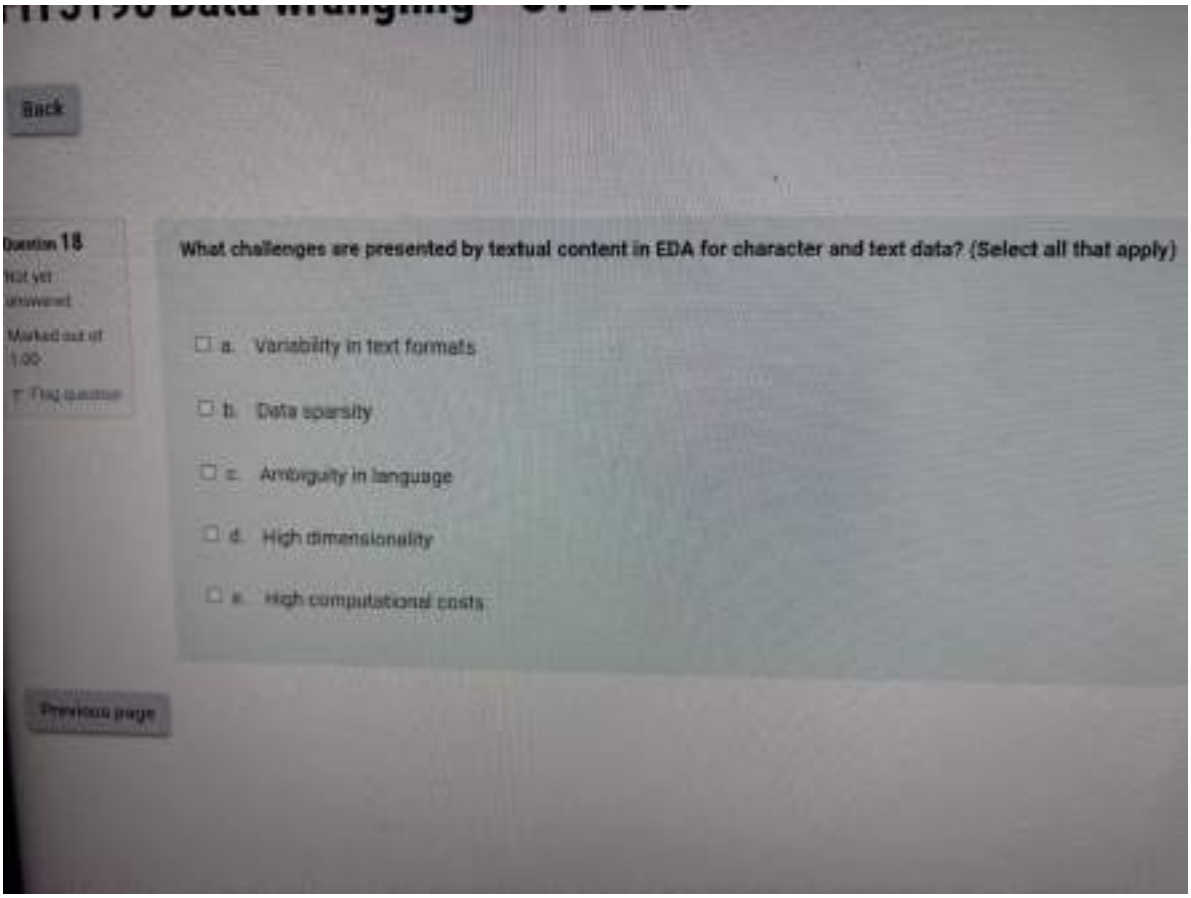
Which of the following are benefits of correctly identifying data types during EDA? (Select that all apply)

a. Guides data cleaning and preprocessing
b. Enables effective visualisation
c. Determines the color scheme of visualisations
d. Assists in feature engineering
e. Allows for appropriate analysis

参考答案： `a, b, d, e`

解析： 正确识别数据类型能指导清洗、可视化、特征工程和后续分析。

Question 149



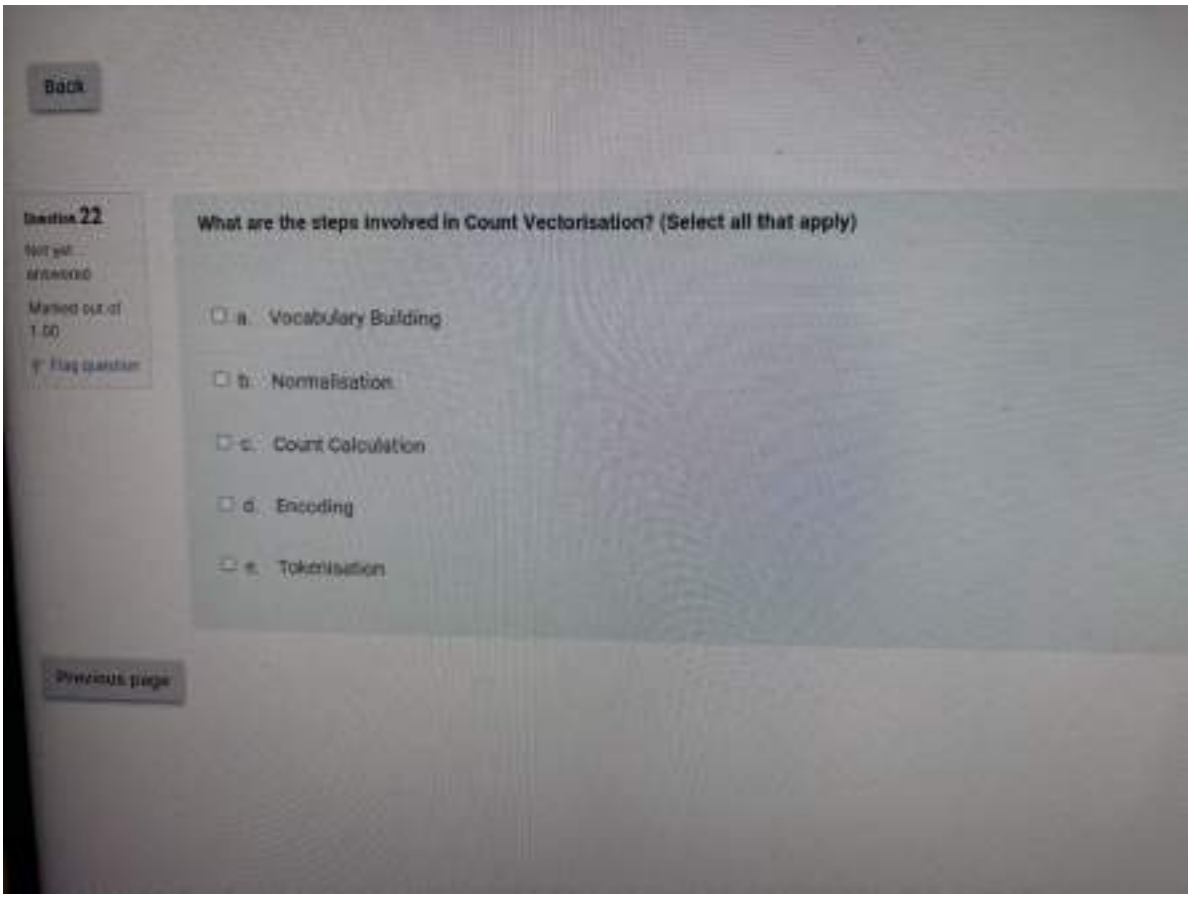
What challenges are presented by textual content in EDA for character and text data? (Select all that apply)

a. Variability in text formats
b. Data sparsity
c. Ambiguity in language
d. High dimensionality
e. High computational costs

参考答案： `a, b, c, d, e`

解析： 和 Q2 同义，文本常见难点包括格式多变、稀疏、高维、歧义和计算成本。

Question 151



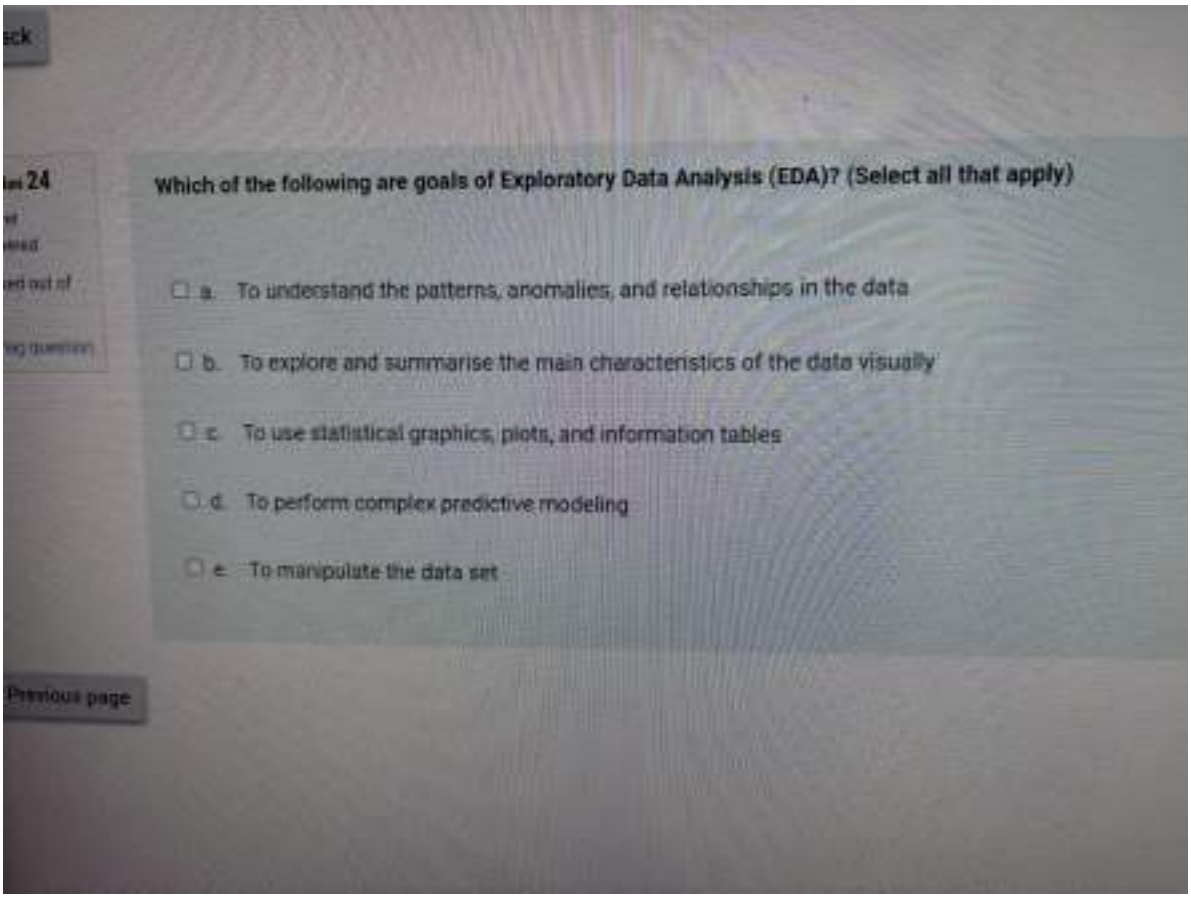
What are the steps involved in Count Vectorisation? (Select all that apply)

a. Vocabulary Building
b. Normalisation
c. Count Calculation
d. Encoding
e. Tokenisation

参考答案： `a, c, e`

解析： count vectorisation 通常先 tokenise、建立词表，再统计每个词出现次数。

Question 153



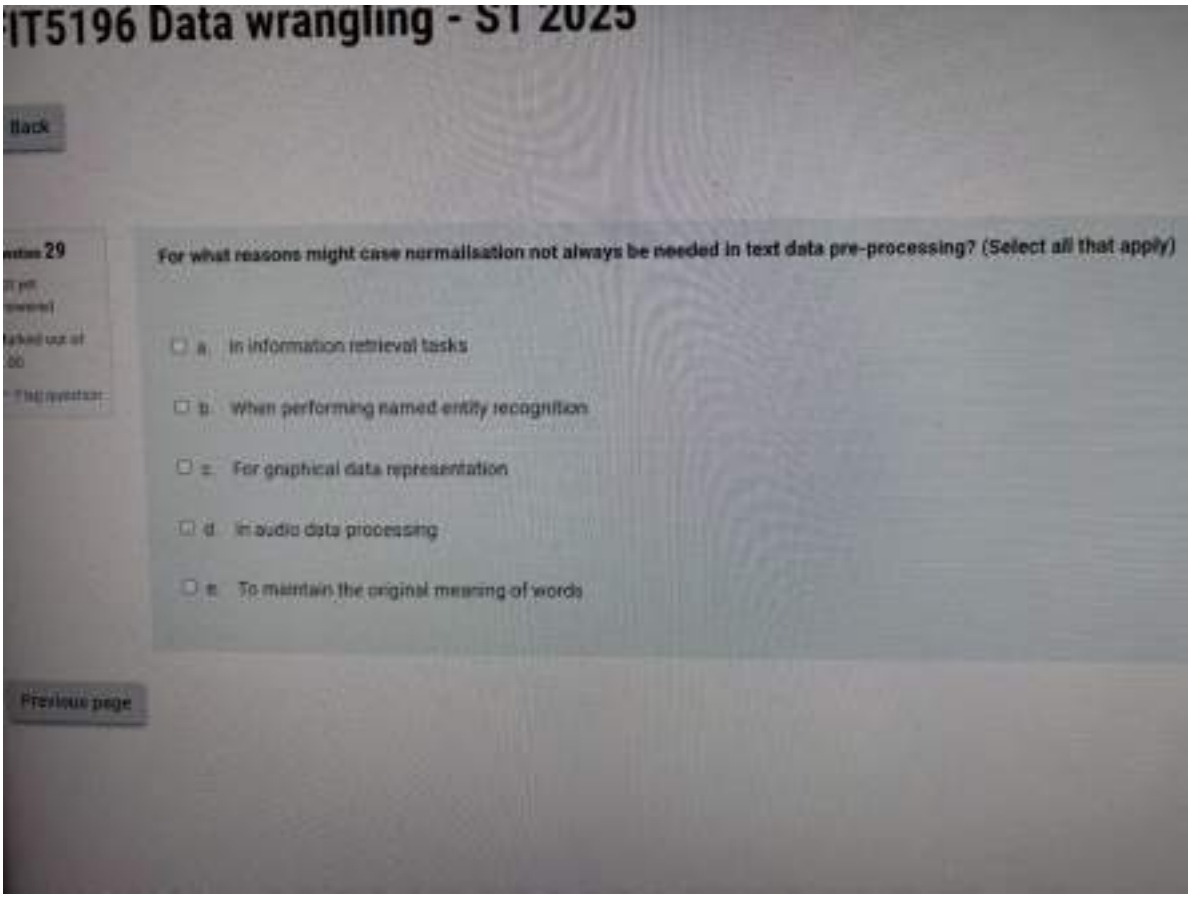
Which of the following are goals of Exploratory Data Analysis (EDA)? (Select all that apply)

a. To understand the patterns, anomalies, and relationships in the data
b. To explore and summarise the main characteristics of the data visually
c. To use statistical graphics, plots, and information tables
d. To perform complex predictive modeling
e. To manipulate the data set

参考答案： `a, b, c`

解析： EDA 目标是理解数据并做可视化总结，不是直接做复杂预测建模。

Question 158



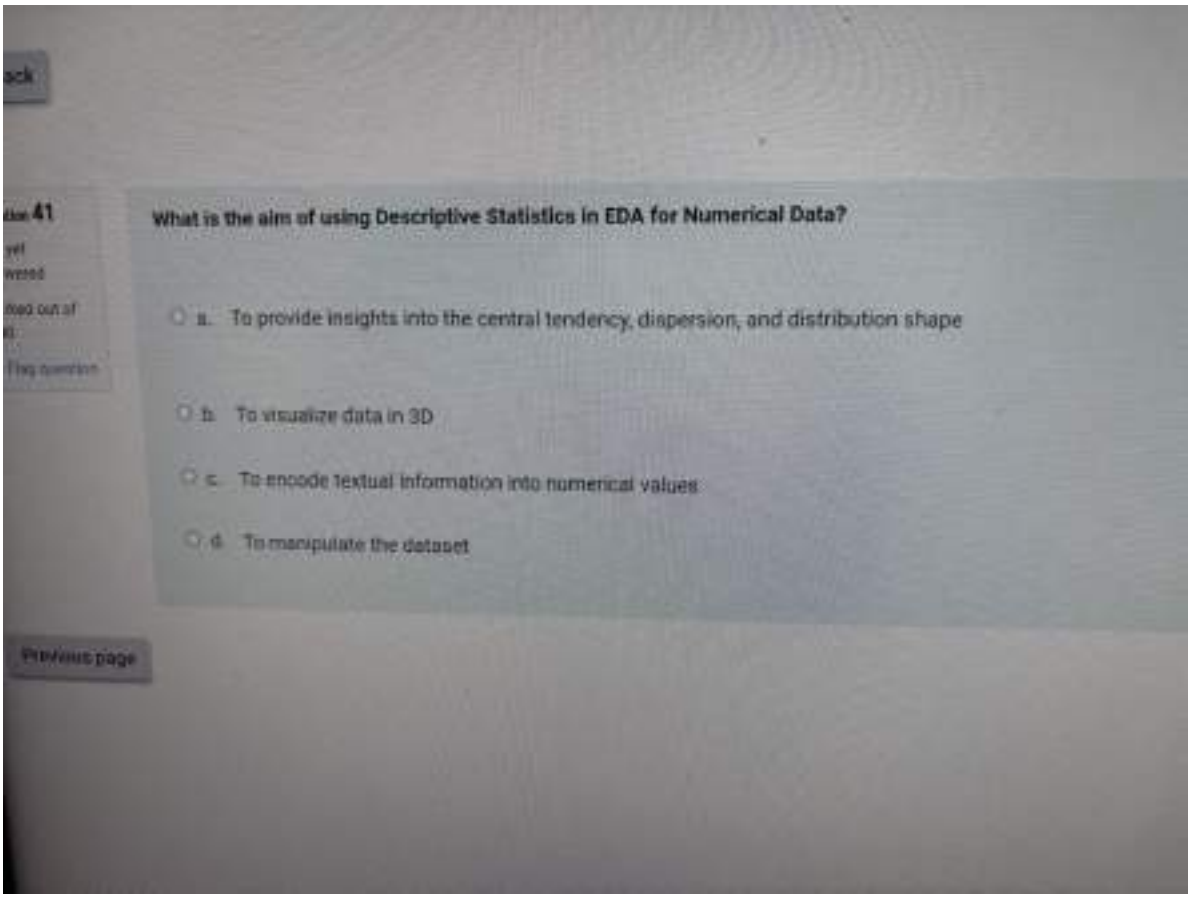
For what reasons might case normalisation not always be needed in text data pre-processing? (Select all that apply)

a. In information retrieval tasks
b. When performing named entity recognition
c. For graphical data representation
d. In audio data processing
e. To maintain the original meaning of words

参考答案： `a, b, e`

解析： 和 Q23/Q24/Q96 同理，IR、NER、保留原始含义时不一定要大小写归一化。

Question 163



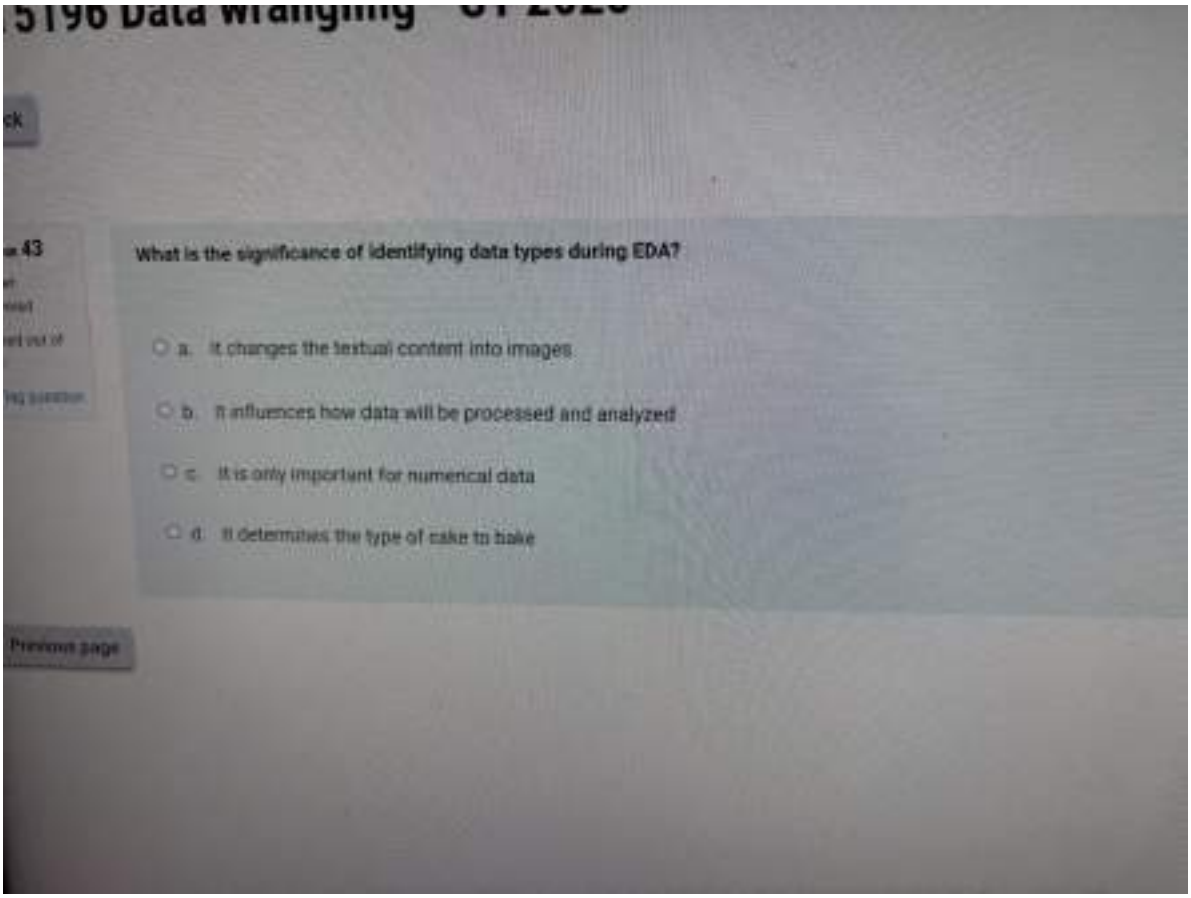
What is the aim of using Descriptive Statistics in EDA for Numerical Data?

a. To provide insights into the central tendency, dispersion, and distribution shape
b. To visualize data in 3D
c. To encode textual information into numerical values
d. To manipulate the dataset

参考答案： `a`

解析： 和 Q123 同义，这里正确项换成了 `a`。

Question 164



What is the significance of identifying data types during EDA?

a. It changes the textual content into images
b. It influences how data will be processed and analyzed
c. It is only important for numerical data
d. It determines the type of cake to bake

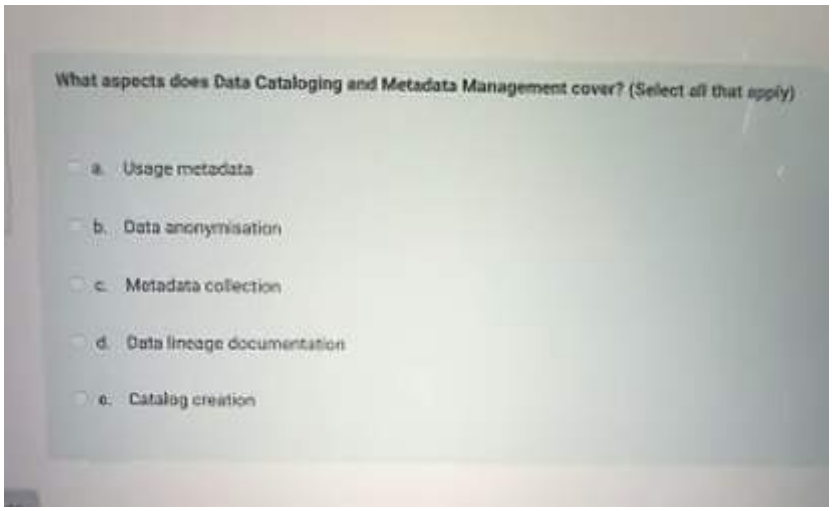
参考答案： `b`

解析： 识别数据类型会直接影响后续怎么处理和分析数据。

Week 5 (25 题)

- 范围：Data Discovery & Collection

Question 10



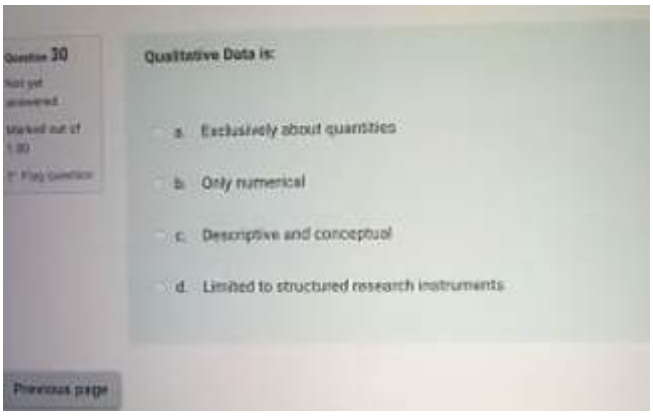
What aspects does Data Cataloging and Metadata Management cover? (Select all that apply)

a. Usage metadata
b. Data anonymisation
c. Metadata collection
d. Data lineage documentation
e. Catalog creation

参考答案： `a, c, d, e`

解析： cataloging / metadata management 关注 metadata collection、lineage、catalog 本身，也会包含 usage metadata；anonymisation 属于隐私保护，不是 cataloging 核心内容。

Question 31



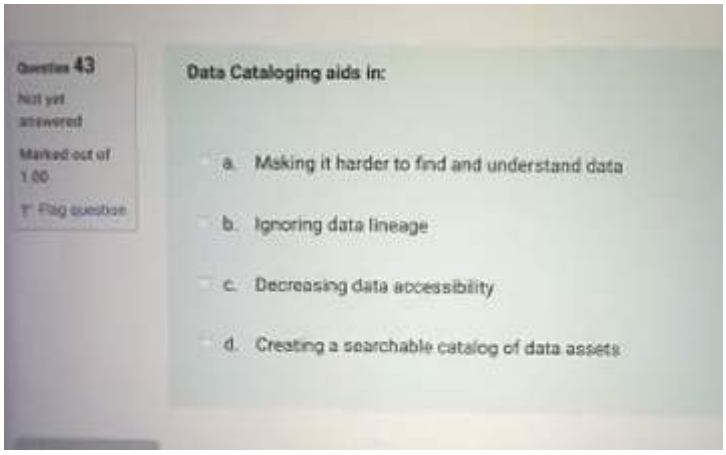
Qualitative Data is:

a. Exclusively about quantities
b. Only numerical
c. Descriptive and conceptual
d. Limited to structured research instruments

参考答案： `c`

解析： qualitative data 强调描述性、概念性，不是纯数量。

Question 44



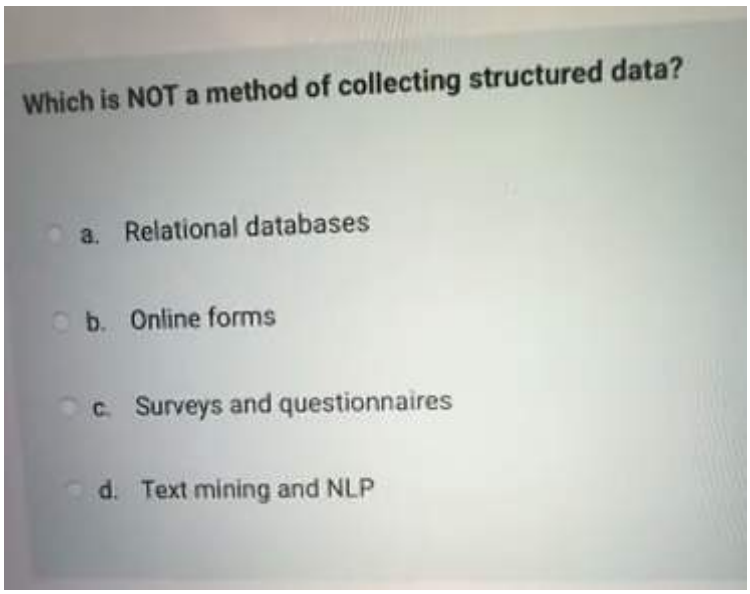
Data Cataloging aids in:

a. Making it harder to find and understand data
b. Ignoring data lineage
c. Decreasing data accessibility
d. Creating a searchable catalog of data assets

参考答案： `d`

解析： data catalog 的作用是让数据资产可搜索、可理解、可发现。

Question 49



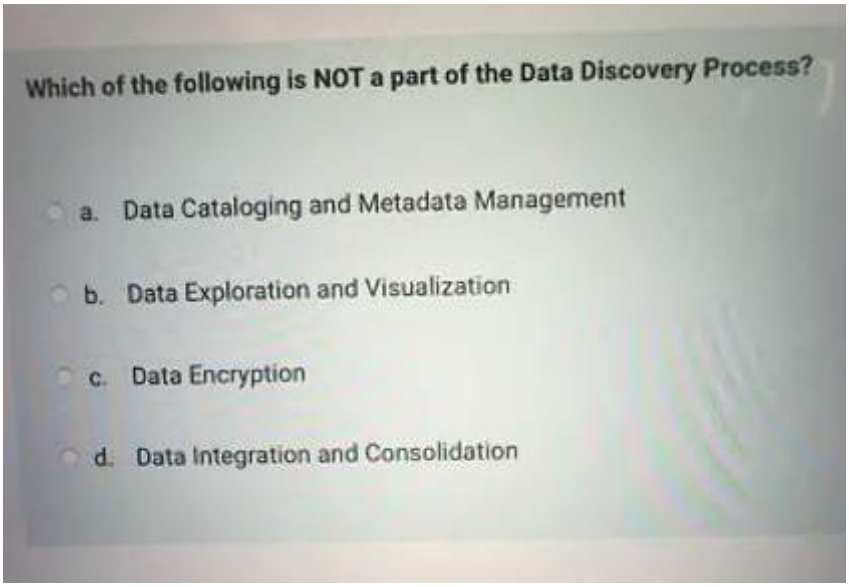
Which is NOT a method of collecting structured data?

a. Relational databases
b. Online forms
c. Surveys and questionnaires
d. Text mining and NLP

参考答案： `d`

解析： text mining / NLP 主要处理非结构化文本，不属于 structured data collection 方法。

Question 50



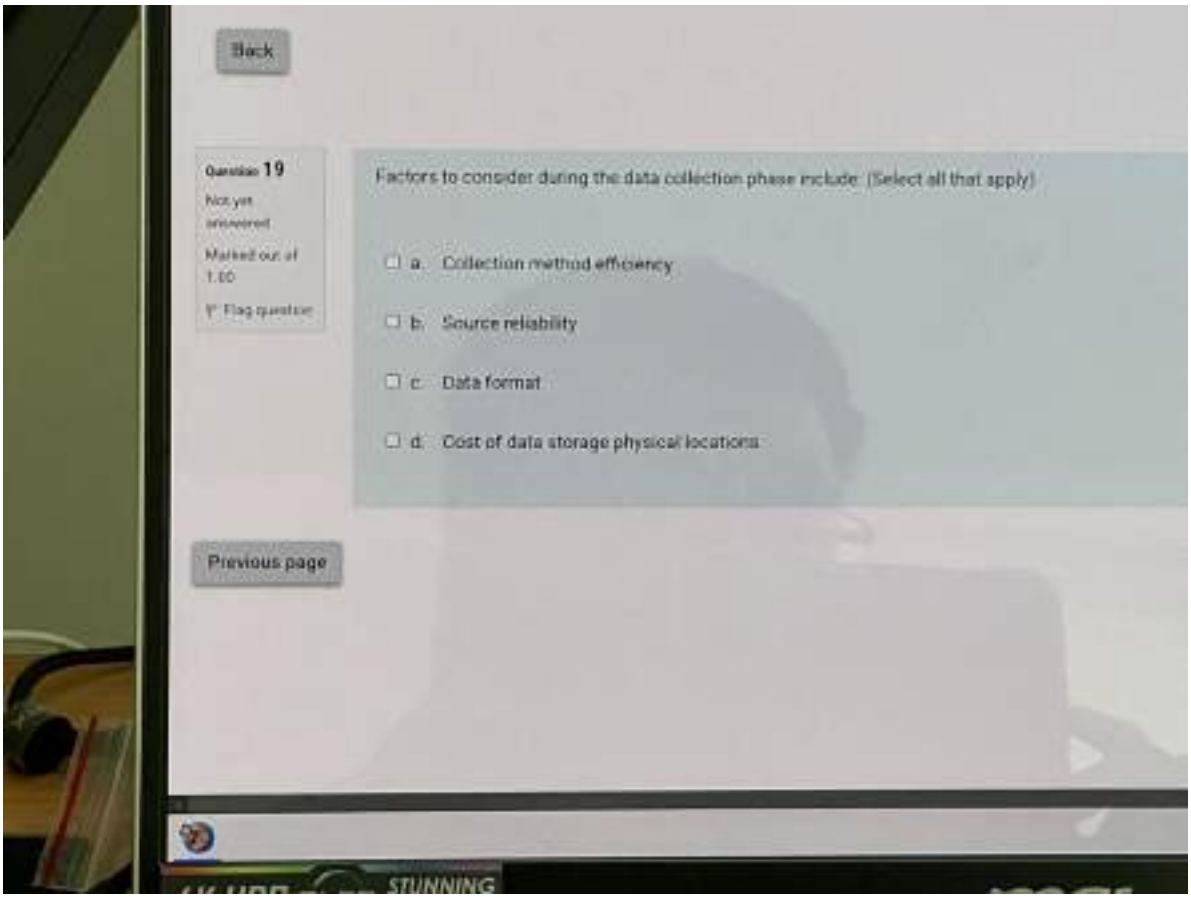
Which of the following is NOT a part of the Data Discovery Process?

a. Data Cataloging and Metadata Management
b. Data Exploration and Visualization
c. Data Encryption
d. Data Integration and Consolidation

参考答案： `c`

解析： data encryption 属于安全控制，不是 data discovery 流程本身。

Question 68



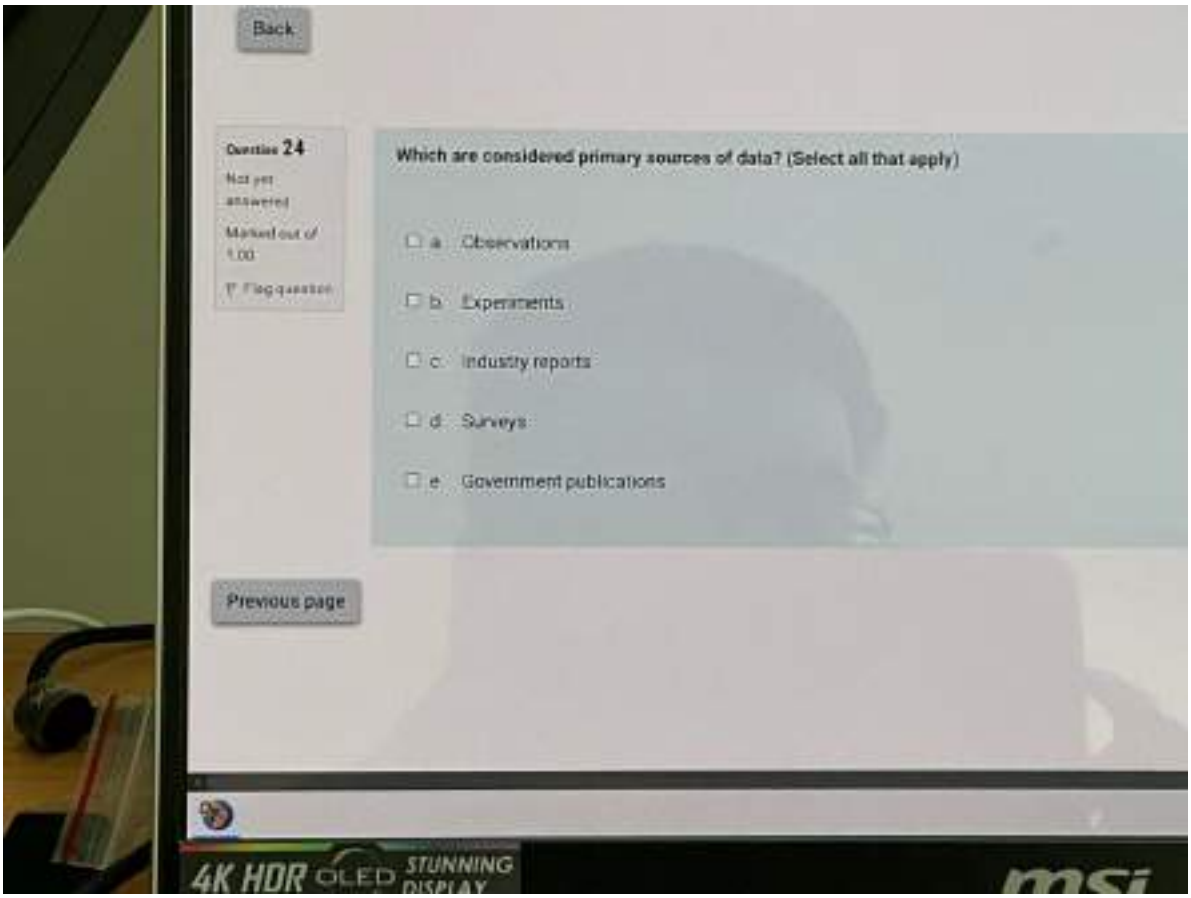
Factors to consider during the data collection phase include: (Select all that apply)

a. Collection method efficiency
b. Source reliability
c. Data format
d. Cost of data storage physical locations

参考答案： `a, b, c`

解析： collection 阶段常看方法效率、来源可靠性、数据格式；“physical locations” 不是这里的核心。

Question 72



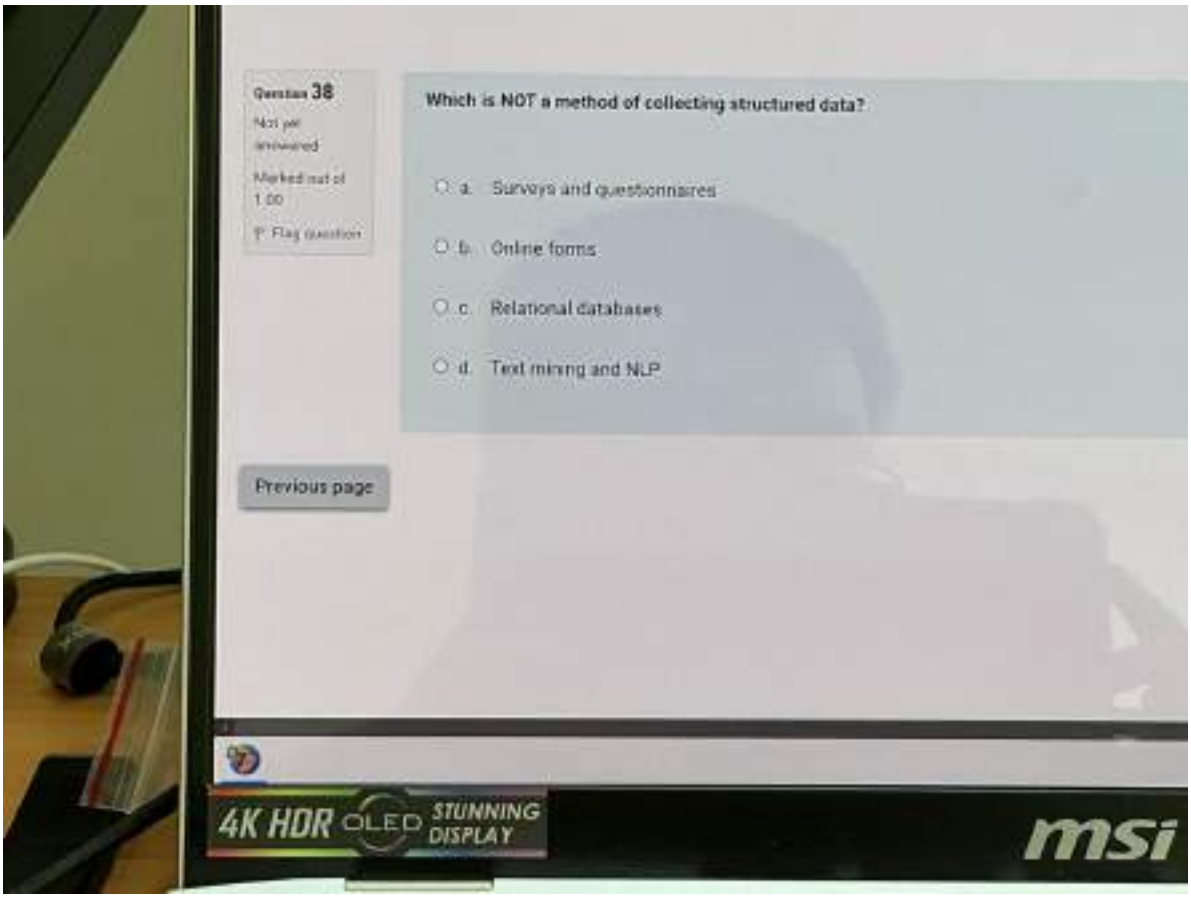
Which are considered primary sources of data? (Select all that apply)

a. Observations
b. Experiments
c. Industry reports
d. Surveys
e. Government publications

参考答案： `a, b, d`

解析： primary source 是直接采集的数据，如 observations、experiments、surveys；industry reports / government publications 更像二手资料。

Question 84



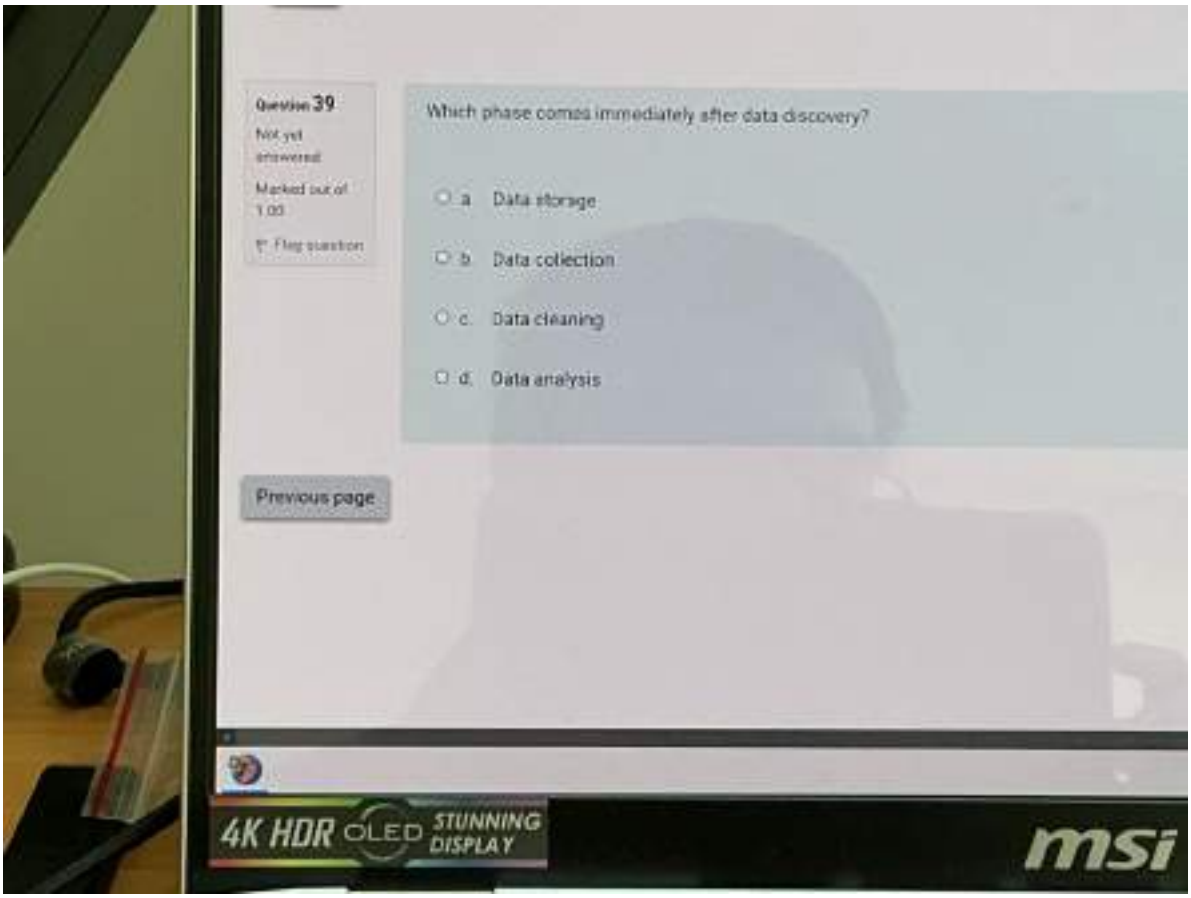
Which is NOT a method of collecting structured data?

a. Surveys and questionnaires
b. Online forms
c. Relational databases
d. Text mining and NLP

参考答案： `d`

解析： 和 Q49 同理，text mining / NLP 不是 structured collection method。

Question 85



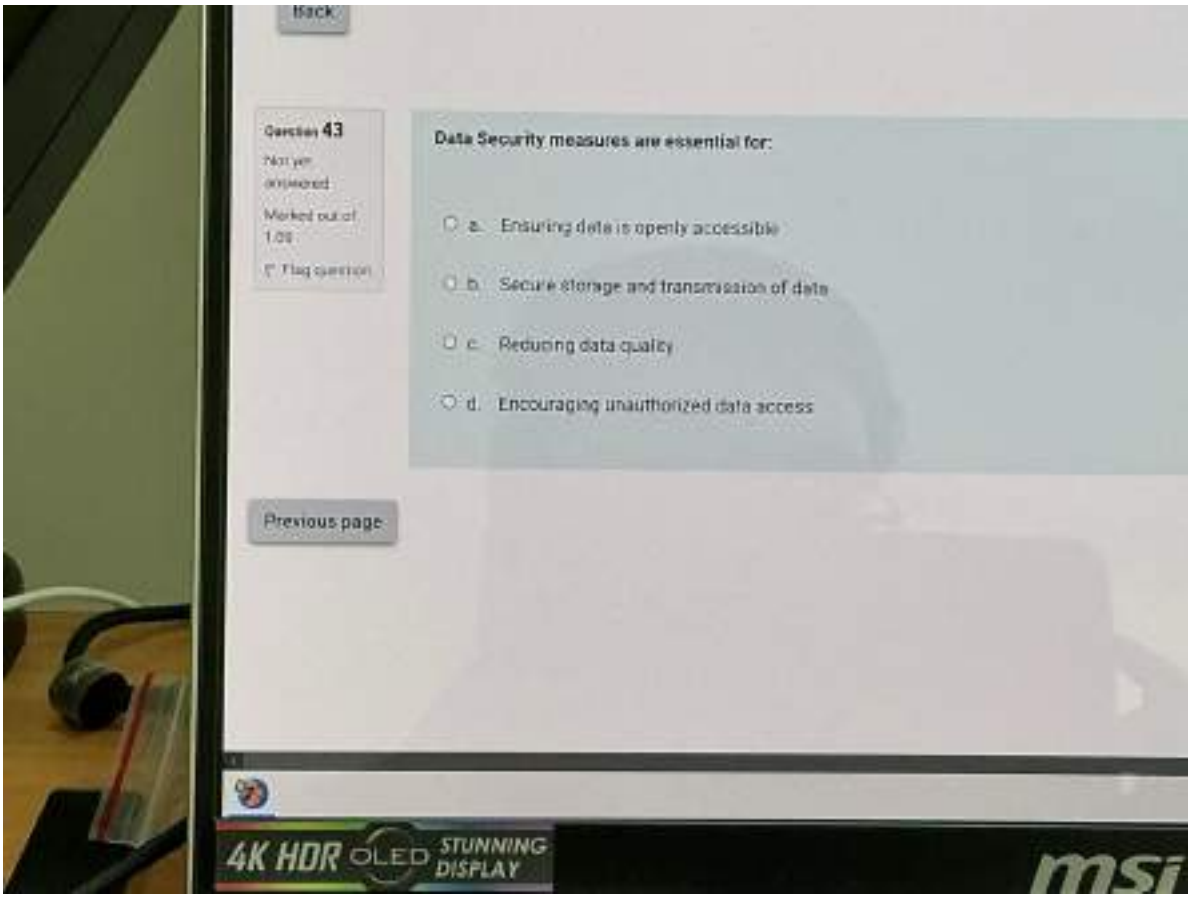
Which phase comes immediately after data discovery?

a. Data storage
b. Data collection
c. Data cleaning
d. Data analysis

参考答案： `b`

解析： discovery 后面紧接着就是 collection。

Question 89



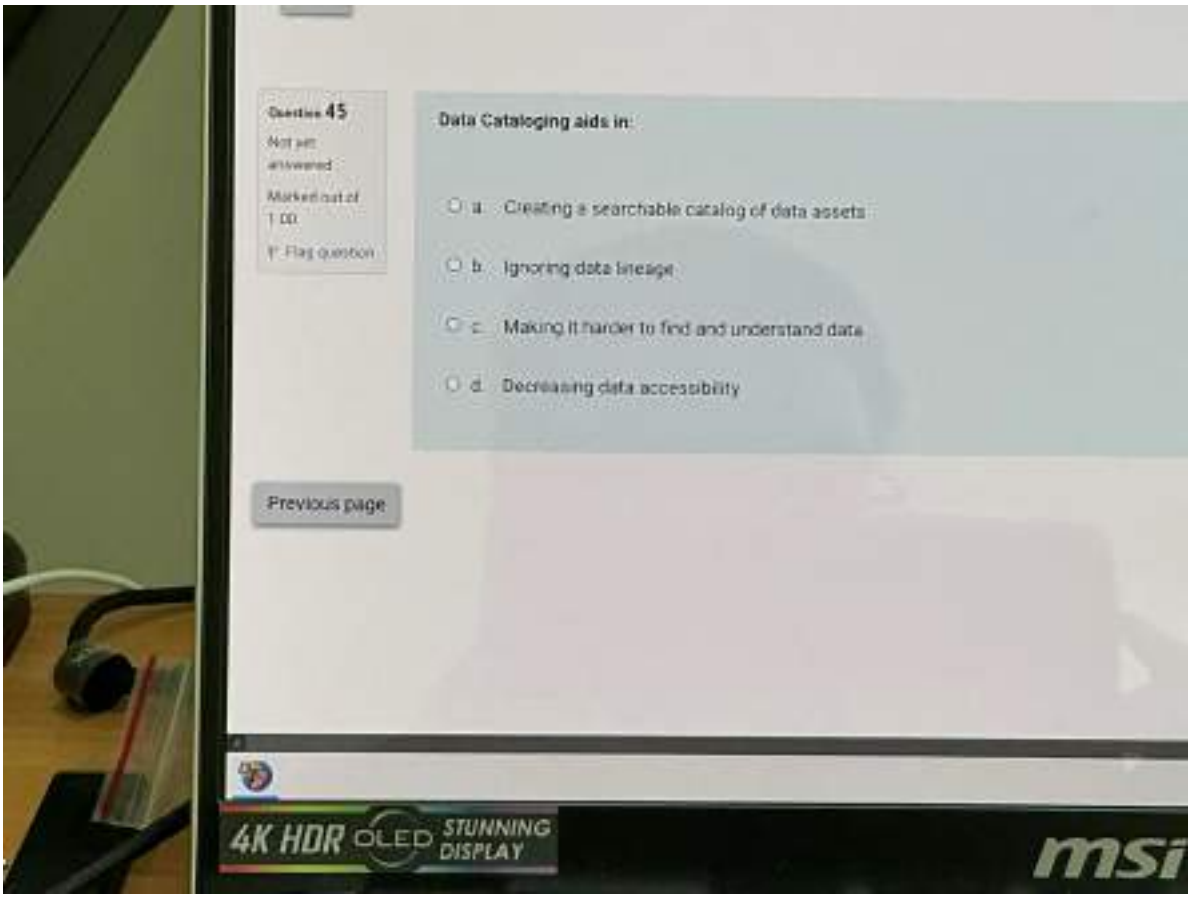
Data Security measures are essential for:

a. Ensuring data is openly accessible
b. Secure storage and transmission of data
c. Reducing data quality
d. Encouraging unauthorized data access

参考答案： `b`

解析： data security 关注安全存储与传输，不是开放访问。

Question 91



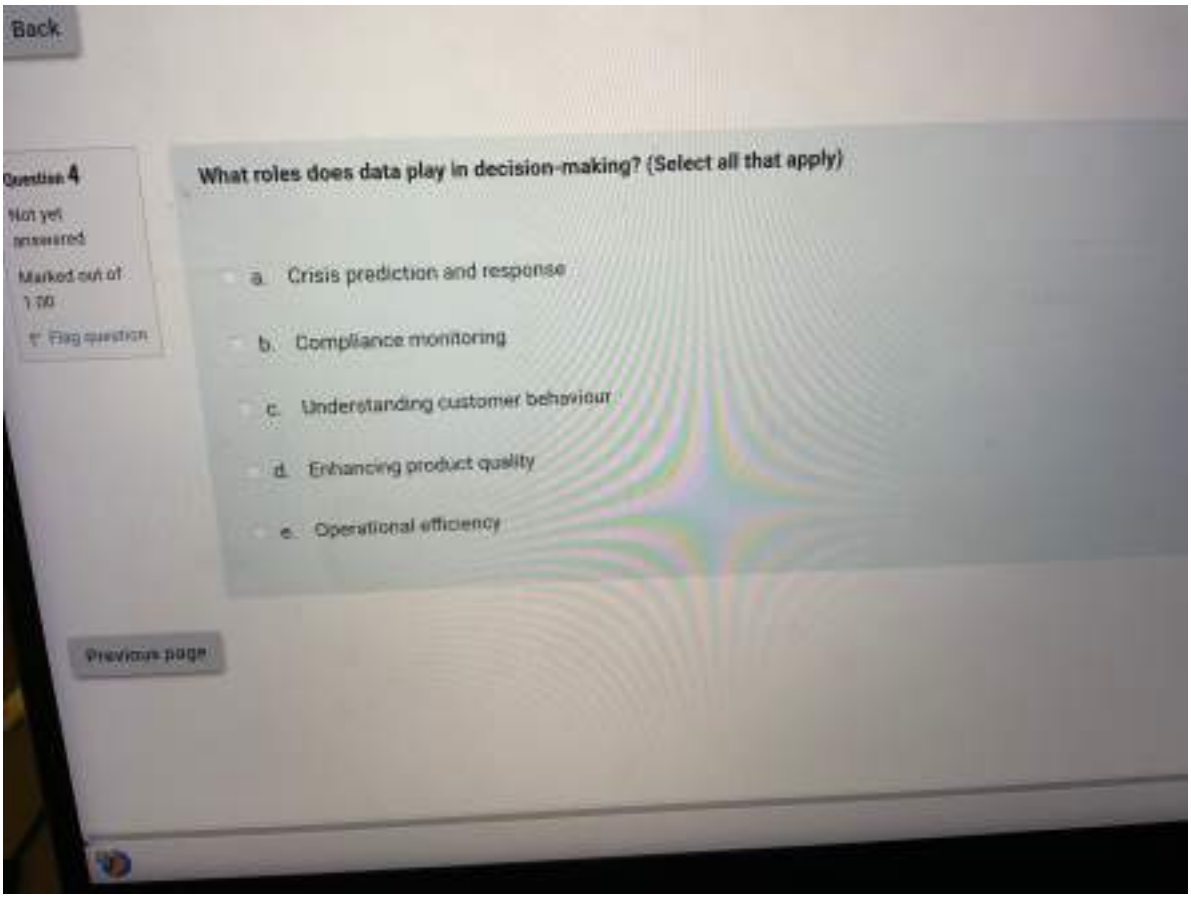
Data Cataloging aids in:

a. Creating a searchable catalog of data assets
b. Ignoring data lineage
c. Making it harder to find and understand data
d. Decreasing data accessibility

参考答案： a

解析： 这题和 Q44 同义，catalog 的目的就是形成可搜索的数据资产目录。

Question 98



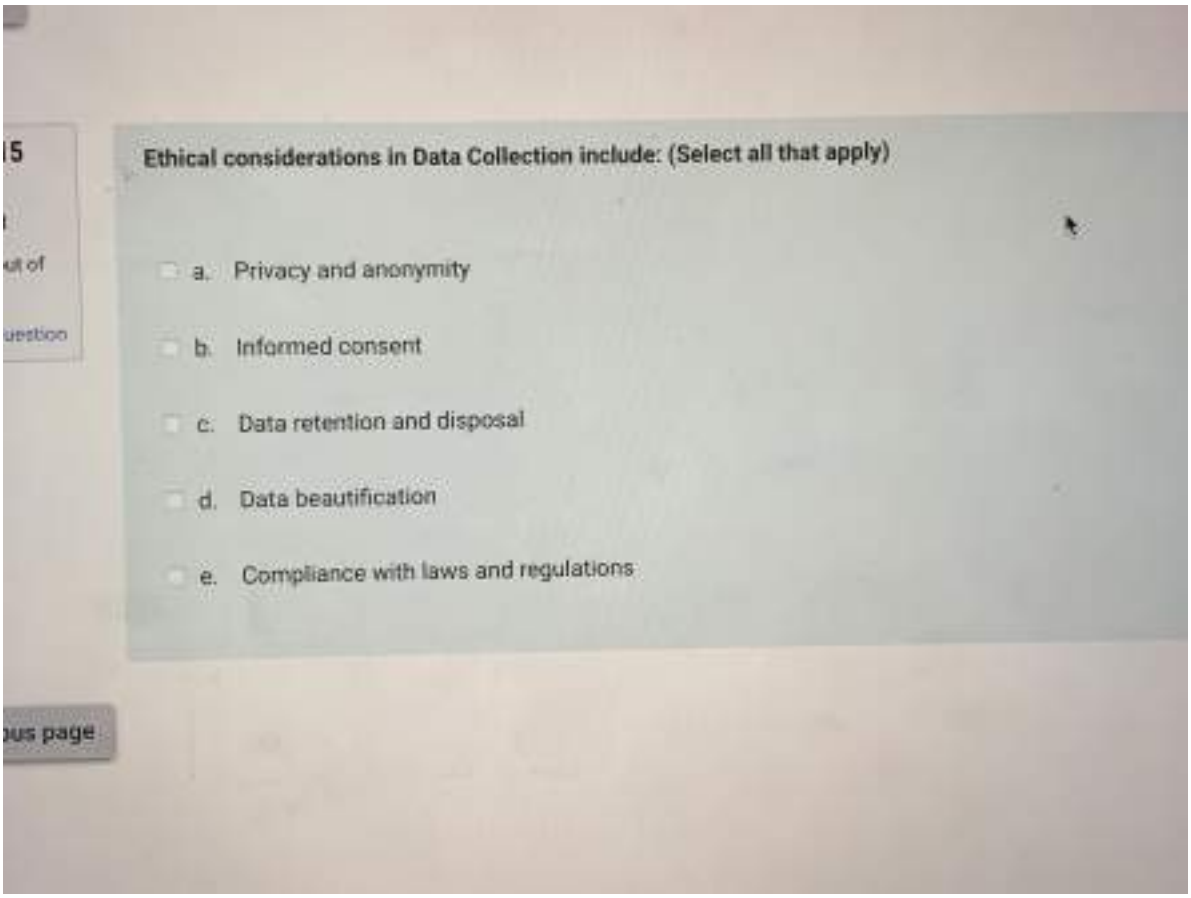
What roles does data play in decision-making? (Select all that apply)

a. Crisis prediction and response
b. Compliance monitoring
c. Understanding customer behaviour
d. Enhancing product quality
e. Operational efficiency

参考答案： `a, b, c, d, e`

解析： 数据在决策里能支持风险、合规、客户洞察、产品和运营优化，所以五项都对。

Question 107



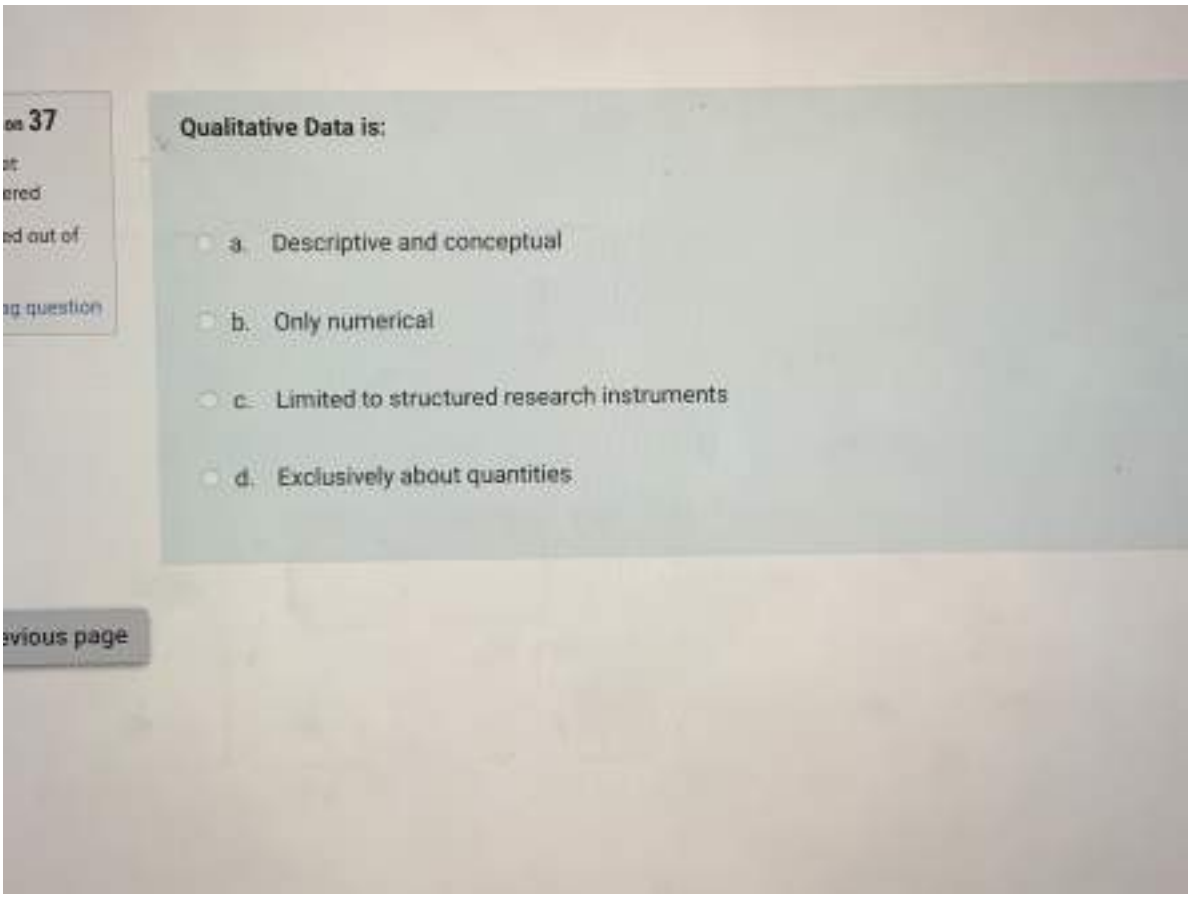
Ethical considerations in Data Collection include: (Select all that apply)

a. Privacy and anonymity
b. Informed consent
c. Data retention and disposal
d. Data beautification
e. Compliance with laws and regulations

参考答案： `a, b, c, e`

解析： 伦理采集通常包括隐私、知情同意、保留/销毁规范和法律合规。

Question 127



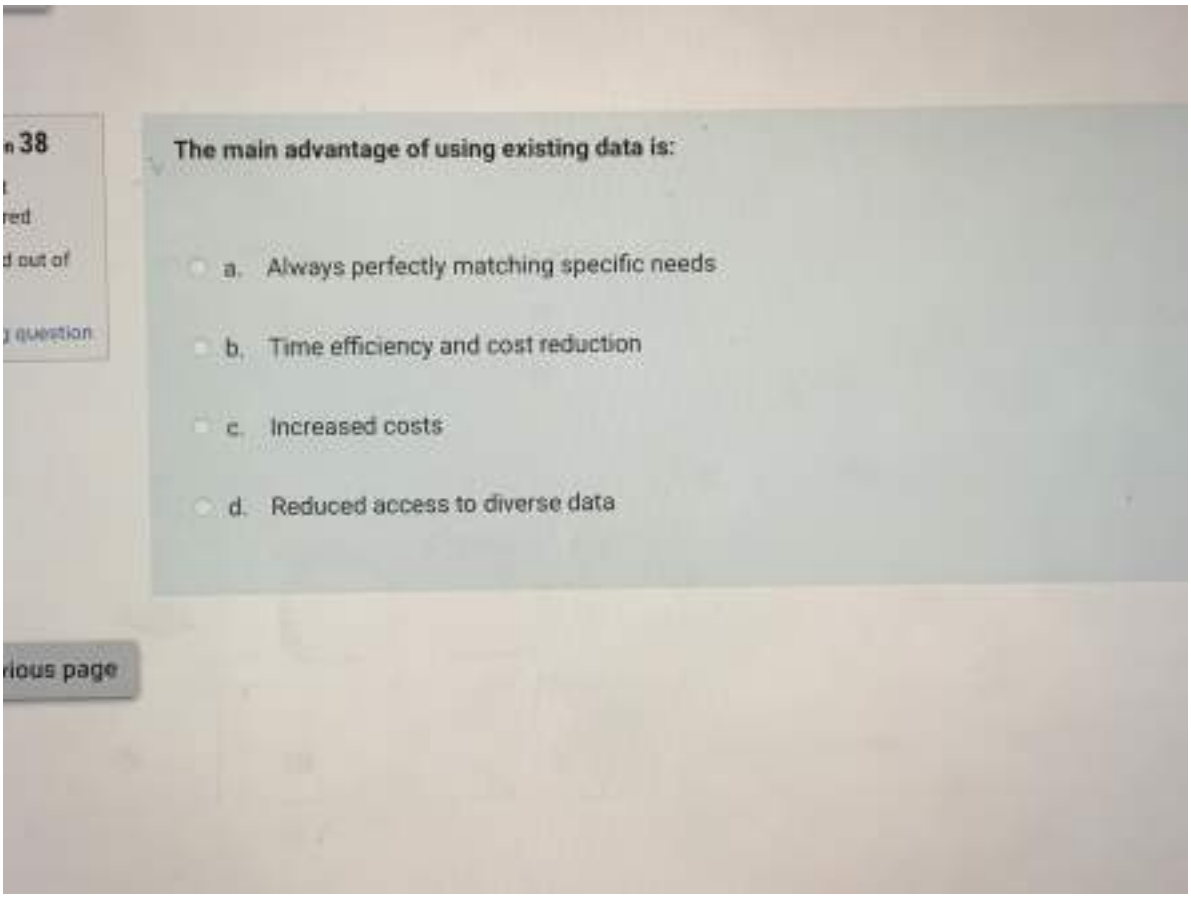
Qualitative Data is:

a. Descriptive and conceptual
b. Only numerical
c. Limited to structured research instruments
d. Exclusively about quantities

参考答案： a

解析： 和 Q31 同义，qualitative data 是描述性/概念性数据。

Question 128



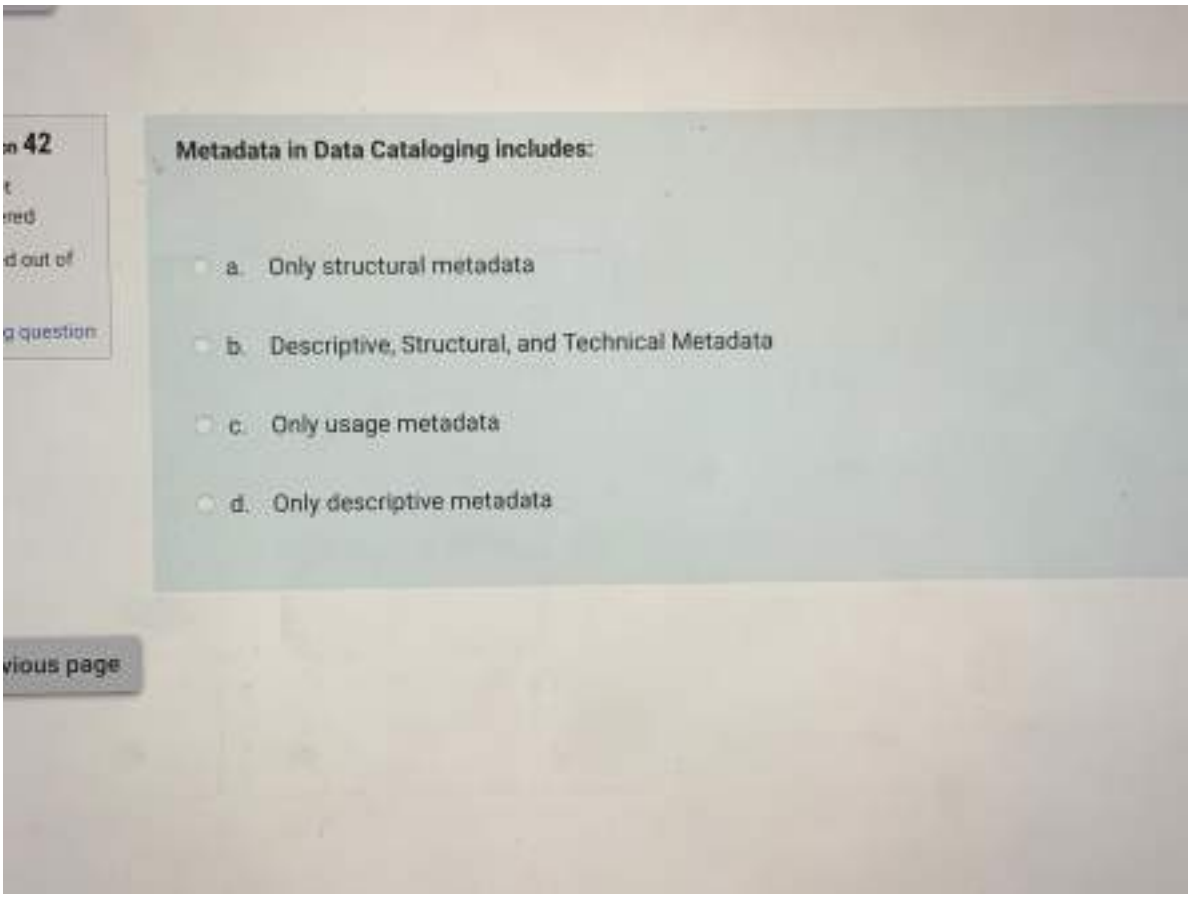
The main advantage of using existing data is:

a. Always perfectly matching specific needs
b. Time efficiency and cost reduction
c. Increased costs
d. Reduced access to diverse data

参考答案： `b`

解析： existing data 的主要优势通常是省时省钱。

Question 132



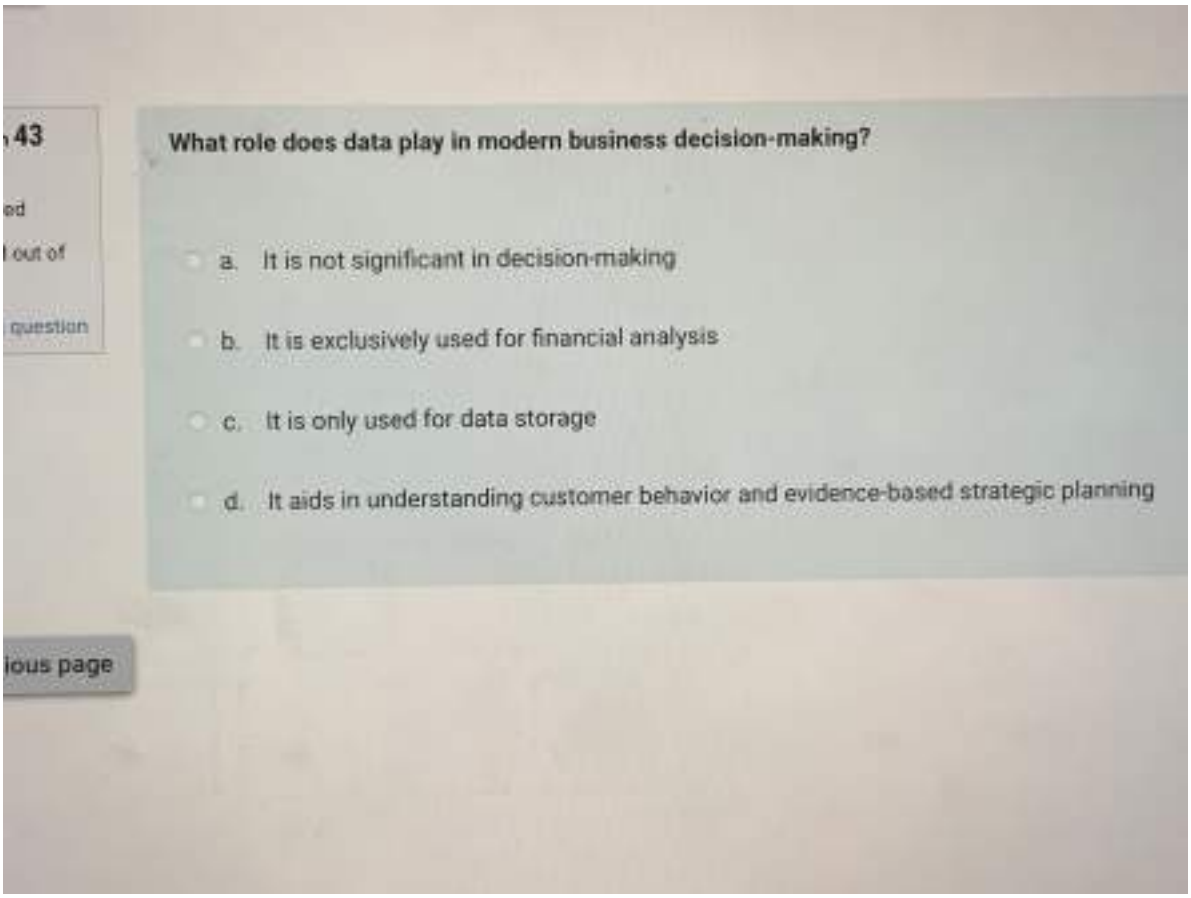
Metadata in Data Cataloging includes:

a. Only structural metadata
b. Descriptive, Structural, and Technical Metadata
c. Only usage metadata
d. Only descriptive metadata

参考答案： `b`

解析： metadata 不只是一种类型，常见有 descriptive、structural、technical 等。

Question 133



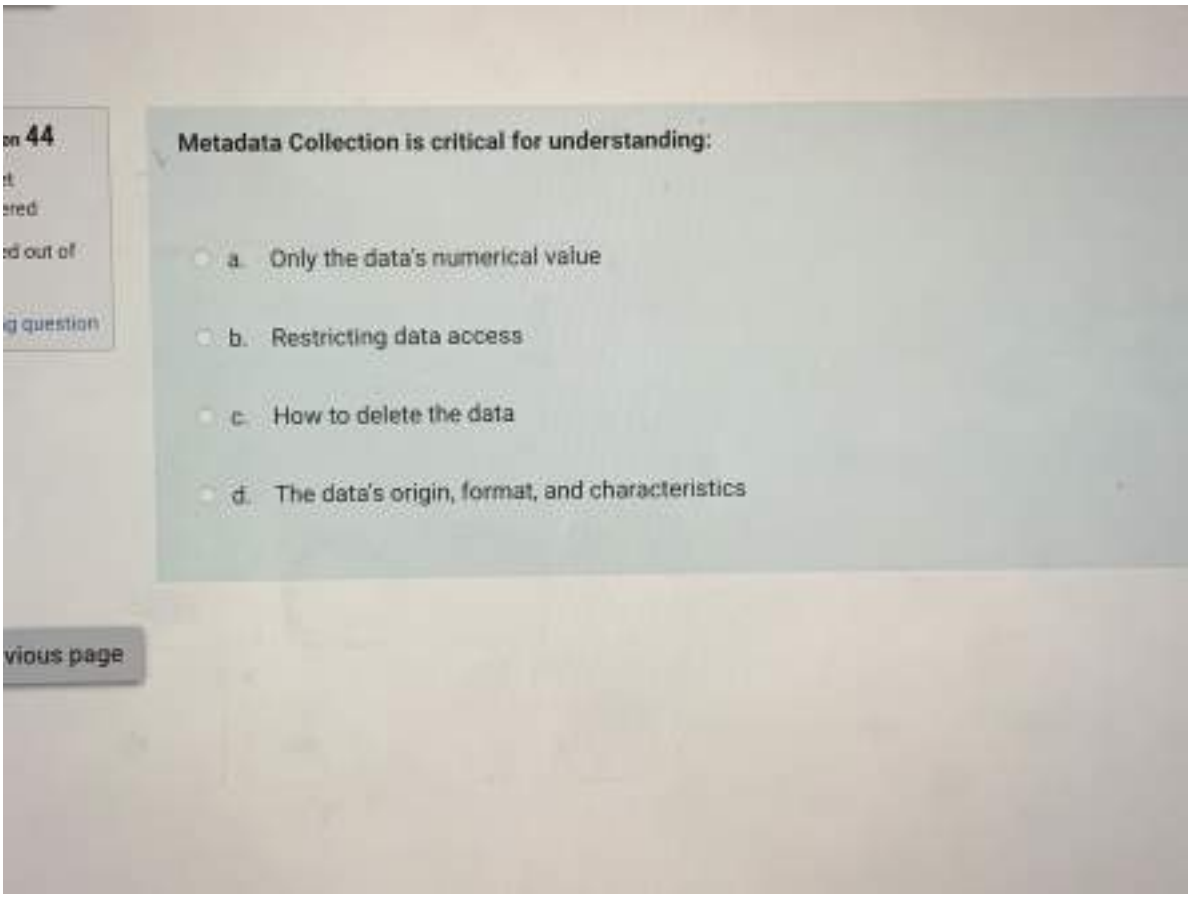
What role does data play in modern business decision-making?

a. It is not significant in decision-making
b. It is exclusively used for financial analysis
c. It is only used for data storage
d. It aids in understanding customer behavior and evidence-based strategic planning

参考答案： `d`

解析： 现代业务决策里，数据能支持客户理解和基于证据的规划。

Question 134



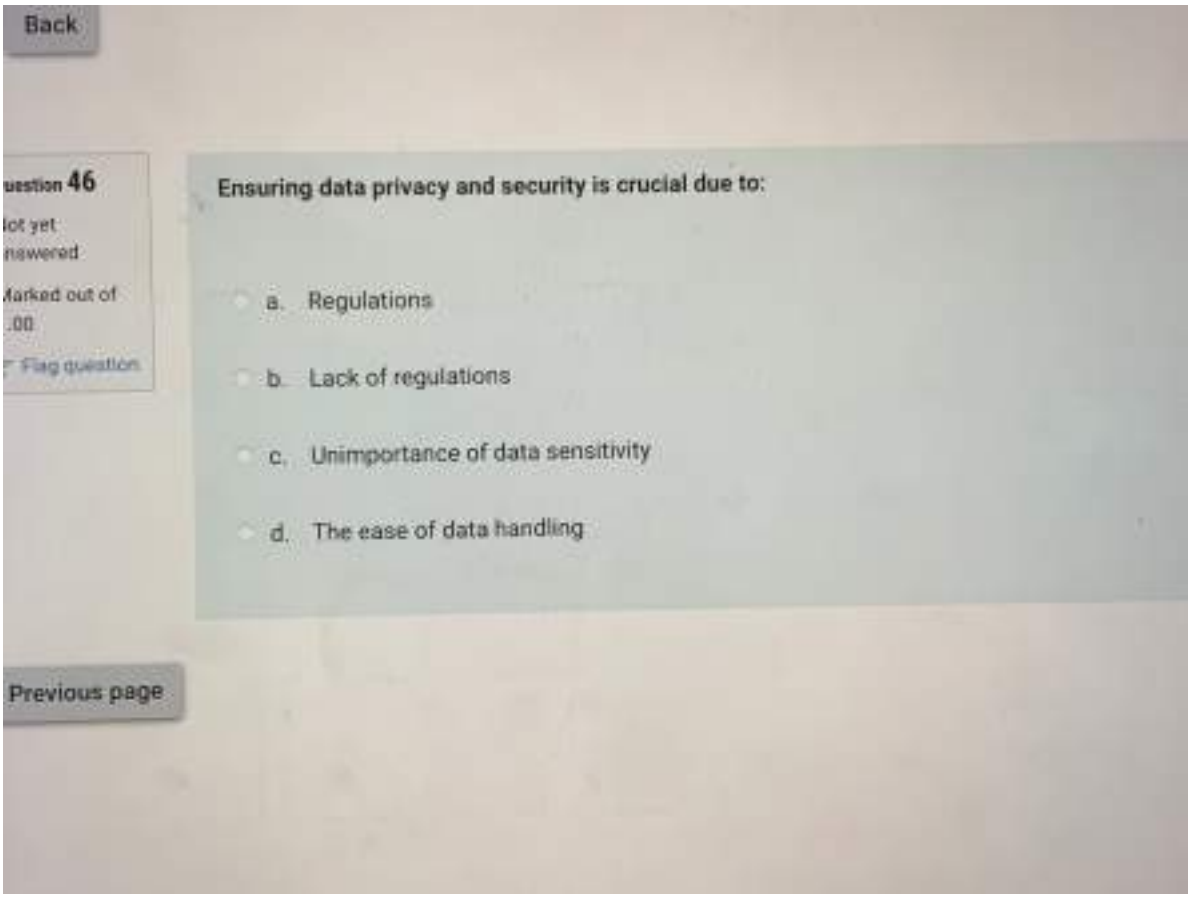
Metadata Collection is critical for understanding:

a. Only the data's numerical value
b. Restricting data access
c. How to delete the data
d. The data's origin, format, and characteristics

参考答案： `d`

解析： metadata collection 帮助理解数据从哪来、什么格式、有哪些特征。

Question 136



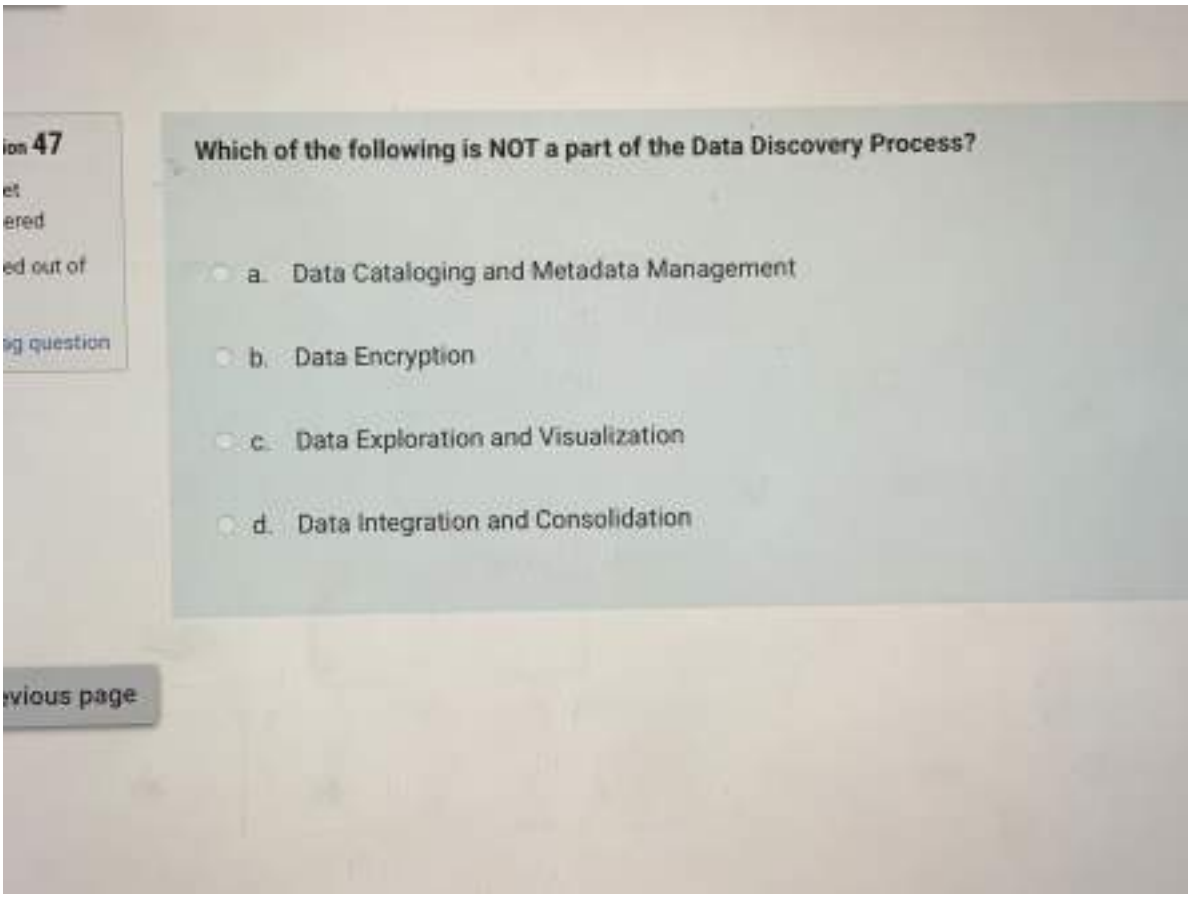
Ensuring data privacy and security is crucial due to:

a. Regulations
b. Lack of regulations
c. Unimportance of data sensitivity
d. The ease of data handling

参考答案： a

解析： 隐私和安全之所以重要，一个直接原因就是法规要求。

Question 137



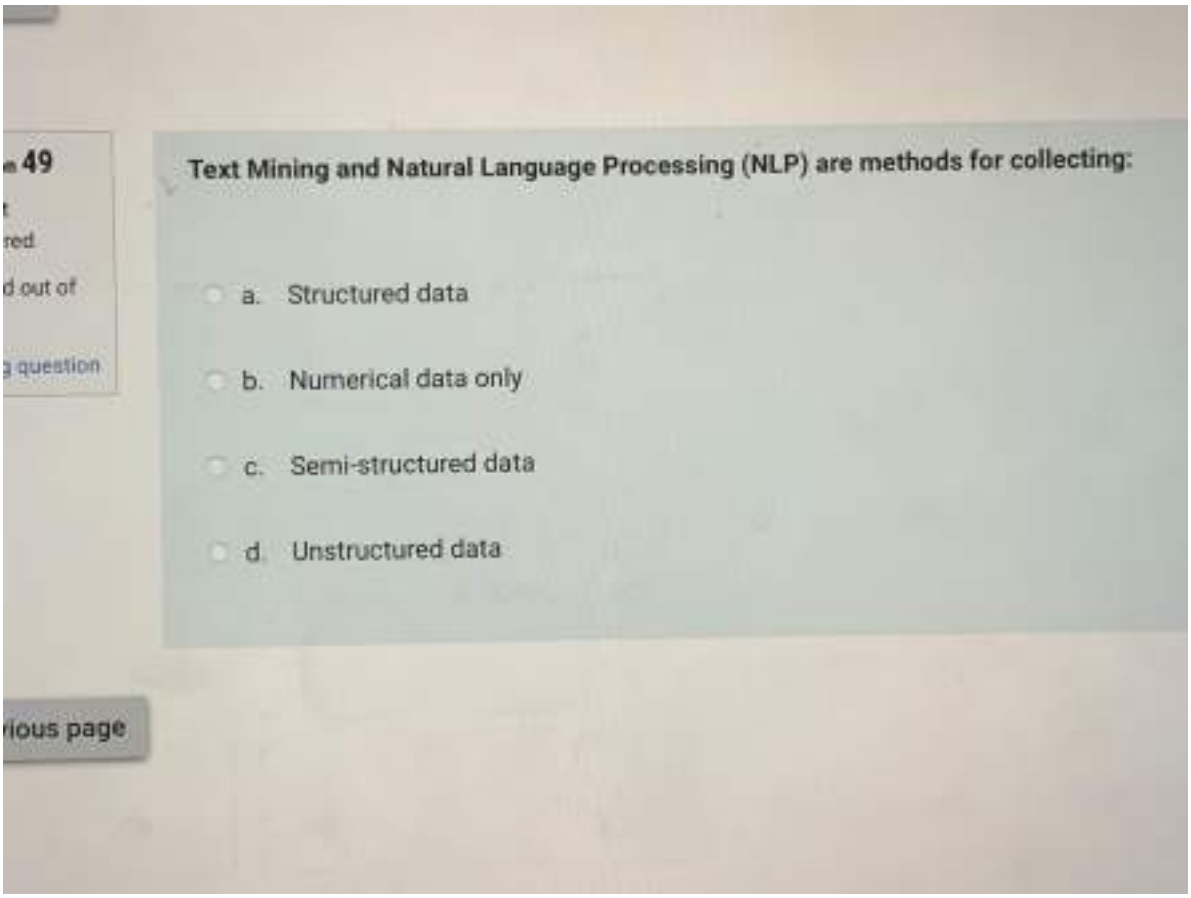
Which of the following is NOT a part of the Data Discovery Process?

a. Data Cataloging and Metadata Management
b. Data Encryption
c. Data Exploration and Visualization
d. Data Integration and Consolidation

参考答案： `b`

解析： 这题和 Q50 同义，只是正确项换成了 `b`。

Question 139



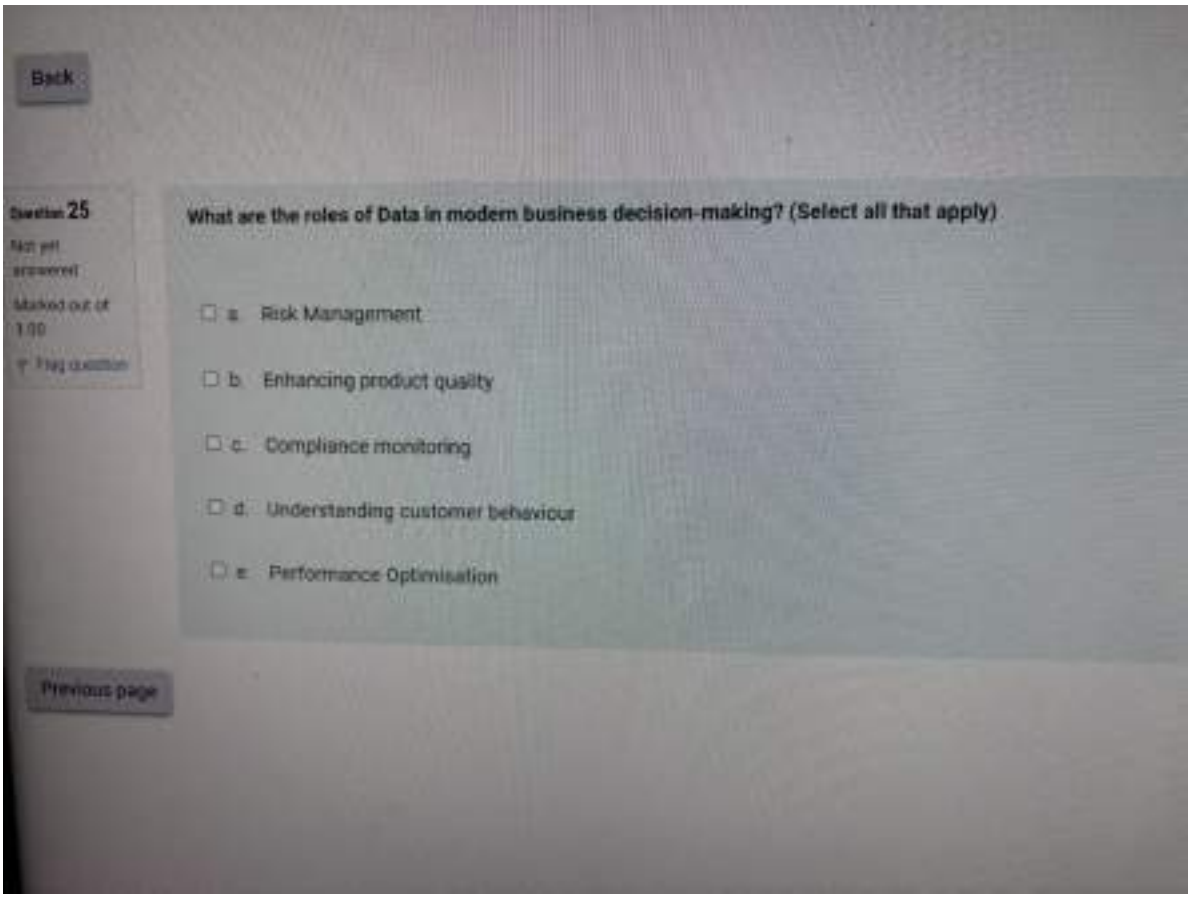
Text Mining and Natural Language Processing (NLP) are methods for collecting:

a. Structured data
b. Numerical data only
c. Semi-structured data
d. Unstructured data

参考答案： `d`

解析： text mining / NLP 处理的是 unstructured text。

Question 154



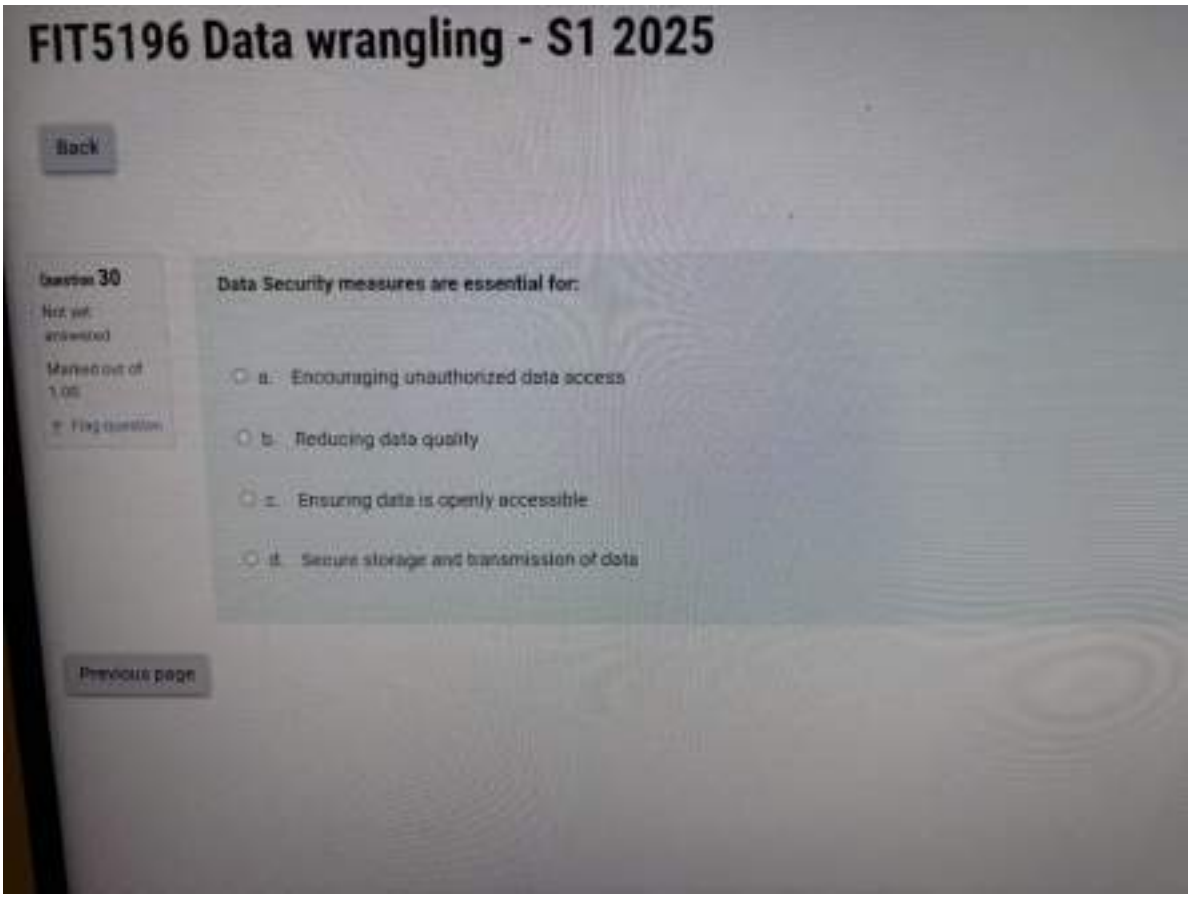
What are the roles of Data in modern business decision-making? (Select all that apply)

a. Risk Management
b. Enhancing product quality
c. Compliance monitoring
d. Understanding customer behaviour
e. Performance Optimisation

参考答案： `a, b, c, d, e`

解析： 这题列出的 risk、quality、compliance、customer、performance 都是数据支撑决策的典型作用。

Question 159



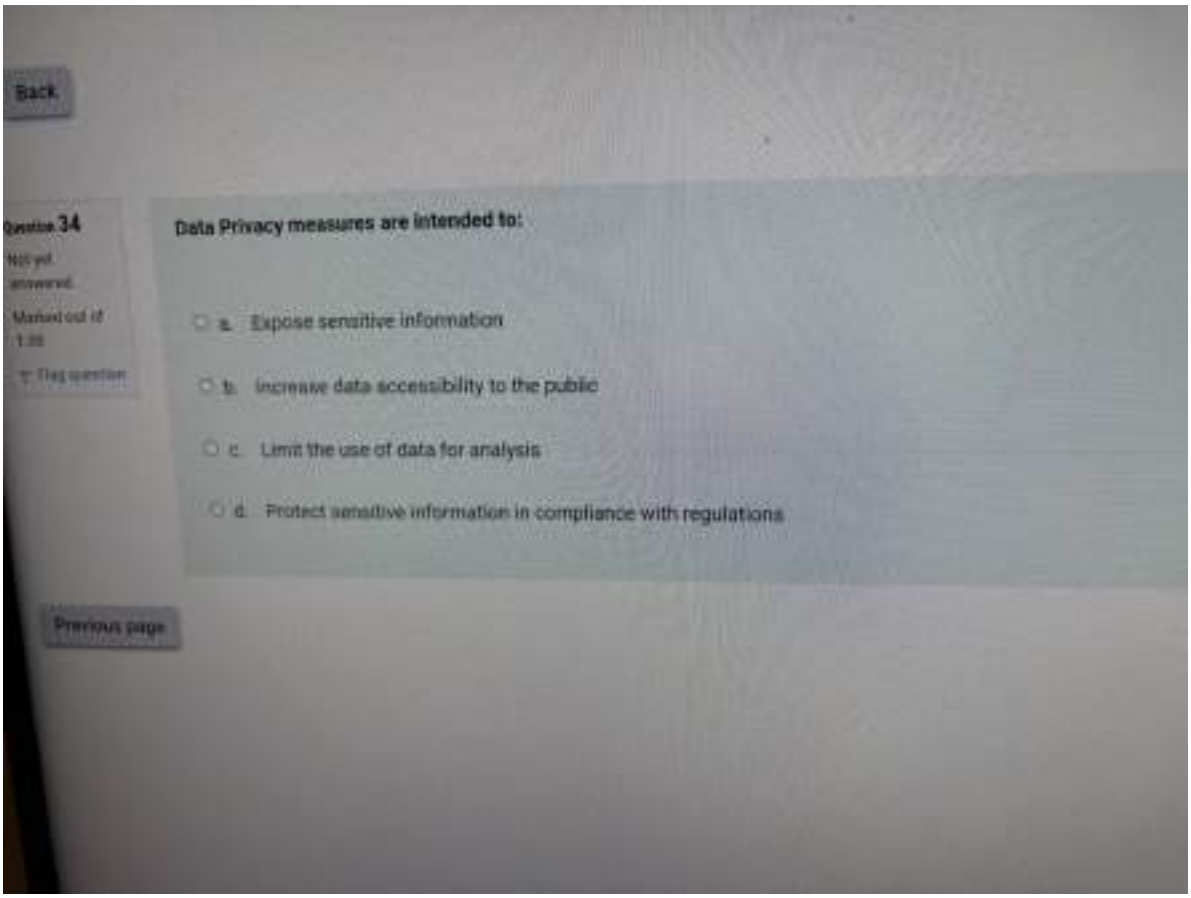
Data Security measures are essential for:

a. Encouraging unauthorized data access
b. Reducing data quality
c. Ensuring data is openly accessible
d. Secure storage and transmission of data

参考答案： `d`

解析： 和 Q89 同义，正确项换成了 `d`。

Question 161



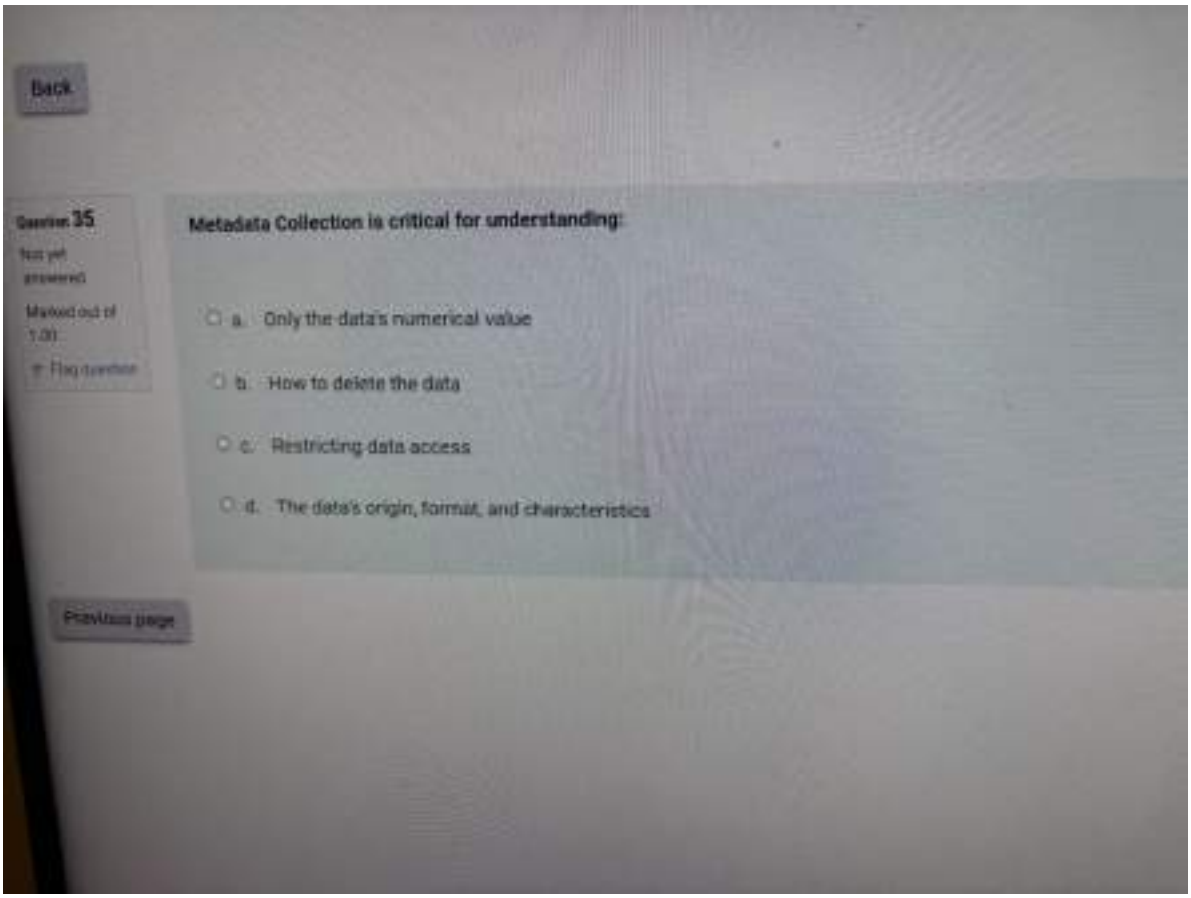
Data Privacy measures are intended to:

a. Expose sensitive information
b. Increase data accessibility to the public
c. Limit the use of data for analysis
d. Protect sensitive information in compliance with regulations

参考答案： `d`

解析： privacy measures 的目标是依法保护敏感信息，而不是公开它。

Question 162



Metadata Collection is critical for understanding:

a. Only the data's numerical value
b. How to delete the data
c. Restricting data access
d. The data's origin, format, and characteristics

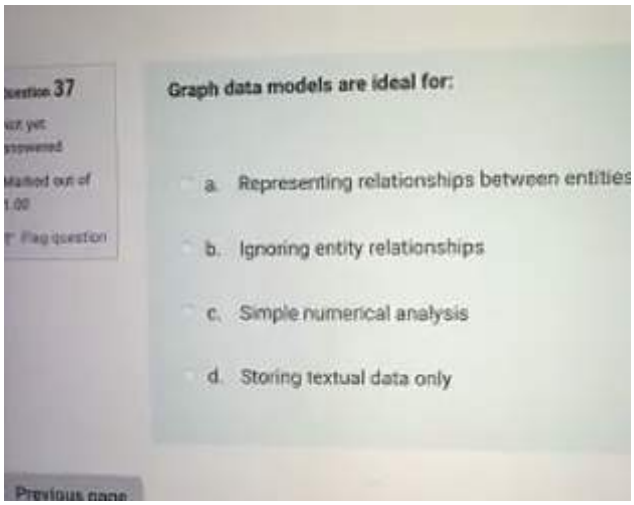
参考答案： `d`

解析： 和 Q134 同义，metadata 帮你理解 origin / format / characteristics。

Week 6 (5 题)

- 范围：Data Structuring

Question 38



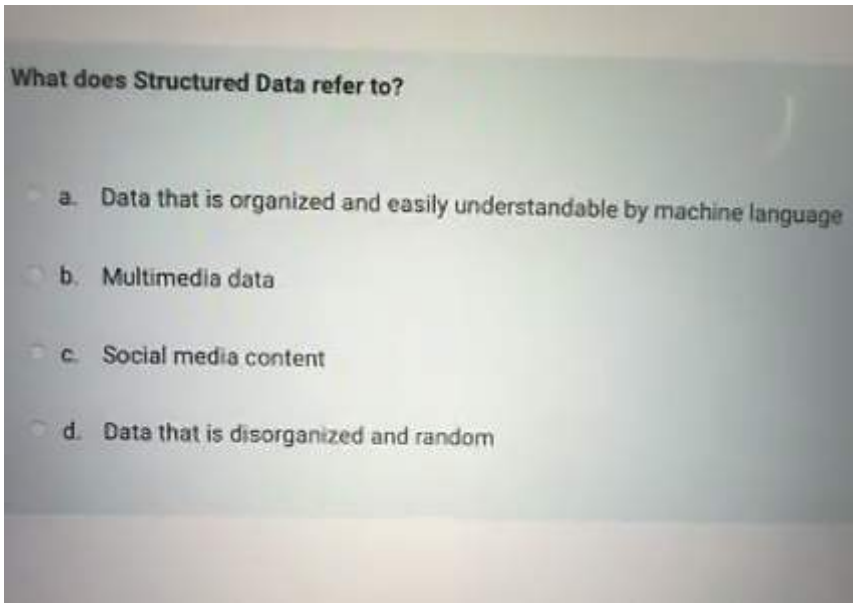
Graph data models are ideal for:

a. Representing relationships between entities
b. Ignoring entity relationships
c. Simple numerical analysis
d. Storing textual data only

参考答案： a

解析： graph data model 最适合表达实体之间的关系。

Question 51



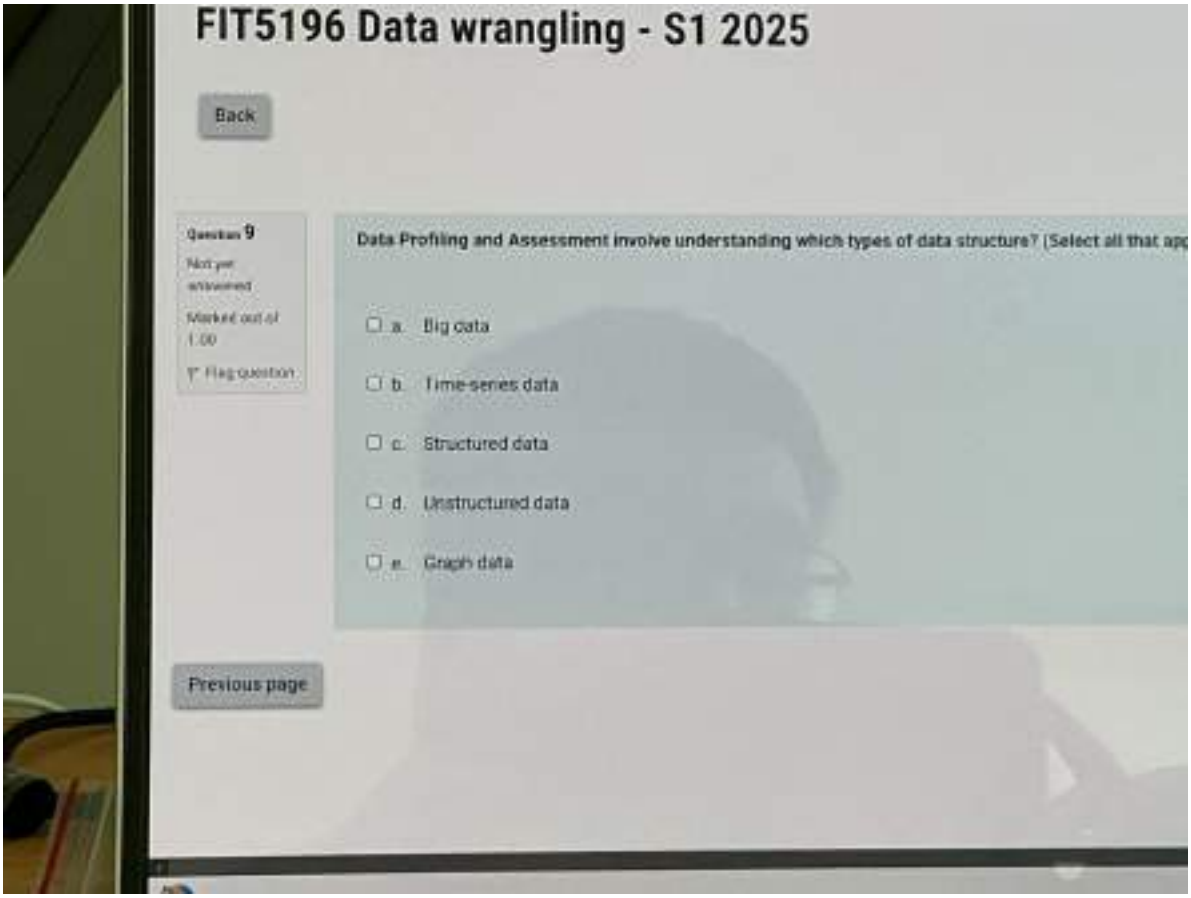
What does Structured Data refer to?

a. Data that is organized and easily understandable by machine language
b. Multimedia data
c. Social media content
d. Data that is disorganized and random

参考答案： a

解析： structured data 是机器容易理解、字段组织清晰的数据。

Question 60



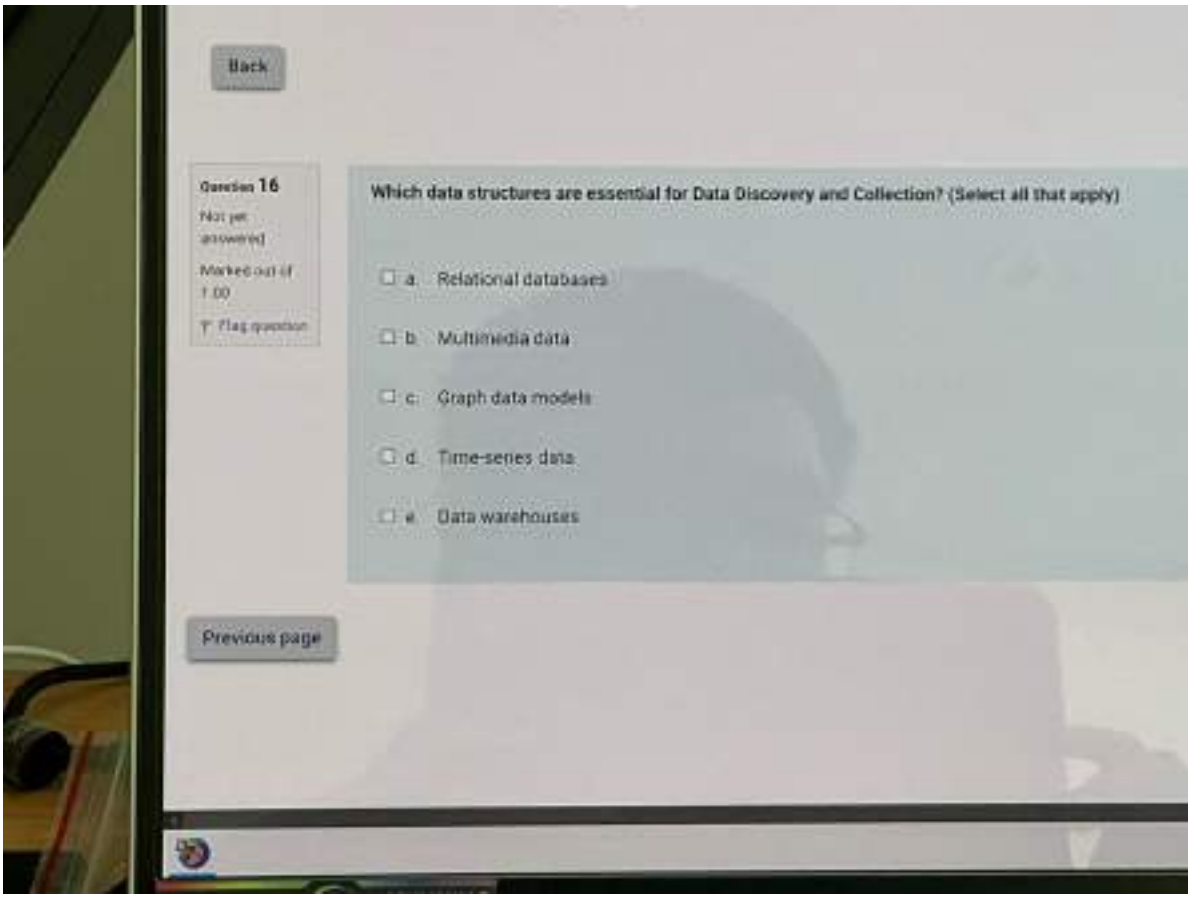
Data Profiling and Assessment involve understanding which types of data structure? (Select all that apply)

a. Big data
b. Time-series data
c. Structured data
d. Unstructured data
e. Graph data

参考答案： `b, c, d, e`

解析： profiling / assessment 需要先看数据是什么结构，因此 time-series、structured、unstructured、graph 都要识别。

Question 65



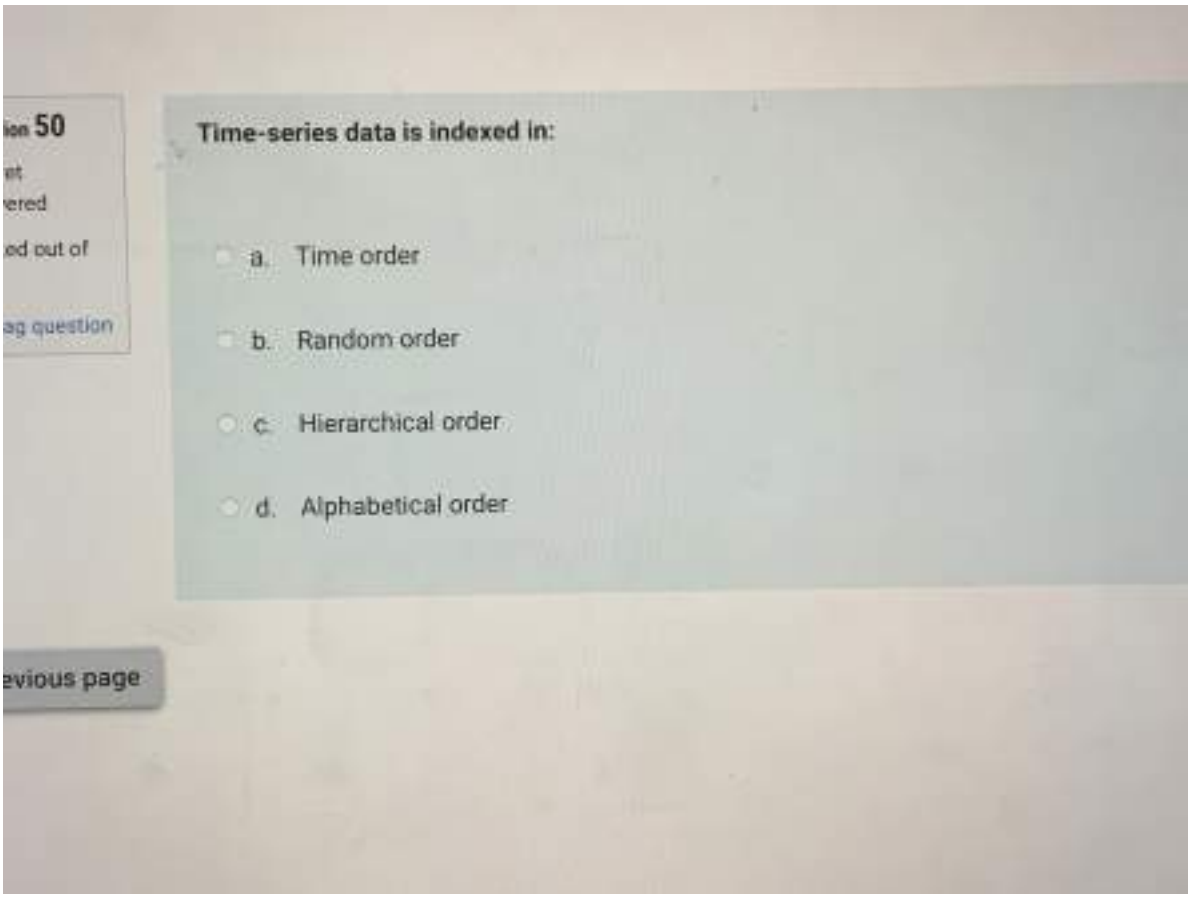
Which data structures are essential for Data Discovery and Collection? (Select all that apply)

a. Relational databases
b. Multimedia data
c. Graph data models
d. Time-series data
e. Data warehouses

参考答案： `a, b, c, d, e`

解析： 这题把“数据结构”和“数据来源/类型”混在一起了；按课程复习时先把这几类都记住。

Question 140



Time-series data is indexed in:

a. Time order
b. Random order
c. Hierarchical order
d. Alphabetical order

参考答案： a

解析： time-series 数据按时间顺序索引。